ECONOMIC SURVEY 2003-04



Government of Goa

Directorate of Planning, Statistics and Evaluation

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CHAPTER 1

General Review

1. Employment Generation Strategy

Creation of adequate work opportunities for the unemployed youth is a crucial requirement of developmental planning. Employment enables a citizen to lead life with self respect. The year 2003-04 assumes special significance as an Employment Generation Strategy was unveiled in the Budget Speech of Hon'ble Chief Minister / Finance Minister of Goa. Though, there have been many schemes launched earlier, it is perhaps for the first time that a comprehensive strategy was announced by the Government. It is expected that the Employment Generation Strategy will be able to trigger the economic growth of the State in the years to come especially in the primary sector which is showing a declining trend.

Retention of employment in traditional sectors such as agriculture and allied activities by making them remunerative

Creation of large scale employment by providing incentives to industries to employ people on a sustainable basis

Encouragement to self employment

While announcing the Employment Generation Strategy, the Finance Minister in his Budget Speech has stated that Goa has performed commendably in terms of various social and economic parameters and the vision is to achieve the standards of a developed nation by 2010. Existence of significant number of unemployed youth can be a stumbling block in achieving this goal. Thus, considering the experience gained in its efforts during earlier two years, the Government intends to implement a three pronged employment generation strategy (see box) to create employment to 25000 youth.

0 Retention

A major problem being faced today is the exodus of rural population to urban areas for employment, especially for white collar jobs. As per 2001 population census 49.77% of Goa's population is in urban areas. It is in this context that the Government of Goa proposes to ensure the sustainability of the village economy by considering the village as the focal point for promoting rural economic activities like agriculture, horticulture, floriculture, dairy development, fisheries, traditional art-crafts etc so that people living in the villages can earn their livelihood in the village itself and do not feel that they have to migrate to urban areas in search of employment.

Statistics reveal that agriculture and horticulture products like rice, sugarcane, vegetables, fruits, flowers etc are being imported from other States in huge quantities. The State has a tremendous demand for these agriculture products. There is a need to improve the economic viability of this sector. The importance of agriculture as a source of gainful employment is declining over the years on

account of various factors including unremunerative prices and lack of proper support systems for storage and marketing. In order to address this problem it is proposed to set up 'Krishi Ghars' which will act as collection, sorting, storage and selling centres for agriculture and horticulture products at various locations covering all talukas in the State. In order to develop a price support mechanism, purchase rates for specified agricultural and horticultural produce will be declared. The mechanism will also take into account factors like transportation, spoilage, storing etc. Initially a chilling unit of around 5 tonne capacity will be set up at the new market in Panaji for sorting and storage of the products. A larger unit of 25 tonne capacity will also be set up subsequently. Retail outlets will be set up through which the goods procured will be sold to government agencies and in the open market thereby eliminating the middleman to the benefit of both the farmers and the consumers. The 'Krishi Ghars' will also be developed as centres of assistance to the farmers to improve the yield and quality of the produce through improved technology and better quality seeds. The agricultural marketing cooperatives and NGOs will be involved in the process. It is also proposed to set up agriculture/horticulture estate to provide gainful employment to landless unemployed youth.

There is a huge demand for milk, eggs, meat etc and their products. Presently, a sizable portion of the demand is being met through import from neighbouring States. It is proposed to revamp the existing scheme of providing milch animals to the farmers by making it flexible with simpler procedures and easy financing. Production based incentives will be extended to the farmers to encourage them to produce more milk. Proper training and other facilities, including provision for artificial insemination will be made available through reputed agencies to improve the quality and yield from the livestock. In order to encourage farmers to go for fodder production, it is proposed to provide financial assistance to reduce their production cost and make them competitive.

The State has 105 kms coast line and 555 kms of inland water ways. The annual fish production is about 70000 tonnes. But the State does not have facilities for fish processing and presently this is being done outside the State. As such, it is proposed to set up a fish processing estate in the State with all infrastructure facilities. It is also proposed to extend concessions in power and water tariffs to the Agriculture and allied activities sector to make it remunerative.

(ii) Creation

Generation of sustainable employment has been made the central focus in the Industrial Policy announced by the Government recently. A balanced industrial development hinges on generation of employment through setting up of new units. The policy has laid down clear direction for industrial development in the State. In order to encourage industries to provide employment to local youth it is proposed to provide incentives to the industrial units to a limited period. The incentive will also eliminate the low wage phenomenon as it is linked to salary to be paid to the employees. It is also proposed to implement Venture Capital Contribution scheme to the existing functional industries to venture out and expand. It is also proposed to provide share capital contribution to local entrepreneurs to start income generating activities. In order to support prospective entrepreneurs to set up SSIs, it is proposed to provide interest subsidy on the loans availed through banks and other financial institutions. Thus, by giving boost to industrial development it is proposed to generate employment opportunities to the unemployed youth.

(iii) Encouragement

It is proposed to encourage the youth to take up self employment. The State has already taken a number of initiatives to provide self employment opportunities to the unemployed youth. The State is implementing the Chief Minister's Rojgar Yojana and Deen Dayal Swarojgar Yojana by providing financial assistance to the unemployed to take up economically viable income generating activities. The Government has also launched a new scheme of awarding works costing upto Rs.4.50 lakh without calling tenders to the unemployed engineers.

It is proposed to launch a new initiative viz 'SWABHIMAN' (self respect) for women to start income generating activities by forming self help groups. It is proposed to create a movement in the State on the lines of 'Lijjat Papad etc' by identifying suitable product. The Government has also launched a 'pre-employment training' scheme to employ persons in Government Departments on stipend basis and consider for absorption against future vacancies.

(iv) Employment Board

The basic problem in the State is mis-match in demand and supply in the employment market and absence of any mechanism to guide the unemployed

youth in getting employment. It is absolutely necessary that the educational system in the State is tailored to the demand for employment. It is in this regard that the Government of Goa has decided to create an **Employment Board** which would work as an agency for providing information on employment opportunities, placement, guidance, training, skill development etc including encouraging self employment. It is proposed to provide training in various semi skilled activities like hair cutting, tailoring plumbing, masonry, tourist guide etc through government establishments like Craftsman Training Institute, Catering College and various other reputed non-government institutions. Training for setting up agricultural plantations will be imparted by the Goa Forest Development Corporation and Goa Horticulture Development Corporation.

Implementation of Employment Generation Strategy

The Government has made sufficient headway in implementation of the employment generation strategy during 2003-04. The projects like setting up of Krishi Ghars under Agriculture sector and 'Kamdhenu Scheme' under Animal Husbandry Sector have been announced and implemented. New Industrial Policy aimed at generating employment has been announced and 7 new schemes have been notified. The Chief Minstister's Rojgar Yojana and the Deen Dayal Swarojgar Yojana have been modified and made people friendly with simplified procedures, low rate of interest, quick clearances etc. Employment Board is being set up. Action is also initiated for setting up of Agriculture Estates, Food Courts etc. The details of progress achieved in this regard have been incorporated in different chapters of this document. To monitor the implementation of the employment generation strategy the Government is in the process of setting up a High Powered Committee headed by the Chief Minister.

2. Goa is No. 1 State in India

Goa has emerged as the most progressive State in the country and its socio-economic indicators are comparable with the developed countries in the world. Many agencies have ranked Goa as No.1 State in the country. Goa is ranked No.1 by the Eleventh Finance Commission as the Best Placed State in the Country in terms of infrastructure facilities (Annexure 1). A study commissioned by India Today magazine has ranked Goa as No.1 State in the Country in terms of over all development. The National Population Commission has ranked Goa as the First State amongst all States/UTs in terms of 12 indicators (composite index) on quality of life (Annexure 2). The CRISIL has declared Goa as the second most fastest growing State in the country. Various human development indicators of Goa are given in Annexures 3 to 5. State-wise comparison on different parameters are given at Annexures 6 to 17.

Astounding progress achieved by Goa during the last four decades is attributable to appropriate development model followed by the State. Goa was fortunate to have a visionary leader in late Shri Dayanand Bandodkar who led the State for over a decade after liberation. His foresight enabled the State to evolve a right development direction and paved the way to its becoming the most progressive State today. (Box 1.1).

3. Goa's Contribution to the National Exchequer

Though, Goa is a very small state both in terms of area and population yet it contributes significantly to the national exchequer. Goa receives over 10% of foreign tourists visiting the country, and thereby generates a lot of foreign exchange. Domestic tourism also gives rise to economic activity and income in the state. Almost 15% (above Rs.1500 crore) of the foreign exchange earnings from tourism are generated in Goa annually. Over 60% of the total iron-ore exported from the country is from Goa alone, which is worth near Rs.1000 crore per annum. Royalty from minerals like iron-ore also add revenue to the central kitty. Approximately, 0.66% of the combined collection of income tax, excise duty and customs are generated in Goa. Per capita contribution of the State from iron ore export, tourism, customs revenue, central excise and income tax is given in table 1.1.

4. State Income

The Gross State Domestic Product (GSDP) of the State at constant prices (1993-94) has registered an annual compound growth rate of 8.7% during the period 1993-94 to 2000-01. The GSDP and NSDP (Net State Domestic Product) at current and constant prices for the period 1993-94 to 2001-02 are given in Annexures 18 to 21. The figures for the

BOX 1.1

Shri Dayanand Bandodkar - a visionary leader

Immediately after liberation of Goa, Daman and Diu in 1961, a Planning Board was constituted to prepare a Plan as expeditiously as possible and to be integrated into the Third Five Year Plan of India. The first Plan (Draft Plan for Goa, Daman and Diu for 1963-1966) prepared by the Planning Board envisaged increasing agricultural production, development of industries, tapping of all available natural resources for generation of hydro-electric power, land reforms, provision of national minimum amenities in rural areas and provision of free and compulsory primary education. The sectoral priorities were also fixed accordingly. The plan proposed an outlay of 39 per cent to power sector followed by Transport, Communication and Building (20 percent), Social services (18 percent), Agriculture and allied activities (14 percent) and other sectors (9 percent). While according top priority for power sector the plan stressed the need for generation of power and recommended setting up of a thermal power station and tapping potential of hydro-electric power.

The Planning Board submitted the draft plan in June 1963 much before the first democratically elected Government took office in December 1963 under the leadership of Shri Dayanand Bandodkar. The vision of a democratically elected government was not embodied in the draft plan prepared by the Planning Board. Shri Bandodkar had a different vision of Human Development which is evident in changed priorities of the government with social services sector receiving highest priority (47 percent) followed by power (16 percent) and Transport (11 percent). As regards power, the Government decided to purchase power from the neighbouring states and develop transmission and distribution network. As power requirement was being easily met from neighbouring states at a lower rate, the power generation programme took a back seat.

Priority given for social services resulted in massive development initiatives and investment in human capital. For instance, the total number of primary schools (including primary sections in non-government secondary schools) had increased from 476 in 1961-62 to 1031 in 1967-68 (including Daman and Diu) resulting an increase of 26.5 % in enrolment. Similarly, the total number of secondary schools (high schools and middle schools) had increased from 91 in 1961-62 to 369 in 1967-68 resulting an increase of over 560% in enrolment Probably, no State in this country has given such an importance for education and achieved phenomenal progress within a short span of six years.

Table 1.1: Small State - But - Contribution to the Nation is significant

		Amount (Rupees in crore)	Per capita (in Rupees)
1	Iron Ore Export - Foreign Exchange Earnings (2001-02)	939.94	6899
2	Foreign Exchange Earnings from Tourism (1999)	1500	11062
3	Customs Revenue (2001-02)	301.48	2243
4	Central Excise (2001-02)	626.98	4665
5	Income Tax (2001-02)	232.60	1731

Table 1.2: Sectoral composition of GSDP at current prices (per cent)

Sl.No	Sector	1970-71*	1980-81	1990-91	2000-01
1	Primary Sector	30.9	24.3	19.9	12.0
2	Secondary Sector	26.5	29.9	28.6	33.4
3	Tertiary Sector	42.6	45.8	51.5	54.6
4	**GSDP**	**100.00**	**100.00**	**100.00**	**100.00**

*Relates to NSDP.

period 1980-81 to 1992-93 for GSDP are given in Annexures 22-23 and figures for NSDP for the period 1970-71 to 1992-93 are given in Annexures 24-25. The per captia income of Rs. 49,673 of Goa in the year 2001-02 is the highest in the country (Annexure 6).

It is pertinent to note that the estimates of State Domestic Product are released in three stages depending on the availability of data. The process has a time lag of two years and figures are made final only after reconciliation with the Central Statistical Organization (CSO), Government of India.

The sectoral composition of GSDP at current prices at different time periods is given in table 1.2. It is evident from the table that the contribution of primary sector to the GSDP is continuously on the decline. The share of primary sector in the GSDP has declined from 30.9% in 1970-71 to 12.0% in 2000-01. At the national level too it has declined from 31% to 24% during 1993-94 to 2001-02.

The value added from agriculture (proper) in GSDP has declined from 16.5% in 1960 to less than 7% in 2000-01. Waning agriculture activity in the State is also evident from the declining trend of work participation in this sector. The percentage of workers in this sector has declined from 60% in 1960 to 27.5% in 1991 and to 16.6% in 2001 population census. This may be attributable to small land holdings and pressure on land due to heavy urbanization leading to less economic viability in relative terms. However, the Employment Generation Strategy unveiled by the Government of Goa during 2003-04, is expected to help in the revival of agriculture and increase in its contribution to GSDP in the State.

The value added from Forestry & Logging in GSDP has been declining in real terms. No logging takes place in Goa. Cutting of trees or use of forest land for any non-forest purposes is totally banned in Government Forests under the Forest Conservation Act, 1980 of Government of India. The Government of Goa has also banned cutting of trees in private forests under the Goa, Daman & Diu (Preservation of Trees) Act which came in to force on 23rd August 1984. The value addition from this sector is on account of dead, dying and fallen trees from government forests salvaged departmentally, major forest products comprising industrial / fuel wood and minor forest products like bamboo, cane etc.

Fish production in the State has almost remained static during period 1998 to 2002. It has declined from 94547 tonnes in 1997 to 71247 tonnes in 2002. The value addition to GSDP too has declined from Rs. 92.31 crore in 1993-94 to Rs. 67.72 crore in 2000-01 at constant prices. Over exploitation and excess mechanization are reported to be the cause for low fish catch. Recent measures like ban on fishing during breeding season and ban on issuing licences to new vessels enforced by the Government has resulted in increased fish production in the year 2003 (Annexure 46).

Contribution from mining & quarrying is mainly from iron ore mining. Mining was an important economic activity at the time of liberation contributing about 17% of the State income. Share of value addition from this sector in the state income at present is less than 4%. On account of continuous exploitation, the iron ore stock in the State is depleting so also its production which has remained almost static for many years. This is a natural resource where there is only depletion and no replenishment unless some new reserves are located. The index of mineral production is also almost static (Annexure 26). As such, a similar trend in value addition to the State income is obvious.

During the last four decades the industrial sector (secondary sector of the economy) in Goa has grown by leaps and bounds. Share of value addition in SDP has increased from about 8% at the time of liberation to over 33% now. At the time of liberation there were very few industries like cashew

processing, fruit and fish canning, laundry soap making, etc. Now the State has over 6000 small scale industrial units and 154 medium and large industries in the organized sector. Value addition from this organized sector is captured through Annual Survey of Industries (ASI). As survey results indicate (Annexure 27), 54% of contribution is from manufacture of chemicals and chemical products followed by 14% from manufacture of food products & beverages and 5% from manufacture of rubber and plastic products. Besides, the State has a huge unorganized sector and the value addition from this sector is only an estimation. On the whole the secondary sector has registered an annual compound growth rate of 11.9% at constant prices (GSDP) during the period 1993-94 to 2000-01.

Tertiary sector is the main stay of economy of the State and tourism is key contributor. The tertiary sector has registered an annual compound growth rate of 9.5% during the period 1993-94 to 2000-01 at constant prices (GSDP). Tourism is not a separate sector in the System of National Accounts. As such, there is no estimate available as regards contribution of tourism to the State income. Studies indicate multiplier effect of tourism in the economy as such its contribution can be linked to most of the sectors of SDP. For instance 'Trade, Hotels and Restaurants' a component of service sector contributes about 15% of value addition to the SDP and it is a fact that most of the hotels and restaurants mainly cater to the tourism industry. Tourism has also activated other sectors like transport, banking, insurance, real estate etc., all of which contribute significant value addition in the SDP.

It is pertinent to note that declining share of agriculture to the GDP is a common feature in some of the major Asian countries. Movement in sectoral shares in GDP at current market prices in some select Asian countries is given in table 1.3. It is evident from the table that share of services sector in GDP of India has increased from 36% in 1980 to 48.8% in 2001 and similar is the trend in other Asian countries except Indonesia and Malaysia. However, it is pertinent to note that the share of industries in the GDP in India is much low in comparison to other Asian countries.

5. Demographic Profile

In India the first population census was held in the year 1872. The period of this census was not the same throughout the country and was held at different points of time. However, from the year 1881 the country adopted a complete and consistent system of population count to be repeated with an interval of ten years. In Goa, the erstwhile Portugese administration was reported to be conducting population census and the history dates back to 1850. The last census conducted by the Portugese administration in Goa was in 1960. From the year 1971 onwards, the population count in Goa has been as per the Indian pattern of decadal census. So far, four population censuses have been conducted in Goa in the years 1971, 1981, 1991 and 2001.

As per the provisional results of the population census 2001, population of Goa is 13,43,998. The decadal growth rate in population has declined

Table 1.3: Movement of sectoral share in GDP at current market prices in some select Asian countries.

(Per cent)

	Agriculture			Industry			Services		
	1980	1990	2001	1980	1990	2001	1980	1990	2001
China	30.1	27.0	15.2	48.5	41.6	51.1	21.4	31.3	33.6
India #	38.1	31.0	24.7	25.9	29.3	26.4	36.0	39.7	48.8
Indonesia	24.8	19.4	16.4	43.4	39.1	46.5	31.8	41.5	37.1
Korea	14.9	8.5	4.4	41.3	43.1	41.4	43.7	48.4	54.1
Malaysia	——	15.2	8.4	—-	42.2	49.6	—-	42.6	41.9
Pakistan#	29.6	26.0	25.0	25.0	25.2	23.0	45.5	48.8	52.0
Philippines	25.1	21.9	15.1	38.8	34.5	31.6	36.1	43.6	53.3
Thailand	23.2	12.5	8.6	28.7	37.2	42.1	48.1	50.3	49.3

For India and Pakistan, data are based on GDP current factor cost
Source: Asian Development Bank (ADB), Key Indicators, 2002.

Table 1.4: Population of Goa and decadal growth rate 1900-2001

		North Goa District	South Goa Disrict		Decadal growth (%)
1	1900	294074	181439	475513	
2	1910	306323	180429	486752	+2.36
3	1921	288039	181455	469494	-3.55
4	1931	313614	191667	505281	+7.62
5	1940	336628	204297	540925	+7.05
6	1950	330874	216574	547448	+1.21
7	1960	349667	240330	589997	+7.77
8	1971	458312	336808	795120	+34.77
9	1981	568021	439728	1007749	+26.74
10	1991	664804	504989	1169793	+16.08
11	2001	757407	586591	1343998	+14.89

from 16.08% during 1981-91 to 14.89% during 1991-2001. The population of Goa and the decadal growth rate during 1900 to 2001 is given in table 1.4. It is evident from the table that prior to liberation the growth in population was very low. However, the first census after liberation held in 1971 indicates large influx of population to Goa. During subsequent censuses the population growth has declined steadily. During this period the birth and death rates too have decreased considerably.

Sex Ratio

The sex ratio in terms of number of females per thousand males for the period 1900 to 2001 is given in table 1.5

Table 1.5: Sex Ratio of Goa 1900-2001

Sl. No.	Year	Sex Ratio
1	1900	1091
2	1910	1108
3	1921	1120
4	1931	1088
5	1940	1084
6	1950	1128
7	1960	1066
8	1971	981
9	1981	975
10	1991	967
11	2001	960

It is evident that till liberation of Goa the sex ratio was in favour of females. However, after liberation the trend has reversed and the sex ratio has been declining in every census in favour of males. The sex ratio of population in different talukas of Goa and of child population in the age-group of 0-6 years is given in table 1.6.

Table 1.6: Sex ratio in different talukas of Goa – Population Census 2001

State / District Taluka	Sex Ratio	
	Total Population	Age-group 0-6 years
Goa	**960**	**933**
North Goa District	**951**	**932**
Pernem	941	902
Bardez	955	916
Tiswadi	968	949
Bicholim	943	926
Satari	962	954
Ponda	937	943
South Goa District	**972**	**934**
Mormugao	893	902
Salcete	1019	950
Quepem	973	917
Sanguem	967	970
Canacona	976	937

It is evident from the table that in 2001 population census only in Salcete taluka the sex ratio is favourable to females and is also the highest in the State. The sex ratio is the lowest in Mormugao taluka. However, it is disturbing to note that the sex ratio of child population in the age-group of 0-6 years is much less than the sex ratio of total population.

(ii) Literacy

As per the Census definition a person aged seven years and above, who can read and write with understanding in any language is considered as literate. Thus, the population in the age group of 0-6 years is considered as illiterate and is also excluded for the purpose of calculation of literacy rate. Literacy rates by sex during 1991 and 2001 census may be referred to in table 1.7. The literacy rate has increased from 75.51 per cent in 1991 to 82.32 per cent in 2001. It means that still there are over 2 lakh people (excluding the population in the age group of 0-6 years) in Goa, who are unable to read and write with understanding.

Amongest the States and Union Territories, Goa is at fourth place in literacy next to Kerala, Mizoram and Lakshadweep. Literacy rates both for male and female population have improved in the year 2001 considerably over the year 1991. However, there is not much improvement in the gap in male and female literacy, which is over 13 percent.

Table 1.7: Literacy rates of Goa by sex in 1991 and 2001

(per cent)

	1991 census	2001 census
Total	75.51	82.32
Male	83.64	88.88
Female	67.09	75.51

Literacy rates in different talukas of Goa by sex and residence are given in table 1.8. It is evident from the table that literacy in 5 talukas viz Canacona, Sanguem, Quepem, Satari and Pernem are below the state average. Bardez taluka has the highest literacy rate of 86.5 % and the lowest literacy rate is 67.7% in Canacona taluka. Male and Female literacy is also lowest in Canacona taluka. Female literacy in Canacona taluka is as low as 54.4% and it is 47.4% in rural areas of Canacona taluka.

(iii) Urbanisation

Post liberation period witnessed heavy urbanisation in Goa. As per 2001 census nearly half the State is urbanised. Urban population data is not available for the period prior to 1950. The trend of urbanisation during the period 1950-2001

Table 1.8: Literacy rates in different talukas of Goa – Population Census 2001(Percent)

State / District / Taluka	Rural			**Urban**			Total		
Goa	**79.7**	**87.7**	**71.6**	**85.0**	**90.1**	**79.7**	**82.3**	**88.9**	**75.5**
North Goa	**82.4**	**90.3**	**74.3**	**86.2**	**91.0**	**81.0**	**84.1**	**90.6**	**77.3**
Pernem	80.9	90.0	71.3	82.1	88.7	75.2	81.1	89.9	71.8
Bardez	87.4	92.6	82.1	85.8	90.2	81.1	86.5	91.2	81.6
Tiswadi	82.2	89.2	75.4	86.9	91.7	81.8	85.3	90.9	79.5
Bicholim	82.0	90.9	72.7	87.1	93.2	80.1	84.1	91.8	75.9
Satari	75.3	84.9	64.8	88.8	94.8	82.6	77.1	86.7	67.2
Ponda	82.5	91.0	73.6	85.4	89.9	80.4	83.4	90.6	75.8
South Goa	**75.1**	**83.3**	**67.1**	**83.8**	**89.1**	**78.2**	**80.0**	**86.6**	**73.2**
Mormugao	79.6	86.5	72.9	84.3	89.9	77.8	83.5	89.4	76.9
Salcete	80.4	86.3	75.0	84.1	88.6	79.6	82.6	87.7	77.6
Quepem	69.7	77.3	61.9	82.1	88.4	75.8	75.4	82.4	68.2
Sanguem	73.6	83.3	63.5	83.2	90.1	76.2	75.2	84.5	65.7
Canacona	62.9	78.5	47.4	80.5	86.4	74.1	67.7	80.1	54.4

P: Person M: Male F: Female

is in table 1.9. It is evident from the table that there has been a steady increase in urban population after liberation. Highest growth was witnessed in 1971 census. However, in terms of absolute increase in urban population, the 1991-2001 decade registered highest ever growth of 1,89,117 persons. The number of towns in the State has increased from 31 in 1991 to 44 in 2001.

Table 1.9: Urbanisation Trend in Goa 1950-2001

Census Year	Urban population as percentage to total population	Decadal growth of urban population (per cent)
1950	12.96	-
1960	14.80	23.12
1971	25.56	132.73
1981	32.03	58.82
1991	41.01	48.63
2001	49.77	39.42

Rural-urban composition of population in different talukas of Goa in 2001 census is given in table 1.10. Mormugao taluka is highly urbanised with an urban share of 83.06% followed by Tiswadi with 65.45%. The less urbanised taluka is Pernem with an urban share of 13.34%. The three Western Ghats talukas viz Satari, Sanguem and Canacona are also less urbanised. It is evident that mainly the coastal talukas of the State are highly urbanised.

Table 1.10: Taluka-wise rural-urban composition of population in 2001

(Per cent)

State / District / Taluka			
Goa	**50.23**	**49.77**	**100.00**
North Goa	**55.08**	**44.92**	**100.00**
Pernem	86.66	13.34	100.00
Bardez	41.51	58.49	100.00
Tiswadi	34.55	65.45	100.00
Bicholim	59.08	40.92	100.00
Satari	86.50	13.50	100.00
Ponda	67.60	32.40	100.00
South Goa	**43.98**	**56.02**	**100.00**
Mormugao	16.94	83.06	100.00
Salcete	41.66	58.34	100.00
Quepem	54.17	45.83	100.00
Sanguem	82.86	17.14	100.00
Canacona	72.90	27.10	100.00

(iv) Workers and non-workers

As per census definition Work is defined as participation in any economically productive activity with or without compensation, wages or profit. Such participation may be physical and / or mental in nature. Work involves not only actual work but also includes effective supervision and direction of work. It even includes part time help or unpaid work on farm, family enterprise or participation in any other economic activity. A person who has worked for 6 months or more, is classified as a 'Main Worker'. If a person has worked for less than 6 months, he is classified as a 'Marginal Worker'. A person who has not worked at all is classified as a 'Non-Worker'. Main and Marginal workers together called as 'Total Workers'. Distribution of main workers, marginal workers and non-workers as per 2001 census may be referred in table 1.11.

(v) Work Participation Rate

The Work Participation Rate (WPR) is defined as the percentage of total workers (main and marginal) to total population. Percentages of total workers, main workers, marginal workers and non-workers to total population by sex during 1991 and 2001 are given in table 1.12.

(vi) Gender gap in workforce

As per 2001 population census, gap in number of working men and women is very high and is higher than the national average. It is evident from the table 1.13 that the gender composition in the workforce is 72% males and 28% females in 2001. The gender gap has increased marginally from 42.8% in 1991 to 44 % in 2001.

(vii) Projected mid year population of Goa from 2002 to 2011

Goa has registered a population growth of 14.89% during 1991-2001. Accordingly, the projected mid year population of Goa for the period 2002 to 2011 is given in table 1.14. Generally, population projection is done as per geometric progression method based on the principle that *Population Begets Population*. This is like calculating compound interest. Population addition to any region comprises natural increase and net of migration. As the state has already reached the level of replacement in population control, the population projection given in table 1.14 is based on the assumption that there may not be significant change in natural growth and net migration during 2002-2011.

5. Uniform Civil Code

Goa is the only State in the country to have a Uniform Civil Code. Under Article 44 of the Directive Principles of the State Policy of the Constitution of India, it is provided that the State shall endeavour to have a Uniform Civil Code for its citizens throughout the territories of India. But, till today neither any State Government nor the Central Government have been able to make any headway in this regard and different communities in this country are being administered by their personal laws. Exception to this is the State of Goa, which inherited the Uniform Civil Code enforced by the Portuguese and the same is continued to govern the citizens of the State even today. The basic features of the Uniform Civil Code are

Table 1.11: Sex-wise distribution of workers & non-workers : Census 2001

Sl. No.	Classification	Persons	Males	Females
1	Main Workers	425702	328349	97353
2	Marginal Workers	96863	47764	49099
3	Total workers (Main + Marginal)	522565	376113	146452
4	Non-workers	821433	309504	511928

Table 1.12: Work Participation Rate : 1991 and 2001

Sl. No.	Persons Males Females	Main workers	Marginal workers	Total workers	Non workers
1	Persons				
	1991	32.8	2.5	35.3	64.7
	2001	31.7	7.2	38.9	61.1
2	Males				
	1991	48.3	1.3	49.6	50.4
	2001	47.9	7.0	54.9	45.1
3	Females				
	1991	16.8	3.8	20.5	79.5
	2001	14.8	7.5	22.3	77.7

Table 1.13: Gender Gap in Workforce (Percent)

		Goa		India 2001
1	Male	71.4	72.0	68.4
2	Female	28.6	28.0	31.6
3	Total	100.00	100.00	100.00

- the Code which contains civil rights of a Citizen is applicable uniformly to all the citizens in this State without any difference regarding sex, religion, caste, creed etc., the right of succession is also uniform to all Citizens without difference regarding sex, religion, caste etc., Children succeed to their parent's estate equally without difference of sex; and whether married or not.

- There is uniform law of marriage for all the citizens under which Registration of Marriage is compulsory and it is limiting monogamy.

Table 1.14: Projected mid year population of Goa 2002-2011

Sl. No.		Projected mid year Population
1	2002	1368674
2	2003	1387807
3	2004	1407207
4	2005	1426879
5	2006	1446825
6	2007	1467050
7	2008	1487558
8	2009	1508353
9	2010	1529438
10	2011	1550818

- Divorce is permissible only through courts and on specific grounds, which are uniform to all citizens.

6. Poverty and unemployment

As per the Planning Commission estimates for the year 1999-2000, poverty of 4.4% in Goa is the 2nd lowest in the country next to Jammu & Kashmir (Annexure 7). Poverty in rural area is 1.35% and that of urban area is 7.52%. This is much below the national average of 26.10%, the rural and urban poverty being 27.09% and 23.62% respectively. It is pertinent to note that the urbanized states including Goa have more urban poverty which may be attributable to emergence of slums. Now, Goa has to strive to achieve zero poverty to become the first State in the country to achieve the feat. Though, in general a very high level of educated unemployment mainly under employment is visible in Goa, there is no accurate figure available on this aspect. There are over one lakh applicants on the live register of the State Employment Exchange (Annexure 28) who are either unemployed or may be under employed.

7. Consumer Expenditure

As per NSS 55th round consumer expenditure survey, the cost of living in Goa is much high than the national average. It is evident from table 1.15 that the average Monthly Per Capita Expenditure (MPCE) in Goa is much high than the national average. There is not much difference in cost of living in rural and urban areas of Goa in comparison to significant differential at all India level. It is also pertinent to note that while in Goa people spend more on food items than non-food items, in urban areas at the national level the expenditure on non-food items is more than food items.

Table 1.15: Average MPCE for Goa and India (in Rupees)

	Rural	Urban
Goa	**869**	**471**
(i) Food	398	1155
(ii) Non-food	593	562
India	**486**	**289**
(i) Food	197	855
(ii) Non-food	410	445

8. International Film Festival in Goa

International Film Festival 2004 is likely to be held in Goa. The Government of India has approved Goa as a permanent venue for the event. So far, Cannes in France has been the venue for International Film Festival. The Government of Goa is taking steps for putting in place requisite infrastructure of international standards like building of multiplexes and auditoriums, setting up of a film market and bringing about overall integrated infrastructure development of main film festival venue around the Kala Academy Complex in Panaji.

9. Gomant Vishwa Sammelan

The Government of Goa has set up the NRI- Goa Facilitation Center mainly to revive and strengthen the roots of the people of Goan origin living outside India with their homeland. One of the activities of the Center is to hold an Annual Convention with a view to provide a platform to representa-

tives of Goan Diaspora, to come together, since such an interaction would help to promote goodwill and co-operation which will eventually be beneficial both to the Goans living in Goa and outside. As per the request of representatives of people of Goan origin settled in various parts of the country , the NRI - Goa Facilitation Center took the initiative to organise the "Gomant Bharati Sammelan" at Ponda in May 2003 with the support of the Government of Goa, which was a grand success. This Sammelan had the participation mainly of Goans settled outside Goa, but within India. Boosted by the success of this Sammelan, the Government of Goa under the aegis of NRI- Goa Facilitation Center held a conclave of World Wide Goans, titled "GOMANT VISHWA SAMMELAN" at Kala Academy Complex, Panaji- Goa from the 3rd to 5th January 2004.

(i) Aim of the convention

"To essentially strengthen the cultural roots of the people of Goan origin and to facilitate their direct and / or indirect participation in ongoing development of Goa in fields of their own chosen interest while removing their apprehension about their alienation".

(ii) Mission

"To explore and develop ways and means for creating and sustaining a vibrant peaceful and prosperous Goa".

The three day Convention had detailed technical sessions of six thrust areas on a) Art & Culture b) Tourism c) Agriculture d) Health care e) Education and f) Industry. Besides, there was a dedicated session for the NRIs and NRGs to voice their concerns. There was also an Interface session which had face to face interaction of the delegates with the Chief Minister, Leader of the Opposition and Chief Secretary.

In order to ensure the success of the convention a 4 tier organisation was constituted

- Organising committee: Consisting of Chief Minister, Cabinet colleagues, MLAs, Industrialists, Secretaries to the Government and key officers of the Corporation of City of Panaji. The committee decided on broad policy matters pertaining to the Convention.

- Executive Committee: The Committee headed by Chief Secretary consisting of prominent citizens was responsible for the detailed programme and implementation of the conference and served as an apex organisation for the various Task Forces and Action Committees.

- Task Forces: Seven task forces, one each for the six thrust areas and the seventh for the Expectations and Concerns Session were formed. The Task Forces included several specialised experts from the concerned field. The Task Forces dealt with the developing, designing and implementation of the details of the relevant technical sessions including the follow up procedure.

- Action Committee supervised the smooth conduct of the convention and was headed by an active member involved in organising the convention successfully.

The Sammelan was inaugurated on the 3rd of January 2004 by His Excellency the Governor and presided by Hon'ble Minister of State for Finance. A commemorative volume containing articles mainly from eminent Goans settled within and outside Goa who have excelled in their respective fields was released on the occasion. Technical sessions on Tourism, Industry, Education and Agriculture were held on 4th January 2004. Besides the sessions, there was a presentation of collage of films by the Tourism Department. In the afternoon session there was a dedicated session on education which was followed by Art & Culture session. The second day ended with a programme titled "GOMANT DARSHAN", which depicted the cultural history of Goa and was appreciated and received the standing ovation by the delegates present. On the 3rd day of the programme, there was also a fruitful interface wherein a face to face question & answer session was held between the Chief Minister, Ministers and the delegates.

The technical sessions and the presentations of the various speakers were broadly based on the following parameters, though not necessarily restricted to the same.

1. Where does Goa stand today?

2. Where does it want to reach say by the year 2020?

3. What are the priority areas / projects which should receive urgent attention?

4. What kind of and what quantity of resources, finance, technology, management expertise, human resources will be required?

5. How much of these will be locally available? How much are to be procured from within the country? How much can be mobilised from the rest of the world?

6. What supportive action is required from the Government and other stakeholders?

7. What kind of contribution will the NRIs & NRGs be able to make for ensuring successful implementation of the identified priority projects?

8. What are the obstacles, risks, threats and what precautions will have to be taken?

9. Can the Task Force activity get translated itself into a mechanism to develop and pursue a time bound, precise and implementable action plan? and

10. How can we ensure continuous assessment of the draft vision in terms of globalization and its effects?

The delegates who participated in the deliberations and interactive sessions expressed satisfaction that the Government and people in Goa are taking their cognizance and they were very enthusiastic to contribute their mite to make Goa a prosperous state without disturbing its noble traditions of harmony and peace. The delegates also showed deep interest in the new concepts like the Goa Education Development Corporation and the Endowment Trust concept of GMC. The points of discussion were regarding laws like Tenancy Act, Mundcar Act, Rent Control Act etc. It is estimated that about 2.5 lakh Goans or people of Goan origin live in foreign countries and an equal number in other parts of the country outside Goa. In the Sammelan it was proposed at this stage not to make effort for attracting the investments from Goan Diaspora, although if anybody proposes to invest in Goa on their own, the Government of Goa will extend all co-operation and help.

OVERVIEW

Goa was fortunate to have a visionary leader in late Shri Dayanand Bandodkar as the first Chief Minister whose foresight laid a strong foundation for the development of the State. While the No.1 status of Goa in the country today has been confirmed by different agencies, the quest to attain the standards of a developed State would require a very high level of investments both in public and private sector and strategic interventions. The Employment Generation Strategy announced in the Budget 2003-04 is definitely a significant step in this direction.

The three-pronged employment generation strategy aimed at addressing the unemployment problem of the State is expected to accelerate the economic growth of the State. The new initiatives like Krishi Ghars, Kamdhenu scheme, Agriculture estates, Food Courts, development of floriculture and horticulture, development of abandoned lakes, ponds and other water bodies, remodeling of notified embankments etc., are expected to create employment opportunities in villages and likely to reverse the declining trend of primary sector in the State Domestic Product in the coming years. The new Industrial Policy 2003 and the resultant new schemes announced by the Government are likely to accelerate the industrial development and create employment opportunities for the unemployed youth. The multiplier effect of all these initiatives along with various other self employment schemes like modified CMRY, modified DDSY, Employment Board etc., would further boost the services sector. These initiatives are also expected to tone up the banking sector and improve the credit-deposit ratio (CD ratio) of banks.

Per capita income of the State is not only the highest in the country but is also nearly three times the national average. Though, a very small State, both in terms of area and population, Goa contributes significantly to the national exchequer in terms of foreign exchange earnings and revenue collection from customs, central excise and income tax. The demographic profile depicts the third stage of demographic transition which is a significant feat of the State since population explosion is the root cause for most of the miseries in the society. The initiative of the Government of Goa in bringing the Goan diaspora under the banner 'Gomant Vishwa Sammelan' is expected to go a long way in utilizing their potential for the development of the State.

CHAPTER 2
Public Finance

From early 1990's Goa started experiencing tremendous pressure on State finances on account of various reasons like introduction of Gadgil-Mukharjee formula for devolution of resources from the Planning Commission to State plans, implementation of Fifth Pay Commission Scales etc. On account of this, borrowings of the State increased substantially. This gave relief in initial years but exerted pressure subsequently due to increase in interest payment and repayment of debt. Though, borrowing per se is not detrimental if investment results in enough returns for repayment and servicing of debt. However, this may be strictly applicable to corporate and business sector and may be difficult in Government as it has to fulfill many social obligations. But, it is necessary that borrowings by the Government are managed professionally. An attempt is made in this chapter to analyse various financial parameters of the State and efforts being made by the Government to improve financial position of the State.

1. REVENUE RECEIPTS

Revenue receipts comprise the following three components.

(i) State's Own Tax Revenue

(ii) State's Own Non-Tax Revenue

(iii) Transfers from Government of India

(i) State's Own Tax Revenue

Various sources of tax revenue are taxes on Sales, Trade etc., State Excise, taxes on Vehicles, taxes on Goods and Passengers, Land Revenue, Stamps and Registration and other taxes and duties on Commodities & Services. It is pertinent to note that Sales Tax alone account for over 70% of Tax Revenue. As such, for the purpose of analysis tax revenue is classified into three categories viz. Sales Tax, State

Table 2.1: State's Own Tax Revenue 1993-94 to 2003-04

(Rs. in crore)

Sl. No.					
1	1993-94	135.51	21.10	31.11	187.72
2	1994-95	159.23	26.01	40.76	226.00
3	1995-96	193.50	26.95	51.25	271.70
4	1996-97	220.30	26.76	55.67	302.73
5	1997-98	258.22	31.99	75.09	365.30
6	1998-99	254.22	35.17	67.82	357.21
7	1999-2000	348.18	36.06	74.24	458.48
8	2000-01	387.82	39.98	87.49	515.29
9	2001-02	401.47	46.13	122.30	569.90
10	2002-03 (Ant)	439.00	47.00	116.00	602.00
11	2003-04 (BE)	565.00	55.00	145.73	765.73

Excise and all other taxes are put under one category as 'Other Taxes'. Trend in State's Own Tax Revenue from 1993-94 to 2003-04 is given in table 2.1.

a) Sales Tax

It is evident from table 2.1 that there was abnormal trend in sales tax collection during 1998-99 and 1999-2000. During the year 1998-99, sales tax collection has declined over the previous year which may be partly due to temporary suspension of activities in few factories like Zuari. During the year 1999-2000, there was a huge growth of 37% in sales tax collection, which is due to rise in petroleum prices and uniform floor rates implementation besides one time settlement of sales tax offered to defaulters. During the period 1993-94 to 2003-04 (BE), sales tax collection has registered an annual compound growth rate of about 15%. Accelerated growth in sales tax in recent years is inspite of the fact that there was decrease in foreign tourist during 2001 due to September 11 incident in US and also decrease in sales tax on naphtha from 18% to 1%. This may be attributable to rationalization of tax structure and various steps initiated to move towards VAT system like bringing many new trades under the net of registration and revision/ enhancement in renewal fees of existing trades.

Various rationalization measures introduced are reduction in sales tax on industrial inputs except petroleum to 2%, concessional rate on furnace oil for industrial purpose, abolition of additional tax except for petroleum companies having turnover of Rs.15 crore and above, reduction in luxury tax during off season for promotion of tourism etc. The cost of collection of sales in Goa is only about 0.55% which is perhaps the lowest in the country. In addition various measures have been taken to strengthen the tax administration like increase in staff strength, additional infrastructure facilities, vigorous monitoring of tax collection system, plugging leakages etc. The State has more than 25000 sales tax dealers and their taluka-wise distribution is given in table 2.2.

Table2.2: No. of registered sales tax dealers and No. of shops licensed to sell liquor as on 31st March 2003

Sl. No.	State/District/ Taluka	No. of sales tax dealers		No. of shops licensed to sell liquor	
		Local	Central	Retail sale of IMFL & CL for consumption	Retail sale of IMFL & CL in sealed bottles
1	**Goa**	**25653**	**20051**	**6931**	**1206**
2	**North Goa**	**14773**	**11397**	**3564**	**813**
	Tiswadi	5754	4956	727	284
	Bardez	4222	3062	1557	299
	Pernem	433	209	286	39
	Bicholim	973	669	273	56
	Sattari	395	127	160	25
	Ponda	2996	2374	561	110
3	**South Goa**	**10880**	**8654**	**3367**	**393**
	Sanguem	966	721	259	32
	Canacona	339	216	321	46
	Quepem	348	181	375	28
	Salcete	6473	5487	1854	207
	Mormugao	2754	2049	558	80

b) State Excise

The revenue under State Excise is generated from Excise Duty imposed on the manufactured liquor, fees collected for import / export of excisable items, licence fees in respect of distilleries / wineries / breweries and liquor shops, shacks, cashew zone auctions, occasional licences, transport permits to carry liquor for personal consumption etc. In order to mobilize the revenue, the excise authorities have been activated time and again and a special flying squad has been formulated which patrol every night and during day time. Strict vigil is being kept on all entry points so that illegal excisable raw material and finished excise goods do not enter or go out of the State. The liquor outlets are regularly checked and appropriate deterrent action is taken so that the illegalities in liquor trade and manufacture are properly dealt with.

Quick action is being initiated to dispose off various matters such as issuing of import / export / transport permits, issuing of liquor licences wherever formalities are complied with by the applicants and disposal of cases wherever illegalities have been detected. Inspection of the places of sale of liquor both in bottles and for consumption are conducted to see that duplicate or spurious liquor is not sold to the consumers thereby endangering their health. Advertisement of liquor goods are banned. It is also proposed to display in each taluka Excise Stations, the requirements for submission of various applications.

During the period 1993-94 to 2003-04 (BE), the annual compound growth rate of state excise was 10%. The State registered an annual compound growth rate of 11.1% during 2000-01 to 2003-04(BE) as against 9.3% during the period 1993-94 to 1999-2000. There are nearly 7000 shops in the State licensed to sell liquor and their taluka-wise distribution is given in table 2.2.

It is pertinent to note that while revenue generation from sale of liquor is an important consideration, the Government is very much concerned about ill effects of drinking. This is evident from the Budget Speech 2003-04 wherein Hon'ble Finance Minister (FM) of Goa has expressed concern over increasing number of cases of liver cirrhosis and other drinking related diseases. The FM noted that drinking is also a major concern for fatal accidents and atrocities on women. There is a need to discourage the root cause and as a restrictive measure, the FM proposed to increase the application fee for retail outlets from Rs.10000 to Rs.25000 so as to minimize the proliferation of new outlets selling liquor. Hon'ble Finance Minister further announced that discretionary powers under section 90(4) will be restricted to places of tourist importance only and such places will be notified and included in the Act itself. Granting of new licenses in villages outside the notified tourist places will henceforth require approval of the concerned Gram Sabhas. Advertisements promoting all liquor products will be strictly banned by amending relevant laws. A health surcharge of 1 percent will be imposed on the excise duty as well as license fees and the proceeds will be channelised to support and encourage setting up of de-addiction centers by voluntary organizations.

(ii) State's own non-tax revenue

Trend in state's own non-tax revenue (net of power and lotteries) during the period 1993-94 to 2003-04 is given in table 2.3. It is evident from the table that non-tax revenue collection has shown phenomenal increase during recent years. During the period 1993-94 to 2003-04(BE) non-tax revenue collection has registered a compound annual growth rate of 21.6%. The annual growth rate during the period 2000-01 to 2003-04 (BE) was as high as 39.1% as against 11.2% during the period 1993-94 to 1999-2000. As per a study conducted by the CRISIL, this is the highest growth rate achieved by any State in the country in non-tax revenue in recent years. This was possible mainly due to successful renegotiation of Power Purchase agreement with Reliance Salgaonkar Power Company Limited (RSPCL), increase in water supply revenues, trading of surplus power, enhancement in fees and user charges on various services and introduction of fees and user charges on various new services.

The Government of Goa entered into a Power Purchase Agreement (PPA) with RSPCL on 10.1.1997. This agreement is for a period of 15 years from August 1999 (from the date of commercial production) to July 2014. As per the PPA, the State is bound to purchase power to the extent of 39.8 MW from RSPCL. Due to over 200% increase in Naphtha price between 1997 to 2000, the cost of power to be purchased from RSPCL increased to over Rs.6.00 per unit. The State has sufficient allocation of power from central grids at a very cheaper cost of just more than a rupee. Thus, purchase of power from the RSPCL proved highly detrimental to the State and Goa suffered loss in sale of power for the first time in the year 2000-01.

In order to reduce the burden of high price of power the Government of Goa renegotiated the PPA with RSPCL twice in the years 2000 and 2002. The PPA

Table2.3 : State's own non-tax revenue 1993-94 to 2003-04 (net of power and lotteries)

(Rs. in crore)

Sl. No.	Year	Power (net)	Water Supply	Others	Total
1	1993-94	26.19	10.68	26.05	62.92
2	1994-95	21.51	11.27	28.98	61.76
3	1995-96	31.16	15.50	37.65	84.31
4	1996-97	48.16	17.65	35.21	101.02
5	1997-98	29.56	19.08	48.54	97.18
6	1998-99	73.59	41.19	43.02	157.80
7	1999-2000	31.96	40.20	47.12	119.28
8	2000-01	-2.29	43.59	61.48	102.78
9	2001-02	30.84	53.13	110.17	194.14
10	2002-03 (Ant.)	60.00	56.00	198.00	314.00
11	2003-04 (BE)	133.00	70.00	243.00	446.00

was successfully amended by granting distribution permission to RSPCL and reducing the sales tax on Naphtha from 18% to 1%. This has resulted in reduction in power purchase commitment from 39.8 MW to 23 MW. Gain due to this reduction in power purchase was more than the loss on account of reduction in sales tax on Naphtha. Thus, the State electricity department started earning profits once again from the year 2001-02. In order to further increase revenue collection, special squads have been formed to check power pilferage and revenue collection is monitored on day to day basis and monthly targets are set for every division. On account of these measures, the State could achieve a very high growth in revenue collection from power during the period 2001-02 to 2003-04 (BE) as against the loss incurred during 2000-01.

Revenue from water supply has registered an annual compound growth rate of 20.7% during the period 1993-94 to 2003-04. As it is evident from table 2.3, there has been tremendous growth in other non-tax revenue sources during the period 2000-01 to 2003-04. The annual compound growth rate registered during the period 2000-01 to 2003-04 was as high as 50.7% as against 10.4% during the period 1993-94 to 1999-2000. This mainly comprises revenue from sale of surplus power and enhancement in fees and user charges on various services and introduction of fees and user charges on many new services. Goa started trading surplus power from the year 2001. Revenue from trading exceeded Rs.80.00 crore during 2002-03. The State is in the process of instituting an on-line IT-enabled trading platform for power trading and commencement of operation is expected shortly. Other measures taken to increase non-tax revenue are introduction of fee for erecting hoardings on government lands, increase in rent for commercial utilization of government buildings, upward revision of rates at various Government accommodation like rest houses, circuit house, eco tourism cottages etc., cess on fluid milk coming into the State, introduction of nominal fee for all type forms prescribed for Government Departments for various purposes etc.

(iii) Transfers from Government of India

Transfers from Government of India may be broadly classified into two categories as share in central taxes and grants under Gadgil Mukharjee formula, centrally sponsored schemes etc. Transfers from the Government of India during the period 1993-94 to 2003-04 (BE) is given in table 2.4.

It is evident from the table that transfer of resources from Government of India to Goa has been quite low. During the period 1993-94 to 2002-03, it has registered an annual compound growth rate of 3.5%. The low growth rate is mainly due to the fact that transfer of resources from Government of India are largely governed by different formulas based on parameters like population, per capita income etc., which are not favourable to Goa. Being a small State with limited population base and high per capita income Goa always gets the least. For instance devolution of central assistance for plan schemes is governed by Gadgil-Mukherjee formula. Under this formula fund allocation is based on weights to different parameters like 60% to population, 25% to per

Table 2.4: Transfers from Government of India

(Rs. in crore)

Sl.No.	Year	Central taxes	Grants	Total
1	1993-94	78.40	61.53	139.93
2	1994-95	86.90	71.04	157.94
3	1995-96	71.05	73.25	144.30
4	1996-97	90.55	69.80	160.35
5	1997-98	96.57	63.80	160.37
6	1998-99	97.12	42.34	139.46
7	1999-2000	95.92	40.12	136.04
8	2000-01	104.85	66.95	171.80
9	2001-02	107.26	59.29	166.55
10	2002-03 (Ant)	114.00	77.00	191.00

capita income, 7.5% to performance and 7.5% is discretionary quota of Dy.Chairman, Planning Commission for special problems. The formula has been subjected to much criticism, as better performing States like Goa feel that performance is not being rewarded. Further, central assistance, formula or non-formula based, devolves to general category states like Goa in the form of 30% grant and 70% loan. As a result though the central assistance triggers increase in share of plan resources but on the other hand also increases the debt burden of the State.

The devolution of central taxes to States is governed by the recommendations of Finance Commission. Present devolution is based on the recommendations of Eleventh Finance Commission and 2004-05 is the last year of its award period. The Twelfth Finance Commission has been set up by the President of India to give its recommendation for the period 2005-2010. The Twelfth Finance Commission had visited Goa in November 2003 and the details of Memorandum submitted by the State are given below.

2 MEMORANDUM SUBMITTED TO THE TWELFTH FINANCE COMMISSION

In its memorandum submitted to the Twelfth Finance Commission the Government of Goa has stated that on account of recommendations made by the successive Finance Commissions, efficiency and performance have been at the receiving end. The incentives for better fiscal management and financial prudence have been virtually non-existent. To add to the woes, smallness of the size of states like Goa has had a further negative impact on devolution resources. In order to tide over the fiscal and financial malaise pervading almost all the states and the center, this trend needs to be reversed. While ensuring that states which are placed at a disadvantage for historical and other reasons are adequately helped, the efforts made by states to improve their financial and fiscal health should be given due recognition and incentives to do even better.

As regards devolution of central taxes (horizontal devolution) the State Government requested for a change in devolution formula. The existing devolution formula (Eleventh Finance Commission) carries weightage of 10% to population, 62.5% to per capita income (distance method), 7.5% to area, 7.5% to infrastructure index, 5% to tax effort and 7.5% to fiscal discipline. It was brought to the notice of the Commission that though a small State its contribution to the national exchequer is significant. Almost 15% (above Rs.1500 crore) of the foreign exchange earnings from tourism are generated in Goa. Nearly 60 percent of the total iron ore exported from the country is from Goa alone. Approximately 0.66% of the combined collection of income tax, excise duty and customs are generated in Goa, but, Goa gets only 0.206% of the total tax devolution to all States taken together. As such, the State Government proposed that at least 40% weightage be given to performance criteria like tax effort, fiscal management, contribution to the national exchequer etc. Moreover, it should be ensured that all States get a minimum of 1 percent share of the total taxes devolved, irrespective of the size or population.

As regards vertical devolution, presently 29.5% of the net tax revenue of the Centre is being devolved to the States as per the recommendations of the Eleventh Finance Commission. In view of the increasing expenditure liabilities of the States to provide for better social and physical infrastructure and the inherent inflexibility to leverage alternative sources of financing, 29.5% of the share in net tax revenues of the Centre is not sufficient for the States. As such, the Government of Goa requested the Twelfth Finance Commission to enhance the share of the divisible pool to 40% of the net revenues accruing to the Centre and also allow States to levy and collect service taxes on items.

As regards non-plan revenue grants, it was pointed out that there exists high degree of subjectivity in the process of devolution and also poor performing States indirectly stand encouraged. The Government of Goa opined that the provision of non-plan revenue gap grant should be stopped totally. Devolution through share in central taxes should be the only source of transfer of non-plan grants to the States and the proposed enhancement of vertical devolution to 40% should compensate for the loss resulting from the discontinuation of non-plan revenue gap grants. Additional 5 percent of the net revenues of the Centre may be kept aside for discretionary non plan grants to be provided in emergent cases to the States. Thus, in totality, 45% of the net tax revenues of the Centre should be earmarked for devolution to the States. However, in case of non utilization of the discretionary quota of 5 per cent, the same will be retained by the Centre. The State Government projected a revenue gap of Rs.2444 crore to be filled in through the formula recommended by the Finance Commission.

As regards restructuring of public finances, it was observed that plan funds devolved by the Planning Commission to the States comprises 30% grant and 70% loan. This ratio acts as severe constraint on the States as the interest burden goes up much more than what is warranted by the creation of assets. This results in a lopsided financial structure leading to higher repayment burden, higher debt servicing burden, increased revenue deficit and also fiscal deficit etc. As such, the Government of Goa recommended rationalization of loan grant ratio of plan assistance from the existing 70:30 to 50:50 with effect from 1997-98 (i.e. beginning of the Ninth Plan period). However, the actual benefit of lower debt servicing may be made effective from the award period of the Twelfth Finance Commission. Further, in case this suggestion is accepted by the Commission, this divisible share in central taxes may be gradually increased from 36% in the first year of the award period to 40% in the terminal year, instead of a flat 40% as proposed earlier.

Other suggestions made by the State to the Twelfth Finance Commission are :-

- Rate of interest on loans given to the States by the Centre should be revised on a yearly basis, based on the market trends.
- Incentive fund set up as per the recommendation of the Eleventh Finance Commission should be discontinued and all the criteria laid down as a precondition to the release of the fund, should be in-built into the performance parameters on which the formula for devolution will be based.
- The Commission may recommend grants to the tune of Rs.75 crore to Village Panchayats and Zilla Panchayats for development works.
- The Commission may provide grants to the tune of Rs. 100 crore to urban local bodies to improve civic amenities.
- The Commission may recommend grants amounting to Rs. 150 crore for upgradation of health infrastructure.
- The Commission may provide grants of Rs. 150 crore for connectivity of tourist circuits.
- The Commission may recommend a grant of Rs. 100 crore as seed money towards construction of the new airport at Mopa.

3. PUBLIC DEBT

As on 31st March 2002, the State has total public debt of Rs. 1784.80 crore. Major component of this is central loan (66%) followed by SLR based market loans (25.6%) and the balance comprises loan from LIC, HUDCO, PFC, NABARD, REC, and NCDC (8.4%). Public debt position as on 31st March from 1995 to 2002 is given in table 2.5.

4. REVENUE EXPENDITURE

Detailed work-study of Government Departments is underway so as to ascertain actual requirement of posts. Accordingly, posts wherever not required are being abolished and additional posts are created wherever essentially required to improve the quality of service to the public. The State has al-

Table 2.5-: Public Debt of the State as on 31st March

(Rs. Crore)

Type of loan	1995	1996	1997	1998	1999	2000	2001	2002
1. Central Loan	485.81	523.72	567.02	627.56	721.10	815.80	988.05	1177.46
2. Market Loans (SLR based)	65.23	82.73	101.98	123.15	213.15	288.16	368.17	457.44
3. Loans from LIC	21.19	25.80	31.44	30.51	35.73	34.45	38.63	36.63
4. Loans from NCDC	0.18	0.16	0.15	0.13	0.11	0.10	0.08	0.06
5. Loans from NABARD	-	4.17	6.85	6.85	5.41	7.67	8.69	19.99
6. Loans from REC	-	-	-	5.09	8.08	10.18	9.71	7.86
7. Loans from PFC	-	-	-	0.09	3.79	14.17	21.93	29.76
8. Loans from HUDCO	-	-	-	3.92	23.66	28.14	51.80	55.60
9. W&M advances	-	-	-	-	-	-	-	-
Total Public Debt	572.41	636.58	707.44	797.3	1011.03	1198.67	1487.06	1784.80

Source :Budget at a Glance 2003-04, Finance Department, Government of Goa.

ready implemented a **Voluntary Retirement Scheme (VRS)** to its employees. Goa is the first State in the country to introduce such a scheme to the Government employees. The scheme was first opened in the year 2000-01. The Government of India has introduced a clause in the Finance Bill 2001 to extend income tax exemption on additional benefits given under VRS. The scheme was opened for the second time during 2001-02 and in all 2500 employees have availed Voluntary Retirement under the scheme. There is ban on creation of new posts, revival of lapsed posts, appointment of work charged employees, which are allowed only in exceptionally rare cases in public interest and with the approval of the cabinet. The State has taken conscious decision to consider closure / merger of schools by considering very low teacher-pupil-ratio in some schools. Already, few schools have been either closed or merged with nearby schools. In some Departments, a number of employees have been declared as surplus and a surplus cell has been created in the Secretariat to redeploy the surplus employees against the vacancies in other Departments. On the whole, it is expected that these measures will go a long way in increasing the productivity in Government departments and containing the growth of expenditure on salaries. Trend in Revenue expenditure under Plan and Non-Plan is given in table 2.6.

The "Task Force for Good Governance" under the chairmanship of Chief Minister has identified, in

Table 2.6: Revenue Expenditure (Rs. in crore)

Year	Plan	Non-Plan	Total
1993-94	66.29	291.05	357.34
1994-95	69.64	319.82	389.46
1995-96	80.48	386.95	467.43
1996-97	88.35	453.76	542.11
1997-98	89.80	540.10	629.90
1998-99	105.67	689.58	795.25
1999-2000	122.44	800.23	922.67
2000-01	141.06	874.83	1015.89
2001-02	182.62	976.48	1159.10
2002-03 (Ant.)	210.13	1089.87	1300.00

the first phase, 20 major Government Departments for total computerization which have direct public interface and huge purchase of inventories. Computerisation of Government Departments is expected to streamline the inventory management system, less use of paper, quick service to the people, and transparency in administration and check on corruption, which is expected to bring down the revenue expenditure of State to a considerable extent.

It is evident from the table 2.6 that annual compound growth in revenue expenditure during the period 1993-94 to 1999-2000 under plan was 10.8% which was much below the non-plan annual compound growth rate of 18.4%. However, during the period 2000-01 to 2002-03, the annual compound growth rate under plan being higher than non plan, the trend has been reversed.

The trend in salaries, debt servicing, pensions etc., under non-plan revenue expenditure is given in table 2.7. It is evident from the table that salary expenditure during the period 2000-01 to 2002-03 has increased at a annual compound growth rate of 4.7% as against 15.5% during the period 1993-94 to 1999-2000. However, huge increase in salaries may be attributed to implementation of Fifth Pay Commission scales. Increase in pension payment in recent years is mainly due to implementation of Voluntary Retirement Scheme by the Government.

Table 2.7: Non-plan revenue expenditure (Rs. in crore)

Year	Salaries	Interest	Pensions	Others	Total
1993-94	117.66	68.17	12.63	92.59	291.05
1994-95	128.09	69.14	15.08	107.51	319.82
1995-96	150.53	89.94	19.78	126.70	386.95
1996-97	171.69	100.50	23.97	157.60	453.76
1997-98	212.18	118.12	29.67	180.13	540.10
1998-99	264.19	143.85	63.20	218.34	689.58
1999-2000	279.57	178.15	68.19	274.32	800.23
2000-01	286.96	212.16	86.43	289.28	874.83
2001-02	313.55	255.97	115.85	291.11	976.48
2002-03 (Ant.)	323.01	292.44	140.00	334.42	1089.87

5. CAPITAL EXPENDITURE

Capital expenditure comprises expenditure incurred under plan for developmental works and mainly debt repayment under non-plan. The trend in capital expenditure under plan and non-plan

Table 2.8: Capital Expenditure (Rs. in crore)

Year	Plan	Non-Plan	Total
1993-94	92.64	35.17	127.81
1994-95	98.45	25.33	123.78
1995-96	131.20	26.59	157.79
1996-97	126.80	31.12	157.92
1997-98	113.28	34.44	147.72
1998-99	126.79	42.93	169.72
1999-2000	129.25	48.27	177.52
2000-01	237.80	64.92	302.72
2001-02	206.29	65.17	271.46
2002-03 (Ant.)	230.66	76.34	307.00

Table 2.9: Deficit indicators of the State (Rs. in crore)

	Revenue Surplus (+)/ Deficit(-)	Fiscal Deficit(-) (Net)	Primary Surplus(+)/ Deficit(-)
1993-94	33.22	-59.95	8.22
1994-95	56.24	-44.53	24.61
1995-96	32.88	-97.35	-7.41
1996-97	22.00	-104.00	-3.50
1997-98	-14.05	-124.64	-6.53
1998-99	-140.78	-269.12	-125.27
1999-2000	-208.86	-344.39	-166.24
2000-01	-226.01	-412.87	-200.71
2001-02	-228.50	-423.53	-167.56
2002-03 (Ant.)	-166.00	-389.00	-101.00

during the period 1993-94 to 2002-03 is given in table 2.8. It is evident from the table that under plan there has been quantum jump in capital expenditure from 2000-01 onwards.

6. DEFICIT INDICATORS OF THE STATE

Various deficit indicators of the State are given in table 2.9. It may be seen from the table that the State plunged into revenue deficit in the year 1997-98 and the same increased substantially in the succeeding years, which may be attributable to implementation of Fifth Pay Commission scales to the Government employees. However, it is expected to decline during subsquent years.

OVERVIEW

The analyses of financial parameters of the State indicate signs of progress. Financial acumen and administrative toughness are visible in the financial management in recent years. Rationalization of tax structure and prudent tax administration has enabled the State to achieve a reasonably good growth in tax collection. The non-tax revenues have achieved unprecedented growth of nearly 40% in recent years, which according to a CRISIL study is the highest in the country. Successful renegotiation of the Power Purchase Agreement with RSPCL has enabled the State to reduce its power purchase commitment from 39.8MW to 23 MW, resulting in revenue savings to the Government. Trading of surplus power and levying/enhancing of fees and user charges on various forms, registration, services etc has also enabled the State to gain substantial non-tax revenue.

Goa has received reasonably good support from the Centre for specific projects like APDRP, Sewerage project etc. in recent years. However, a very low growth in formula based devolution of resources from the Centre is a matter of concern. Goa is adversely affected mainly on account of the fact that most of the devolution of resources from Centre to State are governed by certain formulae which are not favourable to the State. The Government of Goa has strongly recommended to the Twelfth Finance Commission to give at least 40% weightage to performance criteria in the formula for devolution of central taxes to the States. The Government has taken a number of steps to contain revenue expenditure. Goa is the first and the only State in the country to implement a voluntary retirement scheme. This would help the State to bring down its workforce and also the Government will get an opportunity to induct young and talented personnel to the workforce which would increase efficiency and productivity in the administration.

CHAPTER 3

Institutional Finance

1. BANKING

a. Banking Offices

Goa has an excellent banking network spread in every nook and corner of the State. As on 31-3-2003 there were 447 banking offices in Goa, of which 320 were commercial banks and 127 were cooperative banks. Population covered per bank works out to 3062. During the last two decades the number of banking offices has increased by over 70%. No. of banking offices from 1980-81 to 2002-03 is given in table 3.1 below.

Table 3.1: Banking offices in Goa

Sl. No.		No. of banking offices
1	1980-81	258
2	1985-86	297
3	1990-91	306
4	1995-96	375
5	1999-2000	442
6	2000-01	444
7	2001-02	446
8	2002-03	447

b. Deposits

Goa has a very high rate of savings. The Deposits in banks, as on 31-3-2003 were Rs. 10662.40 crore. The per capita bank deposit is as high as Rs. 79333. During the last two decades the deposits have increased by whooping 28 times. Taluka-wise mobilization of deposits (Annexure 29) reveals that about 81 % of the deposits are from the three talukas viz. Salcete, Tiswadi and Bardez. An important characteristic of deposits in Goa is the significant contribution from Non-Resident Emigrants (NREs) which accounts for about 30% of the deposits. Deposits and per capita deposits from 1980-81 to 2002-03 in Goa are given in table 3.2.

c. Credit-Deposit Ratio

The Credit-deposit ratio in Goa was 31 during 2002-03. It has dropped down this year. Though the growth in credit has been reasonably good, the relatively higher growth in deposits has led to low CD ratio. Per capita credit during 2002-03 was Rs.24284. Taluka-wise credit given in Annexure 30 indicates that nearly 43% of the credit is from Tiswadi taluka followed by Salcete (26%) , Bardez (11%) and Mormugao (9%). Credits, per capita credit and CD ratio from 1980-81 to 2002-03 are given in table 3.3.

d. Annual Credit Plan of banks

Sector-wise performance under Annual Credit Plan of banks during 2002-03 and 2003-04 (up to 30th September 2003) is given in table 3.4.

e. Advances by category

Details of advances to priority sector, advances to weaker sections and SC/ST during 2002-03 and 2003-04 (up to 30th September 2003) are given in table 3.5.

2. FINANCIAL INSTITUTIONS

The Indian Economy has a well-knit structure of financial institutions comprising 11 institutions at the national level and 46 at the State level. These institutions provide a variety of customized financial products and services to suit the varied needs of the corporate world. In keeping with the emerging needs of industry in the liberalized environment, these institutions have reoriented their policies and assistance structure with much sharper customer focus, by developing and introducing a variety of products and services.

In the post-reform period, All-India Financial Institutions set up several subsidiaries/associate concerns for offering a wide range of such newly

Table 3.2: Trend in Bank Deposits

Sl. No.		Deposits (Rs. in crore)	Per capita deposits (in Rupees)
1	1980-81	380.92	3779
2	1985-86	800.92	7030
3	1990-91	1636.88	13993
4	1995-96	3714.31	29318
5	1999-2000	7331.01	54288
6	2000-01	8522.75	63413
7	2001-02	9530.98	70915
8	2002-03	10662.40	79333

developed products and services as also for capital market infrastructure development, covering areas such as commercial banking, investment banking, non-banking finance, investor servicing, venture capital financing, infrastructure financing, custodial services, electronic trading in stock exchanges, capital market regulation, registrar and transfer services and credit rating.

Table 3.3: Trend in Bank Credits

Sl. No.		Credits (Rs. in crore)	Per capita Credits (in Rupees)	
1	1980-81	171.18	1698	44.9
2	1985-86	289.11	2537	36.1
3	1990-91	640.94	5479	39.2
4	1995-96	1248.11	9852	33.6
5	1999-2000	2258.24	16723	30.8
6	2000-01	2785.17	20723	32.7
7	2001-02	3151.75	23451	33.1
8	2002-03	3263.70	24284	30.61

The national level institutions comprise five All-India Development Banks (AIDBs), three Specialized Financial Institutions (SFIs) and three Investment Institutions (IIs). At the state level, there are 18 State Financial Corporations (SFCs) and 28 State Industrial Development Corporations (SIDCs). The AIDBs are Industrial Development Bank of India (IDBI), Industrial Finance Corporation of India Limited (IFCI), Industrial Credit and Investment Corporation of India Limited (ICICI Ltd.,), Small Industries Development Bank of India (SIDBI) and Industrial Investment Bank of India Limited (IIBI). The SFIs are Risk Capital and Technology Finance Corporation Ltd. (RCTC), ICICI Venture Ltd (erstwhile TDICI Ltd.) and Tourism Finance Corporation of India Ltd.(TFCI). The Investment Institutions are Life Insurance Corporation of India (LIC), Unit Trust of India (UTI) and General Insurance Corporation of India (GIC).

Of the AIDBs, IDBI, IFCI, ICICI and IIBI provide financial assistance to medium and large indus-

Table 3.4: Sector-wise advances under Annual Credit Plan

Sl. No.		Advances (Rs. in crore)	
			2003-04 (up to September)
1	Agriculture	17.99	17.28
2	Industries	106.41	68.43
3	Services	172.35	103.32
	Total	296.75	189.03

tries, whereas SIDBI caters to the financial needs of small scale sector. AIDBs also undertake promotional and developmental activities. Among the SFIs, RCTC and TDICI provide risk capital, venture capital and technology finance, mostly to start-up companies in the knowledge-based IT and related sectors, while TFCI extends assistance to hotels and tourism-related projects. Among the investment institutions, LIC deals in life insurance business, while GIC provides general insurance cover. LIC and GIC deploy their funds in accordance with the Government guidelines. UTI mobilizes savings of small investors through sale of units and channelises them into corporate investments mainly by way of secondary capital market operations. Besides, the investment institutions also extend assistance to industry through loans and by way of underwriting/direct subscription to equities and debentures. SFCs provide assistance mainly to small and medium enterprises, while SIDCs cater to medium and large scale industries in their respective states. Apart from providing financial assistance, SFCs and SIDCs also undertake promotional and developmental activities.

Table 3.5: Advances to priority sector and others

Sl. No.		Advances (Rs. in crore)	
			2003-04 (up to September)
1	Priority sector advances	1007.73	1057.41
3	Advances to Weaker sections	51.59	123.89
3	Advances to SC/ST	3.81	4.50

3. GOVERNMENT SPONSORED SCHEMES/ PROGRAMMES

For effective implementation of Government Sponsored Schemes/Programmes performance of banks are reviewed by the State Level Bankers Committee, District Level Consultative Committee and Block Level Bankers Committee regularly. The banks, government departments, NABARD and various other institutions represent these committees. Towards ensuring successful implementation of government sponsored schemes and generate self employment opportunities the banks had set

Table 3.6: Assistance sanctioned & disbursed by the Financial Institutions in Goa : 2000-01

(Rs. in crores)

Sl. No.		Amount sanctioned	Amount disbursed
1	Industrial Development Bank of India (IDBI)	81.08	64.48
2	Industrial Finance Corporation of India (IFCI)	4.12	29.55
3	ICICI	38.00	40.53
4	Small Industries Development Bank of India (SIDBI)	53.54	49.87
5	Industrial Investment Bank of India (IIBI)	0.00	0.00
6	Tourism Finance Corporation of India Ltd. (TFCI)	0.00	2.10
7	Life Insurance Corporation of India (LIC)	3.00	3.00
8	General Insurance Corporation of India (GIC)	0.00	2.00
9	State Industrial Development Corporation (SIDCs)	26.15	37.93
10	NABARD (outside SGSY)	13.49	13.49
	Total	**219.38**	**242.95**

Source: IDBI, Report on Development Banking in India, 2000-01.

Assistance sanctioned and disbursed by the financial institutions in Goa during 2000-01 is given in table 3.6.

in the year 2000 a target of forming 734 Self Help Groups (SHGs) in Goa. Up to December 2003 the banks have made steady progress and formed 562 SHGs. Performance under various schemes is as follows.

(i) Swarnajayanti Gram Swarozgar Yojana (SGSY)

This is a holistic programme covering all aspects of self employment such as organization of poor into self help groups, training, credit, technology, infrastructure and marketing. The objective is to bring every assisted family above the poverty line within three years. The beneficiaries may be individuals or groups. Under the programme, four to five key activities are to be identified for each block based on resources, occupational skills of the people and availability of market. Selection of key activities is with the approval of panchayat samitis at the block level and DRDA at the district level. Subsidy under SGSY would be 50% of the project cost up to a maximum of Rs. 10,000/- for SC/STs and 30% of the project cost up to a maximum of Rs. 7,500/- for others. For self help groups, subsidy would be 50% of the project cost subject to a ceiling of Rs. 1.25 lakh. There is no limit on subsidy for irrigation projects. Performance of banks under this programme during 2002-03 and 2003-04 (September 2003) is given in table 3.7.

Table 3.7: Performance of banks under SGSY

Sl. No.		SGSY (Rs. in lakh)	
			2003-04 (up to September)
1	No. of applications sanctioned	517	139
3	Amount sanctioned	117.86	31.72
3	Amount disbursed	106.51	29.59

(ii) Swarna Jayanti Sahari Rojgar Yojana (SJSRY)

The scheme has two components viz (a) Urban Self-Employment Programme (USEP) and (b) Urban Wage Employment Programme (UWEP). The USEP provides assistance to individual urban poor for setting up gainful self-employment ventures, development of women and children in urban areas (DWCUA), assistance to women to set up self-employment ventures in groups and training of beneficiaries for upgradation and acquisition of vocational and entrepreneurial skills. Under self-employment component, maximum loan admissible is Rs. 50,000/- with a subsidy of 15% of the project cost subject to a limit of Rs. 7,500/-. Training for skill development is allowed at a cost of Rs. 2,000/- per trainee. The DWCUA group is entitled to a subsidy of Rs.1,25,000/- or 50% of project cost whichever is less. If this group by itself forms a thrift and credit society, the group is entitled to a lump sum grant of Rs. 25,000/- as revolving fund at the rate of Rs. 1,000/- per member. The UWEP seeks to provide wage employment to beneficiaries living below the poverty line by utilizing their labour for construction of socially and economically useful public assets. Performance of banks under SJSRY during 2002-03 and 2003-04 (September 2003) is in table 3.8.

Table 3.8: Performance of banks under SJSRY

Sl. No.		SJSRY (Rs. in lakh)	
			2003-04 (up to September)
1	No. of applications sanctioned	128	11
3	Amount sanctioned	39.12	2.73
3	Amount disbursed	37.92	2.23

(iii) Prime Minister's Rojgar Yojana (PMRY)

The scheme aims at generating self-employment opportunities for all educated unemployed in the age group of 18-35 years with a 10 years relaxation to SC/STs, ex-servicemen, physically handicapped and women. The scheme covers all economic activities including agriculture and allied activities but excludes direct agricultural operations like raising crop, purchase of manure etc. Project cost is upto Rs.1.00 lakh for business sector and Rs.2.00 lakh for other activities. If two or more eligible persons join together in a partnership, project up to Rs. 10.00 lakh is covered. Subsidy will be limited to 15% of the project cost subject to ceiling of Rs.7,500/- per entrepreneur. Performance of banks under this scheme during 2002-03 and 2003-04 (up to September 2003) is in table 3.9.

(iv) Margin Money Scheme

The scheme envisages providing financial assistance for any industry/business activity (excluding the activities in the negative list) to generate gainful employment in rural areas. Under the scheme an amount up to Rs. 25.00 lakh is given to set up a project. An individual can undertake a project up to Rs. 25.00 lakh. The Khadi and Village Industries Board, Government of Goa operates the scheme. Performance of banks under this scheme (KVIC/KVIB) during 2002-03 and 2003-04 (Upto September) is given in table 3.10.

Table 3.9: Performance of banks under PMRY

Sl. No.		PMRY (Rs. in lakh)	
			2003-04 (up to September)
1	No. of applications sanctioned	253	66
3	Amount sanctioned	180.80	46.30
3	Amount disbursed	156.22	40.84

Table 3.10: Performance of banks under Margin Money Scheme

Sl. No.		Margin money scheme (Rs. in lakh)	
			2003-04 (up to September)
1	No. of applications sanctioned	255	86
2	Amount sanctioned	436.11	156.31
3	Amount disbursed	389.24	138.12

(v) Bio-gas

Under this scheme, a beneficiary is eligible for a subsidy of Rs.1800/- from Central Government and Rs.4000/- from the State Government to set up a bio-gas plant. However, the central subsidy for SC/ST beneficiary, small and marginal farmers and farmers in western ghat areas is Rs. 2300/-. Performance of banks under bio-gas programme during 2002-03 and 2003-04 (up to September 2003) is in table 3.11.

Table 3.11: Performance of banks under Bio-gas programme

Sl. No.		Bio-gas (Rs. in lakh)	
			2003-04 (up to September)
1	No. of applications sanctioned	10	5
3	Amount sanctioned	0.71	1.04
3	Amount disbursed	0.46	1.04

4. NABARD

NABARD is an apex financial institution, established by an Act of Parliament, for promotion of rural development, extending refinance support to banks for financing agriculture and allied activities, non-farm sector activities, Government sponsored programmes such as SGSY, PMRY, KVIC/KVIB assisted schemes etc. Besides, NABARD has been mandated to carry out the inspection of the Goa State Cooperative Bank Ltd. (GSCB), under sections 35(6) of the B.R. Act.

a. Refinance support to banks

The refinance support extended to banks during the last four years is given in table 3.12

While the Goa State Cooperative Bank has been availing refinance to the extent possible to augment its resources to finance various activities, commercial banks, due to comfortable funds position have been availing 'need based' refinance. The major sectors to which refinance was channeled are given in table 3.13.

b. Formation of Self-Help Groups (SHGs)

Besides, NABARD has been supporting formation and linkage of SHGs in a big way. NABARD has extended grant assistance to NGOs, etc. for conducting skill upgradation, orientation and exposure programmes for SHGs. The number of SHGs in Goa (SGSY SHGs excluded), as on 31.12.2003, is given in table 3.14. It is pertinent to note that though 562 SHGs are formed by opening bank accounts, so far only 160 SHGs have availed finan-

Table 3.12 : NABARD - Refinance Support extended to banks

(Rs. in crore)

Agency / Type of Loan Term Loans (Long Term)	1999-2000	2000-2001	2001-2002	2002-2003	2003-04 upto 31st December 2003
Commercial Banks	6.60	8.01	7.84	2.13	—
Goa SCB Ltd.	8.42	9.20	11.26	17.07	1.79
Total Term Loan Refinance	15.02	17.21	19.10	19.20	1.79

cial support from the banks. About 57% of the SHGs are formed by the Banks but their linking with financial support is very low.

c. Loan to State Government under the Rural Infrastructure Development Fund (RIDF) for creation of infrastructure facilities in rural areas

In order to boost up investments in rural areas, creation of adequate infrastructure to support such investments is being financed by NABARD to the State Government out of its RIDF corpus.

The current interest rate on RIDF loan is Bank rate plus 0.5%, which is now 6.00% +0.5% = 6.50%. The repayment period for RIDF loans is 7 years with 2 years grace, which is being worked out separately for each installment of the loan released. Besides, mobilization advance ranging between 10% to 30% is being sanctioned, on case-by-case basis, to facilitate early commencement of works. The various activities which are eligible for loan assistance under RIDF are given below :

- Irrigation – Minor and Major
- Rural Roads
- Bridges

Table 3.13: NABARD - Sector-wise Refinance

(Rs. in crore)

					2003-04 upto Dec 2003
Minor Irrigation	0.522	0.293	0.150	0.057	0.005
Land Development	0.511	0.589	0.542	0.299	0.040
Farm Mechanisation	0.123	0.160	0.204	0.071	-
Plantation / Horticulture	0.120	0.198	0.112	0.504	0.014
Dairy Development	0.423	0.506	0.757	0.510	0.006
Poultry	0.114	0.157	0.165	0.059	-
Sheep / Goat / Pig Rearing	0.002	0.010	0.017	0.003	-
Bio-Gas	0.002	0.007	0.013	0.001	0.001
Fisheries – Marine, Inland, Brackish	0.766	0.595	0.477	0.119	0.031
IRDP / SGSY-FS	-	0.073	0.182	0.018	-
IRDP / SGSY-ISB	0.153	0.107	0.032	0.092	-
Others (incl. SHG, PMRY, etc.)	-	0.189	0.595	0.410	0.033
Total	2.736	2.884	3.246	2.143	0.130
Non-Farm Sector	12.282	14.325	15.856	17.058	1.657
Grand Total	15.018	17.209	19.102	19.201	1.787

Table 3.14: NABARD - Self Help Groups

(Rs. in lakh)

Name of the Self-Help Group Promoting Institution (SHPI)	SHGs promoted	SHGs linked	Bank loan disbursed	Refinance released by NABARD
National Cooperative Union of India (NCUI)	125	83	95.792	46.688
Goa Milk Union (GMU)	39	10	8.082	6.197
Goa State Cooperative Women's Federation(GSCWF)	5	2	1.100	1.048
Jan Jagruti Sanstha (JJS)	5	3	0.750	0.750
Farmers' Clubs under VVV##	16	6	4.220	3.907
Banks	318	47	38.336	23.539
Goa SC/ST Federation	17	-	-	-
GVK	32	8	2.180	0.050
BDO	5	1	0.672	0.672
Total	562	160	151.132	82.851

programme of NABARD

- Water Supply
- Sanitation
- Soil and Water Conservation
- Rural Market Yards
- Education
- Health
- E-governance
- Drainage, Reclamation, Flood Protection etc.
- Inland Navigation
- Fishing Jetty
- Tourism Related Activities
- Strengthening of Power Distribution in Rural Areas

Priority is given to projects which benefit farmers more directly viz. projects relating to irrigation, flood control, agriculture and allied activities, system improvement in power sector with 95% loan assistance, social sector with 85% loan assistance and rural connectivity with 80% loan assistance. The sector-wise sanction of RIDF loans, as at the end of January 2004, is in table 3.15. No loan was availed under RIDF II, III and VI tranches. The position of RIDF supported infrastructure projects in Goa is furnished below :

RIDF-I (1995-96)

Salaulim Irrigation Project : The Government of Goa has availed a loan of Rs.6.85 crore for this project.

RIDF-IV (1998-99)

Chapoli Irrigation Project : The project was sanctioned with loan assistance of Rs.1.85 crore. The project envisages creation of irrigation potential in 550 ha. of land, besides supply of water for industrial and domestic purposes. The project has since been completed and eligible loan amount has also been drawn by the Government of Goa.

Siolim-Chopdem Bridge: The NABARD has partly financed the construction of Siolim-Chopdem Bridge. The State has availed a loan of Rs.4.25 crore from NABARD. The bridge has been completed and opened for public. .

Minor Irrigation Projects : 20 Minor Irrigation Projects for construction of tank, bandharas and lift irrigation, involving total RIDF loan of Rs.2.82 crore was sanctioned to the Govt. of Goa in January, 1999. All the projects have since been completed and brought in 623 ha. under assured irrigation. So far, Rs.2.73 crore have been disbursed by NABARD to the Government of Goa.

Table 3.15: Sector-wise sanction of RIDF loans

(Rs. in crore)

	RIDF-I			RIDF-IV		
	No. of Projects	Amount		No. of Projects	Amount	
Minor Irrigation	-	-	-	21	4.680	4.440
Major Irrigation	1	36.840	*6.85	-	-	-
Bridges	-	-	-	1	4.255	4.255
Total	1	36.840	6.850	22	8.935	8.695

* includes Rs. 1.403 crore called back due to closure/withdrawal of the scheme (Salaulim Irrigation Project)

Table 3.15: Sector-wise sanction of RIDF loans

(Rs. in crore)

	RIDF-VI			RIDF-VII		
	No. of Projects	Amount		No. of Projects	Amount	
Minor Irrigation	5	5.017	0.597	1	4.477	3.432
Roads	37	9.739	6.443	60	10.695	5.883
Bridges	9	2.161	0.377	3	0.618	0.595
Rural Water Supply	9	2.168	1.551	-	-	-
Total	**60**	**19.086**	**8.968**	**64**	**15.790**	**9.910**

RIDF-VI (2000-01)

5 Lift Irrigation Projects (LIS) with a total loan of Rs.5.02 crore was sanctioned in October 2000. However, work only of two projects has been taken up by the Government of which the LIS at Nanora is completed and at Zanodem is in progress. The amount of loan availed is Rs. 0.60 crore. 37 roads and 9 bridge projects with a loan assistance of Rs. 11.90 crore was sanctioned by NABARD. 31 roads and 3 bridges have been completed. The amount of loan availed is Rs. 6.82 crore. 9 rural drinking

Table 3.15 (concluded): Sector-wise sanction of RIDF loans

(Rs. in crore)

	RIDF-VIII		
	No. of Projects	Amount	
Minor Irrigation	-	-	-
Roads	-	-	-
Bridges	-	-	-
Rural Water Supply	2	16.098	10.285
Total	2	16.098	10.285

water projects in South Goa were sanctioned in December 2000 with loan assistance of Rs. 2.16 crore. 7 projects have been completed. The loan assistance availed amounts to Rs. 1.55 crore.

RIDF-VII (2001-02)

A project for construction of 6 bandharas on the river Khandepar to irrigate a command area of 1300 ha. and augment the water supply to the Opa water treatment plant has been sanctioned by NABARD with financial assistance of Rs.4.47 crore. All the bhandaras have been completed. Rs. 3.43 crore assistance has been availed against the sanction. Another project for construction / improvement of 60 roads and 3 bridges with a financial assistance of Rs.11.31 crore has also been sanctioned to the Government of Goa under RIDF-VII of which 53 roads and 3 bridges have been completed. An assistance of Rs. 6.48 crore has been availed.

RIDF-VIII(2002-03)

Two projects with loan assistance of Rs. 16.01 crore were sanctioned for the augmentation of water to the Opa and Assanora water treatment plants. Six bandharas have been constructed on the river Kalay and water will be supplied on need basis by pumping from Salaulim Dam. Similarly, bandharas have been constructed for supply of water to Assanora water works in North Goa. Loan assistance of Rs. 10.28 crore has been availed.

d. Assistance to Cooperative Sector

To support the cooperative sector, grant / interest-free loans have been granted by NABARD. The details are given in table 3.16.

Table 3.16: Assistance to Cooperative Sector (Rs in lakh)

			Interest free loan	
GSCB	Computerisation	14.10	0.65	9.40
	Corporate Plan Preparation	-	23.50	0.65
PACS	Infrastructure – 7 PACS	3.00	-	3.00
	Total	**17.75**	**9.40**	**27.15**

e. Kisan Credit Cards (KCCs)

Special emphasis is being laid by NABARD for issue of KCCs by all banks, especially Cooperative banks. The position of implementation of the scheme, as at the end of November, 2003, is given in table 3.17.

The number of KCC holders covered under the Personal Accident Insurance Scheme (PAIS) is 890 by commercial banks and 891 by GSCB respectively.

Table 3.17 : Kisan Credit Cards

	KCCs issued (cumulative)	Amount involved (Rs. in lakh)
Commercial Banks	1049	231.16
Goa SCB Ltd.	2894	711.06
Total	3943	942.22

OVERVIEW

The State has an excellent banking network and on an average there is a bank for a population of 3000. The State has a very high per capita bank deposits but a low CD ratio. Various credit-linked schemes announced by the Government during 2003-04 like Kamdhenu scheme, Krishi Ghars, interest subsidy scheme, venture capital scheme etc. as per the new Industrial Policy are likely to boost investment climate in the State, which would improve the CD ratio. These new initiatives along with some more new schemes like Swabhiman are expected to boost the formation of Self Help Groups in the State. The formation of Self Help Groups has gained momentum in the State in recent years. But, out of 562 SHGs formed so far, only 160 are linked for financing, which indicate that majority of SHGs have just opened accounts in Banks and may not have started any activity. As such, it is necessary to guide the SHGs to make them functional.

CHAPTER 4
Infrastructure

The Government of Goa has given paramount importance on infrastructure development and on increasing the pace of implementation of various projects. This chapter covers progress made in implementing various infrastructure projects in different sectors of Power, Transport, Water Resources, Sanitation & Sewerage, Urban Development and Public Works.

1. POWER

The Government had executed a Memorandum of Agreement with the Ministry of Power Government of India for implementation of various schemes and other measures for improving the service to the people. This includes benchmarking certain parameters and fixing targets for improvement of these parameters, and implementation of APDRP. The schemes and measures are under implementation.

a. Accelerated Power Development and Reforms Programme (APDRP)

The APDRP, launched by the Ministry of Power, Government of India, aims to prioritize the strengthening/upgrading of sub-transmission and distribution network across the country. This programme is aimed at developing a center of excellence with the S.E.B. system so that the investment in the sub-transmission and distribution network of the State results in quantifiable physical and financial benefits. The programme is funded on 1:1 basis between the Central and State governments. The 50% central share is in the form of 25% loan and 25% grant and the state share of 50% would need to be arranged either from state's own funds or through PFC/REC. The Powergrid Limited has been appointed as advisor cum consultant by the Ministry of Power, Government of India for formulation of schemes, preparation of detailed project report (DPR) and supervision and implementation of schemes.

In Goa, the APDRP is implemented in both North and South Goa Districts. Projects amounting to Rs.159.90 crore have been approved, which comprises Rs. 59.67 crore in North Goa circle, Rs. 53.86 crore in South Goa circle and Rs.46.37 crore for SCADA(Supervisory Control and Data Acquisition) and DMS (Distribution Management System) covering entire Goa. The scope of works under the APDRP include meters and meter testing facilities, R & M of substations, strengthening and upgradation of sub-transmission and distribution system etc. The scope of work approved under APDRP for South Goa circle and North Goa circle are given in Table 4.1 and 4.2.

Projects under the APDRP are under implementation in both North Goa and South Goa circles. Works are undertaken partly through Power Grid Corporation of India Ltd., and also are being partly executed departmentally. During the year 2003-04, works worth Rs. 30 crore on deposit basis to Power Grid Corporation of India Ltd. (POWERGRID) for South Goa Circle have been approved. These works include strengthening of the Sub-transmission and HT network upto 11KV and are critical for improving the power distribution network and improving the quality and reliability of power. Works have been awarded to POWERGRID in November 2003 and POWERGRID has already tendered out these works and also fixed up the agencies and the commencement of work initiated by the laying of foundation stone for 33/11 KV Substation at Konkan Railway (Navelim) on 9.1.2004.

b. Underground network in 5 major cities

Feasibility studies for conversion of overhead network to underground network in 5 major cities of Goa namely Panaji, Margao, Mapusa, Ponda, and Vasco have been carried out. Detailed Project Reports amounting to Rs.153.00 crore have been prepared and submitted to the Ministry of Power, Government of India for sanction under APDRP. The Ministry has so far sanctioned schemes for conversion of overhead to underground network in Panaji and Margao cities at a cost of Rs. 68.26 crore.

Table 4.1: Scope of works under APDRP for South Goa Circle

Sl. No.			Estimated cost (Rs. in lakh)
1	Meters and Meter testing facilities etc		
	i) Meter Testing facilities	2 Nos	20.00
	ii) System (33 & 11 Kv) Trivector meters	1459 Nos	221.15
	iii) Consumer meters (1-ph & 3 -ph)	56077 Nos	1030.66
II	R & M of sub stations		
	R & M of of 33/11 kv Sub- stations	17 Nos	577.64
	R & M of distribution sub- stations	L.S	51.45
III	Strengthening and upgradation of sub transmission and distribution system		
A	Sub-Stations		
	i) New Sub - Stations	4 Nos	392.00
	ii) Sub -Station augmentation	4 Nos	69.00
	iii) Bay Extensions at Sub-Stations	18 Nos	11.70
B	Sub Transmission lines		
	i) New Lines	105.7 ckm	475.65
	ii) Re-conductoring	27 ckm	63.45
C	HT Lines		
	i) New lines	267.8 ckm	562.38
D	LT lines		816.74
	i) New lines	67.6 ckm	67.60
	ii) Reconductoring	691.0 ckm	483.70
E	Distribution sub-stations		
	New & Augmentation	157 Nos	264.18
F	New Service lines	LS	10.00
IV	Modernization works (Indexing)	LS	12.00
V	Supervision Charges @ 5%		256.46
	TOTAL		**5385.76**

c. Resource Mobilisation

The Government has been trading surplus power to other States through the Power Trading Corporation and Global Energy Ltd. After implementation of the Availability Based Tariff (ABT) scheme in the Western region, the Department by proper scheduling of declaration of availability and load management has been earning some revenue by way of pooled Unscheduled Interchange charges (UI) levied on other defaulting States. The revenue anticipated during the year on the above measures is Rs. 134 crore.

By strict implementation of the financial reduction plan through the third Supplementary Agreement executed with RSPCL, the Government would be reducing the financial burden on drawal of power from RSPCL. The present drawal from RSPCL has been brought down from 39.8 MW to 23 MW.

Monthly revenue collection target has been fixed

Table 4.2: Scope of works under APDRP for North Goa Circle

Sl. No.		As per DPR	
			Amount (Rs. in lakh)
1	**METERING**		
	a 33/11 KV feeders meters		0.5
	b DTC Meters	1651	325.63
	c Consumers Meters	37578	570
	d Meters test bench	4	40
	e Computerised data logging system	21	63
	f Computerised billing system	13	260
	g Indexing of consumer network	L.S	10
	h Feasibility study for DMS	L.S	15
	i Improving service connection	L.S	100
	Total		**1384.13**
2	R& M of Substations		
	a Revamping of existing 33/11 Kv S/S	21	190.83
	b Augmentation of Transformer capacity	14	343.9
	c Providing capacitors at 110/33 kv s/s	2x10	183.26
	Total		**717.99**
3	Reconductoring /bifurcaion of 33 kv & 11 kv Feeders		
	a Establishment of new 33/11 kv s/s	1	102.17
	b New 33 kv lines	22km	341.96
	c Re-conductoring/bifurcation of 11 kv lines	260.92 km	1113.34
	d Renovation of 33 kv lines	86.6	312.25
	Total		**1869.72**
4	Renovation of DTC's		
a	Revamping of DTC	LS	270.94
b	New LT lines/reconductoring of LT lines	500km	1178.53
c	Feasibility studies for conversion of O/H n/w into U/G n/w	5	75
	Total		**345.94**
5	Providing Alternate power supply to Panaji		440.51
6	Miscellaneous Works		30.25
	GRAND TOTAL		**5967.12**

for all the O&M Divisions and monitored carefully. Billing and revenue collection, disconnection of installations are observed strictly. Surprise inspection of installation by MRT has been speeded up and it is yielding good revenue, by curtailing the theft of energy. The revenue is being monitored on day to day basis for effective control. Simultaneously, in order to provide quick, efficient and round the clock service to people the Governemnt has initiated mobile van concept to attend to the complaints in short period of time (Box 4.1).

d. EHV Works

The 220KV Colvale to Ponda D/C line has been commissioned in May 2003. Thus power from the Western Region is being fully fed through the 400KV Station to Northern and Central part of Goa. The work of erection of 220/33 KV, 1x50 MVA Substation at Xeldem has been tendered out in two parts. The technical bid is expected to be finalized shortly and thereafter financial bids will be finalized and work order placed by end of this financial year.

e. Measures for energy audit and reduction of Losses

Under the APDRP scheme, the distribution transformer centres are being provided with meters. Already all the feeders at sub-stations upto ll KV have been provided with electronic meters. After completion of the DTC, energy audit will be carried out feeder-wise and losses will be plugged.

2. TRANSPORT

a. Road Transport

i) Motor vehicles

The total motor vehicle population in Goa as on 31-3-2003 is 3,91,430. On an average 25000 new vehicles are registered every year. The distribution of vehicles by type is given at Annexure 31. It is evident that about 72% of vehicles are in the category of motorcycles followed by cars and jeeps including taxis about 18%. The number of new vehicles registered during the period 1992-93 to 2002-03 is at Annexure 32.

Box 4.1
Mobile Consumer Service

In order to provide quick, efficient and round the clock service to the people, the Government of Goa has initiated mobile van service in the major cities of Panaji, Mapusa, Vasco, Margao, Fatorda, Ponda, Curchorem, Bicholim, Canacona, Sanguem, Pernem, Porvorim and Valpoi. Presently, 13 mobile vans equipped with necessary infrastructure, personnel and a wireless set are pressed into service in these cities. Complaints received at the call centres are transmitted to these mobile vans through wireless. One more mobile van is also being deployed to Quepem shortly. Hon'ble Union Minister for Power in his recent visit to the State has promised to provide additional 25 mobile vans so as to extend the service to the villages.

(ii) Subsidy for replacement of buses and minibuses 15 years old and more

During the year 2001-02, Government of Goa introduced a unique scheme for grant of subsidy for replacement of buses and minibuses, which are 15 years old and above, in order to curtail vehicular pollution . The scheme covers the owners of such buses and minibuses having valid regular permits and which are in operation for a continuous period of last 3 years. The scheme provides for disbursement of subsidy ranging from Rs. 1.50 lakh to Rs. 2.25 lakh depending upon the age of the vehicle. Upto December 2003, an amount of Rs. 1.60 crore has been disbursed for purchase of 83 new buses (including 24 buses owned by KTCL). This scheme of the Government of Goa is the first of its kind in the country and has been widely welcomed

by the bus owners, commuters and environmentalists.

(iii) Goa State Subsidy for Yellow/Black Motor cycles, autorickshaws and taxis scheme 2003

Government of Goa has launched a subsidy scheme on 1st August 2003 for replacement and procurement of new yellow/black motor cycles, yellow/black autorickshaws and yellow/black taxis in order to support this small scale self employed vulnerable section of society and to ensure that old, unsafe and polluting passenger transport vehicles are removed from the roads to protect environment and to enhance road safety. The scheme also intends to provide an efficient reliable and comfortable service to the passengers.

The scheme provides for disbursement of 25% subsidy for purchase of new vehicles or for replacement of vehicles subject to the ceiling of Rs. 12,000/- for motorcycle , Rs. 20,000/- for autorickshaw and Rs. 80,000/- for taxi. Under this scheme more than Rs. 13.00 lakh have been sanctioned for purchase of 48 motorcycles, 9 autorickshaws and 13 yellow/black taxis till December 2003. Government has made available the loan amount to the beneficiaries under the scheme at a concessional rate and easy terms of repayment through the Chief Minster's Rojgar Yojana.

b. Airport

Government of India has conveyed in principle the approval for building a new international airport at Mopa. Government of Goa has already initiated steps for expeditious acquisition of land and engagement of international consultant. The mode of financing etc., are being finalized. The airport is expected to be built on BOOT basis. However, the Government of Goa may have to spend about Rs.200 crore towards land and other consequential expenses.

c. Sky Bus Metro

Government of Goa has submitted a proposal to the Ministry of Urban Development, Government of India for introduction of Metro Sky Bus as a pilot project in Goa. The Konkan Railway Corporation is in the process of establishing necessary infrastructure for the test run of this new technology. Regarding technology and other aspects of Sky Bus Metro refer Box 4.2. Many States in India and different countries have shown interest in this technology. The Konkan Railway has received tremendous response from abroad and different States in India and expression of interest indicated by them are given in table 4.3.

d. Inland Water Transport

(i) Maritime School

The Maritime school under the Captain of Ports Department of Government of Goa is the first of its kind in this country imparting Seamanship and Marine Engineering training to the personnel working/seeking employment on inland water and other category of transport mechanised vessels.

The school was started on lst August 1972, in the campus area admeasuirng 5,500 m^2 on the bank of river Mandovi at Britona in Bardez taluka. In terms of infrastructure the school has spacious buildings, staff quarter, pump room, garage, school bus, workshop, hostel facilities for 60 students, library etc. There is also a jetty constructed at the river frontage within the campus of Maritime school, which is utilized for conducting boat practical on the boat belonging to the School. The present modest infrastructure and the facilities available at the Institute can further be augmented in order to upgrade the training standard. The Institute conducts mainly the following two types of courses.

- **New Entrants Training Course**: This is a six months duration course from October to March for DECK and ENGINE ROOM candidates. This course helps the candidate to join as a seamen/sailor or an engine room oilman in Deck & Engine room departments on an Inland vessel. The age limit is 18 years and above and should be minimum VIIIth Std passed. Swimming and being physically fit are the essentials, which has to be recommended by the selection committee of the school. The fees are Rs. 60/- for six months.

- **Advanced Refresher Course:** This is a three month duration course from June to August. This course is prescribed for the candidates who are qualified to upgrade of competency certificate of examination such as lst class engine driver, Inland engineer in Engine room department or IInd class Master and lst class Master in Deck department. The age

BOX 4.2
SKY BUS METRO

All over the world, metro populations are rising and the infrastructure development is not keeping pace. The roads are unable to handle the rising number of vehicles and metro rails face inadequacies in terms of Points and Crossings which prove to be the weak link in increasing the capacity, besides facing the risk of vandalism and derailment. All administrations are in search of an economically viable solution to the transportation problem which is also environment friendly. The present conventional capital intensive technologies are proving to be financially unviable and require heavy subsidy which is not practical.

The new technological solution in terms of Sky Bus metro is based on the concept of Sky Wheels presented in 1989 at World Congress for Railway Research. This innovative alternative transportation solution completely eliminates any probability of vandalism, accident and is virtually maintenance-free. The Technology comprises well proven rail guided system commonly used for normal railway system; Liner Induction Motor Technology or 3 phase asynchronous AAC electrical motive units-well proven and Widely adopted abroad as well as in India; Light weight coaches called "Sky Buses" which are suspended from bogies and travel below rail guides; pre-fabricated latest construction technologies which save time and money resulting in easy execution of the project in busy urban areas without disturbing the existing traffic pattern and information technology tool for economic communications and control. These structural engineering methods are well proven which do not have any project execution risk attached. The project will be of world class standard and will place India in the forefront of providing the much needed alternative transportation solution which is financially viable, environment friendly, synergising well proven existing cutting edge technologies.

Advantages of the Technology would include fast and economical transportation, no land acquisition problems, no demolition problems, no vandalism, fire protection – fastest evacuation in case of fire as compared to underground metros, no capsizing – if at all there is a derailment, the coach keeps hanging and does not fall down, no run over accidents, deep penetration, low capital cost, low operational cost, no interference with normal road traffic, fast execution, it can be built on roads with flyovers, no pollution, no traffic jams, no waiting etc.

Source: Konkan Railway Corporation

limit is 18 years and above and a certificate of competency is essential for the holder of either Deck /Engine room department. The fees range from Rs. 60/- to Rs. 150/- according to the category of the trainees.

Till date, 63 batches of Advanced /Refresher course and 37 batches of New Entrants Training Course have been completed.

(ii) Hydrographic Surveys

The State of Goa has 555 kms of inland waterways. These include certain portion which are used for the plying of the mechanised self propelled vessels/crafts having crafts of more than 2 metres that requires navigational safety measures so as to ply in the inland waterways without any navigational hazards. In order to ensure that the navigable waterways are safe for navigation, the hydrographic surveys of the inland waterways is very essential.

The hydrographic survey is a systemized scientific study of the water body in the rivers, seas and oceans in which the nature of the underwater topography is depicted and the final outcome/results of these hydrographic surveys is the production of the navigational charts through which the mariners can ply their vessels from one point to their destinations safely in the navigable waterways. The professional and special services in the conduct of hydrographic survey involves specialized and trained hydrographic surveyors and certain state of art equipment and vessels.

Table 4.3: Expression of interest for Sky Bus Metro Project in India and abroad

Sr. No			Estimated cost (Rs. in crore)
1	Ahmedabad	63.30	2882.10
2	Bangalore	99.90	4495.50
3	Chennai	118.05	5312.25
4	Coimbatore	28.20	1269.00
5	Delhi	124.90	6620.50
6	**Goa**	**82.60**	**3769.50**
7	Faridabad Gurgaon	84.00	3780.00
8	Hyderabad	42.00	1890.00
9	Kochi	30.00	1500.00
10	Kolkata	26.00	1090.00
11	Lucknow	17.90	824.00
12	Mumbai	111.50	5681.70
13	Noida	52.20	2349.00
14	Pune	87.40	3960.30
15	Thane	14.00	470.00
16	Baghdad (Iraq)	11.60	268.00
17	Damascus (Syria)	34.00	850.00
18	Singapore	34.00	1020.00
19	Dubai to Sharjah (UAE)	12.00	360.00
20	Dubai (UAE)	69.10	2073.00
21	Dhaka (Bangladesh)	23.10	580.00
22	Medina to Mecca (Saudi Arabia)	75.00	2250.00

Source: Konkan Railway Corporation

The last hydrographic survey in the inland waterways of Goa was conducted during the year 1967-76 by the Minor Ports Survey Organisation, Ministry of Surface Transport , Mumbai. Since the surveys were conducted many years ago, it had become outdated and it was also felt dangerous to sell the hydrographic survey charts since they were not realistic due to huge time gap and the siltations that had occurred in the river bed levels. It is also pertinent to mention that under the Hydrographic Survey Committee a perspective plan exists which caters to the hyddrographic surveys in all the major and minor ports of India. It is emphasized in this perspective plan that the hydrographic survey of the major ports in India should be conducted after every five years, while the hydrograpic surveys in the minor ports in India should be conducted after every ten years.

Accordingly, on approval of the Government of Goa, the Ports Department initiated the conduct of the hydrographic surveys in the inland waterways of Goa , in a phased manner. Surveys already conducted are as follows:

- Phase I River Mandovi survey was conducted from 6.3.2000 to 30.4.2001 at a total cost of Rs. 57.97 lakh.

- Phase II River Zuari survey was coducted during the period 11.11.2002 to 24.2.2003 at a total cost of Rs.35.50 lakh.

- Phase III Cumbarjua Canal survey was conducted during the period 10.3.2003 to 24.4.2003 at a total cost Rs. 17.75 lakh.

The hydrographic surveys of the remaining rivers viz. Mapusa river, Chapora river and Sal river are awaiting approval of the Government of India.

(iii) Dredging of Inland Waterways

In order to maintain inland waterways as well as the ferry routes and the ferry ramps for safe navigation, dredging work is undertaken by the Captain of Ports Department. For this purpose the Department is equipped with a pre-liberation grab dredger having grab capacity of 0.5 Cu. Meter and a carrying hold capacity of 30 Cu.Meter. In addition the Department has a hopper barge having a capacity of 300 Cu. Meters. Dredging works executed by the Department during the last three years are as follows.

- Dredging at Cortalim Ferry Ramp was carried out during February-March 2000 and quantity dredged was 717 Cu.M.

- Dredging at Sarmanas-Tonca Ferry route was carried out during March-May 2000 and quantity dredged was 870 Cu.M.

- Dredging at the Mouth of Tonca Canal was carried out during March 2001-June 2002 and quantity dredged was 8000 Cu.M.

- Dredging at Cortalim Ferry Ramp was carried out during October 2002 to March 2003 and quantity dredged was 4300 Cu.M.

- Dredging at the Captain of Ports Jetty, Panaji was carried during April 2003 and quantity dredged was 1200 Cu.M.

- Dredging at the Mouth of River Mandovi across the Aguada Sand Bar was carried out during September 2003 and quantity dredged was 300 Cu.M.

- Dredging at River Navigation Department Jetty at Panaji was carried out (along with a private dredger) during December 2003 and quantity dredged was 1500 Cu.M.

(iv) Catamaran Service to Mumbai

With the starting of Catamaran Service from 29-12-2003, Goa is connected to Mumbai through sea route. This is a private sector venture. The transport vessel has a capacity of 200 passengers in the lower deck and 25 passengers in the upper deck. Duration of the journey is about 9 hours and the tariff is Rs.1580/- and Rs.1980/- for lower and upper deck respectively.

e. Roads and Bridges

The progress achieved under this sector during the year 2003-04 is as follows.

- *Amona Khandola Bridge*: The construction of Amona-Khandola bridge, at a cost of Rs. 3498.00 lakh is in progress. So far 50% of the work has been completed. The bridge is scheduled for completion during 2004.

- *.Bridge at Paular on MDR 25*: Construction of a bridge at Paular on MDR-25 in Pale Constituency is likely to cost Rs. 104.00 lakh and the estimate is being processed for AA & ES.

- *Guleli Padeli Bridge*: A new light vehicle bridge at the sanctioned cost of Rs. 144.24 lakh is being taken up at Guleli–Padeli Madei river. The tender is under process of acceptance.

- *Construction of new Ribander – Patto bridge*: It will be taken up at a cost of Rs. 153.16 lakh. The Government has accorded AA & ES for this bridge. Re-tender is opened on 13-10-03 and is under process of acceptance.

- *Construction of single lane bridge at Akhada-St. Estevam*: It is being commenced at an estimated cost of Rs. 66.04 lakh. Work order has already been issued.

- *Ponda Bye-pass:* The first phase of Southern bye-pass from Dhavali to Farmagudi is completed and the work on phase II i.e. from Dhavali to Curti is taken up. The estimated cost of phase II is Rs. 350 lakh. The construction of bye-pass is completed.

- *Assonora Bye-pass*: The estimated cost is Rs. 145 lakh. Land acquisition formalities are being initiated.

- Black topping of Ponda by-pass road (Phase-II, Part-I) for length of 3.80 Km at a cost of Rs. 104.61 lakh is in progress.

f. National Highways

- Land acquisition for proposed realignment between Km 44 to 66 on NH 17, Ribandar bypass between Km 143 & Km 154 (notification released) and Margao bypass between Km 29/400 & 36/800 (Phase-I) on NH 17 between Km 21/00 & 22/00 on NH 17 are in progress.

- Improvement of accident-prone section by removing bottle necks costing Rs. 219.92 lakh from Km 84/00 to 150/00 on NH 4A is in progress.

- Construction of Mayem bridge approaches completed & revised estimate for bridge construction is with MORT for sanction.

- Construction of Mardol bypass between Km 131/200 & 133/600 on NH 4 is in final stage of completion.

- Realignment between Km 133/600 & 137/00 on NH4 - 90 % work is completed.

- Construction of two lane cable stayed Zuari bridge on NH 17- Tendering of two envelope system is in progress.

- IRQP works between Km 14 & 20 of PMS and Km 1/200 & 2/7 of PPS on NHD & Km 5/0 & 14/00 of PMS & Km 40/0 & 43/75 of PPS are in final stage of completion. IRQP work between Km 84/00 to 93/00 on NH 17 is completed.

3. WATER RESOURCES

a. Development of Madei, Zuari, Galjibag and Talpona River basins

As envisaged in the State Water Policy action has been initiated for development of water resources Madei , Zuari, Galjibag and Talpona River basins. Master Plans for the purpose have been prepared and 237 water resources projects have been identified. Water resources projects identified are 61 in Madei River Basin, 119 in Zuari River Basin, 41 in Talpona River Basin and 16 in Galjibag River basin. The process of preparation of detailed project reports have been initiated as under.

- The work of preparation of details project report for 7 identified hydropower generation projects in Madei River Basin has been entrusted to National Hydro Power Corporation , Faridabad.

- Actions for survey investigation and foundation exploration have been taken up for irrigation and domestic water supply in five minor projects at Gulelli, Charwane, Zarne and Ragada in Sattari taluka and at Karemol in Sanguem taluka in Madei river basin, four minor projects at Gaonkarwada, Kumari, Matse and Kushavati II in Zuari river basin and four minor projects at Karvan I, Karvan II, Gavane and Zilta wada in Talpona river basin.

b. Water Resources Development Programme

The major requirement of water for domestic and industrial need in the State is presently catered in North Goa District by Chandel water works for Pernem taluka, Assonora water works for Bicholim and Bardez taluka and Opa water works for Ponda and Tiswadi taluka and in South Goa District by Salauli water works for Salcete, Marmagao, Quepem and part of Sanguem Taluka and Chapoli water works for part of Canacona Taluka.

Less than average rainfall received during the monsoon of 2001 and 2002 resulted in shortage of water due to reduction of discharges in the river at Assonora and Opa water works in the lean season. In order to ensure availability of water throughout the year, water development programme for augmenting the water sources at Assonora water works and Opa water works was taken up.

The programme of augmentation of water sources at Assonora water works has been implemented including construction of series of six bandharas on Assonora river basin, one bandhara on river Chapora with lifting of water from Chapora river to Amthane Reservoir delivering 25 mld (62.50 lakh Cu.M) and improvement of one bandhara on river Valvonta costing Rs. 410.00 lakh creating the water storage potential of 118.00 lakh Cu.m. This is in addition to existing Amthane dam reservoir having storage of 59.7 lakh Cu.m. This will benefit by providing additional water supply for domestic and industrial use in Bardez and Bicholim Taluka, and ground water recharge along the bank of Assonora River in Bicholim Taluka.

The programme of augmentation of water sources at Opa water works has been implemented including construction of series of seven bandharas on Khandepar river , and five bandharas of Kalay river costing Rs.705.00 lakh creating the water storage potential of 16.09 lakh Cu.m and Lifting of water from Salaulim main canal to Kalay river delivering 50 mld (125 lakh Cu.M) by laying rising main and Gravity main of length 11.865 Kms. costing Rs. 1081.00 lakh.

Apart from this the Water Development programme is under implementation which includes Construction of One Barrage at Opa and two bandharas at Velipwada and Shigao on Khandepar river and one bandhara at Kalay and one Weir cum bandhara at Kharemol on Kalay River at an estimated cost of Rs. 979.00 lakh to create storage of 12.66 lakh cu.m. This will benefit by providing additional water supply for domestic and industrial purpose in Ponda and Tiswadi taluka and ground water recharge along the bank of Khandepar and Kalay river in Sanguem and Ponda taluka.

In addition to above, works have been initiated for construction of one Barrage at Ganjem on Madei River and four bandharas on Kushawati River in Quepem and Sanguem Taluka at an estimated cost of Rs. 848.00 lakh to create storage of 8.26 lakh cu.m. This will provide additional water supply for domestic and industrial use in Ponda , Satari talukas and Quepem and Sanguem taluka respectively and enable ground water recharge along the bank of Madei river in Ponda and Satari taluka and along the bank of Kushavati river in Quepem and Sanguem Taluka.

It is also planned to take up water development works in Canacona taluka by constructing weir at Karvan, bandhara at Ordhofond and Minor dam at Ziltawada to cater to the water supply in water scarce area of Gaondongari in Canacona taluka.

These important measures of the Government are in the direction of optimal utilization of water resources in the State and ensuring the availability of water to the people throughout the year.

c. Hydrology Project Phase II – Externally Aided Project

The need for reliable and easily accessible hydro-meteorological data base for various activities in planning and management of water resources to meet the challenges of ever increasing demand for reliable and good quality water supply for various uses such as domestic water supply systems, agriculture, industrial and power generation etc. is well accepted.

The main objective of the Hydrology project is to improve the institutional and organizational arrangements, technical capabilities and physical facilities available for measurement, validation, collation, analysis, transfer and dissemination of hydrological, hydro-meteorological and water quality data and for basic water resources evaluation within the concerned agencies of Central Government and participating States. The project is aimed at upgrading and expansion of hydrometric and Data Management, Institutional Strengthening with provision for Technical Assistance and training arrangements to support improvement of Institutional and Technical capabilities objectives.

The project was initially launched by the Government of India in December 1995 with International Development Association (World Bank) assistance of SDR 90.10 million. The Government of Netherlands has provided a grant-in- aid of Euro 14.84 million in the form of Technical Assistance (TA) under a bilateral Indo- Dutch agreement. The phase I of the project which was originally scheduled to be implemented in a period of six years has now finally closed on 31.12.2003.

The Phase II of the hydrology project now proposed by the Government of India includes horizontal and vertical extension component. Hydrological Information System (HIS) is proposed to be established in four more states (Punjab, Himachal Pradesh, Goa and Pondicherry) on the similar lines as was done in nine States (Andhra Pradesh, Chhatisgarh, Gujarat, Karnataka, Kerala, Madhya Pradesh , Maharashtra, Orissa and Tamil Nadu) under the Hydrology Project -I. The vertical extension component would involve a vertical shift from collection and processing HIS Data towards the use of such data in the development of decision support system (DSS) for integrated planning and management of water resources in river basins (and sub-basins) and including such activities as early flood warning, drought management, conjunctive use of surface and ground water and integrated operation of reservoirs The Government of India has already forwarded the proposal to the World Bank in December 2003 providing summary details of all participating agencies.

The proposal of the State of Goa in line of above includes the development of HIS for under surface, ground water and water quality sectors. The pro-

posal comprises upgrading 12 existing river gauge station and establishing 18 new river gauge stations and 6 meteorological stations and 2 full weather stations with the establishment of 2 data centers and 2 level two laboratories for water quality testing and 125 observatory wells and 20 piezometer for ground water monitoring . The project further envisages provision for Research and Development Consultancy, Real time study, Salinity and Easturian study, Reservoir siltation study, Remote sensing and GIS study. The project also includes provision for infrastructure development, office automation, computer hardware and software, equipment, vehicles etc. There is also provision for training, operation and maintenance and establishment for a period of six years. The cost of the Hydrology Project II for Goa is Rs. 2569.00 lakh.

d. Augmentation of regional water supply schemes

In order to cater to the ever increasing industrial, commercial, touristic and domestic demand for water supply and to provide drinking water supply facilities on long term and sustainable basis the Government accorded priority for augmentation of regional water supply schemes as given in table 4.4.

(j) Augmentation of Opa Water Supply Scheme

Augmentation of Opa Water Supply Scheme at an estimated cost of Rs. 83.83 crore consisting of con-

Table 4.4: Augmentation of regional water supply schemes

Sr. No.	Name of the scheme	Existing capacity MLD	Proposed augmentation MLD	Talukas covered	Remarks
1.	Opa W.S.S	75	40	Ponda Tiswadi	69% of the work is completed
2.	Assonora W.S.S.	42	30 MLD from 40 MLD WTP at Podossem under Sanquelim W.S.S.	Bardez	30 MLD in progress, 98% of the work is completed
3.	Sanquelim W.S.S.	12	10 MLD from 40 MLD WTP at Podossem under Sanquelim W.S.S.	Bicholim	The main distribution lines to the rural zones is in progress
4.	Salaulim W.S.S.	160	220 (Proposed)	Sanquem, Quepem, Salcette, Mormugao	To be taken up during Tenth Plan External Financial Assistance is being tried.
5.	Canacona W.S.S.	5	5	Canacona	- do-
6.	Dabose W.S.S.	5	10	Sattari	- do-
7.	Chandel W.S.S.	15	-	Pernem	Completed
8.	Madei W.S.S.	-	40 (proposed)	Parts of Bicholim, Ponda & sattari.	To be taken up during Tenth Plan. Feasibility is being worked out.

struction of 40 MLD water treatment plant at Curti, near Ponda, providing and laying 1100 mm dia M.S., 900mm & 700 mm dia D.I. conveying main from Curti to Panaji for a length of 32 kms are in progress. Construction of series of bandharas and Weir are taken up by Water Resources Department. About 69% of work is completed and the whole project is likely to be completed by June, 2004.

(ii) 15 MLD Regional Water Supply Scheme at Chandel, Pernem

The project costing Rs. 30.00 crore is already completed and commissioned. All the reservoirs including MBR have already been constructed. In view of proposed airport at Mopa and IT park near Pernem, it is proposed to augment the capacity of this regional scheme by additional 15 MLD during the Tenth Plan period.

(iii) Augmentation of Sanquelim Water Supply Scheme

Augmentation of Sanquelim Water Supply Scheme at an estimated cost of Rs. 35.00 crore consisting of 40 MLD water treatment plant at Podocem is completed and is under trial run. Providing and laying 500 mm and 450 mm dia D.I. pipes conveying main from Podocem to Assonora for a length of 15 kms and distribution mains for length of 55 kms covering Poriem, Honda, Pissurlem, Cudnem, Surla, Velgue, Pale Mencurem, Sal and Latambarcem villages in Bicholim and Sattari talukas are in progress and final stage. The pipe line work to Podocem to Assonora has been completed and the distribution mains in other areas will be completed by December, 2004.

(iv) Augmentation of Assonora Water Supply Scheme

Augmentation of Assonora water supply scheme at an estimated cost of Rs. 76.00 crore consisting of construction of 5000 cu.m MBR at Assonora Service reservoirs, providing and laying 1100 mm dia M.S. 900 and 750 mm dia D.I. pipes conveying main for a length of 24 kms from Assonora to Porvorim, distribution mains to Revora Nadora and Pirna villages for length of 12 kms is in progress and in final stage of completion. The scheme has been commissioned partially and the distribution reservoir, etc. are in progress in Porvorim.

(v) Madei Water Supply Scheme

The regional water supply scheme at Madei in Sattari taluka of 40 MLD capacity is proposed at a cost of Rs. 75 crore for improvement of service level in nearby villages in Sattari & Ponda talukas and the same is proposed to be taken up during the Tenth Plan period.

(vi) Augmentation of Salaulim Water Supply Scheme

The laying & jointing of conveying mains from MBR to various reservoirs in Margao is in progress. It is proposed to augment the capacity by additional 200 MLD to cope up with additional demands in Salcete and Mormugao talukas during the Tenth Plan period. Efforts are being made to avail external assistance for the purpose.

e. Improvements in existing Water Supply System in Salcete & Mormugao Talukas

It has been observed that due to number of tappings on main Salaulim Water Supply system trunk line, the distribution of water is not equitable. In order to make it equitable it is proposed to construct about 10 storage reservoirs in Salcete taluka as well as to improve distribution system by suitable extensions. As the water supply demand in the Port town of Vasco da Gama as well as industries in and around Sancoale are on the rise, a scheme of 10000 cu.m. capacity MBR with suitable sump & pumps etc. is proposed at Vasco and a 15000 cu.m. capacity MBR is proposed to be constructed at Verna to take care of bulk consumers especially in the high bracket of revenue. These works are proposed to be taken up during the Tenth Plan period.

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f. Augmentation of Rural and Urban Water Supply from Salaulim Impoundage and Strengthening of Sanitation Programme in six talukas of Goa

A concept proposal to augment water supply to South Goa, Ponda & Tiswadi talukas at an estimated cost of Rs. 299 crore on Build, Own, Operate & Transfer (BOOT) basis utilizing Salaulim impoundage at Xelpem to convey 220 MLD of water is under active consideration.

g. Strengthening of vulnerable stretches of Salaulim Water Supply Scheme

The 4.5 Kms stretch is already commissioned and the breakdowns in the said stretch of conveying main have been drastically reduced. The further stretch is also weak and is prone to develop cracks / breakdowns in near future. Hence, another 4.5 Kms of 1400 mm dia PSC conveying main is proposed for replacement with M.S. pipes at an estimated cost of Rs. 8.32 crores in the Tenth Five Year Plan.

h. Water Grid Network

It is proposed to have a water grid network for entire Goa and to have linkages with various Regional water supply schemes. The proposed water grid network will serve as an emergency solution. Under grid network it is proposed to extend:-

- Salauli water supply system to Agasaim & Pillar is in progress.
- Opa / Assonora W.S.S. to Porvorim & other areas
- Chandel W.S.S. to parts of Bardez.

i. Major and Medium Irrigation

(a) Tillari Irrigation Project

Tillari Irrigation Project is a joint venture of the Government of Maharashtra and the Government of Goa across river Tillari near village Tillari Wadi, in Sawantwadi taluka of Sindhudurg district in the State of Maharashtra. The project, after its completion, was expected to irrigate an area of 16,978 ha. (CCA) in Bicholim, Bardez and Pernem talukas of North Goa district of Goa State and 6,676 ha. (CCA) in Sawantwadi taluka of Sindhudurg district in Maharashtra State. The project was also expected to provide 57.43 M. cum of water for domestic and industrial purpose. The water started flowing to Goa only in December 2001 from the Terwanmeddhe pick-up weir in the State of Maharashtra which is expected to create an irrigation potential of 1200 ha. The progress of main components of the project upto December 2003 is given in table 4.5.

There has been an unprecedented cost and time over run in the project. Since the start of the project in 1978-79, there has been heavy urbanization in the State leading to decline in command area of the project. The cost of the project has increased from Rs.44.00 crore in 1978-79 to about Rs.1000 crore at present. There has been no re-appraisal of the project done so far. The practice of making annual financial provision in the budget and releasing share cost payment to the Government of Maharashtra and executing canal works in Goa has continued. The economic viability of the project considering requirement of water in the state was not assessed. Now, for the first time the Government of Goa has taken a major decision to totally review the project and assess the irrigable command area under project. The review process has already been set in motion.

Table 4.5: Progress under Tillari Irrigation Project

Sl. No.	Component	Percentage progress achieved during 2003-04	Cumulative percentage of progress achieved
1	Earth work of main dam	14%	90%
2	LBMC & Link Canal	6%	100%
3	Saddle dam, tail channel, approach channel	2%	100%
4	RBMC in Maharashtra	-	100%
5	RBMC in Goa	-	85%
6	LBMC in Goa	_	84%
7	Distributaries and branches	-	57%
8	Irrigation Potential		
	a. Created	156 ha	-
	b. Utilised	406 ha	-

Being a joint venture of Governments of Goa and Maharashtra there is an inter-state agreement for the project. As per the agreement, the cost of the project towards common works is to be shared in same ratio of requirement of water to each state i.e in the ratio of 73.30:26.70 between the Governments of Goa and Maharashtra respectively. The cost of main canal branches and distributaries of each State is to be borne by that State alone. In view of changed land use scenario in the State, the requirement of water for irrigation may have gone down considerably. As such, spending huge amount of money on the project in the ratio decided over two decades ago does not seem to be logical. The scope of water requirement to Goa is being reviewed and discussed with the joint venture partner, the Government of Maharashtra. However, any decision on the project would require consent of the Government of Maharashtra in view of inter-state agreement.

(b) Salaulim Irrigation Project

The project was approved by the Planning Commission, Government of India in December 1971 at a cost of Rs. 9.61 crore. The latest estimated cost of the project based on 1989 rates is Rs. 88.30 crore. The latest revised cost of the project is likely to be about Rs. 160.00 crore. The project is aimed to create an irrigation potential of 9686 ha (CCA) (14326 ha.ultimate) of which, potential of 5229 ha (CCA) has already been created. The project also has water supply component of 404 MLD for domestic and industrial use, of which 160 MLD has already been created and being utilised. The project is at an advanced stage of execution.

The Government has decided to curtail the distributory D3 canal of Salaulim Irrigation Project at ch. 8950 mts at Navelim before crossing river Sal and allot the balance stored water for domestic and industrial use in South Goa. The progress under various components of the project upto December 2003 is given in table 4.6.

j Minor Irrigation

Irrigation potential created under Minor Irrigation during the year 2003-04 is 63 ha. Thus, cumulative irrigation potential created under minor irrigation is 22,950 ha. and utilization is 18,446 ha. Action has been initiated to undertake a comprehensive review of lift irrigation schemes for renovation, expansion, and modernization in order to substantially augment the irrigation potential of the State.

k. Waiver of loans/hire charges

The Government of Goa has waived the principal and the interest on loan granted to project affected people of Salauli and Anjunem Irrigation Project for construction of houses, renovation of model houses etc. In all 486 project affected people of Salaulim Irrigation Project and 344 of Anjunem Irrigation Project were benefited and the amount waived was Rs.113.12 lakh. The Government has also waived the principal and interest and penal interest on the hire charges of the Bull dozers given to the farmers for developing land in the Command

Table 4.6: Progress under Salaulim Irrigation Project

Sl. No.	Component	Percentage progress achieved during 2003-04	Cumulative percentage of progress achieved
1	Earth work of main dam	14%	90%
1	Main Dam	-	100%
2	Main Canal	-	100%
3	Distributories	-	97%
4	Branch Canal	-	100%
5	Minors	-	100%
6	Irrigation Potential		
	a. Created	4128 ha(ULT)	1293 ha (CCA)
	b. Utilised	11634 ha (ULT)	3993 ha (CCA)

Area of Anjunem/Salaulim Irrigation project amounting to Rs.61.15 lakh, which benefited 608 farmers.

4. SANITATION AND SEWERAGE

a. Rural Sanitation

The programme of constructing single seater pour flush water seal latrines for rural population is being implemented since 1985-86. Efforts are being made to provide low cost toilets below poverty line in rural areas through Sulabh International, a non Government Organisation. Already 4190 toilets (till November) have been constructed during 2003-04. Thus, the total number of latrines constructed from 1985-86 to 30th November, 2003 is 63285.

b. Sewerage

(i) Environmental Upgradation of Panaji City

The whole Panaji city is supplied with treated water. The present total supply is of the order of 12 MLD which is just sufficient to meet the demands from various sections. The waste water flow is of the order of 10.50 MLD and thus the existing Sewage Treatment Plant (STP) at Tonca, Panaji is most inadequate and has technically reached its saturation point. As a result, the treatment of waste waters is ineffectual and factually non–existent for part sewage of the city. Considering possible environmental hazards due to flow of partly treated/untreated effluents to Mandovi Estuary, a project proposal for Environmental upgradation of Panaji city has been approved by the Ministry of Environment and Forest, National River Conservation Directorate, Government of India. The project cost of Rs.14.10 crore is to be shared between the central and state governments in the ratio of 70:30. The project envisages the following components:-

- Additional Sewage Treatment Plant at Tonca, Panaji for 12.5 MLD.
- Renovation & remodelling of the existing STP at Tonca, Panaji.
- Laying & jointing 500mm D.I. and HDPE effluent disposal pipeline to the Mandovi Estuary.
- Conditioning of the final effluent for re-use for irrigation & gardening
- Extension of sewerlines to leftout areas of Panaji city.
- Construction of community toilets
- Construction of staff quarters for sewage treatment Complex at Tonca

Work order has been already issued to M/s H.N. Bhat & Co on 9th October 2003. The stipulated date of completion is 16-02-2005.

(ii) Extension of Sewage and Sewerage Treatment Facilities to Taleigao, Durgawadi, Dona Paula and Caranzalem

The project at an estimated cost of Rs. 30 crore is formulated for extension of sewage and sewerage treatment facilities to Taleigao, Durgawadi, Donapaula and Caranzalem. The project is proposed for implementation during the Tenth Five Year Plan. Survey is completed and estimate is under preparation.

(iii) Margao Sewerage Scheme

Margao Sewerage scheme with the latest revised cost of Rs. 18.68 crore is partially commissioned in the year 2000. The North zone of Margao City covering the areas like Ambaji, Fatorda (part), Comba, Pajifond etc. has been completed and connected to the sewerage treatment plant. The work of providing network to the Central Zone consisting of areas like Fatorda (part), Borda , Gogol, Old Market etc., work is taken up in the year 2000 and expected to be completed by March 2004. An amount of Rs. 86.04 lakh has been spent on this during the current financial year and anticipated expenditure upto March 2004 is Rs. 239.29 lakh.

On completion, a population of about 30000 can be benefitted by providing house sewerage connections. The revised estimate sanctioned by the Government for both North and Central Zone including construction of sewage treatment plant is Rs. 18.68 crore.

(iv) Providing Sewer Network in South Zone of Margao City

The South Zone of Margao City consisting of thickly populated areas like Khareband, Shirvodem, Navelim, Pajifond etc. is yet to be covered with the sewer network. The estimate in South zone is roughly costing Rs. 50 crore. In order to obtain financial grant from the Govt of

India a proposal was submitted to the Ministry of Environment (Government of India) and being re-modified as per the recent guidelines of the Ministry.

(v) Vasco Sewerage Scheme

Various extension of sewer network proposed to be taken under Vasco Sewerage Scheme is as follows. :

- Extension of Sewer lines to Adarsh Nagar area of Vasco City.
- Improvement of renovation of STP at Baina Vasco replacement of Grit separator.
- Improvement of renovation of STP at Baina Vasco providing bye pass arrangements to installations of STP.
- Providing sewerage net work in Driver hill Orulem and Belabai area of Vasco town Phase II.
- Extension of sewer line in Bogda Phase II in Mormugao.
- Extension of sewer line Near Maruti Temple at Mangor Vasco.
- Improvement to the existing sewerage treatment plant at Vasco with bye pass arrangement and modifications of the various units as may be required.

5. URBAN DEVELOPMENT

a. Integrated Development of Major Towns

The scheme envisages construction of markets, shopping centers, parking lots and such other infrastructure facilities for the overall improvement of the urban areas. The Goa State Urban Development Agency has taken up a number of such projects in various urban areas in the State. Important projects and their present status are as under.

(i) **Multipurpose Truck Terminus (Transport Plaza), Margao**: Land acquisition for an area of 1,35,463 sq. mts, is in the final stage. Selection of a Project Management Consultant is in progress.

(ii) **Utility Centre (Pragati Plaza), Margao**: Land acquisition for an area of 37,995 sq.mts is in progress

(iii) **Transportation Centre at Quepem Road , Margao:** Land acquisition for an area of 29,331 sq. mts. opposite KRC station Margao and selection of Project Management Consultant are in progress.

(iv) **Vegetable Market at Mapusa**: Construction work of a modern market to accommodate over 150 vegetable vendors presently scattered in the market area shall commence before March 2004

(v) **Market at Curchorem - Cacora-** Work of a modern market with 202 shops will commence before March 2004

(vi) **State Level Renewable Energy Awareness Education Park Margao:** Land Acquisition for an area of 17,844 sq. mts. opposite SGPDA market complex at Margao is in progress

(vii) Other projects such as Mini Market at Tilamol, Quepem, acquisition of premises for an auditorium at Vasco, development of parking lots, construction of gardens, improvement of pavement etc. are also in the planning stage.

b. ACC Marg Technology for road works

A MOU was entered into on 18.7.03 between the Government of Goa and the A.C.C. Limited for resurfacing, strengthening and construction of roads using ACC Marg Technology in the State of Goa. The Government decided to adopt this technology for the Corporation of the City of Panaji, Mormugao Margao, Mapusa, Ponda and Curchorem for total length of 20 km. A.C.C. will guarantee the performance of the A.C.C. Marg for a period of 6 years, and shall execute the contracts under their overall responsibility.

c. Pay Parking

Pay Parking System in Municipal Area is proposed to be introduced. A committee consisting of the Commission, Corporation of the City of Panjim and Chief Officer of Margao and Ponda Municipal Council is constituted. The Committee has submitted a report for introduction the system of Pay Parking in Municipal areas of the Corporation of the City of Panaji

d. Abolition of Octroi

The Government has discontinued the system of collection of octroi on diesel and petrol by Municipal Councils/Corporation with effect from 1.4.2001. Instead of octroi, presently sales tax is being collected . The Municipal councils are being compensated for the loss of revenue on this account. An amount of Rs. 1,30,31,446/- has been released to Municipal Councils/Corporation as first instalment.

e. Urban Reform Incentive Fund

For carrying out urban reforms, the Government of India has formulated a scheme viz. 'Urban Reform Incentive Fund". Government of Goa has signed a MOU with the Ministry of Urban Development to carry out reform in the following sectors.

- Repeal of the Urban Land Ceiling and Regulation act at the State Level by resolution
- Rationalisation of stamp duty in phases to bring down to 5%
- Reform of Control Laws to remove rent control so as to stimulate private investment in rental housing
- Introducing of computerised process of registration
- Reform of property Tax so that it may become a major source of revenue of urban local bodies and arrange for its effective implementation so that collection reaches at least 85% by the end of Xth plan period
- Reform of Property Tax so that it may become a major source revenue of urban local bodies and arrange for its effective implementation so that collection efficiency reaches at least 85%
- Levy of reasonable user charges by urban Local Bodies , with the objective that full cost of O & M (operation & maintenance) is collected by end of the period
- Introduction of double entry system of accounting in Urban Local Bodies

f. Unified Building Bye Laws & Zoning Regulation

The Unified Building Bye-Laws & Zoning Regulation to be applicable to the entire State has since been finalized by the Committee constituted by the Government. The Committee is expected to meet shortly to approve the draft report.

g. Fire Services

Important achievements of the year 2003-04 are as follows.

- The provision of Fire Station at Porvorim has been included in the project being executed through the private agencies near the Police Station. The Government is also considering the proposal for setting up of a modern Fire Station at Curchorem. A plot of land is being acquired for the same.
- A modern Industrial Fire Station at Verna was inaugurated on 30.04.2003. A modern Fire Station Building at Ponda was completed and inaugurated on 26th October, 2003.
- A modern Fire Station building at Mapusa is nearing completion . A plot of land has been identified at Bicholim for building a Modern Fire Station. Action is being initiated to acquire the land after obtaining approval from Government.
- Fire Service is being geared up to tackle forest fires by imparting training to Fire / Forest personnel. Necessary implements for tackling forest fires are proposed to be acquired.
- The Government has created 95 posts of various categories to augment the existing strength.
- As regards induction of women in Fire Service, a B.Sc. Fire Technology Degree Course is expected to be started shorlty, wherein seats will be reserved for women candidates at the first instance. A proposal for re-designation of Firemen to Fire Constable and incorporation of the same in the R.Rs. and disperse with gender bias is under consideration of the Government in order to accommodate women candidates
- The Goa Fire Service has now been identified as Regional Training Centre by the Ministry of Home Affairs, Government of India for imparting professional career courses on All India basis and one batch of XIth External Sub-Officers course was conducted from July to December, 2003.
- Two Fire Personnel were awarded President's Fire Service Medal for meritorious services on 15th August, 2003. Three Fire Personnel were awarded Chief Minister's Fire Service Medal for meritorious service on 19th December, 2003.
- During the period of 7 months from April, 2003 to October, 2003, the Fire Service have attended 1149 fire and emergency calls during which 156 lives and property worth Rs.45.87 crore were saved / salvaged.

6. PUBLIC WORKS

Progress made during 2003-04 is as follows:

- Construction of Secretariat Complex at Porvorim has been awarded at an estimated cost of Rs. 2200.00 lakh in 2002. The civil work is under completion stage. Remaining items of work such as interiors, AC and land-scaping have been taken up.

- An amount of Rs. 3.50 crore has been ear-marked by the Eleventh Finance Commission, Government of India under special problems grants for the existing Secretariat Building in Panaji. Proposal for this purpose is being initiated for approval from Government of India.

- Construction of Jogger's Park at Althino-Phase II is sanctioned for Rs. 67.80 lakh and the work is in progress.

- Construction of additional Office Building at Quepem at an estimated cost of Rs.1.39 crore is nearing completion.

OVER VIEW

Priority accorded for development of infrastructure is visible in terms of huge investment envisaged by the Government. Under the Power sector projects amounting to over Rs.150 crore are under implementation mostly under the Accelerated Power Development Programme. Underground network in five major cities of Panaji, Margao, Mapusa, Ponda and Vasco amounting to Rs.153 crore has also been planned and the Government of India has already sanctioned schemes for conversion of overhead network in Panaji and Margao cities at a cost of Rs.68.26 crore. In order to provide quick, efficient and round the clock service to the people, the Government has initiated mobile van service in the major cities. Presently, 13 mobile vans equipped with necessary infrastructure, personnel and a wireless set have been pressed into service and one more Van is being deployed to Quepem. Hon'ble Union Power Minister has promised to provide additional 25 mobile vans so as to extend the service to the villages.

Towards reducing vehicular pollution in the State, the Government has implemented a unique scheme of providing subsidy for replacement of buses and minibuses, which are 15 years old and above. The scheme has been extended to yellow / black motor cycles, auto rickshaws and taxies with effect from 1st August 2003. The Government has already initiated steps for expeditious acquisition of land and engagement of international consultant for construction of a new Airport at Mopa. The State has submitted a proposal to the Ministry of Urban Development, Government of India for introduction of Metro Sky Bus as a pilot project in Goa. The Konkan Railway Corporation is in the process of establishing necessary infrastructure for the test run of this new technology. In order to ensure safe navigational standards, the Government has undertaken dredging of inland waterways and is also in the process of revising the hydrographic survey of inland waterways conducted over three decades ago. The construction of Amona-Khandola bridge is in progress. Construction of bridges at Paular, Guleli - Padeli and Ribander-Patto are also planned.

The national level concept of linking of rivers is being done in Goa to a considerable extent. In order to ensure availability of water throughout the year augmentation of the water sources at Assonora water works and Opa water has works has been accomplished by constructing series of bandaras. Project reports are under preparation for 7 hydropower generation projects in Madei River Basin. Action for survey investigation and foundation exploration have been taken up for irrigation and domestic water supply in Madei and Talpona river basins. The State has submitted a proposal to the Government of India for participation in Hydrology Phase II project aided by the World Bank. Augmentation of regional water supply scheme of Opa, Assonora and Sanquelim is in full swing. The Tillari Project is under review and the Salaulim Irrigation Project is expected to be completed shortly in all respects. Environmental upgradation project of Panaji City costing Rs.14.10 crore is under implementation. The Government has prepared plans for integrated development of major towns. Action has already been initiated for Transport Plaza, Utility Centre, Transportation Centre and State Level Renewable Energy Park at Margao, Vegetable Market at Mapusa and Market at Curchorem-Cacora. The State has signed a MOU with ACC Limited for construction of roads using ACC Marg Technology in the State.

CHAPTER 5
Industries and Minerals

1. NEW INDUSTRIAL POLICY, 2003

The Government of Goa announced its new Industrial Policy during 2003-04. The Mission of the Industrial Policy is to ensure accelerated Industrial Development, catalyze Economic Growth, ensure balanced Regional Growth, protect Environment and above all create sustainable Employment to local youth of the State.

The policy document strives to achieve overall economic growth of the State through accelerated industrial development. The policy document focuses on the creation of sustainable employment opportunities for the people of the State. This will be achieved by providing industry an access to high quality infrastructure, extending institutional support, technology upgradation, deregulating the business environment for an efficient, proactive and transparent administrative framework and catalysing the entrepreneurial as well as creative capabilities of the human resources and providing market support. The policy document aims at ensuring a facilitative regime that explores and unleashes the energies of the private sector to create an environment in which industry, both existing and new, can prosper.

The mission can become a reality with clearly laid down objectives that will provide both focus and direction. The document lays down the following major objectives :

(i) Promote industries specifically identified as "Thrust Areas".

(ii) Promote industries which would consume locally available raw materials, have consumption pattern within the State and neighbouring areas.

(iii) Develop Goa as the 'Export / Import Hub' and Encourage export oriented industries.

(iv) Develop self-employment opportunities for the local youth especially in rural and semi urban areas.

(v) Promote and encourage the agro-based industries to give a boost to the rural economy.

(vi) Promote and encourage development of handicraft products to give boost to local artisans.

(vii) Create a healthy climate for the growth and promotion of small-scale and cottage industries.

(viii) Promote and encourage the process of Technological upgradation / automation in the existing units as well as new industries so that need of migrant labour is gradually reduced, thereby arresting the influx of migrant labour.

(ix) Create cluster development centres in rural areas around which traditional livelihood earning professions can develop and thrive.

(x) Encourage industries which can convert existing and generated industrial waste such as mining rejects, slag etc., into useful products.

(xi) Encourage participation of women entrepreneurs in the industrial development of the State.

(xii) Ensure balanced growth to address regional economic and social disparities in the State.

(xiii) Encourage promotion of all such industries that are environment friendly and do not indulge in wasteful consumption of resources.

(xiv) Facilitate revival and rehabilitation of sick industrial units by devising suitable schemes.

(xv) Promote Research and Development (R&D) to enable the industry to have access to state of art technology.

(xvi) Encourage the participation of Industry Associations in decision making process and support their initiatives in promoting industrial development.

(xvii) Encourage artistic talents amongst local youth.

The policy document identified the following as thrust areas for focussed attention :

- Pharmaceuticals, Drugs and Biotech Industries.
- Food processing and Agro based Industries.
- IT and IT enabled services.
- Eco tourism / Heritage tourism / Adventure tourism / Event tourism / Medical tourism.
- Entertainment Industry.

In order to ensure that thrust areas receive focused attention, the policy document envisages setting up Pharma park, Food park including Wine park, Software Technology Park (STP), Agro Economic Zone, Biotech park, Apparel park, Special Economic Zone and Film City.

In line with New Industrial Policy, the Government has issued Notification announcing seven new schemes. The salient features of the schemes are given below.

The Capital Contribution Scheme

The objectives of the scheme are to provide financial support to local entrepreneurs, sick units for revival, expansion of existing functional units and promote locally developed technology.

The scheme has the following eligibility criteria.

- Units which are in operation for a period of at least 3 years will be eligible except those covered under Sick Unit Revival Policy
- Partnership firms and private limited concerns are eligible under this scheme except those covered under Sick Unit Revival Policy.
- Only those units which are permanently registered with the Directorate of Industries, Trade & Commerce or cleared by High Powered Co-ordination Committee will be eligible.

The quantum of contribution admissible under the scheme is as follows.

- The quantum of contribution shall be subject to the budgetary provisions and increase of sick units under Revival Plan. The applications not considered during the fiscal year shall lapse and fresh applications are to be filled during the next financial year
- Maximum capital contribution upto Rs. 1.00 crore per unit will be available subject to the condition that the contribution of the promoters should be equal or more than the capital contribution by the government. However, this shall not be applicable to sick units.
- The capital contribution will be for a period of 5 to 10 years as may be indicated in the sanction order with guaranteed return of minimum 6% or actual profit.
- There is no need of any collateral security. In case of failure to pay , recovery shall be done under the law in force including the recovery provision under the Land Revenue Code 1968

The scheme envisages preference to the units in small scale sector, units involved in research and development, technical oriented units and the units run by women entrepreneurs. In case of sick unit under Revival Plan, the capital contribution will be in proportion as indicated in the sick unit revival policy and subject to

- the unit generating net revenue in VAT/ST on any other tax of Government of Goa
- The actual generation of net tax revenue during the preceeding year (in proportion) as indicated in the sick unit revival policy

The scheme prescribes time limit for clearing the applications received under the scheme. The Directorate of Industries Trade & Commerce shall scrutinize and consider the applications within 3 months and shall convey its decision within 4 months from the date of receipt of application. In case of approval the capital contribution shall be released within 60 days from the date of approval.

(ii) Share Capital to Local Entrepreneurs and self-employed Scheme, 2003

The main objective of this scheme is to encourage local youth to start income generating activities and to encourage self-employment. Eligibility criteria prescribed under the scheme are as follows

- Individuals who are born and residents for 15 years in Goa shall be given preference under this scheme, though the scheme does not bar others, who are resident of the State of Goa for at least 15 years prior to the application under this scheme and have passed graduation or XII or X or schooling from the State of Goa. In case of group of individuals, all persons in the group should satisfy the above criteria. Self help groups of women and disabled persons shall be given priority.

- This scheme shall also cover those who are covered under subsidy scheme framed for black and yellow taxi-motorcycle/rickshaw drivers/owners for balance 75% of requirement. The age criteria will not apply for those who are applying for replacement of old taxis/motorcycle/rickshaws. Age criteria is also not applicable in case of self-help groups.

- Applicant shall be below 40 years of age to be eligible under this scheme. In case of group (except in case of registered Self-help Group), all the individuals shall be below 40 years of age. However, in case of women, disabled person, Scheduled Caste/Scheduled Tribe and Other Backward Class person, the age limit shall be relaxable by 5 years.

- Applicants shall be minimum VIIth pass. However, this requirement may be relaxed in deserving cases by a specific order of Director of Industries, Trade and Commerce.

- The income of the applicant along with spouse shall not exceed Rs.80,000/- per annum.

- The benefit under this scheme shall not be available if the applicant has availed subsidy of more than 15% of project cost under any other scheme, and/or if any scheme debars him specifically for receiving benefit under this scheme. Those who can be covered under Kamdhenu scheme shall not be covered under this scheme except for setting up of dairy/chilling unit.

- Priority will be given to women entrepreneurs to the extent of 30% of budgeted outlay on the scheme in case the applicants exceed total budgetary provision.

- All beneficiaries covered under CMRY scheme are automatically covered under this scheme.

Quantum of Capital Contribution admissible under the scheme

- Contribution under this scheme shall be Rs.1.00 lakh per individual subject to the condition that it is 50% of total project cost. However, in case of applicant having professional qualification, degree or diploma in Engineering including qualification like I.T.I., Vocational XII or any other Government recognized Certificate course/training programme, the contribution may be increased up to 2.00 lakh per individual, provided the business activity shall be linked/connected to his qualification in broad sense and subject to the condition that the contribution does not exceed 50% of the project cost.

- The scheme does not debar individual/group of individuals from availing additional loan from Economic Development Corporation or banks but the same shall be separately applied for and approved by Economic Development Corporation or banks concerned.

- In case of group of individuals, the maximum extent of capital contribution shall be Rs.10.00 lakh subject to condition that the individual limit per head is not exceeded. There is no restriction on number of members of the group.

- Those trained by the Goa Handicrafts Rural and Small Scale Industries Development Corporation or by Khadi Village and Industries Board or Training cum Production Centre or from approved Non-Government Organisations or under scheme of Directorate of Education and have obtained Certificate of successful completion of training shall be covered under this scheme and will be treated as professional for the purpose of benefit under this scheme.

- Applicants who are eligible for assistance under CMRY will be provided capital contribution under this scheme subject to the

condition that the contribution under this scheme does not exceed 50% of the project cost. The eligibility criteria for such persons shall be governed by the CMRY scheme.

Security

The self guarantee by the applicant / parents / spouse / parents of spouses along with one guarantor owning a residential accommodation or plot of land or any other property in the State, shall be adequate security for the purpose of availing benefit under this Scheme. However, where the assets financed are fixed assets as tangible security and where the capital contribution is below 50,000/- only personal guarantee by the applicant and additionally by his parents or spouse or parents of spouse (even though they do not have any fixed assets or property) is adequate. In such cases, first charge of mortgage / hypothecation of fixed or current assets will be held by Director of Industries, Trade and Commerce / Economic Development Corporation or as decided by the Government.

Repayment

The Capital Contribution given under this scheme should be paid back to the Government in equal monthly instalments within 10 years with moratorium as may be decided not exceeding a period of one year and the first instalment shall be due after the date of release of the capital contribution to the beneficiary. However, the Directorate of Industries, Trade and Commerce reserves the right to reduce the total repayment period up to 5 years depending on the nature of business. If the beneficiary fails to pay the instalments within the stipulated time period, then the amount will be recovered as per laws in force including arrears of land revenue. The capital given under this scheme will be interest free.

Insurance cover

There will be a special insurance cover wherein the beneficiary shall be absolved of his outstanding dues / payments in case of death or permanent disability. In such case, the Director of Industries, Trade & Commerce, with the approval of Government shall write off such balance outstanding dues. For purpose of this scheme, permanent disability will be one that makes the beneficiary incompetent to carry on the business for which he has availed the benefit. Insurance premium at the rate of Rs.200/- per lakh or part thereof once before disbursal shall be recovered from the beneficiary.

(iii) Preferential Purchase Incentives for Small Scale Industries Scheme, 2003

This scheme is intended to encourage and give boost to the Small Scale Industrial Sector. Under this scheme, Small Scale units registered in the State of Goa shall be given special treatment in any tender floated by the Government Departments or any purchases made by any Government Department, as under :

- Small Scale Unit shall be allowed to match to lowest tender price floated by non Small Scale units provided its quoted price is within 15% of lowest quoted price floated by Non-Small Scale Industrial unit.
- In case of contractual tenders for installation, commissioning of machinery / items, etc. the preference of quoted price, the pro-rata percentage over and above the lowest quotation shall be granted to the Small Scale Industrial units for being qualified to match the lowest one.
- For the purpose of availing benefit under this Scheme, the Small Scale Industrial unit shall be required to match standard of quality as required by the State Government Department.
- The Scheme shall be applicable even to purchase of computer hardware and other electronic items except in cases of purchases of the Education Department under their Schemes.
- The tender documents shall be supplied to Small Scale Industrial units at the rate of maximum up to Rs.200/-.
- Earnest money shall be required to be deposited and the same shall be to the maximum of Rs.500/-.
- No other security deposit is required.
- Priority in payment shall be ensured to Small Scale Industrial units and the payment shall be effected by the concerned Department within 60 days from the delivery of goods or completion of tendered work, failing which the Department shall be liable to pay an interest at the rate of 0.75% per month.

Only those Small Scale Industrial units having turnover not exceeding Rs.5.00 crore per annum

for last preceding 3 years and registered in the State with the Directorate of Industries, Trade and Commerce shall be eligible for the benefit under this scheme.

(iv) Interest Subsidy Scheme, 2003

The scheme intends to give boost to the Small Scale Industries and tiny units in the State by providing incentives in the form of interest subsidy. The eligibility criteria under the scheme are as follows

- Only new small scale industries and tiny units which go into commercial production after the commencement of this scheme shall be eligible to avail facilities under this scheme.
- Only those units engaged in the manufacturing activities and registered with the Directorate of Industries, Trade and Commerce shall be eligible under this scheme.
- Units are eligible for the benefit, for a period of 5 years that is 20 quarters subsequent to the quarter in which the commercial production has commenced.
- This Scheme is applicable to industries covered under "Green" list and specified entries of "Orange" List.
- This scheme shall not be applicable to units under revival plan as defined under sick unit revival policy.

The scheme is applicable to the units which have availed loan as term loan and working capital from nationalized banks or Scheduled Bank or Co-operative Bank or Economic Development Corporation Ltd. or any other Financial Institution notified by the Government of Goa.

Quantum of subsidy will be to the extent of 1 $^{1}/_{2}$% of the total net turnover or 30% of the interest paid by the units, whichever is less, subject to a ceiling of Rs.5.00 lakh per annum.

(v) Goa State Financial Incentives to the Industries for Certification and Patenting Scheme, 2003

The scheme proposes to encourage the industrial units to obtain national and international certification and patent rights in order to maintain requisite standards of quality of products / processes. The units that obtain Indian Standard Institute, International Standard Organisation certification or any other national / international certification for quality standards and / or patent right on product and / or processes are eligible to avail financial incentives under this scheme, provided the industrial unit is under the category of small / medium and large scale industry or service industry falling under Green / Orange Category list.

Following units are covered under this scheme

a) Units registered with the Director of Industries, Trade and Commerce will be eligible for the grant under this Scheme.

b) Units not registered with the Director of Industries, Trade and Commerce but those, which are cleared by the High Powered Co-ordination Committee or any Committee or Authority formed to grant such clearances for investment in the State.

c) Service Industries not covered under sub-clause (a) and (b) above are also eligible to take benefits under this scheme provided they fall under the following category.

i. Hospitals or any other medical Institutions registered with or recognized by the Directorate of Health Services.

ii. Educational Institutions recognized by the Directorate of Education / Directorate of Higher Education / Directorate of Technical Education or any other educational institutions affiliated to Goa University / Goa Board / Goa Board of Technical Education or any other institution covered under the Goa Education Development Corporation or Human Resource Development Foundation.

iii. Co-operative Banks other than Multi State Co-operative banks registered with Registrar of Co-operative Societies under Government of Goa.

iv. Any other service sector specially notified for this purpose by the Director, Directorate of Industries, Trade and Commerce.

Those units who have already obtained Indian Standard Institute, International Standard Organisation or any other international certification or any patent right but have not availed any financial assistance from the State Government for the purpose are also eligible to apply for incentive

under this scheme for additional certification, if any, provided the certification is qualified for the benefit under this scheme. Those units who have availed of central assistance for this purpose will also be eligible but only to the extent of balance amount after deducting the central assistance availed.

Quantum of assistance admissible under the scheme is as follows.

- Financial assistance/subsidy to the extent of Rs.2.00 lakh or the actual expenses incurred whichever is less will be granted per unit.
- Financial incentives can be availed once in a lifetime of the unit, irrespective of change in ownership/constitution or product provided that after such a change in ownership/constitution/product; if the applicant becomes ineligible to use such certification under which the benefit is availed than the unit will once again be eligible to avail the benefit under the scheme. Provided that the loss of certification due to inadequacies in the unit shall not be reason enough to avail this benefit for the second time.

(vi) Incentives to Women Entrepreneurs Scheme, 2003

The objective of this scheme is to encourage women entrepreneurship and encourage women to start industry for self-employment.

The scheme has the following eligibility criteria

- Those units which have gone into commercial production on or after the date of publication of this scheme shall be eligible subject to the fulfilment of eligibility criteria under the appropriate schemes under which special benefit is granted.
- Only partnership and proprietory concern are eligible provided the ownership is with women, to the extent of 100% in case of proprietory concerns and 51% in case of partnership firm with condition that the balance share of 49% is not entirely held by husband or father or brother or son and there is at least one additional partner other than husband or father or brother or son.

Incentives admissible under the scheme are as follows

- 5% additional benefit under Local Employment Subsidy Scheme. This is over and above what is eligible under the original Scheme.
- Preference in Capital Contribution and Special Capital Contribution Schemes.
- In case of Interest Subsidy Scheme prescribed limit of 1 1/2% of turnover will be increased to 2% and 30% of interest paid will be increased to 35%, however, subject to overall ceiling of Rs. 5.00 lakh.

(vii) Early Bird Offer Scheme, 2003

The scheme envisages to encourage setting up of industries in Goa. Those units who have gone into commercial production within one year from the date of publication of this Scheme shall be eligible for benefit under this scheme subject to the fulfilment of eligibility criteria under the respective schemes. Incentives available under the scheme are as follows.

- 5% additional benefit under Local Employment Subsidy Scheme. This will be over and above the eligibility limit laid down under this scheme.
- Preference in Capital Contribution and Special Capital Contribution Schemes.
- In case of Interest Subsidy Scheme, prescribed limit of 1 1/2% of turnover will be increased to 2% and 30% of interest paid will be increased to 35%, subject to overall ceiling of Rs.5.00 lakh.

2. OTHER IMPORTANT INITIATIVES

A facilitation counter is being opened at the Directorate of Industries for the public. The registration process of SSI units is computerized. It is also proposed to open an Investment Promotion Board by enacting suitable legislation. It would act as a facilitator for investment and serve as a single window clearance authority for medium and large scale industries. It is also proposed to start a single window system for granting permanent registration for all SSI units.

3. FOREIGN DIRECT INVESTMENT (FDI)

During the period August 1991 to October 2002, the total number of FDI proposals approved by Goa stands at 191 involving investment of Rs. 970.72

crore. This forms 0.34% of total FDI of Rs. 284716.99 crore approved by all States in India. Share of different States in foreign direct investment during the period August 1991 to October 2002 is given in table 5.1.

4. REVIVAL OF SICK UNITS

The Government has constituted an Empowered Committee to formulate guidelines for grant of relief and concessions to various industrial and commercial units. The committee has submitted its report suggesting suitable relief and concession to sick industrial units . Board for Industrial and Financial Reconstruction (BIFR) is examining around 12 cases of sick medium and large industries for the period from 1.4.2002 to 31.12.2003.

Table 5.1: Foreign Direct Investment approved by the major States/UTs August 1991 to October 2002

Sl. No.		No. of approvals			FDI approved (Rs. in crore)	% to Total
		———	———	——		
1	Maharashtra	1199	3089	4288	49457.96	17.37
2	Delhi	259	1914	2173	33993.99	11.94
3	Tamil Nadu	568	1732	2300	23610.93	8.29
4	Karnataka	460	1659	2119	23606.99	8.29
5	Gujarat	511	578	1089	18501.58	6.50
6	Andhra Pradesh	248	834	1082	13166.27	4.62
7	Madhya Pradesh	70	161	231	9270.88	3.26
8	Goa	60	131	191	970.72	0.34
9	Other States / UTs	1336	2425	3861	36606.24	12.86
10	State not indicated	2517	3390	5907	75531.43	26.53
	Total	7228	15913	23241	284716.99	100.00

Source: Website of TIDCO

5. SMALL SCALE INDUSTRIES

During the year 2003-04 upto December 2003, 101 SSI units have been permanently registered with an investment of Rs. 696.10 lakh providing employment to 618 persons. In addition, 582 SSI unit have been registered provisionally with an investment of Rs. 4109.20 lakh generating employment to 4588 persons. Trend in registration of small scale units, employment generation and investment during the post-statehood period of Goa is given in table 5.2. The office of the Development Commissioner, SSI, Government of India has identified 60 most important clusters to be considered for financing by banks. Both the districts of Goa have been identified for focussed development of pharmaceutical industry under SSI.

6. THE 3RD ALL INDIA CENSUS OF SSI

The third all India Census of SSI was launched in the State at the instance of the office of the Development Commissioner, (SSI), Ministry of SSI, Government of India , New Delhi. It is related to the reference year 2001-02. It covers the SSI units permanently registered upto 31.3.2001 and sample survey of un-registered units which were not permanently registered till March 2001. The frame list contained 4842 permanently registered units in Goa. The quick results of the Census have already been published in the month of September 2003 by the office of DC, SSI, New Delhi.

7. MEDIUM AND LARGE INDUSTRIES

The State has a High Powered Co-ordination Committee to accord permission for setting up of medium and large industries in the State. The Committee so far has cleared 345 industrial projects in Medium & Large Industrial sector. There are 154 Medium and Large Industries in the State with an investment of Rs. 2368.26 crore generating an employment to 22,560 persons. Pharmaceutical indus-

Table 5.2: SSI units registered in Goa : 1987-2002

				Fixed investment (Rs, in million)
1	1987-88	3271	22657	508
2	1988-89	3602	24438	603
3	1989-90	3924	26136	719
4	1990-91	4120	27328	807
5	1991-92	4344	28459	900
6	1992-93	4558	29672	999
7	1993-94	4787	30918	1126
8	1994-95	4995	32042	1224
9	1995-96	5118	33136	1344
10	1996-97	5278	34472	1576
11	1997-98	5488	36734	1842
12	1998-99	5765	39432	2191
13	1999-2000	5949	40797	2215
14	2000-01	6157	42312	2235
15	2001-02	6469	44222	2856
16	2002-03	6714	46163	3027

try has emerged as a major component in the industrial development of the State. Current status of Pharma Industry in India and also in Goa is given in Box 5.1.

8. SUBSIDY

While the State has discontinued the State Investment Subsidy Scheme, the Government is in the process of clearing the backlog in subsidy disbursement to industrial units. During the year 2003-04, upto December, Rs.3.30 crore subsidy has been released to 47 industrial units.

9. PRIME MINISTER'S ROJGAR YOJANA

During the year 2003-04 (upto December 2003) 144 applications have been received out of which 105 applications have been recommended by the Task Force Committee to the various banks. An amount of Rs. 83.44 lakh has been sanctioned by the banks.

10. CHIEF MINISTER'S ROJGAR YOJANA

The Chief Minister's Rojgar Yojana has been modified during the year 2003-04 and procedures have been simplified for faster clearances. The scheme has low rate of interest of 3.5% per annum and comfortable repayment period of 10 years. The modified scheme has received tremendous response and so far about 500 beneficiaries have been covered. The target is to cover 1000 beneficiaries by March 2004.

11. DEEN DAYAL SWAYAMROJGAR YOJANA

Under this scheme of providing financial assistance for self-employment, up to March 2003, 800 applications were received and 95 eligible candidates were selected for assistance. So far 89 allotment letters have been issued, of which 62 beneficiaries are running their activities and 27 are in the process of obtaining necessary licenses like NOC from Municipalities, Trade licenses etc. The scheme has been revised in the Month of April / May 2003. Under the revised scheme, about 4000 applications were issued and 1300 applications dully filled in have been received. So far 700 applicants have already been interviewed. It is proposed to allot 200 kiosks by March 2004.

BOX 5.1
Development of Pharma Industry In India And Goa

The pharmaceutical industry in India in Post Independence period, began to take shape around 1956-66 and has an impressive track record. The country which was importing medicines from painkillers to antibiotics at the time of independence, today can meet 90% of the needs of the domestic market and also export even to the most developed countries. The capital investment which was Rs. 140 crore in 1965-66 has now increased to Rs. 3,400 crore - almost 224 times. During the same period production of bulk drugs has increased from Rs. 18 crore to Rs. 5,440 crore i.e. 300 times more. Production of formulations has similarly grown almost 140 times from Rs. 150 crore to about Rs. 20,000 crore. From negligible exports of Rs. 3 crore in 1965-66 exports have registered an impressive growth of Rs. 10,000 crore. Presently, in India, 90% of the units are located in states like Maharashtra, Gujrat, Andhra Pradesh, Tamil Nadu and Goa. Today there are nearly 10,000 pharmaceutical units and it is expected that within next 5 years production will be doubled to 40,000 crore. The industry today is backed with a high technical manpower which is available in the country and also good infrastructure . Some of the top Indian Pharmaceutical giants have set up their industries abroad and they have state of the art plants with USFDA and MCA certifications.

In Goa, major growth in Pharma sector started in late 1990's. The 5 Years Tax Holiday for Goa announced in the Union Budget in the year 1993 gave further impetus for development of Pharma sector in Goa. From 1995 onwards an all round development took place and today pharmaceutical industries has over 120 registered units employing approximatly over 18,000 people. Apart from the incentives and tax holiday , Goa is selected because of its ambience and pollution free nature which is conducive to the Pharma industry. A number of auxiliary units such as packaging , printing and allied industries have started to meet the requirement of the pharmaceutical industry where the employment is also quite large.

Extracts from the speech of Shri Ramnath Kare, Planning Board Member of Government of Goa, at the Goamant Vishwa Sammelan held in Panaji in January 2004.

12. GHRSSIDC

New emporium at Directorate of Industries premises, Panaji was opened in the month of July 2003 to give impetus to the sale of Goan Handicrafts. New mini emporium was opened at St. Monica Jetty during the year. The renovation of most of the emporia were undertaken. Training programmes were organized in collaboration with Jute Manufactures Development Council, Kolkatta at Pernem, Curchorem , Poinguini and Canacona. Also training in improvement of skill of the artisans in clay products was organized at Pottery Centre Bicholim. To popularize local handicrafts among the student community, exhibitions were organised at various schools and colleges. Training in jute manufacturing was conducted with the objective of making the NGO's work commercially. The Corporation has decided to implement a scheme for Home Processing / Repackaging units for day to day household consumption items thereby engaging individuals / NGO's / SSI units.

13. MINERALS

Mining is an important economic activity in Goa (Box 5.2). While it has triggered the economic activity, it has also created a degraded environment and is therefore a matter of concern to the Government.

During the year 2003-04 special attention was paid in identifying the sources of pollution in order to adopt abatement measures to minimize the adverse effects. The efforts were directed towards

- *siltation of rivers/nallahs*: The catchment areas are periodically inspected and necessary directions are issued to the mining companies to take up adequate protective measures to control the turbidity of the water flowing into the river / nallahs. The catchment areas of river Khandeapar, Kushawati, Selaulim are closely monitored from time to time.

- *dust pollution*: a meeting was convened on 4.12.03 along with the Suptd. of Police and Transport authorities. The mining companies are to ensure that all the tippers are not subjected to overloading and that they are properly covered with tarpaulins.

- damage to agriculture / horticulture lands in the vicinity affected due to siltation were identified.

The Directorate of Mines has issued 433 licenses / leases. During the year 2003-04 (upto Dec 2003) 83 quarrying leases for different type of minor minerals were granted and an amount of Rs. 39.38 lakh was collected towards royalty. In all 28 cases of illegal extraction on minor minerals were detected on which an amount of Rs. 2.75 lakh was recovered towards compounding of offences. Similarly 270 cases of illegal transportation were detected on which an amount of Rs. 11.74 lakh was realized towards compounding of offences. The Directorate of Mines took part in the world Mining Congress and Expo 2003 held at Pragati Maidan, New Delhi and was awarded a special commendation Award by the organizers.

BOX 5.2
MINING IN GOA

The mining belt of Goa covers an area of approximately 700 sq.kms. and is mostly concentrated in four talukas namely Bicholim, Salcete, Sanguem and Quepem. Mining has been very important element in the economic history of modern Goa. It has provided the trigger to boost economy of the mining talukas. Goa is a major iron ore exporting State and over 60% of country's iron ore export is from Goa. In terms of foreign exchange earnings it amounts to nearly Rs.1000 crore per annum. Mining in Goa is done by open cast method which necessitates the removal of over burden overlying the iron ore formations. On an average about 2.5 - 3 tonnes of mining waste is obtained in order to produce a tone of iron ore. The average annual production of iron ore is of the order of 15-16 million tonnes, in the process removal of which about 40-50 million tonnes of mining waste is generated. Such a huge quantity of mining waste only creates a problem for its storage but is also a source of environmental pollution.

OVERVIEW

The Government of Goa has announced its new Industrial Policy aimed to accelerate industrial development, catalyzing economic growth, balanced regional growth, environment protection and creating sustainable employment to local youth of the State. In pursuance to the Policy, the Government has notified seven new schemes for implementation. The Government is in the process of clearing the backlog of payment of subsidy to industrial units. The chapter gives an overview of development of Pharma Industry in India and Goa. In this regard it is pertinent to note that the office of the Development Commissioner, SSI, Government of India has identified 60 most important clusters to be considered for financing by banks. Both the districts of Goa have been identified for focussed development of pharmaceutical industry under SSI. The modified Chief Minister's Rojgar Yojana and Deen Dayal Swarojgar Yojana have received tremendous response. In mining sector, the Government has paid special attention to identify the sources of pollution in order to adopt abatement measures to minimize the adverse effects. The State received special commendation award in the World Mining Congress and Expo 2003 held in New Delhi.

CHAPTER 6
Information Technology

The spread of Information and Communication Technology "ICT" brings hope that Government can transform and, indeed, efforts everywhere are to use technology to improve delivery systems. The objective is to create a fully e-governed, information-rich state with real-time access to information, on-line commercial transactions, state-of-the-art communications, and industrial infrastructure. The USP for Goa would be top of the line Infotainment Industry, which would require extensive IT Education at all levels to provide the necessary manpower and the Government's commitment to a transparent and responsive administration, through on-line smooth e-governance. Thus, the ultimate aim is to make Goa an "Intelligent State".

For giving the maximum benefits to Citizen of the State and to harness the advances in IT and to restructure government - citizen - business interfaces with the objective of good governance, the Government of Goa has constituted a "Task Force for Good Governance". It lays down the policy framework, ensures flow of sufficient funds for IT infrastructure, suggests administrative reforms and initiatives on e-governance for benefits to common masses. Efforts are being made for promoting IT awareness in the State by way of conducting technical seminar / workshop / annual events, etc. To visualize the concept of 'Enabling and Empowering' the people, the main issues which relate to spreading of ICT to masses are in terms of making available appropriate IT infrastructure, developing suitable e-Governance software for all Government Departments, training to staff, IT Awareness / Promotion, providing precise incentives / support to IT industries, etc.

In line with these objectives, various initiatives have been taken such as infrastructure development for IT industries, communication infrastructure for connecting all Government Departments, training, IT promotions, etc. The details are as under:

1. E-GOVERNANCE

a. Dharani

Dharani project is an integrated IT solution for Managing Land Records and Empowerment of Common People. The project has been designed with the objective of maintaining up to date land records on line, so that certified copies of record of rights and survey plans can be issued to the public across the counter. In Goa, computerization of land records of Tiswadi Taluka was inaugurated by Hon'ble Shri Venkaih Naidu, Union Minister for Rural Development on 23-10-2001. Since then the Dharani project has made a significant progress and Goa has emerged as the first State in the country to computerize 100% land records. Highlights of Dharani project are as follows.

Dharani

100% LAND RECORDS COMPUTERIZED



- Goa is the first State in the country to computerize 100% land records
- DHARANI implemented in all taluka offices and city survey
- Land records (form I & XIV and form D) issued from computers at all the offices
- Land records are now available to the farmers instantly
- Land records are clear and accurate
- Original land records scanned and preserved for resolving disputes

Dharani

CONNECTIVITY

- All taluka offices and city survey offices interconnected through GoaNET
- On line connectivity through 64 KBPS Leased Lines
- Few village panchayats also connected to Goa NET through Dial up lines
- Remaining village panchayats are being connected
- Land records of any village or city can be issued from any taluka office, city survey office, village panchayat or mahiti ghar



Dharani

ON LINE MUTATION



- Entire mutation process computerised
- On line mutation implemented in all the taluka offices and city survey offices
- Local database at taluka offices
- Central database at secretariat, updated simultaneously
- Process is transparent
- Close monitoring and speedy disposal
- Role based access
- Bio metric authentication
- Data encrypted
- Tampering of records not possible

Dharani

MAHITI GHARS (INFO KIOSKS)

- Mahiti Ghars for providing on line services to citizens
- Citizen-Government interaction made simple and fast
- Ten mahiti ghars already operational and many more at rural areas being set up
- Operated by private agencies with support from the government
- Open from 8.00 am to 8.00 pm including holidays
- Acts as extension counters of Government departments



For the purpose of issuing record of rights i.e. Form I & XIV and survey maps to citizens of Goa on a trial basis, two information kiosks under the name "*Mahiti Ghar*" were made operational from 24th October 2001. There were demands to open such kiosks in various parts of the State and also to extend the operational activities to cover more and more departments. The Government is therefore in the process of setting up another 30 Mahiti Ghars. Currently, there are ten Mahiti Ghars operational to provide services like, issue of Land Record form I & XIV, Drivers Learners license. More services are being added.

Dharani

SELF OPERATED ENQUIRY COUNTERS



- Touch screen kiosks in all taluka offices
- Direct access to public
- Konkani Marathi and English interface
- Step by step easy to use instructions
- Public can access Form I & XIV, maps and form D

b. GoaNET- Wide Area Network

An important ingredient of effective e-Governance is an existence of a Wide Area Network (WAN) connecting all the Government departments / offices to provide various Governmental services to its villages and citizens. Government has successfully set up an alternative backbone in position with higher capacities and better technology connectivity to the next logical levels of the Government offices, called as GoaNET. Each department's LAN will in turn be connected to the Central Hub through a dedicated leased line as and when the department is computerized. The GoaNET is functioning since 30th May 2003. Currently, Collectorates of North and South Goa Districts, 11 Taluka offices, 4 City Survey Offices, Land Records, 10 Mahiti Ghars and 5 RTOs have been connected to the GoaNET Central Server through 64 KBPS leased lines. Presently, Form I & XIV, Form D, Driving Learners Licence etc are issued through GoaNET. All Municipal Councils are proposed to be connected to the GoaNet Server shortly for providing centralised Birth & Death Certificates at any Point of Sale. Other citizen services planned on-line through GoaNET are collection of taxes like sales tax, excise, property tax etc., collection of utility payments like water bills, electricity bills etc., registration of vehicles, acceptance of application forms etc.,



Dharani

INTEGRATION OF TEXTUAL DATA AND MAPS

- Textual and map data integrated
- Maps digitised
- Certified plans / maps issued from computer within a short time
- Maps are updated regularly
- Data encrypted
- Bio metric authentication
- Data archival
- Previous version of data / maps are maintained
- Standardized GIS software



c. Other important e-Governance initiatives

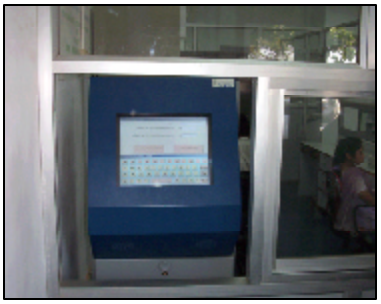
- Computerisation of Transport Directorate to issue online drivers learner license, Vehicle Registration Certificate, etc. in a well planned manner from the e-RTA Portal.



- Government has identified standard GIS engine for all Government Departments for development of customized GIS applications. Government has agreed in principle to set up a Centre of Excellence for GIS, which will have common facilities for all Government Departments.

- Major initiative has been taken up for setting up of Touch Screen Kiosks at all Taluka Head Quarters. Necessary efforts are being made for identification of site, preparation of site and to provide suitable connectivity to it. Currently, two Touch Screen Kiosks are running successfully at Panaji and Margao

- With a vision to provide single window services to the Citizen of Goa, it is essential to integrate all e-Governance software solution on the Standard platform and data structure. Metadata Structure is finalised for integrating all citizens' data, which shall be linked to the Centralized Metadata Repository, which shall be accessed by the Citizen oriented services. This shall also help in interchanging the data between the e-Governance Applications, which have been already developed, and for future upcoming applications for development.



d. Task Force for Good Governance

The Government of Goa has constituted the "Task Force for Good Governance" under the Chairmanship of Hon. Chief Minister and it shall lay down policy framework, ensure flow of sufficient funds and initiatives on e-Governance benefits to common masses and suggest administrative reforms etc. It has identified twenty departments for immediate and complete computerization. These include the important departments of Revenue, Sales Tax, Police, Transport, Registration, Electricity, Health, Tourism, Municipal Councils, Panchayats, Animal Husbandry, Water Resources, Labour, Woman & Child, Sports, Archives, Agriculture, Secretariat etc. The balance departments will be taken up as soon as work gathers pace on the aforesaid departments. The Government has identified 20 departments for computerization on priority.

The status of computerization in these departments is given in table 6.1.

For successful deployment of the e-Governance, the Government has already identified officials for IT implementation Cells, which shall oversee development of IT in their respective Departments as the coordinator to implement the e-Governance solution. Technical persons are being deputed to the identified departments to assist in computerisation.

e. Computerization projects undertaken by the National Informatics Centre

(i) Municipal Administration Software (MAS)

MAS has been designed with a view to computerise various functions in urban local bodies, so that citizen services can be provided across the counter. MAS consists of different modules like registration of births and deaths, house tax, lease/rent, trade licences and accounts. MAS has been implemented in all the 12 municipal councils and Panaji Municipal Corporation. It is planned to inter-connect all the urban local bodies and issue birth and death certificates on-line from all offices and Mahiti Ghars.

(ii) Integrated Finance Management System

Integrated Finance Management System (IFMS) has been implemented in Directorate of Accounts, Panaji and its South Goa office at Margao. IFMS consists of modules like bill processing, cheque printing, treasury receipts and payments, works audit, GPF, advances etc.,

(iii) Info Gram

Info Gram is a comprehensive IT Solution, covering all activities in the Village Panchayats. **Info Gram** consists of various modules like inward register, outward register, registration of births and deaths, accounts, taxes, fees, issue of licences, NOCs, certificates, important works, panchayat details etc. **Info Gram** has been implemented in five Village Panchayats viz. Shiroda, Navelim, Merces, Sanquelim and Chicalim. Info Gram is being implemented in Cortalim, Assagao, Benaulim, Cansaulim and Loutolim Panchayats.

(iv) Other computerization activities undertaken by the NIC are:

- **State Secretariat**: Cabinet Decision Monitoring System, Budget Information System, Vehicles Management System, Bills Processing System and Quarters Allocation Management System
- **High Court/District Courts**: Registration of Cases, Preparation of Daily/Weekly Cause Lists, Daily Orders, Judgements etc. are done on computer.

Table 6.1: Status of computerization in 20 identified departments

Sr. No.		
1.	Revenue Department (Collectorates, Taluka offices)	(i) Hardware has been delivered/ installed in Collectorates and Mamlatdar offices in the first phase of computerisation. (ii) Software Development of few software modules have been completed and fine tuning of this is under progress.
2.	Municipal Offices (Directorate of Municipal Administration, Municipal Councils)	(i) SRS has been completed. (ii) Software Development designed /documentation is being prepared & Software Development will be completed shortly. (iii) Municipal Administration Software (MAS) developed by the NIC is already in implementation in all the municipalities.
3.	Panchayat Offices (Directorate of Panchayats, Village Panchayats)	(i) The computerisation of Village Panchayats is under implementation under the project named INFOGRAM developed by the NIC.
4.	Transport	(i) The computerisation of RTO is under progress. One Mahiti Ghar and one RTO counter at K.T.C. Bus Stand, Panaji have already started functioning to issue Driving Learner's Licence. Permanent Driving Licence will be issued shortly. The remaining RTO Counters at various RTO Offices will be set up in a phased manner.
5.	Land Records	(i) Land Records have been computerised in all the 11 Taluka offices and computerised form I & XIV are being issued by Mamlatdar of respective Talukas and also in Mahiti Ghars. (ii) Multi Lingual Language Interface Touch Screen Enquiry Counter has already been set up in Mamlatdar office, Panaji and started functioning from 30/05/2003.
6.	Registration Offices	(i) SRS completed (ii) Complete deployment plan & hardware requirement is being prepared under the project 'GOA ON LINE' financially supported by Ministry of IT & Communication, Government of India, New Delhi.
7.	Electricity Department	(i) Power Grid Corporation is identified as a consultant to Electricity Department. They have submitted a detailed report, which includes all aspects of Electricity Department. The proposal is under consideration.
8.	P. W. D.	(i) The project proposal for complete computerisation had been submitted by C-DAC is under consideration. The detailed discussion is held with the concerned officials of PWD.

Sr. No.		
9.	Sales Tax Department	(i) Tenders were floated for Software Development of VAT. The Software Development Agency has been identified and MOU is being finalized to take up the Software Development of the Department.
10.	Police	(i) The detailed SRS for Police Department has been completed and subsequently the notice for expression of interest was issued. The technical proposal has been scrutinized for computerisation of Police Department.
11.	Tourism	(i) Web site has been developed and complete computerisation is under consideration.
12.	Secretariat	(i) The detailed SRS for complete computerisation of the Secretariat has been completed by C-DAC and software Development has been undertaken under the project Goa On-line financially supported by Ministry of IT & Communication, Government of India, New Delhi.
13.	Personnel Department	
14	Directorate of Accounts	(i) The various modules have been completed and deployed by the NIC and are being used by Accounts Department. Necessary efforts are being taken up to make the Accounts procedure end-to-end solutions.
15	Health Services Goa Medical College	(i) SRS of GMC has been completed and the presentation was made to all concerned on Hospital Information System and is ready partially for deployment at GMC under the project "GOA ONLINE" financially supported by Ministry of Information Technology & Communication, Government of India, New Delhi.
16	Information & Publicity Department	(i) The total web site has been developed and deployed for providing to the Public Services like daily news, Search engine, Archival Engine and other related services.
17.	Excise	Being takenup
18.	Rural Development	
19.	Civil Supplies	
20.	Employment / Labour	A fresh SRS is complete. The complete end –to-end solution will be ready shortly.
21.	Few more Departments such as Water Resources, Animal Husbandry, Sports, etc. have been taken up for complete computerisation.	

- Water Billing and HT billing
- Election Information System comprising polling booths, nomination, staff allotment, polling day, counting etc.
- Pay roll has been implemented in 50 offices.
- Internet / email connections have been given to all departments. Video conferencing facility has been established at the NIC Centre, to facilitate Video Conferencing between Centre and States.
- Website of various departments have been developed.

f. Training of Government servants

Training of the Government servant is one of the key components for successful deployment of e-Governance project. In this aspect, nearly 4000 Group 'C' Government employees have been trained by the e-Governance Cell under the Directorate of Technical Education. The software application training and handholding support is taken up in a phased manner in tune with software application development. The Department is making efforts to identify 1-2 training institutes at each Taluka level to train staff of the departments. In addition, the National Informatics Centre is conducting every month two training programmes of one month duration. During the year 2003-04, 388 officers / officials have been trained by the NIC.

2. DEVELOPMENT OF INFRASTRUCTURE FOR IT INDUSTRY

Various projects initiated for IT infrastructure in the State are as follows :-

a. Setting up of Software Technology Parks of India (STPI)

An office of Software Technology Parks of India (STPI) has been setup in December 2002 in Goa to facilitate approvals required by IT Units from the Government of India. Land has already been allotted to STPI for high-speed data communication facilities in Goa. MOU is being finalized to be executed between Government of India and Government of Goa and after signing of MOU, operation of STPI services shall commence within 18 months (Box 6.1).

BOX 6.1

SOFTWARE TECHNOLOGY PARK (STP) SCHEME

The 100% Export Oriented Unit scheme (STP scheme) is for setting up of software development and IT enabled services firm in India for 100% Export. The STP scheme is administered by the Directors of STPI. Benefits of STP scheme are:

- Income tax holiday as per section 10 A of the IT Act.
- 100% customs duty exemption on imports of capital equipments
- Equipment can also be imported on loan or lease basis
- All relevant equipment / goods including second hand equipment can be imported (except prohibited items)
- 100% excise duty exemption on indigenous items procurement.
- Central Sales Tax reimbursement on indigenous items procurement.
- Green card enabling priority treatment for government clearances / other services.
- 100% foreign equity investment in the companies permissible.
- Sales in the Domestic Tariff Area(DTA) up to 50% of the foreign exchange earned by the STP / EHTP unit.

Software Technology Parks set up in different parts of the country under the aegis of Software Technology Park of India have been playing a significant role in placing the country on world IT map. Statistical data reveals that there has been a significant increase in software exports from STPI members. During the period 1991-92 to 2002-03 the software exports by STP units has increased from Rs.17.00 crore to Rs.37,176 crore registering an increase by over 2000 times.

During the year 2001-02 exports through software parks has registered a growth of 47% . During the year 2002-03, software exports increased to Rs.37,176 crore compared to Rs. 29,523 crore during 2001-02, an increase by 26% (Box 6.2).

BOX 6.2

STPI posts 26% growth in software exports at Rs. 37,176 crore

Software Technology Park of India units have posted a 26% growth in software exports in 2002-03 at Rs.37,176 crore compared to Rs. 29,523 crore in 2001-02. This constitutes about 80% of the country's total software exports.

Karnataka led the march with a software export revenue of Rs.12,350 crore during 2002-03, posting a growth rate of 25%. Tamilnadu followed with an export of Rs.6305 crore, posting a growth rate of 26% and Maharasthra Rs.5508 crore, a growth rate of 20%. Andhra Pradesh posted software export revenue of Rs.3638 crore (30% growth rate) during the period. Haryana exported software worth Rs.2734 crore and Uttar Pradesh touched Rs.2541 crore in 2002-03. The growth rates of both the states were 28 and 27 per cent respectively. Delhi touched Rs.2065 crore in software exports with a growth rate of 18% and West Bengal posted the highest growth rate of 99% at Rs.1200 crore. While Kerala posted a growth rate of four per cent at Rs165 crore, Orissa had a growth rate of 22 % at Rs.260 crore. Other states registered a total software export of about Rs.410 crore.

More than 7400 units are registered with STPI of which 3544 units exported software in 2002-03.

Source: The Economic Times dated May 22, 2003

Development of Software Technology Park in Goa is at nascent stage (Box 6.3). However, in view of priority being accorded by the Government for the development of Information Technology in the State it is expected that Goa STP would also emerge as a major export house in the country in the years to come.

BOX 6.3

SOFTWARE TECHNOLOGY PARK AT VERNA, GOA

Three buildings (STP -I, II, & III) having an area of 3600 sq. mts each has been sold to M/s Control Net Inc. Pvt. Ltd. STP -III has been sold to M/s Condor Polymeric Pvt. Ltd. This group has a number of companies in the business of Hospitality, software Development , Call centre etc. STP - II was sold to M/s Information Technologies India Ltd (ITIL) and M/s Burr Brown India Ltd (BBIL). M/s ITIL was in software development business while M/s BBIL were in Medical Transcription business. Infrastructure by way of water supply, power supply, roads, street lights, standby generators, international connectivity (VSNL) have been provided at this Park.

b. Special Economic Zone

Pre-feasibility study for Special Economic Zone for Pernem Taluka was prepared by M/s. Pricewater House Coopers Limited (PwC), New Delhi. The report envisages a total development of 19200 Hectares of land in Pernem Taluka in three phases. The report indicates that area under consideration for SEZ is commercially viable and further detailed feasibility study would be taken up. As far as projected revenues are concerned, it is suggested to follow the broad guidelines issued by Government of India on setting up of SEZ which stipulates that infrastructure may be provided to occupants on a commercial basis. Therefore, it is envisaged that all infrastructure facilities including power, water, sewerage, telecommunications, education, health services, hotels, etc. would be provided with user/service charges so as to operate on a financially viable basis. It also suggests that the developed plots and built up space in the SEZ should also be sold to prospective buyers on a profitable basis. SEZ could be worked on PPP/Joint Venture model to make it successful revenue model.

The pre-feasibility study report has been accepted by the Cabinet subject to the SEZ being limited to a smaller area. An Empowered Committee has been constituted to oversee the establishment of SEZ for speedy implementation of the project.

c. Hi-Tech Habitat at Dona Paula, Goa.

A Hi-Tech Habitat Project is proposed to be undertaken in a plot of land admeasuring 70 acres. 45 acres of this land will be developed through a Joint Venture project. Selection of the Consultants for the project is under progress, who shall be providing necessary consultancy support for identification of JV partner.

Government will earn revenue by way of dividend earned on this equity through ITG besides generating gainful employment to a large number of educated youth and help to improve the economy in the State.

d. IT Park at Mandrem –Morjim in Pernem Taluka.

The project for creation of IT Park is being planned in an area of approx. 500 acres located at Mandrem-Morjim. The Land Acquisition proceedings have already been initiated. This IT Park would also help in attracting large investments in IT and provide employment to about 20,000 to 30,000 educated youth.

3. IT AWARENESS / PROMOTION

The Government has been supporting the promotion of IT through conducting seminars/conferences for attracting the best practices in IT into the State. In 2002-03, the Government had organized "Goa Agenda" an annual event, in association with the Goa Chamber of Commerce and Industry and more than 100 CEOs from various IT firms have attended the conference for delivering their thoughts to make Goa as an IT hub. Such events shall be held regularly to invite industrial houses to set up their business in Goa. In this regard, Government proposes to introduce incentives/relaxation for setting of IT industries units, such as exemption of various taxes/duties, priority for power connection, etc.

OVERVIEW

The Government has accorded priority for e-Governance projects in the State. The DHARANI project aimed at managing land records and empowerment of common people has made significant progress. Goa has emerged as the first State in the country to computerize 100% land records. Clear and accurate land records are now available to the people instantly. Entire mutation process has been computerized. On line mutation has been implemented in all the taluka offices and city survey offices. Resurvey of entire State is also in progress. Ten Mahiti Ghars have already been operational and many more are being set up including rural areas. Touch screen kiosks have been installed in all taluka offices. The Government has already set GoaNET, an intranet work of the State aimed at connecting all the Government Departments/offices to provide various Government services to the people online. The Government has constituted a 'Task Force for Good Governance' under the Chairmanship of Hon'ble Chief Minister for immediate and complete computerization in 20 identified Departments and considerable progress has been achieved in this regard. Projects like Info Gram covering all the activities in the village panchayats and MAS (Municipal Administration Software) covering all the activities of urban local bodies are under implementation. Development of infrastructure for IT industry in Goa is at a nascent stage. An office of Software Technology Park of India (STPI) has already been set up to facilitate approvals required by IT Units from the Government of India. Setting up of Software Technology Scheme of STPI is also being implemented and few units have been setup in Verna Industrial Estate. Pre-feasibility study for setting up Special Economic Zone in Pernem taluka has been completed. Setting up of Hi-tech Habitat at Dona Paula and IT Park at Mandrem-Morjim are also gaining momentum. It is expected that with the development of IT infrastructure facilities the State would emerge as an important software development and export house in the country.

CHAPTER 7
Tourism

Tourism is the most important component of State's economy and plays an important role in the socio-economic development of the State. Since a large multitude of people in Goa are dependent on tourism and related activities, a decisive promotional thrust and reworking of appropriate tourism model have been identified as key elements in placing the potential of the State on a higher growth orbit. It is in this context that the budget for tourism in recent years has been increased to manifold laying special emphasis on creating brand identity to Goa, developing infrastructure and diversifying from beach centered tourism to other forms of tourism like heritage tourism, hinterland tourism, eco-tourism, business tourism, health tourism, adventure sports etc.

Goa a tiny emerald on the western coast of India is indeed a place of tranquility and the most sought after safe tourist destination for the visitors from all over the world. This tiny State, which made its beginning with two lakh tourists in 1978 could succeed in galloping the tourist traffic to the extent of 20 lakh in 2003 which is more than its population. Taking into consideration the ever increasing tourist traffic, the State Government has constituted an autonomous body "Goa Tourism Development Corporation Ltd.", which has a large fleet of luxury tourist coaches, boat cruisers and comfortable 'Residency' establishments to cater to the middle income and higher income strata of the tourists. Subsequent to declaring Tourism as an Industry, the Government of Goa has offered ample facilities through reduction of taxes on power, water, luxury taxes, etc. to the travel trade.

1 TOURISM PROMOTION

As a part of tourism promotion, the Government of Goa launched its aggressive campaign to accelerate its promotional movement, by strengthening its marketing strategies through organizing road shows, participations in Travel & Tourism Marts in India and abroad since 2002. In addition to this, extensive promotional campaign was launched through print and electronic media which resulted into tremendous tourist traffic to the State. As a result the State received 2029738 tourists as compared to 1596941 in 2002, showing an increase of 27% tourist traffic to the State. For the year 2003, the targeted growth is 30%. During the 2002-03 Goa received 384 charter flights and this year 584 charter flights have been confirmed which include Russia and few new destinations. Details of tourist arrivals in Goa from 1985 and foreign tourist arrivals by charter flights are given in Annexures 33 to 35. The State has bagged many prestigious awards in tourism sector including the Galileo Express Travel & Tourism Awards by the Indian Express Group for the most innovative marketing campaign (Box 7.1)

As a part of the promotional activity, the Tourism Department has participated in various tourism related events in India and abroad, such as Travel & Tourism fairs held at Kolkata, Hyderabad, Ahmedabad, Delhi, Bangalore and Chennai; India International Tourism Mart (IITM) at Mumbai; South Asia Travel & Tourism Expo at New Delhi; Travel & Tourism Explore at Nagpur, Indore; Explore at Bangalore, Chennai, Goa; World Travel India at Mumbai; Goa Travel Meet was organised in association with Goa Chamber of Commerce & Industries. Overseas participation were at International Tourismus Borse (ITB), Berlin, Moscow International Travel & Tourism (MITT), Moscow, Arabian Travel Mart (ATM), Dubai, AUTIDIK at Moscow-Russia; World Travel Mart in London, Travel Fair at Portugal and FITUR in Spain-Madrid.

Goa for the first time participated in the Travel Fair in Lisbon, Portugal and FITUR in Spain which evinced tremendous response from the trade in these regions which may result into direct charters to Goa in the forthcoming season. The Tourism Department has also organised special promotions by organising the Road Shows within the country at Jaipur, New Delhi, Ludhiana, Baroda, Pune, Mumbai where special presentations were made before the members of the media and the travel

BOX 7.1

The Most Innovative Marketing Campaign Award to Goa

Galileo India, India's leading Global Distribution Services company for the airline and travel industries, along with Express Travel & Tourism from the Business Publications Division of the Indian Express Group, have announced the 'Galileo Express Travel & Tourism Awards' to honour organisational and individual excellence among travel industry professionals. Nominees for the Award included Andhra Pradesh, Australia, Chattisgarh, Dubai, Goa, Haryana, Himachal Pradesh, Hong Kong, Karnataka, Kerala, Maharashtra, Malaysia, Mauritius, Orissa, Rajasthan and Sri Lanka.

The nomination criterion was based on creativity in vision and enhancement in work and was measured against stringent qualitative and quantitative industry benchmarks in order to determine the highest standards of excellence practiced. Based on nominations and relevant material sent by all International and State Tourism Boards, the Jury selected '**GOA TOURISM & Its Campaign – GO GOA' for the Most Innovative Marketing Campaign Award**. Goa's latest campaign with 'Go Goa' as the logo and tag line of 'Go Goa 365 days on a Holiday' and its road shows in India & abroad, new website : www.goatourism.org with several competitive campaigns both in print and electronic media, touch screen kiosks, quality brochures, presentations in more than 25 cities and 5 countries have made this achievement possible.

trade. In order to widen its foreign tourist base Goa went to the extent of exploring Russian market which was not covered by India Tourism, although it has got vast tourism potentials which was proved with the tremendous tourist traffic generated through charter flights to Goa.

2 BEACH DEVELOPMENT AND SAFETY MEASURES

Sandy beaches of Goa are major tourist attraction. The beaches are divided into different stretches such as (i) Nerul-Sinquerim-Candolim-Calangute-Baga (ii) Anjuna–Ozrant-Vagator (iii) Morjim-Mandrem-Arambol (including lake area)–Keri (iv) Siridao-Bambolim (v) Bogmalo-Baina (vi) Palolem-Agonda-Galgibag-Canaginim-Betul (vii) Cavelossim-Varca-Fatrade-Benaulim (viii) Colva-Betalbatim-Majorda-Utorda-Velsao. All remedial & preventive measures have been taken to ensure that the beaches are kept neat and clean. Private contractors have also been engaged for cleaning, sweeping, collection / lifting and disposal of garbage including litters, plastic bottles / bags, waste etc. and also garbage generated by shacks from the beach shacks including sand dunes / vegetation area, parking area / approach roads / access to the beaches etc. Cleanliness drives are also carried out with the help of schools, NGO's on most of the beaches of Goa which are more frequented with tourists and beach goers. A committee has been constituted to closely monitor and check the cleanliness on beaches comprising of NGO's, members of local jurisdiction such as representatives of Panchayats / Municipalities, Hoteliers / Shack Owners, Officials of Department etc. The Committee members are authorized to certify the bills of the contractors who have been engaged for cleaning purpose after total satisfaction of their cleaning work.

Tourism Department is the nodal authority to grant permission for erection of temporary shacks on the notified beaches of Goa for the tourist season starting from October to 15th June every year. The application for shacks are invited through local dailies and the permission is granted after following the formalities and prescribed guidelines for allotment. Tourist booths have been installed on all the notified beaches of Goa for safety and security of the tourists / beach goers. The tourist police posted on the beach provide information to the tourists who prefer swimming, bathing and assist the Lifeguards on duty during the time of emergencies.

In order to ensure safety of beach goers / tourists, 59 Lifeguards have been deployed on the beaches of North & South Goa. The Lifeguards are provided with latest sophisticated equipment such as

Rescue boards, Rescue tubes, Oxygen rescue citators, Whistles, Binoculars, Loud hailers, Rescue ropes, Rescue lifebuoys, First aid Kits, swimming goggles, uniforms, flags, caps etc. so that the tourists/bathers are secured while swimming. Lifeguard stands are installed on most of the notified beaches which are more frequented with tourists, so that the Lifeguards can have a clear view of swimmer/bathers in the sea.

3. HERITAGE CONSERVATION-GOA HERITAGE HOUSE TOURISM SCHEME 2003

This is a land mark decision of the Government to launch a scheme for Heritage Conservation linking it to promotion of tourism in the State. Though, the Government of Goa published a report on 'Conservation of Buildings and Sites of Historical and Aesthetic Importance' way back in 1991, this is for the first time that a scheme has been announced for Heritage Conservation. The scheme is in operation from 31-7-2003.

The objective of the scheme is to conserve the houses of heritage value, which may get destroyed in course of time if no remedial action is taken and to promote cultural tourism using these heritage houses. A heritage house means any house that

a. has architectural, cultural, social, archaeological, historic or aesthetic value or has value for the community and /or is associated with a person and/or event of local , regional or national importance and

b. complies with the minimum age if any fixed by the Heritage Tourism Committee to be declared as a Heritage House and

c. is declared to be a Heritage House By the Heritage Tourism Committee on the recommendations of the Heritage House Committee.

The Department of Tourism shall assist the owners/managers of the Heritage House for the conservation of the Heritage House by providing the following financial assistance

i) **Soft Loan** not exceeding Rs. 5 lakh at an interest rate of 7.5% p.a. and with the requirement of repayment of the loan amount within a period not exceeding 10 years from the date of release of the first installment of the loan;

ii) **Hard Loan** not exceeding Rs. 5 lakh at an interest rate of 7.5% p.a. and with the requirement of repayment of the loan amount within a period not exceeding 5 years from the date of release of the first installment of the loan

iii) **Grant in aid:** Not exceeding Rs. 1.50 lakh subject to the condition that the Heritage House cannot be withdrawn from the scheme for a period of 3 years from the date of receipt of the grant in aid and also subject to fulfillment of all other terms and conditions of the scheme. The grant-in-aid sanctioned shall not be disbursed in cash to the owner / manager of the Heritage House but applied directly by the Department of Tourism to the cost of conservation project of the Heritage House.

The Department of Tourism will assist those owners/managers of the Heritage House who are interested in providing entertainment to visitors with an interest free Cultural Entertainment Loan not exceeding Rs. 50,000 with the requirement of repayment of the loan amount within a period not exceeding 2 years from the date of release of the amount.

Other salient features of the scheme

- The scheme shall be administered by the Heritage Tourism Committee
- The Heritage Tourism Committee shall frame detailed rules and regulations to implement the scheme. Technical panels to advise in the matters of architecture, construction, conservation, heritage and art shall be appointed by the Committee. Each panel shall consist of i) senior architect ii) conservation specialist and iii) expert in folk art and regional art history
- The technical panel shall carry out the professional work of preparing drawings , specifications, estimates, tenders, etc. for the conservation of each house separately. It will also make site inspections, certify the bills, lay down the norms of maintenance etc. Professional fees for the architectural experts will be as per standard professional fees recommended by the Council of Architecture constituted under the Architects Act, 1972 or the Indian Institute of Architects , a professional body registered under the Societies Registration Act, 1860 subject to the ap-

proval of the Finance Department. Professional fees to conservation specialist and folk art specialist will be decided by the Heritage House Committee with the approval of the Finance Department. Professional fees to the technical panel shall not exceed 10% of the cost of conservation project for a Heritage House.

After the conservation project for each Heritage House is prepared by the Technical Panel, the same shall be presented to the Heritage Tourism Committee for final clearance. The owner/manager of the Heritage House shall be invited to take part in such presentation and offer his/her views.

After the terms are finalised with the owner/manager , the Director of Tourism shall execute a Memorandum of Understanding (MOU) with the respective owner/manager of the Heritage House

The payment to the executing agency shall be released directly by the Director of Tourism based on the certification by the Technical Panel, and after obtaining confirmation from the owner/manager of Heritage House regarding satisfactory completion of work

The Department of Tourism through its Director shall have the right to inspect the selected house and verify the quality of services offered and heritage value from time to time and in case the entertainment provided in the Heritage Houses is not compatible with States current and traditional values and standards he has the right to ask the owners to cease the form of entertainment.

The Houses for viewing shall be kept open for visitors on days and at times specified by the owners and a donation box may be allowed to be kept for voluntary contribution from the visitor. The Heritage House for accommodation has to be registered as a 'hotel' under the provisions of the Goa Daman and Diu Registration of Tourist Trade Act , 1982 and conform to the provisions thereof. Sale of food and beverages to the residents will be permitted.

4. GOA DAMAN & DIU ANCIENT MONUMENTS AND ARCHAEOLOGICAL SITES AND REMAINS ACT 1978

During the year 2003-04, the Directorate of Archives and Archaeology has declared 10 archaeological sites/ monuments as protected sites/monuments under the Goa Daman & Diu Ancient Monuments and Archaeological Sites and Remains Act 1978 (table 7.1). Thus, in all there are 52 archaeological sites/monuments under the State Act.

Table 7.1: Archeological sites / monuments declared as protected sites / monuments during 2003-04

Sr. No	Name of the site	Village/locality	Taluka	Area	Ownership
1	The Fort of Reis Magos	Reis Magos	Bardez	10,000 m2	Govt. of Goa
2	The fortress of Khorjuve	Khorjuve	Bardez	20,000m2	-do -
3	The Cave of Sidhanath at Tar Surla	Surla	Bicholim	5000m2	-do -
4	The mosque and tank at Tar Surla	Surla	Bicholim	10,000m2	Private
5	British cemetry at Dona Paula	Dona Paula	Tiswadi		Private
6	Site of fortress at St. Estevam	St. Stevam	Tiswadi	10,000m2	Govt of Goa
7	The fort of Mormugao	Mormagao	Mormugao	5,000m2	Govt.of Goa
8	The site of Kaivailya math at Consua	Consua	Mormugao	4,000m2	Govt.of Goa
9	The cave at Shigao	Shigao	Sanguem	5,000m2	Private
10	The site of Narayandev at Vichundrem	Vichundrem	Sanguem	10,000m2	Private

5. CONSERVATION AREAS IN PANAJI – FONTAINHAS FESTIVAL

Experiments are being made to develop certain conservation areas as tourist attraction. Fontainhas festival in Panaji is an attempt in this direction. In this regard it is pertinent to note that the School of Planning and Architecture, New Delhi had conducted a comprehensive study in Panaji long time ago and identified Campal, Riverfront, Fontainhas and Portais as the possible major conservation areas in Panaji. These areas comprise approximately 62 hectares. Fontainhas, Portais and St. Tome area forms part of the road between the Ourem Creek and Altinho hillock. Some of the unique features of this are the informal layout of streets, winding lanes and stepped pedestrian accesses, informal layout of lanes with buildings abutting on narrow streets, uneven heights of different buildings, verandahs, sloping roofs, window details, mouldings and various other architectural details are different in each building but are nevertheless within the same general local style, colour of the natural pigments (brick red, pale yellow and ultramarine blue) with white mouldings and cornices, compactness of built-up forms owing to the high degree of ground coverage etc.

6. THE GOA BOAT RIDES AND WATER SPORTS ACTIVITIES GUIDELINES 2003

In order to regulate water sports activities in the State, the Government of Goa has framed guidelines for registration and regulation of boat rides and water sports activities on rivers, lakes in the State of Goa and also on sea along the coast of Goa.

Permission for running boat rides / water sports activities

a. Every operator of boat rides or water sports activity in the rivers, inland waters and lakes shall obtain due permission from the Captain of Ports.

b. Every operator of boat rides or water sports activity in the sea along the coast shall obtain permission from the Mercantile Marine Department for operation of the vessel in the sea.

Other Requirements

(i) No objection certificate shall be obtained from the Mormugao Port Trust if any boat ride or water sports activity is to be operated within or has to transit through the jurisdiction of Mormugao Port Trust

(ii) No objection certificate shall also be obtained from the Department of Fisheries and / or the Department of Water Resources as the case may be if any boat ride or water sport activity is to be operated either in the river, lake or sea along the coast. NOC / Licence from concerned local body if required by law shall also be obtained.

(iii) Those water sports equipment not requiring licence or certification either from the Mercantile Marine Department or the Captain of Ports will require certification from National Institute of Water Sports (NIWS) Caranzalem or any other agency approved by Government of Goa, Department of Tourism

(iv) The operator / applicant shall produce to the Department of Tourism, Government of Goa proof regarding insurance of the passengers as per the capacity of the boat equipment to the extent of at least Rs. 2.00 lakh per passenger

(v) All life saving equipment as required by the competent authority must be made available on board or at the bank / beach as the case may be.

(vi) The applicant shall produce a certificate issued by N.I.W.S, or Captain of Ports or M.M.D. as the case may be, showing that the crew have been trained for (a)manning and operating the boat / equipment; (b) life saving techniques.

(vii) Registration Number plate for a specific boat / equipment shall be obtained through the Department of Tourism at the cost of the operator and prominently affixed at a place on the boat / equipment to be specified by the Department.

(viii) Capacity of the boat as certified by the Mercantile Marine Department / Captain of Ports / National Institute of Water Sports shall be mentioned in the registration Number Plate and complied with strictly.

(ix) Area of operation shall be clearly specified by the operator as per permission obtained from the Captain of Ports or Mercantile Marine Department and the same shall be strictly followed for the activity.

(x) The water sports activity shall be operated in the specified / earmarked area only and within visual range.

(xi) The registration certificate shall be made available on board at all times.

(xii) The registration shall be valid for one year only.

(xiii) All safety norms prescribed by the Competent Authority and all the norms specified in the booklet "Safety Norms for Water Sports" issued by the Ministry of Tourism, Government of India shall be strictly adhered to at all times.

(xiv) The period and hours of operation as specified in the permission shall only be allowed.

The water sports activities allowed for operation are

- Boat rides / water scooters from jetty or ramp where embarkation and disembarkation of passengers can be carried out safely.
- Parasailing in specially marked areas, the beach / sea / river.
- Jet Ski in well defined areas beyond normal swimming and watersports areas.
- Water skiing & fun rides using powerboats outside the swimming areas of the beach / water body.
- Windsurfing & Dinghy, Kayaking, Canoeing, and Sailing in clearly marked area and within the visual range of the rescue boat driver.

7. GOA REGISTRATION OF TOURIST TRADE ACT, 1982

The Goa Registration of Tourist Trade Act, 1982 has been enacted in the State and is in force since April, 1985. Under the Act persons dealing with tourists and matters connected therewith have to register their activities with the Directorate of Tourism. The activities registered are Hotels, Travel / Excursion Agencies, Dealers in Handicrafts, Boats, Tourist Taxis, etc. The Hotels are further categorized under A, B, C & D category.The requirements for registration are N.O.C. from respective Panchayats / Municipalities, N.O.C. from Directorate of Fire Services and other licenses or certificates required to be held under the law from the competent authorities. The registration fees, which have to be paid annually are given in table 7.2.

The revenue collected under this Act during the year 2002-03 was Rs.10.84 lakh.

Table 7.2: Annual registration fee for tourism related trades / activities

1	Hotel Keeper	—	Rs.600 / - per year for 'A' category
		—	Rs.400 / - per year for 'B' category
		—	Rs.300 / - per year for 'C' category
		—	Rs.200 / - per year for 'D' category
2	Travel Agents	—	Rs.300 / - per year
3	Excursion Agents	—	Rs.200 / - per year
4	Dealers	—	Rs.200 / - per year
5	Tourist Taxi	—	Rs.120 / - per year
6	Tourist Boats	Tourst Boats & Launches carrying	Fees
		1 – 5 pax	Rs. 1000 / -
		6 – 12 pax	Rs. 1500 / -
		13 – 50 pax	Rs. 2500 / -
		51 pax and above	Rs. 5000 / -

8. OTHER ACHIEVEMENTS DURING 2003-04

- Aesthetic façade illumination of Shree Vijaydurga Saunsthan Keri- Ponda, Naga Masjid at Ponda, O.L. of Piety Church – Diwar and Shree Shakti Vinayak temple - Diwar has been completed.
- High Mast Lights at Utorda, Varca and Bambolim have been erected and commissioned.
- Construction of Sulabh Shauchalaya complex at Shree Shantadurga Kuncolienkarin temple , Fatorpa has been completed.

9. PROMOTION OF ART & CULTURE

In pursuance to Tourism policy of diversification to other forms of tourism, paramount importance is given for the promotion of Art & Culture in the State. Progress made during 2003-04 in this regard is as follows.

- Action has been taken for construction of Ravindra Bhavan at Fatorda , Margao and work will be commenced shortly.
- Action has been taken to set up Mini Kala Bhavans at Marcela, Bicholim, Mandrem, Canacona and Mapusa.
- Setting up artists village ' Kala Gram' at Xelvona in Quepem taluka on the line of 'Shilpgram' is being taken up.
- Under the scheme of Kala Sanman financial assistance is given to the persons distinguished in literature and arts. The quantum of assistance under the scheme has been enhanced from Rs. 500/-p.m for general category and Rs. 800/- p.m. for State Awardees to Rs. 1250/- p.m. and Rs. 1600/- p.m. respectively. During the year 2003-04, 499 artists are being covered under the scheme.
- During the year 2003-04, 14 awards will be presented to outstanding personalities in the field of culture under the scheme Presentation of State Cultural Awards. It is proposed to enhance the cash awards from Rs. 5000/- to Rs. 20,000/- and it is also proposed to institute an Youth Award. Besides, it is also proposed to confer Awards to the artists residing outside Goa State for their contribution in the field of Goan culture .
- 157 schools are receiving grants ranging from Rs. 50,000/- to Rs. 75,000/- to promote performing arts in schools. 23 cultural organizations are also receiving grants from the Government. 75 Bhajani mandals and 11 Government schools are being supplied with cultural equipment.

OVERVIEW

The Government has laid special emphasis on creating brand identity to Goa, developing infrastructure and diversifying from beach oriented tourism to other forms of tourism like heritage tourism, eco-tourism, adventure sports etc. The Government has launched aggressive campaign to accelerate its promotional movement by strengthening its marketing strategies through organizing road shows, participations in travel & tourism marts in India and abroad. In addition, extensive promotional campaign has been launched through print and electronic media. On account of these initiatives the tourist in flow to the State has increased considerably and also the State has bagged the 'Most Innovative Marketing Campaign Award' in the year 2003. The Government has taken remedial & preventive measures to ensure that the beaches are kept neat and clean. Lifeguards have been deployed on the beaches with sophisticated equipment to deal with any mishaps at beaches. The Government has also notified guidelines to regulate water sports activities in the State.

Towards Heritage Conservation and promoting Heritage tourism, the Government has launched Goa Heritage House Tourism Scheme 2003. Experiments are being made to develop certain conservation areas as tourist attraction. Fontainhas festival in Panaji is an attempt in this direction. The Government has declared 10 archaeological sites / monuments as protected sites / monuments under the Goa Daman & Diu Ancient Monuments and Archaeological Sites and Remains Act 1978. The State has taken significant steps in promotion of Art & Culture by taking up various schemes like construction of Ravindra Bhavan at Fatorda, Margao, Mini Kala Bhavans, setting up artists village 'Kala Gram' Kala Sanman scheme etc.

CHAPTER 8
Social services

The Government of Goa has started, in recent years, a number of initiatives like Cyberage Student Scheme, Infrastructure Development Loan scheme, Goa Scholar Scheme, Education Loan scheme, Interest Free Loan Scheme, Dayanand Social Security Scheme etc. This chapter covers all such initiatives and other major achievements under the sectors of education, health, social security and welfare, labour welfare and housing.

1. EDUCATION

A. School Education

The Government of Goa is giving huge emphasis on younger generation. Education being the foundation for human development some new programmes have been implemented with an aim at revamping the education system in the State. The programmes envisage infrastructure development in schools and making computer education an essential part of the education system. Also, the focus is on educationally backward talukas viz. Pernem, Bicholim, Sattari, Quepem, Sanguem and Canacona. The details of the programmes and the progress so far are given below. The details of enrollment in schools, educational institutions and drop out rates are given in Annexure 36 and 37.

(i) Infrastructure Development Loan

The scheme aims to provide interest free loans to the non-Government Aided Educational Institutions for infrastructure development with 50% grant from the Government. Infrastructure developments include buildings, equipment, toilet, water supply, library facilities, play ground etc. However, purchase of furniture like desks, tables, cupboards etc. purchase of laboratory equipment, library books, C.D. Rom, are also allowed with a proviso that any one item should not exceed 25% of the total loan amount. It also includes the cost of land for the purpose of construction of school building and playground not exceeding 50% of the total loan amount with prior permission of the department. The expenditure on salary, consumables like chalks, dusters, laboratory material like glass, plastic, chemicals is however, not covered.

A unique feature of the scheme is emphasis on educationally less developed talukas of Pernem, Bicholim, Sattari, Quepem, Sanguem and Canacona. The loan amount is disbursed through the Goa State Infrastructure Development Corporation. To become eligible for loan, the schools have to fulfill various eligibility criteria like minimum enrollment, passing percentage in SSCE etc.

The scheme is open to aided educational institutions and it should be at least five year old from the date of recognition. Schools which are more than 10 years old shall be eligible for a maximum loan of Rs.10 lakh at primary level and Rs.20 lakh each at secondary and higher secondary levels. Schools which are 5 to 10 years old shall be eligible for a maximum loan of Rs. 5 lakh at primary level and Rs. 15 lakh each at secondary and higher secondary levels. Eligibility of composite schools will be as per addition of each category. The scheme has comfortable repayment period of 15 years in equal monthly installment of Rs.278/- per lakh.

Under this scheme, so far Rs.15.62 crore has been sanctioned as infrastructure loan to 138 schools.

(ii) Cyberage Student Scheme

This is a new initiative of the Government aimed at providing extensive practical exposure to students in computer education. Goa is perhaps the only State in the country to start such a novel initiative. The scheme envisages supply of one personal computer for home use along with one UPS and necessary software to all students in standard XI and computer stream vocational course students.

A student has to apply in prescribed format with a registration fee as decided by the Government ev-

ery year. The fee may be refunded at the end of two years or after the student passes out, whichever is earlier, provided the computer with accessories are returned in good working condition. The application has to be routed through the school where the student is enrolled. A concession in fee to the extent of 50% shall be admissible to students of SC / ST & OBC communities (excluding creamy layer as notified) and also to students of higher secondary schools located in less developed talukas of Pernem, Bicholim, Sattari, Sanguem, Canacona and Quepem.

The equipment supplied shall carry a valid warranty of one year. Maintenance beyond the warranty period would be the responsibility of the student's parents / guardian. The equipment will not include printer, scanner and other additional accessories. The equipment supplied shall be the property of the Government for a period of two years from the date of supply and cannot be used for any commercial purpose or sold during the period.

The scheme has received tremendous response and during the year 2002-03, 2806 computers have been distributed. During the year 2003-04, over 10000 computers are being provided to students.

(iii) Report on status of Government school buildings

In order to provide good buildings to Government schools, the Government has prepared a status report of all the Government school buildings. School buildings have been classified as very bad, bad, average, good and very good in condition. Provision of drinking water and toilet facilities in all the schools is also being ensured.

(iv) Scheme for minor repairs to Govt. Primary and Middle School buildings through Parents Teachers Associations.

As regards the scheme for minor repairs through PTA information from all the Taluka ADEI's regarding forming of PTAs and opening of Saving Accounts in the bank is being collected. It is proposed to release funds to PTAs to carry out minor repairs.

(v) Training and Employment Board

The Government is in the process of constituting a Training and Employment Board to provide training / guidance to unemployed youth aged between 18 to 35 years who otherwise cannot be employed due to the requirement of education / training or experience. Under this project, training will be provided in various skilled and semi-skilled activities through Industrial Training Institutes and other Institutions. The Board will also have a link with Chief Minister's Rojgar Yojana under which trainees will be provided loans for setting up their own business after successful completion of training period.

(vi) Vocational Education

The scheme of vocational education is conducted at 10+2 stage with a view to provide an alternative to higher education and bring work and education closer. Various commerce based, agro based, home science based and technical education vocational courses have been introduced in the State in 40 higher secondary schools. Cent per cent area based opportunities are provided to students for training in vocational education by identifying suitable courses and imparting skills to the students in vocation that is able to provide employment opportunities.

(vii) Integrated Educational Complex at Bambolim

Land is being acquired for setting up an integrated educational complex at Bambolim. It was notified under section 4 of the Land Acquisition Act, 1894 in the Official Gazette dated 26th November 2002. Notification under section 6 of the said Act indicating that the said land is required for public purpose has also been issued in the local newspapers dated 4th January, 2004.

(viii) Mid School Meal Scheme

The Mid School Meal Scheme has been implemented from 9.1.2003 on pilot basis in Sattari taluka by providing cooked ready to eat food items. The scheme successfully continued in Sattari taluka till 12.4.2003 during the last academic year 2002-03. About 3828 students were covered under the scheme in the taluka of Sattari.

During the year 2003-04, the Mid School Meal Scheme is implemented with effect from 11.11.2003 in Sattari taluka in the first instance by providing cooked ready to eat food items like Kachori, Kadi ladu, groundnut chikki, Parle Glucose, Shankar Pali and Parle Marie. The scheme has been extended to Canacona and Bicholim talukas. The remaining three talukas of Pernem, Quepem and

Sanguem are also proposed to be covered during this academic year.

(ix) Education for disabled children

Under Disability Act 1995, there is a statutory responsibility on the part of the Central and State Governments to ensure that all disabled children receive free education in an appropriate environment till the age of 18 years. In Goa, there are five schools to cater to the need of disabled children which provide specialized education to the disabled children. Disabled students are also admitted in general schools without any discrimination.

B. Higher Education

Towards promoting higher and technical education in the State, the Government has undertaken many new initiatives during 2003-04. The details are as follows :-

a. Interest Free Loans for Higher Education.

(i) Objectives

The main objectives of the scheme are to promote pursuit of higher and technical education by younger population of Goa, to ensure that economic conditions and financial difficulties do not come in the way of such pursuit and, to assist eligible candidates by way of interest free loans to undertake approved degree and diploma courses at undergraduate and post graduate levels, in India or abroad.

(ii) Salient features

- The scheme provides for exemption from the payment of interest charges, as long as the loanees adhere to repayment schedule strictly. For this purpose, three different financing plans are offered. In the event that the situation faced by a particular candidate requires minor modification in the financing plan, the scheme provides for flexibility to make adjustments. Failure to comply with the repayment schedule, however, will attract the liability of payment of interest at a prescribed rate.

- The scheme enumerates the eligibility criteria to be followed for the selection of candidates, criteria for economic/financial constraints and details of approved courses that can be undertaken.

- The Government will establish a 'Higher Education Promotion Fund'(HEPF), from which all loans that are sanctioned under this scheme will be disbursed and to which all recoveries made under the scheme shall be credited. The Government will make arrangements for management of HEPF and its audit. (Box 8.1).

BOX 8.1
Higher Education Promotion Fund

The Government of Goa has established a fund viz. 'Higher Education Promotion Fund'(HEPF) through a initial contribution of Rs.2.5 crore. All loans sanctioned under the Interest Free Loan Scheme and all recoveries effected from the beneficiaries during the 'pay back period' shall be credited to this fund. The Government will take steps to replenish the quantum of funds available with HEPF from time to time. All scholarships sanctioned under the Goa Scholar Scheme will also be paid out of this fund.

Under the Interest Free Loan Scheme all loans disbursed during the year 2002-03 will be deemed to have been sanctioned out of HEPF. All candidates who availed of this scheme during 2002-03 must furnish an undertaking to the effect that they shall comply with this scheme before the next quantum of disbursal of loan is made to them.

For management of HEPF, Govt. may entrust the work to any department, public sector undertaking or corporation, financial institution, co-operative Bank or commercial Bank. Any management fees charged by such appointed managing agency/bank shall be paid out of HEPF itself.

HEPF will be subject to audit by Comptroller and Auditor General of India every year.

(iii) Eligibility criteria for grant of loans

Any person below the age of 30 years who was born in Goa and/or has been domiciled in Goa for at least 10 years shall be entitled to apply for, and receive, loans under the scheme provided he fulfills all of the following conditions: -

- Has passed the qualifying examination (XIIth class examination for undertaking undergraduate and post graduate diploma or degree programmes and Xth or XIIth class examination for undertaking diploma programmes) from an institution located in Goa.

- Has obtained 60% or more marks in the qualifying examination. (For candidates belonging to SC / ST / OBC categories, the qualifying marks will be relaxable by 10%).

- Meets the prescribed eligibility conditions relating to economic / financial constraints.

- Undertakes any of the courses approved under this scheme.

(iv) Criteria for economic/financial constraint

- Applicants whose total family income, including the income of both the parents, is up to Rs.2,00,000 per annum shall be eligible to apply for loans under this scheme for undertaking approved courses within India. In the event that a brother or sister of the applicant is also pursuing studies at higher / technical education level (whether or not such sibling has applied for, or availed, loan under this scheme), the eligibility limit for family income will be raised to Rs.3,00,000 per annum.

- For pursuing higher or technical education in an institution located outside India, the family income limit for eligibility to apply for loans will be Rs.4,00,000 per annum. In the event that a brother or sister of the applicant is also pursuing studies at higher / technical education level, within India or abroad, (whether or not such sibling has applied for, or availed, loan under this scheme), the relevant eligibility limit for family income shall be raised to Rs.5,00,000 per annum.

(v) Approved courses

Loans under the scheme can be availed of for pursuing any higher / technical education courses, at undergraduate or post graduate level in the fields of Engineering (degree and diploma), Dentistry, Medicine, Pharmacy, Architecture, Finance including Chartered Accountancy and Cost Accountancy, Law, Fine Arts, Home Science, Management, Environment and Computer Science.

The following conditions must be fully met:-

- The course requires full time attendance.

- The tuition fees payable to the institution for undertaking the course exceed Rs.10,000/- p.a. or on pro-rata basis, e.g. Rs.5,000/- per semester of six months duration.

- The course is recognized by any of the following institutions / authorities:-

  - University Grants Commission (UGC)

  - All India Council of Technical Education (AICTE)

  - Medical Council of India(MCI)

  - Dental Council of India (DCI)

  - Architectural Council of India (ACI)

  - Board of Technical Education, Goa

  - Any other recognized authority approved by Government of India or Government of Goa.

In respect of educational courses offered by institutions located outside India, the Government will maintain a list of approved Universities / Courses that can be pursued by loanees under this scheme. The Selection Committee constituted under the scheme shall have the powers to add names of additional Universities / Courses on this list after obtaining, and assessing, the necessary details relating to that particular University / Course.

(vi) Guidelines for reservation

Efforts will be made to accommodate all eligible applicants for grant of loan. In the event that the number of candidates has to be restricted in any particular year owing to the financial situation of the HEPF, it shall be ensured that not less than 2% of the number of selected candidates belongs to Scheduled Caste, 7.5% to Scheduled Tribe and 19.5% to Other Backward Classes. Further, 30% of the selected candidates in each category (General, SC, ST and OBC) shall be women. In case the number of otherwise eligible candidates among reserved categories is less than the percentage prescribed above the slots remaining vacant shall be offered to General category candidates.

(vii) Amount of Loan

Table 8.1 provides information regarding the maximum amount of loan that can be availed of by a candidate in a year.

Table 8.1: Maximum amount of loan that can be availed by a candidate in a year under interest free loan scheme

Sr. No.	Tuition fee charged by the Institution	Maximum Loan per year
1.	Rs.10,000 – Rs.20,000	150% of the tuition fee charged
2.	Rs.20,000 – Rs.50,000	125% of the tuition fee charged or Rs.30,000/- whichever is more
3.	Rs.50,000 – Rs.1.00 lakh	100% of the tuition fee charged or Rs.62,500/- whichever is more
4.	More than Rs.1.00 lakh	Rs.1.00 lakh per annum
5.	For studies abroad	150% of the actual tuition fees charged or Rs.4.00 lakhs p.a. whichever is less (exchange rate as on 1st July of that year shall be applicable).

Any amount received by the selected candidate as scholarship or benefit under any other scheme shall be deducted from the maximum loan permissible under this scheme. It shall be the duty of the selected candidate to inform relevant details in his application.

It shall be open to the applicants whether to avail loans at the maximum level of entitlement or to seek a lesser amount that would fulfill actual needs and will entail easier repayment schedule.

(viii) Financing Plans

Three different financing plans are envisaged under the scheme covering periods of 5 years, 7 years and 9 years. In the event that specific needs of an applicant are not met within any of the packaged financing plan, a modified plan may be worked out by the Selection Committee for such candidate after obtaining and assessing full details of the case and recording the reasons for non-adherence to one of the packaged financing plan.

The 5 year financing plan is suitable to the needs of those applicants who wish to pursue approved courses of upto 2 years duration, or are in the midst of their course and have upto 2 years of studies left.

- The 5 years financing plan envisages disbursement of approved amount of loan at the beginning of first and second years. Year 3 shall be the period of moratorium and Year 4 and year 5 shall comprise the 'Pay Back Period'(PBP). During the PBP, loanee shall repay monthly installments equal to 1/12th of the amount disbursed to him annually (in the year 1 and 2 above). For example, a candidate who received a loan of Rs.30,000 per year in the first and second years of the financing plan will have to pay back Rs.2500 per month during the PBP i.e. Year 4 and Year 5. Such monthly installment shall become payable on 1st day of every month and must be paid by 10th day of each month in order to claim exemption from payment of interest.

- It shall be open to the candidates to seek different quantum of loan amount in first and second years as long as the amount sought in a particular year is within the maximum prescribed limit. In such cases the monthly repayment installment shall be equal to an amount arrived at by dividing the total loan taken by 24 i.e. the number of months in Pay Back Period.

- No interest will be charged as long as the loanee strictly adheres to repayment schedule during PBP i.e. the year 4 and year 5.

- Failure to make regular and timely repayments in accordance with the repayment schedule during PBP will make the loanee liable to payment of interest. Such interest will be charged at a prescribed rate (not less then 10%) compounded monthly upto the end

of moratorium period, and then calculated as equated monthly installments at the same rate of interest.

The 7 year financing plan is suitable to the needs of those applicants who wish to pursue approved courses of upto 3 years duration, or are in the midst of their course and have upto 3 years of studies left. Applicants who are in the last year of their current course and desire to undertake an advanced level course after that may also wish to avail of this plan.

- The 7 years financing plan involves disbursement of approved amount of loan at the beginning of first, second and third years. Year 4 shall be the period of moratorium and Year 5, year 6 and year 7 shall comprise the 'Pay Back Period'(PBP). During the PBP, loanee shall repay monthly installments equal to 1/12th of the amount disbursed to him annually (in the year 1, 2 and 3 above). For example, a candidate receiving a loan of Rs.30,000 per year during the first three years of the plan will have to pay back Rs.2500 per month during the PBP i.e. Year 5, 6 and 7. Such monthly installment shall become payable on 1st day of every month and must be paid by 10th day of each month in order to claim exemption from payment of interest.

- It shall be open to the candidates to seek different quantum of loan amount in different years as long as the amount sought in a particular year is within the maximum prescribed limit. In such cases the monthly repayment installment shall be equal to an amount arrived at by dividing the total loan taken by 36 i.e. the number of months in Pay Back Period.

- No interest will be charged as long as the loanee strictly adheres to repayment schedule during PBP i.e. the year 5, year 6 and year 7.

- Failure to make regular and timely repayments in accordance with the repayment schedule during PBP will make the loanee liable to payment of interest. Such interest will be charged at a prescribed rate (not less than 10%) compounded annually upto the end of moratorium period, and then calculated as equated monthly installments at the same rate of interest.

The 9 year financing plan is suitable to the needs of those applicants who wish to pursue approved courses of up to 4 years duration, or wish to combine the remaining portion of their current course with a subsequent advanced level course.

- The 9 years financing plan involves disbursement of approved amount of loan at the beginning of first four years. Year 5 shall be the period of moratorium and Year 6 to year 9 shall comprise the 'Pay Back Period'(PBP). During the PBP, loanee shall repay monthly installments equal to 1/12th of the amount disbursed to him annually (in the first 4 years above). Such monthly installment shall become payable on 1st day of every month and must be paid by 10th day of each month in order to claim exemption from payment of interest.

- It shall be open to the candidates to seek different quantum of loan amount in different years as long as the amount sought in a year is within the maximum prescribed limit. In such cases the monthly repayment installment shall be equal to an amount arrived at by dividing the total loan taken by 48 i.e. the number of months in Pay Back Period.

- No interest will be charged as long as the loanee strictly adheres to repayment schedule during PBP i.e. the year 6 to year 9.

- Failure to make regular and timely repayments in accordance with the repayment schedule during PBP will make the loanee liable to payment of interest. Such interest will be charged at a prescribed rate (not less than 10%) compounded annually upto the end of moratorium period, and then calculated as equated monthly installments at the same rate of interest.

(ix) Application procedure and calendar

- Applications under the scheme will be invited in a prescribed format every year and will be received till a prescribed due date. All candidates, whether applying for the first time or for sanction of 2nd or subsequent installment of loan, will be required to submit their application with full details within the due date. Applicants who fail to submit the prescribed form duly filled in and complete with other necessary documents, shall not be eligible to be considered for grant of loan that year.

- All applications received under the scheme shall be registered and a unique registration number shall be generated and provided to the applicant. The registration number must be quoted in all subsequent correspondence and documentation.

- Applications for loans under the scheme will generally be decided upon by the Selection Committee by 31st August every year. Applicants may be required to submit additional details, or to appear for a personal interview before the Selection Committee, should such a need arise.

- Any applicant who furnishes false or incorrect information or fails to furnish any necessary information, shall be disqualified from obtaining any benefit under the scheme. In the event that some loans have already been disbursed to such disqualified applicant, the loans shall become payable at once and will attract the liability of payment of interest at prescribed rate.

- In the event that the applicant has applied for a course that involves deadlines in relation to payment of fee and other charges, it shall be the responsibility of the applicant himself to arrange adequate finances for meeting such deadlines. In such cases, the loan amount granted under the scheme can be utilised to pay back any such bridging loan obtained by the applicant from relatives, friends or Banks.

(x) Selection procedure

Activities relating to scrutiny of applications and sanctioning of loans shall be undertaken by the Selection Committee constituted by the Govt. The Committee shall frame, and be guided by, appropriate rules of procedure for smooth operation of the scheme. Such rules of procedure can be changed from year to year but generally will not be modified after notice inviting applications has been issued for that year.

(xi) Documentation required

- On approval of the application, and before the first disbursal of the sanctioned loan amount, the beneficiary applicant shall execute a bond in favour of the Govt. of Goa for repayment of the loan amount in accordance with the terms and conditions of the relevant financing plan. A similar bond will also need to be executed by one of the parent of the beneficiary applicant. In the event of neither of the parents are in a position to execute the bond, the same can be executed by a relative or friend who is of major age and, is a income tax payer or, owns a house or a plot of land exceeding 1000 sq.mts. in Goa.

- In the event that both the parents of the approved beneficiary applicant are deceased or special personal conditions exist, Selection Committee or the Govt. may permit that the bond may be executed only by the beneficiary applicant.

(xii) Situations not covered under the scheme

The Selection Committee constituted under the scheme or the Govt. may take necessary steps to remove the difficulties in response to a peculiar situation faced by an otherwise eligible or deserving candidate. In all such cases, reasons for making modifications that deviate from the scheme shall be recorded.

(xiii) Response to the scheme

The scheme has received overwhelming response. During the year 2003-04, up to December 440 students have been granted loans under the scheme.

b. Goa Scholar Scheme

(i) Objectives

The main objectives of the scheme are to promote, pursuit of Post Graduate studies by younger population of Goa, to ensure that economic conditions and financial difficulties do not come in the way of such pursuit and to assist meritorious candidates by way of scholarship to undertake post-graduate studies in institutions of proven excellence, in India and abroad.

(ii) Salient features

- The scheme envisages selection of a specified number of "Goa Scholars" from among the younger population of Goa and to provide them with financial assistance in their pursuit of Post Graduate studies. Intellectual distinction is necessary but not a sufficient condition for selection as a 'Goa Scholar.' Selection committee will seek among candidates excellence in qualities of

mind and in qualities in person that offer the promise of effective service to the world in the decades ahead. The Goa scholarships, are investments in individuals rather than in project proposals.

- The scheme enumerates the procedures to be followed for the selection of Goa scholars and details of approved institutions of excellence where these scholars can pursue post graduate studies.

- The scheme will be managed "by the Higher Education Promotion Fund" (HEPF) that has separately been established.(Box 8.1).

(iii) Eligibility for selection as Goa Scholar

Any person below the age of 30 years who was born in Goa and/or has been domiciled in Goa for at least 10 years shall be entitled to be selected as a Goa Scholar provided he/she fulfils all of the following conditions.

- Has passed the qualifying under -graduate examination from an institution located in Goa.

- Has achieved a meritorious rank in the qualifying examination.

- Possesses qualities of mind that offer the promise of effective service to the world in the decades ahead.

- Has obtained admission to an institution of proven excellence for undertaking post-graduate studies, as approved under this scheme.

(iv) Reservation of Candidates

In order to provide adequate opportunities to candidates belonging to different communities, the selection of Goa Scholars will follow a 200 point roster. Further, the selection process will provide equal opportunity to women candidates and will ensure that at least 30% of Goa Scholars selected in any particular year are women. In case an otherwise eligible candidate among reserved categories is not available for selection on a roster point, the slot shall be offered to a General category candidate.

(v) Preference to candidates facing economic/financial constraint

The scheme envisages promotion of merit. However, other qualifications and claim for selection being similar, preference will be given to applicants whose total family income, including the income of both the parents, is up to Rs.2,00,000 per annum for national scholarship and Rs.5,00,000 per annum for international scholarship.

(vi) Approved courses

Selected Goa Scholars can pursue post-graduate studies in any institution of proven excellence in India or abroad, in the fields of Engineering, Dentistry, Medicine, Pharmacy, Architecture, Finance, Law, Fine Arts, Home Science, Management, Environment and Computer Science, Administration or any other field of importance to the State. The following conditions must be fully met

- The course requires full time attendance.

- The institution concerned is of proven excellence and has been included in the list of approved institutions under the scheme.

- The course offered by the institution is recognized by any of the following bodies/authorities.

  - University Grants Commission (UGC)

  - All India Council of Technical Education (AICTE)

  - Medical Council of India (MCI)

  - Dental Council of India (DCI)

  - Architectural Council of India (ACI)

  - Any other recognized authority approved by Govt. of India or Govt. of Goa

In respect of courses offered by institutions located outside India, the Govt. will maintain a list of approved Universities/Courses where Goa Scholars can pursue studies under the scheme. The Selection Committee constituted under the scheme shall have the powers to add names of additional Universities/Courses on this list after obtaining and assessing, the necessary details relating to that particular University/Course.

(vii) Amount of Scholarship

The amount of financial assistance granted to the selected Goa Scholars (National) shall be Rs.2,00,000 per year for two years, payable on a half-yearly basis. Selected Goa Scholars (International) shall be granted a one time financial assistance of an amount equal to US Dollars 15,000. Disbursement of financial assistance shall be made in Indian Rupee in all cases.

(viii) Selection procedure

Selection Committee constituted by the Government shall undertake all activities relating to scrutiny of applications and selection of Goa Scholars. The Committee shall frame and be guided by appropriate rules of procedure for smooth operation of the scheme. Such rules of procedure can be changed from year to year but generally will not be modified after applications have been called for a particular year.

(ix) Situations not covered under the scheme

The Selection Committee constituted under the scheme or the Government may take necessary steps to remove the difficulties in response to a peculiar situation faced by and otherwise eligible and meritorious candidate. In all such cases, reasons for making modifications that deviate from the scheme shall be recorded.

(x) Scholarship granted during the year 2003-04

The scheme received a tremendous response. In all 48 applications were received for various post graduation courses, out of which 23 students were short listed by the Screening Committee. Upon interviewing these students and also considering admission and other parameters like analytical, admission, initiative, creative, exra-curricular, motivation and overall qualities, the following four students were granted scholarship.

(i)	Mr. Neil Rodrigues	-	Engineering (International)
(ii)	Mr. Ninad Laud	-	Law (International)
(iii)	Dr. Gayle Rebello	-	Medicine (International)
(iv)	Miss Bilwa Navathe	-	Engineering (National).

c. Financial assistance to non- Govt. Aided Colleges for maintenance and upgradation of infrastructure facilities

(i) Objectives

The objective of the scheme is to support the efforts of the management of the non- government aided colleges for provision of better infrastructure facilities and creation of conducive atmosphere for learning.

(ii) Coverage

The scheme shall be applicable to all non government aided colleges in Goa. However special preference will be given for colleges from underdeveloped talukas such as Canacona, Pernem , Satari, Sanguem Quepem and Bicholim

(iii) Eligibility

All aided non government colleges including professional colleges shall be eligible to apply for the loan subject to the following conditions.

- The minimum enrollment shall be 200 in the underdeveloped talukas and 300 in other developed talukas
- The College should be at least 5 years old from the date of recognition. Priority will be given to the older institutions
- The average passing percentage of all the streams for the last 3 years should not be less than 50% in less developed talukas and 60% in developed talukas. The colleges with higher passing percentage will be given priority

(iv) Amount of Loan

Colleges with 5 to 10 years of standing shall be eligible to receive maximum loan of Rs. 40.00 lakh. Colleges with more than 10 years shall be eligible to receive maximum loan of Rs. 50.00 lakh

(v) Purpose of Loan

The loan is meant for improvement of infrastructure facilities such as construction of new buildings/extension of existing buildings/renovation of existing buildings /provision of toilets, construction of play grounds etc. 50% of the loan amount

sanctioned could also be utilised towards the cost of land provided prior permission of the Government is obtained. Expenditure towards consumables shall not be covered.

(vi) Disbursement of Loan

The loan amount shall be disbursed through the Goa State Infrastructure Development Corporation in the following manner:

- 50% of the amount sanctioned shall be released immediately on the issue of sanction order
- 45% of the loan amount shall be released on the production of a certificate from Government registered engineer and chartered accountant to the effect that the first installment has been fully utilised
- The balance 5% of the loan amount shall be released subject to production of the certificate that the previous amount sanctioned was utilised , that the project is completed and completion certificate is produced by the management, and the occupancy certificate is issued .

(vii) Repayment of loan

- Repayment of the loan shall not attract payment of interest as long as the loanee institution adheres strictly to the repayment schedule
- Repayment of loan would start from the 7th month from the date of sanction irrespective of as to whether the project is complete or not. Work should start within 6 months failing which loan sanctioned will be recovered from grants
- Loan shall be repayable in 15 years in equal monthly installments of Rs. 556 per lakh per month.
- The installment will become payable on 15th of every month till the loan amount is fully repaid
- The loanee institution shall be entitled to claim half the amount of installment due as grant in aid if the payment for all the previous period as well as the current period had been made within the due date of payment.
- In the event of default to repay the loan within 15 days of the due date, the loanee institution shall not be eligible to seek the grant in aid component . In case of default of 3 consecutive installments, the entire amount outstanding shall be liable for penal interest @ 12%. The Minister of Education may grant relief in delay of payment for a period not exceeding 3 months under exceptional circumstances
- In case of default on the part of management to repay, the loan installment shall be recovered from the payment of non-salary / salary grants payable to the College by the Directorate of Higher Education. The Government also reserves the right to call back the entire amount of loan with penal interest of 12% per annum.
- The payment will be made by challan.
- The repayment of loan along with interest if any to the Infrastructure Development Corporation shall be settled as mutually agreed upon.

(viii) General Conditions

- Any earlier outstanding loan shall be adjusted against the loan to be disbursed under the scheme.
- Assets acquired shall not carry any rental value for the purpose of reimbursement at any time in future.
- The college shall not be eligible to get similar benefits under any scheme of Government of Goa.
- The college shall follow all financial rules and regulations as laid down by Government from time to time.
- The Government reserves the right to accept or reject any application depending on any assessment.
- The loan availed shall not be clubbed with any of the scheme of the UGC. The Government may relax any of the eligibility criteria.

(ix) Process of application

The eligible colleges desirous to avail the loan facilities shall apply in the prescribed format. The application shall be supported by an approved plan and the cost estimate of the project dully certified by a registered engineer and chartered accountant. The application should also be supported by latest audited statement and documents of sale deed/long lease deed or any other documents which indicate the ownership of the land where the building stands.

During the year 2003-04, up to December, six valid applications have been received seeking financial assistance under the scheme.

d. New schemes in the offing

The Goa Cyberage Student scheme is proposed to be made applicable to higher education also. A scheme for granting financial assistance for organizing and attending conferences, seminars, workshops on the subject of educational importance is also contemplated. Details of both the schemes are being worked out.

C. Technical Education

a. Directorate of Technical Education

(i) MOU with IIT Bombay

- The Government of Goa has signed a Memorandum of Understanding with Indian Institute of Technology Bombay in July 2001. This MOU envisages upgradation of Technical Education in the State and giving competitive edge to people of Goa in the global scenario.

- A Conference " Strategic Directions for Technical Education in Goa" held in November 2001 discussed the existing scenario of technical education in the state, future prospect and policy formulation for achievement of the goals.

- Senior Professor of IIT, Mumbai is deputed to give his service as Professor in Charge and Principal of College of Engineering , Goa at Famagudi, in pursuance of upgrading the academic activities at the college. 52 positions of teaching staff at the level of Professor/Assistant Professor/Lecturer including 22 new posts are created and taken up for filling. 10 existing faculty members are deputed for undergoing Ph.D/M.Tech programmes.

- In order to improve the faculties, more than 50 temporary teachers are appointed to supplement the existing regular teaching staff for smooth and effective conduct of teaching process.

- The overall facelift project at Goa College of Engineering campus costing Rs. 37 crore has been taken up. This project encompasses repairs, maintenance and modifications in the existing building structure and construction of new buildings for hostel , IT laboratory, library extension etc.

- The MOU has made it possible for the engineering students from Goa to be admitted to M.Tech programme at IIT Bombay from outside GATE admission process.

- Expert Lectures of highly qualified faculties from IIT Bombay have been organized in Goa College of Engineering for the benefit of the students. Some final year students have been permitted to take up the project work at IIT Bombay under the guidance of their faculty.

- Dr. S.L. Narayanamurthy, Ex- Dean Resource Mobilization & Professor in Chemical Engineering , IIT Bombay has been appointed as Advisor, Governemnt of Goa on Technical and Higher Education. His experience and expertise in the field of Technical Education is expected to have a remarkable impact on the technical education in the State of Goa. He is also entrusted with the additional responsibilities of Director of Technical Education.

- Leveraging Goa Government-IIT Bombay MOU for Quality Enhancement in Food Technology Department at Government Polytechnic Panaji, steps have been taken to upgrade the department. As IIT Bombay has a significant expertise in Food Processing and related areas through execution of major research and development consultancy and extension project, a recent visit by experts to Govt. Polytechnic, Panaji will help the Department in upgrading infrastructure, laboratories and equipment thus offering opportunities for faculty members to do consultancy projects with the food industry in Goa which will also be beneficial to the students for placement after completion of the Diploma Programme

(ii) Setting up of IIT Bombay Campus in Goa

As regards setting up of an IIT Bombay campus in Goa (Box 8.2), recently an interim report of Committee nominated by the Director, IIT-Bombay was submitted to the Government with recommendations and observations of an action plan for identifying the academic programme of interest for the Goa Campus of the Institute, which will take into account the requirements of the State and in the context of national priorities.

It is proposed that the activities of the campus in the initial phase will focus on short courses catering to local needs and interest. In the second phase Postgraduate level programmes such as IIT-Diploma, Master's Degree level programmes certificate courses, doctoral programmes will be developed and conducted primarily in view of faculty and research activities in selected areas considering local aspirations, National Level Policies and International Scenario. The requirement of infrastructure, land and utilities location, budgetary provisions are presently being worked out.

BOX 8.2

IIT Bombay and BITS Pilani in Goa

As per the MOU signed with the IIT Bombay, the Government of Goa has already initiated steps for setting up of an IIT Bombay campus in Goa. In private sector, the Birla Institute of Technology and Science, Pilani is also in the process of setting up an extended campus in the State.

Virtual extended campus of IIT- Bombay has been set up at Goa University Campus. This has created on line distance education facilities for the benefit of students of Goa in general and students of technical education in particular.

(iii) Goa Education Development Corporation

The Government of Goa has set up a Goa Education Development Corporation to promote and assist in the rapid and orderly establishment, growth and development of educational institutions, service and facilities in the State of Goa in order to develop competitive flexible and value-based education system in meeting the individual, institutional and socio-cultural developmental need of the people of Goa. Functions of the Corporation include

- To establish and manage educational estates at places selected by the State Government,
- To develop educational areas selected by the State Government for the purpose and make them available for educational institutions to establish themselves,
- To promote schemes for preparation of graduates for the knowledge based economy and society,
- To develop educational testing services and make them available to educational institutions for selection of candidates,
- To develop facilities relating to information technology, such as broadband connectivity and wide area networks, and make them available for educational institutions,
- To develop learning and training materials, including e-learning software and technology,
- To promote schemes to increase literacy, in particular that relating to information technology, at all levels,
- To implement schemes for the benefit of students, such as loans and scholarships, libraries and book banks, publication of books, involvement of parents in education process, to assist them in their pursuit of vocational, higher and/or technical education,
- To encourage development of research facilities,
- To hold conferences, workshops, panel discussions etc. in educational field,
- To undertake schemes or works either jointly or on agency basis, with other corporate bodies or institutions, or with the Government, in furtherance of the purpose for which the Corporation is established and all matter connected therewith.

Long term objectives of the Corporation are

- To act and serve as the interface between the education system industry and the student
- To identify and bridge the gap between unfulfilled needs and demands of each sector of the education system and industry
- To transcend the artificial boundaries that de-limit schools, junior / senior colleges and technical institutions by collectively engaging them in the process of moulding and encouraging the student to pursue that career for which he / she is best suited.
- To equip the education system with alternative modes of delivering education through the marketing of education related services
- To promote schemes that will prepare graduates for knowledge - based economy and society
- To update and enable the faculty of various technical institutions to keep abreast with latest development in related fields by organizing conferences, workshops , seminars etc.
- To generate awareness regarding alternate career paths

Short term objectives of the Corporation are

- To facilitate career building by ensuring a 'right fit' between an individual's chosen vocation and personality traits through counselling and aptitude testing
- To develop the competitive edge of students in Goa , commensurate with that of their counterparts in other States in India

b. Goa Engineering College

Major achievements during the year 2003-04 are as follows :

- The student intake capacity has been increased from 150 to 320 since 2001 due to which it is decided to expand infrastructure and get a new building constructed for Information Technology department. The construction work of a shopping and utility complex for the use of students and campus residents is to start in the near future. The renovation of building for Medical Centre, to set up medical facility to students, staff and campus residents of this College is in an advanced stage. A book bank has been developed at the College library to help needy students in which approximately 2000 volumes are added annually.
- Consultancy rules have been sent to the Government for revision, which would make it possible to increase internal revenue generation. Teaching faculty is encouraged to submit proposal with AICTE for modernization of laboratories.
- The basic amenities like water and electricity are in a better state of supply during this year to the College campus. The planned shopping complex will give relief to the students, staff, visitors and campus residents. The transport facility to the College campus is in a very good state as all 3 old College buses are renovated and one new mini bus and TATA Sumo is purchased recently. The KTCL has started direct buses to this College one from Panaji and one from Vasco, both in the morning and evening. Medical facilities to the College campus are about to start shortly.
- The Academic session 2003-04 started on 21st July, 2003. 55 Lecturers were recruited and appointed as contract / visiting faculty well in time causing no delay or disruption of teaching and examination work. 22 new teaching posts and 11 non-teaching posts have been created for this College in this financial year. The process of appointing permanent faculty on regular basis through GPSC has already started. A well equipped computer centre consisting of 100 PC's, a new EPBAX system, internet and cable connectivity facility for the College campus have been established. Further the internet connectivity is in the process of getting expanded from 64 to 128 KBPS.
- The campus recruitment has proved to be the best in this year and has reached as high as 90% placement. This substantially leads towards the college development.

c. Government Polytechnic, Panaji

Achievements during the year 2003-04 are as follows :

- The Institute has been selected as "Replication Institution" under Canada India Industry Institute Program

- The Institute arranged training in Computer Graphics for deaf and dumb students in the Computer centre set up for 'Persons with Disabilities'. These students have shown remarkable progress in developing necessary knowledge and skill in this area. Some of them are offered jobs on the basis of their competency.

- The Institute provides "Value added and employment oriented education" particularly by arranging industrial training to some of the students and also by offering need based courses as optional.

- A separate computer Lab is set up for training persons with disabilities and also rural youth.

- To promote increased interaction between industry and educational institution and thereby equip the students to meet the demand of the jobs market, the Institute has set up Industry Institute Partnership cell funded by the Ministry of Human Resource Development.

- The polytechnic has generated revenue of over Rs.12.75 lakh through internal resource generation activities such as production work, testing, consultancy, training etc. Certificate course in 'Interior Design & Decoration' is offered by the Department of Architectural Assistantship. The Department of Civil Engineering has offered consultancy service to various organizations in 'mix design' and has carried out survey projects.

d. Government Polytechnic, Bicholim

Major achievements of the institute during the year 2003-04 are given below :

- Steps are taken towards close collaboration between parents and teachers for inculcating right social, moral and cultural values in the young minds of the students of this polytechnic. Open day and parents meet is planned in February 2004. Series of lectures by eminent personalities were held in the field of personality development, attitude formation, moral values etc., for the benefit of students.

- Web site of Government Polytechnic, Bicholim was launched with domain name www.gpb.nic.in.

- Internal Revenue Generation during the period April to November 2003 increased to Rs.52,262 as compared to Rs.38,313 during the same period of the last year.

e. Goa college of Architecture

Major achievements of the College during the year 2003-04 are as follows :

- The Government of Goa has accepted the proposal to establish 'Goa Institute of Design' as an umbrella institute for the growth of various design schools in the State. Resources to the tune of Rs.12.00 crore are required over a period of five years to establish the Institute.

- It has been decided to invite eminent scholars from different countries and international institutes to spend some time (from one semester to two years) in residence at Goa and teach at the College of Architecture. This is intended to expose the in-house teachers to global trends and better teaching methods. One Professor from the University of Canberra, Australia is already in residence at present.

- University of Birmingham (UK) and University of Canberra (Aus) are actively negotiating with the College to start a joint exchange program.

- Under exchange programme, the University of Porto (Portugal) has sent a student to pursue his final thesis.

- The consultancy cell " Research and Design Cell" is already functioning and is dealing with four projects viz. KTC Terminus Panaji, Patradevi Memorial, IFFI and Goa University landscaping.

- The College has initiated a programme to undertake study of Goa's urban environment and propose solutions. A project for upgradation of the KTC transport terminus is developed. A similar study is under way for development of Vasco city.

- The college became the first college to offer a "Placement Program" for its graduates. Out

of a class of 31 students, 10 graduates were placed in various professional offices through campus interviews.

f. Goa College of Pharmacy

A new post graduate course, M. Pharm, in Pharmacology has been started with an admission capacity of 5 seats. The number of degree seats has been increased from 40 to 60. The College is selected as Replication Institute of Canada- India Institute Industry Linkage Project (CIIILP).

2. HEALTH

a. Health Infrastructure in Goa

Goa has one of the most extensive health systems in India. At present, the Directorate of Health Services has a network of 172 sub-centres, 29 RMDs, 19 PHCs and 5 CHCs. In addition to this, the two District Hospitals viz. Hospicio Hospital, Margao in South Goa District and Asilo Hospital, Mapusa in North Goa District, and three other specialized / general hospitals viz. Leprosy Hospital, Macazana; T.B. Hospital, Margao; Cottage Hospital, Chicalim; also serve as referral Hospitals. In all there are 1234 beds in these health institutions, of which 424 beds are attached to CHCs / PHCs. There are 18 Dental clinics, two Homeopathic Dispensaries (one attached to UHC, Panaji and the other attached to CHC, Pernem) and three Ayurveda Dispensaries. There are four Urban Health Centres in the four major towns of Panaji, Margao, Vasco and Mapusa and one Medical Dispensary at Sada, Vasco and one STD clinic at Baina, Vasco.

Goa has 152 hospitals both in private and public sector. In all, there are 2070 allopathic doctors registered under the Goa Medical Council and 466 dentists registered under the Goa Dental Council as on 31.12.2003. Thus, the doctor-population ratio works out to 1:636 and the population served per bed works out to 275. The dentist-population ratio is 1: 3213.

Out of 172 sub-centres only 35 are functioning in Government buildings, 129 are in rented buildings and remaining are operating from rent-free panchayat / voluntary society buildings. Government is planning to construct new buildings for the sub-centres. At present, construction of buildings of two sub-centres at Verna and Rivona has been approved by the Government. The construction work of buildings for the two sub-centres at Varca and Bhatpal is proposed to be taken up during 2004-05.

In order to ensure availability of doctors in sub-centres, a new scheme to employ Residential Private Practitioners on consolidated remuneration of Rs. 6,000 / - per month has been evolved. They have to conduct OPDs at the sub centres and have to give an undertaking that they will be available round the clock for service. However, they are allowed to do private practice after office hours.

The State has a shortfall in number of PHCs as per the Government of India norms. As such, it is proposed to establish 5 Primary Health centres at Porvorim, Chimbel, Colva, Chicalim, & Mayem in a phased manner. Out of 19 PHCs, 18 are operating from own buildings and one PHC in Chinchinim is functioning in a rented building. Land has been acquired for construction of building for PHC, Chinchinim. As regards CHCs, land for construction of building for CHC, Canacona has been identified and the work has been allotted to GSIDC. The upgradation of PHCs to CHCs at Sanquelim, Sanguem, Bali and Bicholim is under process. It is also proposed to take up construction work for setting up of Kidney Specialty Hospital with dialysis facilities at Canacona shortly.

b. New Buildings of District Hospitals & Institute of Nursing Education

The Government is determined to construct a full-fledged building for District Hospitals Asilo and Hospicio. The Government has allotted the work to the GSIDC to construct 5 storied building for Asilo Hospital at Peddem, Mapusa. As regards the Hospicio Hospital at Margao, the land belonging to T.B. Sanatorium situated at Monte Hill, Margao has been identified for construction of new building. An area of 15 hectares at Bambolim, near Panaji for construction of Institute of Nursing Education has been identified and construction of the same has been entrusted to GSIDC.

c. Viral Hepatitis E (Jaundice)

An outbreak of Viral Hepatitis E (Jaundice) came to light mainly in Panaji during August – September 2003. Though, cases were reported from all over Goa, majority of them were from Panaji with most of them giving a history of having eaten at a particular restaurant. The total number of cases reported during the outbreak was 453 and only one death was reported.

The State Administration acted swiftly to contain the spread of disease. Apart from tracing the contaminated pipeline and stopping the water supply

through it, extensive measures were taken such as IEC through newspaper advertisements, talks on Doordarshan, All India Radio, private news channels and cable TV, posters, pamphlets, school rallies etc.

Regular meetings were held to monitor the situation. All Health personnel were alerted and private doctors were contacted and asked to look out for cases. Surveillance data was pooled from all Health Centres, Medical College and private doctors. Teams were formed of personnel from District Collectorate, Health, Municipalities, Police, FDA etc. to verify hygienic conditions in restaurants and to act on the spot to prevent any outbreak. The strategies worked very well and the disease was controlled within a reasonable time with minimum damage. A high level Task Force has been set up to investigate on the factors that led to the outbreak and to suggest the measures to be taken to prevent any future eventuality in this regard.

d. SARS

In the year 2003, a new disease viz. SARS emerged as a serious threat to humankind. In India, the first case of SARS was suspected in Goa on 10th April 2003(Box 8.3). However, the State had already geared up for the new disease much earlier. On 2nd April 2003, the then Health Secretary, Government of Goa had personally supervised institution of various measures for tackling the disease, should it be detected in Goa. Goa Medical College and Directorate of Health Services were collaborating with the Port and Airport authorities and all passengers arriving from the 13 SARS affected countries were screened before entering the State. Isolation facilities were arranged at GMC as well as Cottage Hospital, Chicalim and ID Hospital, Ponda. Doctors were advised to be alert and refer suspected cases to GMC.

From the time before the first case was suspected till the time the threat of SARS receded from the world, the State has acted proactively in the best interest of the people of the State and taken all necessary steps to prevent as well as deal with SARS. Guidelines laid down by international agencies like WHO and national authorities like the National Institute for Communicable Diseases were meticulously followed. In fact, Goa was the first State in the country to take measures against the disease and set up a screening mechanism.

BOX 8.3

FIRST CASE OF SARS IN INDIA WAS SUSPECTED IN GOA

A private doctor from Vasco da Gama referred a patient to Goa Medical College on 10th April 2003 with suspicion of SARS in view of fever with history of travel to the SARS affected country. He was admitted to the isolation room at GMC and investigated. His samples were sent to the National Institute of Virology (NIV), Pune. He was discharged on 12th April since he was afebrile and without symptoms, as per the guidelines from WHO. The NIV, Pune declared the case as SARS on 16th April 2003 based on the serological tests conducted by it. He was readmitted for tests and follow-up and discharged after 2 days. He has recovered fully. Another case admitted at Cottage Hospital, Chicalim was investigated and found to be negative for SARS and discharged after treatment.

Various measures taken by the State to deal with SARS included setting up of a High level Task Force, creation of SARS Isolation Wards, Port & Airport Health check-up of passengers, 24-hour SARS Information Centre at GMC, Contact Tracing & Epidemiological investigation etc.

e. Malaria

Though statistics indicate declining trend in incidence of Malaria, the disease continues to be one of the important public health concerns in the State. To combat this disease the State administration is working tirelessly with active involvement of various Departments like Directorate of Health Services, Municipalities, Panchayats, PWD, Registrar of Co-operative Societies, Town and Country Planning Department etc. Important measures taken to tackle this include: (i) Anti Parasite Measures, (ii) Anti Vector Measures and (iii) Public Health Education through IEC Activities.

Under Anti Parasitic Measures, early diagnosis and prompt treatment of Malaria cases through Active, Passive, Mass and Contact parasitic surveillance of weekly interval in Malaria problematic localities like Porvorim, Panaji, Candolim and Margao areas and fortnightly interval in all other areas were carried out. Additional man-power was also deployed for surveillance work. During this surveillance, a total of 278647 patients were screened

for malaria (upto December 2003). Under Anti Vector Measures, regular/recurrant antilarval operations(Physical, Biological and Chemical Measures) and fogging with Pyrethrum 2% extract were carried out.

Various public health education activities like public meetings, talks to public/students, programmes on AIR and Doordarshan, press notes/releases and distribution/display of printed posters, leaflet on various aspects of prevention/control of the disease etc., were carried out to create/increase public awareness and to encourage community participation in eliminating mosquitogenic/malariogenic conditions.

On account of these concerted efforts, during the year 2003 (January to December), the total number of malaria cases and falciparum malaria cases have declined to 11370 and 1503 cases respectively as compared to 16818 total malaria cases and 3348 falciparum cases during the corresponding period of previous year.

essary measures by the community and 17666 health cards to labourers working at the construction sites were issued and 6556 cards were renewed. Fines to the tune of Rs. 4,18,400/- were imposed and Rs.1,41,800/- was collected as fine against defaulters during 2003. The distribution of patients of communicable diseases is given in Annexure 38.

f AIDS Control Programme

The most serious health challenge the entire world facing today is HIV/AIDS. Controlling its spread and taking good care of its victims has become an inevitable task. Rigorous efforts are being made to prevent the spread of HIV/AIDS at initial stage itself by involving NGOs and the community. HIV prevalence rate among STD attendants in different sentinel sites in Goa from 1994 to 2002 is given in table 8.2. Some of the strategies adopted to prevent and control HIV/AIDS in Goa are as follows.

Table 8.2: HIV positivity among STD patients in Goa from 1994 to 2003*

	Prevalence rate (%) among STD attendants	
1994	12.64	STD clinic Baina
1995	21.65	STD clinic Baina
1996	20.27	STD clinic Baina
1997	15.79	STD clinic Margao
1998	19.61	STD clinic Baina and Margao
1999	13.47	STD clinic Baina and Margao
2000	12.02	STD clinic Margao and GMC
2001	14.23	STD clinic Margao and GMC
2002	13.15	STD clinic Margao and GMC
2003	14.67	STD clinic Margao and GMC

* sentinel surveillance

During the year 2003, appropriate epidemiological investigations of reported malaria cases were done and it was observed that the disease is prevalent at construction sites. Out of 11370 cases of Malaria, 8664 cases were detected among workers in hotels and construction sites. Accordingly, appropriate action plans were drawn with particular emphasis on affected localities.

As per the provisions under the Goa Public Health Act, pertaining to malaria/mosquito breeding prevention/control, 496 notices for undertaking nec-

With no cure, the only way to control the spread of HIV infection being promotion of behavioural changes for adoption of healthy life styles, several innovative communication strategies have been undertaken to raise awareness, improve knowledge and understanding, attitudes and skills needed to bring above changes in behaviour, safer sexual practices, negotiating schemes, etc. in the general population, particularly the youth and other marginalized groups.

To reduce transmission among the high risk groups

such as, sex workers at Baina, migrant labourers, homo sexuals, hotel staff and tourism related workers; targeted intervention (TI) projects have been undertaken through the NGOs with financial assistance from the Goa State Aids Society. During 2003-04 upto 31st December 2003, 10 TI projects have been funded and an amount of Rs. 63 lakh has been sanctioned to NGOs.

To minimize the risk of transmission of HIV through blood and blood products, all the Government Blood Banks have been modernized and suitably strengthened with requisite equipment, manpower, consumable, etc. Testing for transmissible diseases like Hepatitis B&C, Syphilis, Malaria and HIV has been made mandatory. During the year 2003-04 (upto September), 7717 blood units were collected in the three Government blood banks, of which voluntary donations accounted for 3415 units and 4302 blood units were collected as replacement. Out of these, 90 blood units (1.17%) were found to be HIV reactive.

The Family Health Awareness Campaign was conducted in the months of May – June 2003 and all health centres under DHS were involved in awareness raising, detection and prompt treatment of STDs. A total of 10053 people from the target group (15-49 years) attended the camps held at various Health Centres. Voluntary counseling and testing centres (VCTC) have been set up at GMC and Hospicio Hospital. During the period January to October 2003, 9705 blood samples were tested for HIV in these centres of which 877 were found HIV positive.

g. Mediclaim scheme

The scheme envisages providing financial assistance to every citizen of Goa for availing super specialty treatment that is not available in the State of Goa, whose annual income is less than Rs. 1.50 lakh. The Government has enhanced the assistance limit to Rs. 3.00 lakh for super specialized treatment like cancer, kidney transplant, neuro surgery and open heart surgery including post operative care. It is proposed to introduce a system of compulsory registration of beneficiaries with a payment of nominal fee of Rs. 100/- per member of the family. The proposal of registration under mediclaim is under process. During the year 2003-04, upto 31st December 2003, an expenditure of Rs.6.00 crore has been incurred covering 740 beneficiaries.

h. 'Pre-Natal-Diagnostic Techniques' (PNDT) Act, 1994

In order to check female foeticide the 'Pre-Natal-Diagnostic Techniques' (PNDT) Act, 1994 came into operation from 1st Jan 1996. The Act is being enforced and steps have been taken to register all Diagnostic Centre/clinics functioning in Goa. Upto 31st December 2003, 87 diagnostic centres have been registered.

i TB control Programme

Globally, it is estimated that 3 million people die due to Tuberculosis every year, majority of them in developing countries. TB kills more adults in India than any other infectious diseases. More than 1000 people a day, one every minute die of TB in our country. The number of TB cases detected in Goa from 1997-98 to 2002-03 is given in table 8.3.

Tuberculosis Control Programme is likely to undergo a remarkable change as Goa has been included in the Revised National Tuberculosis Control Programme (RNTCP). The Goal of RNTCP is to cure at least 85% of new sputum patients detected and to detect at least 70% of all such patients. One of the components of RNTCP is the Directly Ob-

Table 8.3:The Detection of TB cases from 1997-98 to 2002-03

	No. of TB cases detected	No. of sputum examination done	No. of sputum positive detected
1997-98	2866	16193	672
1998-99	2688	17388	607
1999-2000	2570	14699	674
2000-01	2443	16780	653
2001-02	2480	11152	684
2002-03	2241	10713	663

serve Treatment Short Course (DOTS) Strategy. The RNTCP provides for standardized and tested drug regimen, which has proved highly successful wherever applied. The total number of dose will be reduced, thus promoting patient compliance. DOTS will ensure effective treatment.

j Trauma unit at Hospicio Hospital, Margao

The Government of India has sanctioned and released an amount of Rs. 142.00 lakhs for setting up of a Trauma unit at Hospicio Hospital, Margao with an objective to upgrade and strengthen the emergency facilities in the hospital. A scanning unit has already been installed. Two ambulances fully equipped with necessary facilities are also to be purchased shortly. Some civil works, equipment and furniture are also planned for this unit.

k. Goa Medical College and Hospital

(i) Academic achievements

The Medical Council of India (MCI) has approved the continued admissions of 100 seats at Goa Medical College for the third year in a row. The College has also been permitted by the MCI to start postgraduate course of M.S. ENT from the year 2004 granting intake of 2 seats per year. The MCI has inspected Department of Neurosurgery for starting M.Ch. Neurosurgery and the proposal is under consideration. Several new textbooks and National as well as International journals were added to the library of GMC. Several teaching posts which have fallen vacant were filled. Teachers from several disciplines were deputed for conferences, continuing Medical Education Programmes and Workshops in India and abroad so that their knowledge and skills are improved with the ultimate benefits of the students and the patients. Research programmes in different departments were initiated and continued in basic as well as applied research.

(ii) Center of Clinical Excellence in GMCH by Ranbaxy Laboratory Limited

The Department of Preventive and Social medicine in collaboration with Directorate of Health Services, Indian Council of Medical Research and National Institute of Occupational Health took up the research study project to identify the causes for kidney failure cases in Canacona taluka. In recognition of the research potential of the Institute, Ranbaxy Laboratory Limited has offered to set up a Center of Clinical Excellence in GMC for advanced research on new drug molecules.

(iii) Establishing super specialty facilities and improving other hospital facilities

Several steps have been taken to improve the existing hospital facilities and to provide new ones. The project establishing super specialty facility (Cardiology and Cardio-surgery) through public and private participation is in the advanced stage and short listed agencies have made their presentations. Hip and knee joint replacements surgeries are now being done routinely. Computerization of hospital services as well as College Departments has started with provision of computers to many units. A pilot Plasma Pyrolysis plant to deal with biomedical waste will be installed soon. Incinerator has already been commissioned. Blood component separation unit is now ready for commissioning and approval of FDA is awaited. Order has been placed for the new CT Scan Machine. Plans are afoot to acquire a Colour Doppler machine and three Hemodialysis machines. Government of India is assisting Goa Medical College Hospital in acquiring an MRD and in establishing Regional Cancer Center, EYE Bank and also upgrading the Trauma Unit. Maintenance work of college and hospital building was taken on priority basis and all the operation theatres in the hospitals have been refurbished.

(iv) New Medicine Block of 450 beds

The Goa State Infrastructure Development Corporation has been entrusted with the task of building a new Medicine Block of 450 beds including auditorium of 1000 capacity and library block with all modern amenities.

i. Institute of Psychiatry and Human Behaviour

- It is proposed to start 'After Care Home Services for rehabilitation of chronic patients' shortly. Facilities at the "Child and Adolescent Guidance Clinic " are being upgraded considering increase in attendance of children at the clinic.

- The MCI inspection for recognition of P.G. Courses was carried out in the month of June 2003 and IPHB has been recognised for MD & DPM courses. During the year 2003, 3 Post graduate students have passed MD in Psychological Medicine and 2 Post graduate students have passed D.P.M.

(m) Dental College

Government decision to give sports marks for selection of Post Graduate M.D.S students has been implemented. It is proposed to prepare a website and take up computerisation of all departments including patient record. The first post graduate batch in 'Oral Medicine and Radiology' and 'Periodontics' will appear for final examination of the Goa University in April/May 2004.

3. SOCIAL SECURITY AND WELFARE

a. Dayanand Social Security Scheme

The Dayanand Social Security Scheme initiated by the Government in the year 2001-2002 under the 'Freedom from Hunger' Project has received overwhelming response from the people of the State. The scheme envisages providing pension to single women, disabled persons, senior citizens and economically weaker sections of the society like toddy tapers, coconut pluckers, rickshaw drivers, taxi drivers, motorcycle pilots, traditional fishermen, agricultural labourers, bus/truck drivers, bus conductors and bus/truck cleaners. This is a most unique social security initiative in the country. A member under the scheme will get financial assistance of Rs.500/- per month in the first year, which would increase at the rate of 5% annually. The financial assistance will go to the spouse and children on death of the member. The scheme has become operational from January 2002, and so far 55,200 persons have been receiving pension regularly.

b. Welfare of Senior Citizens

Senior citizen are entitled for various welfare measures like 50% concession in fare in KTC buses, 30% in railway fare to those who have completed 60 years and 50% concession on air travel to those who have attained the age of 65 years and above. They are also entitled for 1% higher rate of interest on fixed deposits in banks.

c. Other Social Welfare Programmes

In order to ensure flow of benefits to SCs, OBCs and Disabled, the Government has revised the eligibility criteria under various schemes. The income ceiling has been raised from Rs.11,000/- to Rs.25,000/- per annum. The validity period of income certificate has been raised from 1 year to 3 years. The rates of stipends and scholarships have been enhanced. The income limit under the scheme of 'Awards for Marriage with Disabled' has been enhanced from 24,000/- to Rs.50,000/- per annum. During the year 2003-04 upto December 2003 under Education Programme, 1810 OBC students, 100 SC students and 175 disabled students have been covered for stipends and scholarships. Five beneficiaries have been covered under the programme 'Awards for Intercaste Marriages' and ten beneficiaries under the programme 'Awards for marriages with disabled persons', 59 students studying in Special Schools for Disabled have been assisted for books, stationery and uniforms.

d. Goa State Scheduled Caste and Other Backward Classes Finance and Development Corporation Limited

The Goa State Scheduled Castes and other Backward Classes Finance and Development Corporation Limited, a Government of Goa undertaking was registered under the Companies Act, 1956 on 2nd April, 1990. Its main objective is to promote economic and development activities for the benefit of Scheduled Castes and Other Backward Classes and to assist the poorer section of these classes and generate self-employment opportunities.

The Corporation has been appointed as Channelising Agency of National Scheduled Castes and Scheduled Tribes Finance and Development Corporation (NSFDC), National Backward Classes Finance and Development Corporation (NBCFDC), National Handicapped Finance and Development Corporation (NHFDC) and National Safai Karmacharis Finance and Development Corporation (NSKFDC) for implementation of various self-employment schemes.

As channelising agency this Corporation is implementing various schemes for SCs, OBCs, Physically Handicapped etc. under which loans are being disbursed for undertaking business activities and also for the purpose of higher education. In the year 2003-04, under NBCFDC scheme, an amount of Rs.45,59,688/- was disbursed as loan amount for self-employment to 25 beneficiaries and Rs.1,41,851/- was given as education loan scheme as part payment to 5 beneficiaries from Other Backward Classes community. Under NSFDC scheme an amount of Rs.10,90,490/- was disbursed as loan for business to 3 beneficiaries from Scheduled Caste community. Likewise under Micro Finance scheme 1 beneficiary from Scheduled Caste was given Rs.10,000/-. Under NHFDC scheme an amount of Rs.1,00,000/- was given for Service Centre to one handicapped beneficiary.

e. Women and Child Development

The Supplementary Nutrition Programme envisages supplementing nutritional intake of children, pregnant and nursing mothers by providing nutritious food. Presently, 46000 beneficiaries are availing the benefit of the scheme, of which 37,143 are children and 8,857 are women. During the year 2003-04, grants have been released to 14 voluntary organisations for taking care of children in need of care and protection. About 865 children are benefited under this programme

4. LABOUR AND LABOUR WELFARE

a. The Goa Employment (Condition of Service) & Retirement Benefit Act 2001

This Act has been brought into force with effect from 19th September 2001. The provision of this Act require every employer to employ any person only if such person possesses the prescribed identity card and no employer can continue to employ any person in his establishment if such a person does not possess this card.

The provision of section 3 (2) of this Act also require every person seeking employment in the State of Goa to possess the said Identity Card and section 5(2) provides that he shall produce the identity card before the employer at the time of employment. Therefore, the persons on the live register of the employment exchange and who are job seekers will have to possess such Identity cards issued by the Competent Authority notified under the Act. Secretary (Labour) Government of Goa has been notified as the competent authority under the Act for the purpose.

Under the provisions of the said legislation the employer in the establishments namely

i) a motor transport undertaking as defined under the motor Transport Workers Act 1961 (Central Act 27, 1961)

ii) a plantation as defined under the Plantation Labour Act 1951 (Central Act 69, of 1951)

iii) an inland vessel mechanically propelled including a barge

iv) a mine, quarry or oilfield and

v) an establishment in which any work relating to the construction, development or maintenance of building , roads bridges or canal, or relating to operations connected with navigation, irrigation, or the supply of water or relating to the generation , transmission and distribution of electricity or any other form of power is being carried on are required to get identity Card to their employees in the prescribed manner as notified by the Government vide notification No. CL/ P.A/ 107 /2001/ 4151 dated 20.9.2001 issued from the competent authority appointed under this Act. Besides, the factories have already been covered under this Act. It is also proposed to extend the provisions of the Act to Shops and Establishment Act 1973.

So far, 55,000 Identity Cards have been issued. It is expected that the task will be completed on 31.3. 2004. Thereafter, contribution at the rate of 5% of total wages in respect of casual / temporary employees will be recovered from the employers which will be paid to the concerned workmen on their attaining the age of fifty years or his dependent, heirs, as the case may be in case of death of the workman before the age of fifty years, whichever is earlier.

b. Retrenched Workers Assistance scheme

In order to help the people retrenched from employment due to unavoidable circumstances the Government has evolved 'Retrenched Workers Assistance Scheme' extending various benefits. The scheme being implemented through Goa Labour Wefare Board is under implementation from 1.4.2003.

The retrenched workman will be eligible for monetary benefit as under:

a.	upto the age of 40 years	Rs. 2000/- per month or 50% of the gross earning whichever is less
b.	40 years to 50 years	Rs. 2500/- per month or 50% of the gross earnings whichever is less
c.	Above 50 years to 58 years	Rs. 3000/- per month or 50% of the gross earnings whichever is less

Duration of Benefits

a. Length of service of Minimum 4 ½ years- Benefits upto one year from the date of retrenchment

b. Two months for every additional year above 5 years (a) + (b) shall be subject to a maximum of 30 months

Eligibility

1) The retrenched workman or the workman affected by closure claiming the benefit under the scheme should have at least 4 1/2 years of continuous service

2) No retrenched workman shall be eligible to the benefit on completion of the age of 58 years irrespective of eligibility as above

3) Retrenched workmen having a minimum 5 years of service in more than one Company shall also be eligible for the benefits under the scheme- burden of proof lies on the workman

4) Workman who has resigned or abandoned the service on his own accord or has enjoyed the benefits of Voluntary Retirement scheme shall not be eligible to claim the benefits

5) The retrenched workman shall cease to enjoy monetary benefits under the scheme the day he becomes gainfully employed. He shall also give an undertaking to voluntarily disclose his employment.

6) The retrenched workmen claiming this benefit shall have to give an undertaking that in the event of benefit of back wages in a settlement arrived at in the course of conciliation or by an award of the Industrial Tribunal he shall be liable to refund the amount paid by the Board forthwith. In such cases the employer shall be at liberty to calculate the amount and deduct the same at source and deposit it with the Goa Labour Welfare Board within one month from such settlement / award.

7) The retrenched workman should possess social security card issued under the Goa Employment (Condition of service) and Retirement Benefit Act 2001

8) Application for the benefits under the scheme shall be approved by the Board or any committee constituted by the Board.

So far, 60 applications of retrenched workers have been approved by the Board claiming benefits under the scheme. However, the same are kept in abeyance pending amendment of Goa Labour Welfare Fund Act 1986. The proposed amendment of the Act will include revision of employers and employees contribution as well as starting Government Contribution to the Board.

c. Amendment to Goa Daman and Diu Shops and Establishments Rules 1975

The Goa Daman and Diu Shops and Establishments Rules 1975 have been amended to provide for renewal of registration certificate within 30 days before the expiry date. This has removed the inconsistency in respect of provisions pertaining to renewal of certificate in the Act and the Rules

d. Increase in minimum wages

Minimum wages have been revised from 1.7.2003 in the following :

a. employment in construction or maintenance of roads or in building operations, stone breaking, stone crushing and maintenance of building

b. employment in any residential hotel , restaurant or eating house

c. employment in any shop and commercial establishment other than a residential hotel, restaurant or eating house

d. employment in automobile repairing workshops and garages

e. employment in agriculture

f. employment in private motor transport undertakings.

5. HOUSING

a. Goa Housing Board

The Board has prepared the draft State Housing and Habitat policy and submitted to the Government for approval. It has taken initiatives to amend the Act & Rules to simplify the process of allotment to make it more people friendly. It is also intended to remove the domicile clause wherever applicable.

Construction of 24 Duplex Bungalows at Porvorim, 60 LIG flats and 24 shops at Davorlim and land development work at Farmagudi, Ponda and Ambaji and Fatorda have been completed. Construction of 10 LIG tenement at Davorlim, 34 LIG flats at Tivim and land development work at Camurlim, Phase I and allotment of plots is under progress.

b. Indira Awaas Yojana

Under Indira Awaas Yojana, during the year 2003-04, up to November, 67 new houses have been constructed and 15 old houses have been repaired.

c. Housing Activities in Private Sector

Building works sanctioned both in urban and rural areas during the year 2001-02 may be referred to in table 8.4 and 8.5 respectively.

d. House Building Advance to Government Employees

The innovative housing scheme for Government employees started in collaboration with the HDFC has evoked significant response. So far, a loan of Rs.16.17 crore has been sanctioned to 462 employees. The Government by amending the provisions of the Goa School Education Act/Rules has extended this scheme to the employees of non-Government Aided schools.

Table 8.4: Building works sanctioned in private sector in urban areas, 2001-02

Sl. No.		No. of Plans	Plinth area (Sq. mts.)	Floor area (Sq. mts.)	Estimated cost (Rs. in lakh)
1	*Residential buildings*				
	a. New Constructions	139	35800	48916	2033.05
	b. Alterations & Additions	51	9766	11261	768.16
2	*Industrial Buildings*				
	a. New Constructions	3	3742	7104	423.65
	b. Alterations & Additions	0	0	0	0
3	*Commercial Buildings*				
	a. New Constructions	37	7393	13707	836.32
	b. Alterations & Additions	12	1506	1168	160.59
4	*Institutional Buildings*				
	a. New Constructions	5	4445	8957	524.45
	b. Alterations & Additions	2	2988	3286	44.34
5	*Other buildings*				
	a. New Constructions	11	1850	1533	76.05
	b. Alterations & Additions	9	1974	1386	63.57
6	*Total*				
	a. New Constructions	195	53230	80217	3893.52
	b. Alterations & Additions	74	16234	17101	1036.66

Table 8.5: Building works sanctioned in private sector in rural areas, 2001-02

Sl. No.		No. of Plans	Plinth area (Sq. mts.)	Floor area (Sq. mts.)	Estimated cost (Rs. in lakh)
1	*Residential buildings*				
	a. New Constructions	522	67009	102273	5303.00
	b. Alterations/ Additions	97	14476	18355	1465.87
2	*Industrial Buildings*				
	a. New Constructions	17	21110	24852	1769.98
	b. Alterations/ Additions	5	3434	5520	46.21
3	*Commercial Buildings*				
	a. New Constructions	32	12063	6569	943.81
	b. Alterations/ Additions	6	1860	2958	160.24
4	*Institutional Buildings*				
	a. New Constructions	8	4222	7872	331.14
	b. Alterations/Additions	6	872	1016	42.21
5	*Other buildings*				
	a. New Constructions	8	1836	3106	129.24
	b. Alterations/Additions	9	1084	1515	56.22
6	*Total*				
	b. New Constructions	587	106240	144672	8477.17
	b. Alterations/Additions	123	21726	29364	1770.75

OVERVIEW

Emphasis of the Government on younger generation is clearly visible in various new initiatives like Cyberage Student Scheme, Goa Scholar Scheme, Interest Free Loan Scheme etc. Under the Cyberage Student Scheme more than 10000 computers are being given to standard XI students at their homes. The scheme is also being extended to Higher Education. The Goa Scholar Scheme is to promote talented students from Goa to pursue higher education both in India and abroad. In the year 2003, four students have been selected for scholarship under the scheme. The interest free loan scheme for pursuing higher education has evoked tremendous response. The Government is also implementing Infrastructure Development loan scheme for development of various infrastructure facilities in non-Government aided schools and colleges and the scheme has received a very good response from the managements. In order to improve the infrastructure in Government schools, the Government has prepared a status report as regards the condition of Government schools.

For the development of Technical Education in the State the MOU with IIT Bombay is being implemented scrupulously. Its impact is so encouraging that the campus recruitment at the Goa Engineering College has proved to be the best and has reached as high as 90% placement. As regards setting up of IIT Bombay campus in Goa, the Government is in the process of acquiring land and finalizing action plan for the purpose. Goa is also being attracted by top notch institutions like BITS Pilani in the private sector. In order to promote and assist in the rapid and orderly establishment, growth and development of educational institutions in the States, the Government has formed Goa Education Development Corporation.

The year 2003 witnessed problems of SARS and Jaundice which the Government handled diligently. Action has been initiated for construction of new buildings for district hospitals Asilo and Hospicio and also for Institute of Nursing Education. The Government has also initiated action on setting up 450 bedded new Medicine Block and establishing super specialty facilities at the GMCH. Under the Dayanand Social Security Scheme so far over 55000 beneficiaries have been covered.

CHAPTER 9

Agriculture and Allied Activities

Towards achieving the goal of developing a sustainable village economy the Government of Goa initiated many new schemes like Krishi Ghars, Agriculture Estates, Food Courts, Kamdenu etc. during the year 2003-04. The objective is to generate employment opportunities in rural areas by encouraging farmers to take up traditional primary activities. The details of these new initiatives and progress made under various ongoing programmes are given below :

1. AGRICULTURE

0 Establishment of Krishi Ghars

In order to provide ready market and good price to the producers of vegetables, fruits, flowers and other horticulture crops the concept of Krishi Ghars has been evolved for the first time in Goa during 2003-04. The scheme aims at generating employment opportunities in rural areas. The concept involves developing a well-knit network of collection, transportation, preservation and marketing mechanism for Agriculture / Horticulture produce.

The scheme is being implemented with the active involvement of cooperative societies with financial support from National Cooperative Development Council and the State Government. The cooperative societies entrusted with this work are i) M/s Adarsh Cooperative Society Ltd., for areas of Salcete, Quepem and Canacona ii) M/s Goa Bagayatdar Cooperative Society for areas of Ponda, Valpoi, Sanguem and Mormugao iii) M/s Krishi Bazaar has been entrusted with the areas of Tiswadi, Bardez , Pernem and Bicholim. These Societies are entrusted to create the necessary infrastructure for collection, grading and sale outlets for vegetable, fruits and flowers.

The process of registration of vegetable growers is being done with the assistance of Extension Officials appointed by these Societies and locally produced vegetables are procured at various locations of Krishi Ghars and collection centers of respective agencies. During the year 2003-04, eight Krishi



Ghars have been established at Kadamba Bus stand at Panaji, Mapusa and Margao and in the market places in Candolim, Chapora and Mapusa in Bardez taluka. In addition, 3 collection cum grading centres for vegetables have been established by these agencies at Margao, Ponda and Canacona. In all it is proposed to establish 107 outlets and 5 procuring centres within two years.

A 30 tonnes cold storage has been commissioned at new market complex, Panaji. The same is being hired to the entrepreneurs and farmers for storing milk , fruits, flowers and dry fruits.

(ii) Promotion of Agricultural Estates

The scheme envisages development of Agricultural Estates with all facilities for culture of high value flowers and vegetables through unemployed youth. The concept is to provide land in the Agricultural Estates to the unemployed youth on lease basis to cultivate high value crops. Marketing of products is proposed to be further linked to Krishi Ghars. The Horticulture Corporation is in the process of finalizing modalities of the scheme. Presently, two sites at Sanguem are under consideration for establishment of Agricultural estates.

(iii) Development of model farm at Chimbel

The Horticulture Corporation is developing a farm at Chimbel with an area of 1.2 ha. as a model farm mainly for production of organic manure, which is expected to generate an annual income of about Rs.1.40 lakh.

(iv) Food Court at Verna

Steps have been initiated for setting up of Food Court at Verna. This would comprise sales outlets for ethnic products from Goa, tourism related items and handicrafts, restaurants and fast food centres, conference hall, auditorium, public utilities, exhibition hall etc. While the Government will play the role facilitator, the project will have major participation from private sector.

(v) Promotion of Tender Coconut Carts

The scheme aimed at employment generation envisages providing tender coconut carts to interested unemployed youth. Carts are provided upon fulfilment of the terms and conditions of the scheme like domicile of 10 years, age group of 20 to 40 years, willingness to pay part payment of the cost of the cart etc. Registered self help groups and the co-operative societies are also eligible for benefit under the scheme. Each cart will be provided a subsidy of 50% of the cost or a maximum of Rs.9,000/- at source by the Government of Goa. The balance cost of the cart would be paid by the beneficiary to the Horticulture Corporation before releasing the cart. Considering the present cost of the cart, the beneficiary has to pay an amount of Rs. 9,000 including the incidental and handling charges. The cart would be released as per the availability of the stock and after verification of the fulfilment of terms and conditions of the selected beneficiaries will be required to execute a bond on a stamp paper of Rs. 20. The scheme started during the year 2003 has evoked a good response and 2 carts have been financed so far and 5 more carts will be financed shortly.



(vi) Marketing of fruits & vegetables by the Horticulture Corporation

The Horticulture Corporation is supplying fruits and vegetables to various Government institutions like Goa Medical College, Institute of Psychiatry and Human Behaviour, Asilo Hospital Mapusa, Hospicio Hospital, Margao, Central Prison, Aguada, Provedoria Orphanage, Chimbel and Old Age Home, Goa Velha. The total supply to these institutions is about Rs.18.00 lakh per year. It is proposed to cover more such institutions in future and increase business turnover of the Corporation. The Horticulture Corporation also sells vegetables and fruits at retail in Panaji Municipal Market and the annual sale is about Rs. 21 lakh. Considering impressive performance of the outlet, it is proposed to upgrade the same to increase the sales to Rs. 70 lakh per annum.

(vii) Rainfall

The total rainfall received during the year 2003 was 2682.9 mm. This is much higher than the last two years rain fall of 2128.1 mm and 2270.4 mm during the year 2001 and 2002 respectively. Though the rain fall pattern during the year 2003 was quite satisfactory, this was much below the average rain fall of over 3000 mm during the last decade 1991-2000. The trend in rain fall in different years may be referred to at Annexure 39.

(viii) Hybrid Rice cultivation in the State

Hybrid rice cultivation is emerging as an answer to boost per unit paddy production and productivity. About 90% of the area, which is under high yielding variety with the average production of not more than 4000 kgs per hectare has been stagnating for the last few years. The hybrid rice, which has got the potential to increase the production, was introduced from last two to three years. The hybrid rice has potential to give double the yields as compared to the high yielding varieties. The trials undertaken by introduction of hybrid KRH-2, PHB 71, PA 6222, Sahayadri during the last three years has shown encouraging results. The front-line demonstrations on hybrid rice and the minikits has shown adaptability by the farmers. The farmers have started slowly switching over from high yielding varieties to hybrid rice.



(ix) Crop forecast for 2003-04

Crop forecast for the year 2003-04 is given at in table 9.1. The trend in agriculture production and land use pattern in Goa are given at Annexures 40 and 41 respectively.

(x) Floriculture

Floriculture has received the much needed boost in the State of Goa for the promotion of high value flowers like orchids, gladioli, gerberas and anthuriums. The State Government has taken a number of steps like seminars, trainings etc for promotion

Table 9.1: Crop forecast : 2003-04

Sl. No.	Crop	Kharif 2003 (Provisional)		Rabi 2004 (Anticipated)	
		Area (Ha.)	Production (tonnes)	Area (Ha.)	Production (tonnes)
1	Rice	34300	95000	16500	55000
2	Pulses	500	350	10500	8500
3	Groundnut	325	450	2150	4050
4	Total	35125	95800	29150	67550

of floriculture in the State. Two resource persons from the Kerala Agricultural University were invited to guide farmera on floriculture development especially growing of orchids in May 2003. About 200 invitees attended the programme. In June 2003, about 150 progressive farmers and officials attended a training programme wherein 2 floriculturists growing orchids and anthuriums shared their experiences. A demonstration unit housing 1000 orchids of different varieties of Dendrobium



genera was set up at Ela Farm, Old Goa in August 2003. Extensive efforts were undertaken by the Agriculture Department to screen the prospective growers for selection of sites etc. and in order to train and guide them further on actual setting up of units, 50 farmers were tentatively selected. The selected farmers were made to undergo a one day orientation cum training programme in August 2003 at Farmers Training Centre, Ela under the guidance of resource person from the Kerala Agricultural Univerisity. Further screening of the prospective orchid growers was carried out and 30 farmers actually opted to set up units during this year.

An interactive session of prospective orchid growers and known sources offering their services for supplying various inputs and setting up of units was held in October 2003 at Farmer's Training Centre, Ela, Old Goa. This helped the farmers to overcome the hurdles and initiate the setting up of their units. The concept of forming of a Floriculturists Cooperative Society was discussed in length as the members would benefit in terms of sale of their produce and for procuring various inputs for their units. Another orchid unit housing 1000 plants has been set up at Codar farm.

The process initiated has shown positive results and proposed to be continued further. As a result of this exercise, initially two units, one each of gerbera and orchid were set up in Salcete taluka and then another 6 units were set up. M/s Goa Flora is setting up a unit with a capacity of 15000 plants at Sal, Bicholim. Similarly many more units are coming up. The farmer who has set up the gerbera unit at Navelim in the month of August 2003 has gained sufficient confidence and has already initiated the process to expand to an additional 4000 gerbera plants.

As regards marketing, at present flowers produced are absorbed by the local market. Opening an outlet at the Airport by the Floriculturist Cooperative Society is beig explored. It is expected that Krishi Ghars being set up in different parts of the State would also assist the floriculturists in marketing their produce.

(xi) Agricultural implements/machinery

Towards mechanization of cultivation of crops, the Agriculture Department extends custom services to the farmers by providing tractors, power tillers, bulldozers and paddy reapers. The department has a fleet of 45 tractors, 6 bulldozers, 4 power tillers

and 17 paddy reapers, which have provided 20293 hours of custom service to the farmers during the year.

(xii) Crop Insurance

The Government of India has launched the new National Agricultural Insurance Scheme (NAIS) from Rabi 1999-2000 season. The State Government has been implementing this scheme in the State of Goa from the Rabi 1999-2000 season at taluka level in collaboration with General Insurance Corporation of India.

The scheme broadly covers paddy, pulses, groundnut, ragi and sugarcane crops. Under the scheme, insurance coverage is compulsory for all loanee farmers availing Seasonal Agricultural Operations (SAO) loans from Financial Institutions for notified crops in notified areas upto the full loan amount. Additional coverage under the scheme is also available (at the option of farmer), beyond the loan amount upto the value of 150% of average yield. (Average yield of the State based on yield of past 3 years in case of paddy and past 5 years in case of pulses, groundnut, ragi and sugarcane crops). The maximum value of additional coverage is equivalent to State average yield multiplied by Minimum Support Price (MSP) announced by the Government of India of the Market Price (MP) where MSPs are not announced in the last season / year.

For loanee farmers, in such cases where amount of crop loan availed works out to be more than either the value of threshold yield or 150% of average yield, normal premium rates (lowest of flat rates or actuarial rates) shall be applicable on the full amount of loan availed, as full amount of loan is to be compulsorily insured. Coverage is also available for all non-loanee farmers, on optional basis.

The Financial Institutions shall compulsorily cover all crop loans disbursed, for notified crops, through Kisan Credit Cards and banks shall maintain necessary registers and control for smooth and effective coverage of loan.

Performance under the scheme from 1999-2000 to 2002-03 is given in table 9.2.

(xiii) Market Intervention

In order to provide support to the farmers to sustain falling prices, support price has been given to sugarcane and arecanut growers as detailed below.

Support price to sugarcane: Support price of Rs.100 / - per tonne has been provided to sugarcane growers in order to encourage them to bring more area

Table 9.2: Performance under National Agricultural Insurance Scheme

Season	Crop	No. of Farmers	Area in Ha.	Sum insured (in Rs.)	Premium (in Rs.)	Benefit to farmers / claims (in Rs.)
Rabi1999-2000	Paddy	52	50.25	96940.00	1163.60	9963.62
	Sugarcane	27	15.43	254817.00	2549.00	
Kharif 2000						
	Paddy	972	2145.92	1397928.00	30055.45	7944.00
	Sugarcane	21	39.14	587035.00	5870.35	
Rabi 2000-01						
	Paddy	63	26.74	79231.00	911.16	
	Sugarcane	150	136.56	1138475.00	11384.75	
Kharif 2001	Paddy	669	1641.81	995102.00	21892.25	205042.21
Rabi 2001-02	Paddy	27	35.00	50700	659.10	
Kharif 2002						
	Paddy	703	1477.80	1148536	28136.65	
	Sugarcane	375	365.35	7338327	106405.64	
Rabi 2002-03	Paddy	21	28.00	43500	598.13	
	Sugarcane	133	146.71	2408258	49764.54	

Source: Directorate of Agriculture, Government of Goa.

under the crop. During the year an amount of Rs.60.59 lakh has been paid as support price for sugarcane.

Support price to arecanut: The support price scheme approved in the year 2001-02 has been continued during 2003-04. An amount of Rs. 65.54 lakh was paid as support price for arecanut during the year. The support price of maximum of Rs. 15/- per kg., or to the extent of actual difference between the cut off price of Rs. 60/- and actual market price received from the farmer was paid.

(xiv) Watershed Development Programme

The National Watershed Development Programme in Rainfed Areas (NWDPRA) is implemented at Warkhand in Pernem taluka, Maulinguem in Bicholim taluka, Savoi Verem & Kavlem-Betoda in Ponda taluka, Morpilla & Cavrem -Pirla in Quepem taluka. During the year, 20 Watershed Associations were registered and 166 Self Help Groups and 109 User Groups were formed. The programme is being executed through participatory approach and planning is done by Watershed Associations. Under the Western Ghat Development Programme, the watersheds selected are in Canacona, Sanguem and Sattari taluka. Integrated approach is followed to develop agri-horticultural based cropping system. Also soil and water conservation measures are undertaken to preserve the environment and ecosystem and increase the water table in the watershed areas. From the year 2003-04, some activities are extended to the non-selected watershed areas also.

(xv) Desilting/repairs of Ponds/Lakes

The Government has taken a significant decision to revitalize the ponds and lakes that are either abandoned or not being put to optimum use. An identification survey carried out by the Directorate of Agriculture has located 745 ponds/lakes in the entire State. Out of this, 438 are of very small size of less than 100 sq.mts., 164 are of the size 100 to 500 sq.mts, 25 are of 500 to 1000 sq.mt., size, 53 are of 1000 to 10,000 sq.mts., size and 65 are of the size of about 10,000 sq.mts.

It is proposed to desilt/repair these ponds/lakes, which is expected to create an additional potential of about 2000 ha. benefiting about 21,400 farmers. It is proposed to involve user groups, NGOs and consultant agencies for the purpose. A High Powered Committee under the chairmanship of Hon'ble Minister of Agriculture has been formed for working out modalities and monitoring the scheme.

(xvi) Remodelling of notified protective bunds

The State has about 18000 hectares of low lying agricultural land which is locally called as 'Khazan' land. There are about 500 land pieces of khazan land which have been reclaimed by constructing narrow embankments along river creeks. The reclaimed khazan lands have always to be protected from sea and creek water intrusion by repairing and maintaining protective embankments (bunds). Due to various reasons like seawater tides, flood waters, waves created by movement of barges and activities of marine borers etc., the bunds erode, settle and receive several types of damages. Occasionally breaches occur in the bunds. The bunds constructed in the State are narrow and delicate which require very prompt and high level maintenance and repairs. Due to changed socio-economic status of the farming community, the Tenants Associations who are responsible for the maintenance and repairs under the Agricultural Tenancy Act are facing difficulties to maintain such narrow bunds to the required maintenance level. As a result, due to fear of losing the crops due to inundation of their fields, the farmers have hesitation to concentrate on cultivation of their lands.



In view of this the Government formed a new scheme of remodelling of embankment for protecting khazan agricultural land. The notified embankments are renovated by the Government by adopting improved design with wider and structurally stronger sections. The material used is good quality of red moorum against the traditional use of river silt. The width of the bunds is made 4 meters to facilitate movement of inspection vehicles, trucks for transportation of materials etc., so that periodic repairs and maintenance could be done speedily and economically. In the traditional method, materials were transported manually or by the use of canoes. Under this scheme the work of remodelling of embankments are done by the Government on 100% initial investment. The beneficiaries contribution is 10% of the cost incurred to be paid in maximum ten annual instalments. The Government contribution is upto 90%. There is a state level high power committee to monitor the scheme. During the year 2003-04, five works are being taken up.

(xvii) Repairs to protective embankments

Under this scheme the Tenants Associations execute the work of repairs to notified bunds protecting khazan agricultural land at 100% initial investment. The Government provides 50% cost of repairs, subject to a maximum of Rs. 6000/- per hectare of area benefited, to the Tenants Associations/ beneficiaries after completion of works. During the year 2003-04, till 15th January 2004, 15 notified bunds have been complete. By March 2004, it is proposed to complete in all 30 works spending about Rs.28 lakh towards 50% of cost to be paid by the Government.

2 ANIMAL HUSBANDRY

Towards tapping a very high potential for Dairy Development in the State the Animal Husbandry sector received a major boost during the year 2003-04. The Government has launched a major scheme 'Kamdhenu' to achieve self-sufficiency in milk production and generate substantial employment opportunities to the people. The details are as follows :

(i) Kamdhenu Scheme

Objectives

1. To help the farmers to get financial assistance for purchase of Cross bred Cows and Improved She Buffaloes for his Dairy Unit
2. To promote self employment in the Dairy sector
3. To boost milk production in the State
4. To strengthen the Dairy Co-operative movement in the State.

Eligibility

- Scheme is open to individuals, dairy cooperative societies and groups of unemployed youth if they register Producer Cooperative Society
- At least 15 years domicile and some knowledge or experience in Cattle or Buffaloes farming.
- Preference to unemployed
- The applicant should possess a cattle shed / construct a cattle shed before submitting application
- The applicant should have proper infrastructure for hygenic maintenance or housing of animals and installation of equipment
- The applicant should have sufficient space for storage of feed and fodder.
- Those who have availed or applied for benefit under any other scheme for purchase of animals will be disqualified from applying under this scheme. Those who have already purchased animals under 1-10 milch animal scheme can purchase balance animals under this scheme. Those who have availed benefit under 1-10 milch animal scheme can avail benefit under Kamdhenu scheme only after 3 years. Those who have availed benefit under Modern Dairy scheme will become eligible to Kamdhenu scheme after 5 years.

Salient features and procedures

a) Easy availability of loan with a low rate of interest.

b) Guarantee in case of loss of animal due to death

c) Operation of the scheme through Bank.

d) Government guarantee to Bank as regards subsidy component and repayment of loan in case of loss of animal due to death.

e) To be applied in prescribed application form

f) Non-refundable processing fee of Rs.100/- is to be collected at the time of accepting the application at the Government Dispensary/ Hospital.

g) Persons applying through cooperative societies are exempted from processing fee if the society gives an undertaking that it will deposit the monthly instalments of the farmers to the financial institution.

h) Processing period is 15 days. Within this period the Hospitals/Dispensaries shall submit the applications of all eligible farmers to the Directorate of Animal Husbandry and Veterinary Services to place before the Selection Committee. The Selection Committee may verify facts given by the applicants in their applications if necessary by deputing Officer/Expert.

i) The Selection Committee comprises Hon'ble Minister for Animal Husbandry as Chairman and Secretary (AH), Dairy Union Representative, Representative to be nominated by CM, Representative of the GSCB Ltd., and Nodal Officer of the AH & VS Department as members and the Director Animal Husbandry as Member Secretary.

j) The list of farmers approved by the Selection Committee has to be sent to the Goa State Cooperative Bank Ltd, along with details like number of animals, maximum amount of loan to be financed etc.

Guidelines for purchase of animals

a) Farmers are free to purchase animals on their own or take the assistance of the Veterinary Doctor designated/ appointed for the purpose.

b) Animals may be purchased from any recognized market outside Goa or approved breeders in Goa or through Cattle and Buffalo Mellas arranged in Goa by the Directorate of Animal Husbandry and Veterinary Services or from outside Goa.

c) In case an animal is purchased from a Dairy Farmer in Goa such purchase will have to be only from farmers who have registered their names and given the detail report in the prescribed 'Declaration Form' with the Directorate of Animal Husbandry & Veterinary Services.

d) The eligible participants under the scheme can buy up to 10 animals and the cost of each animal is restricted to maximum of Rs.16,000/- for the purpose of finance under the Scheme. The beneficiary will have to deposit 10% of the cost of each animal as margin money and the remaining 90% cost of the animals shall be the loan amount. The loan shall be restricted to Rs.14,400/-, which shall be financed through the Bank. In case of SC/ ST/OBC the margin money of 10% will be reduced by Rs.800/-. The loan amount will be repayable in 42 monthly instalments, which shall include an interest rate of 9% per annum while calculating the basis of equated monthly instalments (EMI). The beneficiary would need to deposit only 50% of the EMI as long as his payments are regular and timely, since the Government will release the remaining 50% of the EMI to the financial institution on a monthly basis. The Government is committed for payment of 50% of EMI, which is equal to the instalments paid by the farmers. In other words, the beneficiary will get the subsidy of 50% and his animal will be insured for only Rs.10/- per month as contribution to the Insurance Fund. In case the beneficiary defaults and does not make payment towards his loan account for 4 months consecutively, he shall be sent a warning letter and a last chance for payment. In case he continues to be irregular up to six months his remaining loan amount shall be frozen and his animal becomes the property of the Department.

e) The beneficiaries availing of this scheme shall be eligible for an infrastructure and working capital loan, which is restricted to a maximum of Rs.5,000/- per animal. The farmer pays 75% of the loan and interest and the Government ensures the payment of 25% of the equated monthly instalments, which includes the interest at 9% per annum. This infrastructure and working capital loan is provided to the farmers so as to enable them to repair cowshed or purchase the initial stock of feed and milking equipment etc.

The scheme launched recently has evoked tremendous response and 550 farmers have already availed benefit under the scheme.

(ii) Incentive to milk producers

In order to encourage the farmers in the dairy sector, the Government decided to give assistance to the milk producers by way of incentive to the tune of 5% to 9% of the value of the payment received by the farmer for the milk supplied to the dairy and till date 7500 farmers have benefited.

(iii) Animal Diseases

Measures have been taken to control Anthrax and Foot and Mouth Disease by resorting to vaccination of cattle. Goa is free from Rinderpest diseases and efforts are now being made to eradicate the foot and mouth disease.

(iv) Cess Collection

The Animal Husbandry Department collects cess on fluid milk coming from neighbouring States. During the year 2003-04, up to November, an amount of Rs. 23 lakh has been collected.

(v) Control of Stray Animals

Four NGO's working in the field of Animal Birth Control have been identified to look after stray dogs and cats and one NGO to look after the welfare of the cattle which are old, wounded and homeless. These NGO's are associated with the implementation of the Prevention of Cruelty Act.

(vi) Goa Meat Complex

Since the year 2001-02 the Goa Meat Complex is running in profit. During the year 2002-03 the Company has paid dividend @ 5% to its shareholders. During 2003-04 also it is expected that the Company will continue to earn profit and pay dividend to its shareholders (Box 9.1).

Various measures have been taken to improve the infrastructure facilities and service to the people. A Wet Rendering Plant has been established for utilization of slaughterhouse waste, to produce by products like blood meal, bone meal, meat meal etc. A new mini-slaughterhouse has been constructed to improve the slaughtering services to the people by encouraging local meat traders. An effluent Treatment Plant of 156.25 cum/day capacity is being put up to treat the effluent and thereby take utmost care of the environment and conserve and economically utilise the waste water.

The Company also proposes to carry out the following modernisation works to the slaughterhouse so as to bring it at par with European Standards and increase export of its products.

- Construction of holding hall to hold the animal carcasses after slaughter for 24 hours
- Construction of an additional processing hall
- Construction of a cold storage of at least 100 tonnes capacity
- Increase in the size of Lairage
- Establishment of a Bio-methanation plant
- Procurement of modern equipment for slaughter house

The distribution of livestock and poultry population as per the live stock censuses held in the years 1992 and 1997 are given at Annexure 42.

BOX 9.1

GOA MEAT COMPLEX IN PROFIT

During the year 2002-03, the Goa Meat Complex has earned a gross profit of Rs.28.68 lakh and net profit of Rs. 14.84 lakh and proposed dividend of 5% to its share holders. After the signing of an Agreement with Al-Nafees Frozen Food Exports Ltd., New Delhi, for leasing spare capacity of the Modern Abattoir for minimum guaranteed 120 animals per day @ 356/- per animal since August 1999, with 12% increase every year, the Company has come back on rails and earned profit. Further, the Government of Goa is endeavouring to stop illegal slaughter of bovines, in the municipal and other areas and also prevent the entry of unhygienic & clandestine import of beef from other States. These efforts of the Government of Goa are expected to earn additional income of about Rs.20.00 lakh per annum without adding much amount to operational costs.

3. FORESTS

a. Area under Forests and Forest Produce

The State has 38.5% of its land under forest cover, which is much higher than the national average. In absolute terms, 1424.38 sq.kms. of area is under forests, of which 1224.38 sq.kms. is Government forest and 200 sq. kms. is private forest. Physiographically, the State of Goa consists of three narrow strips viz. the coastal belt, the middle plateau and the uplands. Of these the uplands constitute the maximum extent of forests, which fall in the Western Ghat areas running through almost parallel to the western coast of the peninsula. The most thickly populated talukas of Salcete, Tiswadi & Bardez have practically negligible forest area while the taluka of Sanguem tops the list in the extent of forests. Taluka-wise distribution of area under forest and distribution by species are given at Annexures 43 and 44 respectively.

The Government of Goa has adopted a conservationist policy in the management of forests. Green felling has been banned and harvesting is confined to the salvaging of dead, dry, fallen or diseased trees. Trees from privately owned areas are controlled by enforcement of provisions of the Preservation of Tree Act and also promoting planting of trees in private lands, communidade lands, roadsides etc. through a variety of packages. Forest produce by item and quantity during the year is given at Annexure 45.

b. Eco-tourism

Eco-tourism development plan comprises development of appropriate infrastructure, organizing eco-tourism trips, mass awareness programmes etc. For easy mobility and with a view to observe wildlife, nature and manmade articrafts, the roads leading to various eco tourism spots have been developed such as Deer Safari Park, Watch Towers at Bondla Wild Life sanctuary, Doodsagar water falls and Cotigao Wild Life Sanctuary, sunset points at Bhagavan Mahavir Wild Life sanctuary at Mollem, National Park, Boat and Cannoe excursion at Salim Ali Bird sanctuary. About 2000 m trekking trails have been laid out in forest areas with a view to promote trekking expeditions. Eco tourism guides have been trained and authorized by the Forest Department so that tours are conducted to interesting places / features and local employment generated. Management plans for Cotigao and Bondla wildlife sanctuaries are being prepared. The number of tourists visited the eco-tourism areas has increased from 70942 in 2001-02 to 76505 in 2002-03.

c. Urban beautification and recreational forestry

This is being given special emphasis. One new garden at Navelim and 3 new gardens in Municipal areas of Curchorem, Panaji and Pernem have been taken up for development during 2003-04 in addition to maintenance of existing 6 gardens. Plantation along the road side has been taken up wherever road widening activity has been completed. 118.50 ha. of degraded communidade land has rehabilitated by raising plantations of indigenous species.

d. Non-invasive use of forest resources

The Forest Department has intensified the non invasive use of forest resources and launched two projects on medicinal plants. These projects are fully funded by the Central Government. A third project formulated by the CIMAP, Lucknow is also under consideration of Government of India for approval and funding. Cultivation and harvesting of medicinal plants is a significant step towards augmenting revenue from forests without harming the forests.

e. Joint Forest Management

For the first time, the Government of Goa has launched the Joint Forest Management scheme during 2003-04. The scheme envisages management of forest jointly by the Forest Department and local community through protection, conservation and regeneration and in turn both shall share the responsibilities and reap the benefits. So far, 30 JFM committees have been constituted involving villagers who are in the vicinity of the forest. On experimental basis, only 10% of the total area of 8990 ha under cashew plantations is envisaged to be given to the local communities under the scheme, for an initial period of 5 years. Local communities in the form of the cooperative societies in collaboration with the Goa Forest Development Corporation will tend these plantations and reap the cashew nuts and apples etc. In turn, they will pay fixed amount to the Corporation in suitable instalments. This is intended also to generate local self employment.

f. Botanical Garden cum Eco-Recreational Park at Salaulim Dam Site

The Goa Forest Development Corporation (GFDC) has taken up development of Botanical Garden cum Recreational Park at Salaulim Dam Site. This is proposed to be developed on the lines of Vrundavan Garden at Mysore in Karnataka. Initial work of setting up of the Nursery to produce ornamental plants and development of landscaping with the help of experts in the field has been undertaken. The project is expected to generate self employment to a large number of people in the surrounding villages.

g. GFDC started producing V-4 grafts

Hitherto, the GFDC used to buy V-4 grafts from private nurseries approved by the Government. From the year 2003-04, the Corporation has started producing V-4 grafts to cater to its requirement as well as to supply these grafts to the Agriculture Department. About 63000 V-4 grafts were supplied to Agriculture Department @ Rs.12.50 paise per graft. Due to this, the Corporation could save about Rs.5.00 lakh on procurement of grafts and at the same time it succeeded in meeting the requirement of local people by supplying them with genuine grafts of improved varieties.

h. Resource mobilization

The Forest Department has taken various measures to generate additional revenue by hiking fee for transit pass, saw mill fee, carpentry fee for small & big carpentry unit, accommodation rate for the rest houses and cottages at various sanctuaries and National Park, depot licence fee with effect from April 2003. A fee of Rs.100/- per tree for felling licence has also been introduced from 14/7/2003. The table rate for various forest produce has also been hiked to mop up additional revenue.

4. FISHERIES

a. Ban on mechanised fishing within 5 km area

During the year 2003-04, mechanized fishing within an area of 5km from the coast was totally banned to protect the interest of traditional fishermen. To facilitate various enforcement measures, the Fisheries Department is in the process of acquiring a new patrol boat. Similarly, one fishing vessel is also proposed to be acquired. The Department has not issued NOC for construction of new vessel, but only replacement of vessels are allowed for which quota of only five has been fixed. The details of fish catch and their export are given at Annexures 46 and 47.

b. Census of Fishermen

In order to ascertain socio-economic condition of fishermen the Directorate of Fisheries has undertaken a Census of all active fishermen in the age group of 18 to 60 years. Information is being scruitinized and compiled. Results of the Census are expected to help the Government in implementing various welfare schemes.

c. Achievements under different schemes

- Infrastructure development works completed during the year 2003-04 are construction of fishing jetty, net mending shed, workshop, sulabh toilet at Cutbona jetty complex and construction of sulabh toilet at Malim jetty complex.

- Various infrastructure development works in progress are construction of pavement, retaining wall, workshop and net mending shed at Malim jetty , construction of overhead tank at Cortalim fish landing centre, and construction of overhead tank and repair of jetty at Cutbona.

- Financial assistance being given to fishermen under various welfare schemes are subsidy up to Rs. 2.00 lakh for construction of hull and purchase of engine, subsidy of Rs. 12,000/- for purchase of inboard motor and Rs.10,000/- for outboard motor, subsidy of Rs. 5,000/- for purchase of sea gill nets, subsidy of Rs.20000/- for acquiring new wooden/FRP crafts, reimbursement of excise duty on diesel, group accident insurance cover to active fishermen, subsidy for purchase of insulated boxes to fishermen engaged in selling fish and financial assistance of Rs.30,000/- per hectare upto a maximum of Rs.3.00 lakh for development of shrimp farms.

- The Prawn Hatchery at Benaulim operated by BFDA has shown reasonable improvement in prawn seed production. During the year 2002-03, the Hatchery produced 8.25 million seeds generating a revenue of Rs. 31.20 lakh.

The quality of seeds produced was quite good and appreciated by the farmers. During the year 2003-04, the hatchery is expected to produce 10 million prawn seeds which wood fetch a revenue of about Rs. 40 lakh. The scheme of giving financial assistance in the form of subsidy to fresh water fish farmers has been implemented and has gained momentum during the year 2003-04. Various undergroup of fresh water bodies have been identified for fresh water fish culture and their ponds are being stocked.

5. COOPERATION

a. Goa Cooperative Societies Act 2001 and Rules

The Goa Cooperative Societies Act 2001, encompasses unique features like curtailment of Arbitrary Powers of the Registrar, more autonomy to Cooperative Institutions, removal of quasi judicial powers of the Registrar, formation of representative general body for institutions with large membership, establishment of cooperative authority for deciding disputes etc. Necessary rules under this Act have been framed and are published in official gazette to elicit public opinion. The same are expected to be laid before the Goa Legislative Assembly in the ensuing session.

b. Reform initiatives

Cooperative audit system has been streamlined by evolving guidelines and norms for awarding Audit Class so as to grade the performance of societies in a rational and impartial manner. Guidelines have been issued to ensure that the cooperative institutions make adequate provision towards non performing assets / bad debt so that a true and clear picture of financial status is reflected in their statement of accounts. Towards streamlining the functioning of Urban Credit Cooperatives and to avoid indiscreet competition in mobilizing deposits, advancing loans etc. by them, necessary instructions / norms have been laid down to have uniformity in interest rates on deposits / loans and maintaining required ratios between resources and advances. Also, opening of new branches is restricted so as to avoid overlapping in their respective area of operation.

c. Mapusa Urban Cooperative Bank Ltd.

To salvage Mapusa Urban Cooperative Bank Ltd. from its present crisis and restoring public confidence in the Bank, the Government has initiated various steps. A Committee of Administrator appointed by the Government has taken over the charge of the Bank from 1.10.2002. Besides, financial support of Rs. 200 lakh has been provided by way of share capital. Further, funds of various Government undertakings / autonomous bodies / institutions have been deposited with the Bank. The Reserve Bank of India, after taking review of status of the Bank, has permitted the Bank to advance loan upto Rs.50,000 / - in each case.

d. Support to Sanjivani Sahakari Sakhar Karkhana Ltd.

The Government has been making all possible efforts to support the Sanjivani Sahakari Sakhar Karkhana Ltd. which is in loss since inception mainly due to inadequate production of sugar cane within the State. An amount of Rs. 300 lakh by way of share capital and Rs. 125 lakh by way of loan has been released to the sugar factory, in the larger interest of sugarcane growers, to enable it to operate the crushing season during 2003-04. Further, a package of initiatives is mooted to rejuvenate its working which include Voluntary Retirement Scheme, Block Government guarantee of Rs. 2500 lakh, subsidy on export of sugar, increase of quality and acreage of local cane, modernization of outdated machinery and equipment, promoting bye products etc.

e. NCDC support for cooperatives in Goa

The Government of Goa is giving priority for infrastructure development of various cooperative institutions to improve their functioning through NCDC support. The NCDC has approved two cold storages and ice manufacturing projects one each in North and South Goa. Further, infrastructure development for procurement, processing and marketing of local fruits and vegetables through "Krishi Ghars" set up by three marketing processing cooperatives at various places in Goa is also being implemented. This will encourage the local farmers to grow fruits and vegetables by bringing additional area under cultivation and thereby provide them gainful auxiliary occupation. Four proposals for setting up three textile projects and one cashew processing project have been forwarded to the NCDC for approval.

f. Other achievements in cooperative sector (1.4.2003 to 31.12.2003)

- The Goa State Cooperative Milk Producers Union Ltd. Curti Ponda has procured 11826933.5 kg. of milk amounting to Rs. 12,84,17,458.
- The Goa State Cooperative Bank Ltd. Panaji advanced loans of Rs. 11,308.51 lakh for agricultural and non agricultural purposes.
- 41 new cooperative societies have been registered.
- 14 short term training programmes/courses were organized through the training centre run by the Goa Rajya Sahakari Sangh.
- The Government has granted 50% concession to women in compliance of the criteria prescribed for registration of Urban Cooperative Credit Societies with a view to encourage more participation of women in cooperative movement and to elevate their economic conditions.
- The details of distribution of cooperative societies and their important characteristics are given at Annexure 48 and 49.

OVERVIEW

The Government is making sustained efforts to revitalize Agriculture and Allied activities in the State. The Employment Generation Strategy announced in the Budget 2003-04 aims at encouraging rural youth to take up agriculture and allied activities in their villages. The Government is in the process of creating a network of collection, distribution and marketing infrastructure for agriculture produce of farmers. Few Krishi Ghars and Collection cum Grading centres have already been established. The scheme is being implemented with the active involvement of cooperative societies. Action has already been initiated for setting up agricultural estates and food courts to engage unemployed youth in these activities. The Horticulture Corporation is already in the process of developing a farm at Chimbel as a model farm mainly for production of manure. The Government has made significant progress in development of floriculture by arranging series of trainings, orientation camps, demonstrations etc. by involving experts from Kerala Agricultural University. There has been enthusiastic response from Goan farmers to take up high value floriculture. The Government has provided support price to sugarcane and arecanut growers as committed. The Government has made sufficient progress in remodeling of notified embankments protecting khazan lands and desilting/repairs of abandoned ponds, lakes etc.

Towards making Goa self-sufficient in milk production the Government has introduced a new scheme “Kamdhenu” to provide milch animals to individuals, groups, societies etc. The scheme aims to cover about 20000 animals. The scheme operated through GSCB is fully guaranteed by the Government. The Government is also encouraging the milk producers by providing incentive to the tune of 5% to 9% of the value of the payment received by the farmers for the milk supplied to the dairy. It is significant to note that the Government has been able to put the Goa Meat Complex back on rails and the institution has earned profit in the year 2002-03.

In order to promote tourism, the Government is in the process of setting up a botanical garden at the Salaulim dam site. Eco-tourism projects are also under implementation. For the first time the Government of Goa has launched the Joint Forest Management scheme during 2003-04. The Government continued with various welfare schemes for the benefit of traditional fishermen and development of fisheries sector in the state. The Government has framed the rules under the Goa Cooperative Societies Act 2001 and the same is expected to be laid before ensuing Assembly session. The Government has also initiated various reform measures for the development of cooperative institutions in the State. The Government has initiated a number of steps to salvage the Mapusa Urban Cooperative Bank Limited from its present crises. In the overall interest of sugarcane growers in the State the Government is providing all possible support to the Sanjivani Sahakari Sakhar Karkhana to enable it function smoothly.

CHAPTER 10
Rural Development

1. RURAL DEVELOPMENT

a. Self Help Groups

In order to create awareness about various rural development schemes, awareness camps are organized by DRDA at village level, wherein people also get an opportunity to express their economic problems and difficulties. Up to November 2003, 14 such camps have been organized. This enabled to form various Self Help Groups and increase number of beneficiaries under different schemes. The number of beneficiaries under different schemes has almost doubled during the last two years. It is proposed to cover all the BPL households within next three years. Success story of 'Devi Sateri Self Help Group' in Ponda taluka formed by the Rural Development Department is given in Box 10.1.



b. Swabhiman

The issue, which is uppermost in most peoples' mind today, is that of employment and livelihood. In the case of women, it has been found that the single biggest factor leading to their empowerment is economic independence. That is why Self Help Groups (SHGs) are internationally accepted and gaining popularity. Thus, the Hon'ble Chief Minister announced this new initiative titled 'SWABHIMAN' (self respect) for women, the mentally and physically disabled and senior citizens in the Budget Speech 2003-04 to enable them to be independent and earn a regular, decent income. The scheme is being drafted. It is tentatively proposed to follow the following modalities by taking a leaf from the success stories of similar initiatives like Amul doing a pilot project in Andhra Pradesh with rehabilitated sex workers and Lijjat Papad etc. where the entire co-operative movement is undergoing a radical change as concepts of professionalism and private marketing techniques are gradually being incorporated.

- To involve a private agency to identify products having a market demand and which each SHG can manufacture. The agency would advise on branding issues and quality control etc.
- To set up a Women's Development Corporation to manage the initiative. The Corpora-

BOX 10.1

Devi Sateri Self Help Group

This is a success story of women from Ananwadi ward in Shiroda village of Ponda Taluka, in North Goa District. Ananwadi is nearly 60 kms away from the capital city of Panaji. Main occupation of the people in Shiroda village is agriculture. The Sarpanch of the village organized a meeting of villagers belonging to below poverty line families to explain various schemes implemented by the Rural Development Department for their upliftment. The meeting was attended by the Gram Sevika of RDA. As a result, a group of SC women from Ananwadi ward of this village formed a group. Literacy rate of this ward so also their income levels is very low. But, they had some basic knowledge of bamboo works. Thus, it was decided to provide financial assistance to the group under SGSY and to further train the group in bamboo works. The group was named as "Devi Sateri Self Help Group". In the first meeting itself, they contributed Rs.50/- p.m. and they opened an account in the Goa State Co-operative Bank, Shiroda. With the amount thus contributed, they started interlending amongst themselves as per their needs. The group took up the activity of making bamboo items. Since the group was functioning well, after six months they were sanctioned the revolving fund amounting to Rs.10,000/- The group was taught the habit of thrift and internal lending to the members. Since the group had already gained experience, they wanted to increase their production and hence applied for a loan of Rs.60,000/- out of which an amount of Rs.30,000/- was released as subsidy. The group started marketing their products in their own village every Tuesday, which is a "Bazaar Day" in Shiroda.

The group is now well versed in making bamboo products such as fruit baskets, flower baskets, supali etc. Their livelihood has improved a lot compared prior to formation of group. Earlier, the group members used to make bamboo products, but the little money they used to earn out of it was just sufficient to get them two square meals to their family. They were not even aware that schemes for rural development (self employment) existed. But, after they formed the group, they have a small amount of money as their group savings in their account. The formation of the group gave them a boost to increase their income and also incur expenditure for their children's education and purchase of raw material to carry on their economic activity. This group has a good bank record and they have been regularly paying their loan installment to the bank. The group helps solving many problems that affect any group member or in the community as a whole. The success of this group inspired other villagers and thus 8 self help groups were formed in Shiroda village.

tion will set up collectives of women, disabled persons and senior citizens who could be SHGs, Mahila Mandals, Cooperatives etc. It will handle aspects like training programmes, quality control and marketing of the products, loans and subsidies, tie-ups with major companies or industries who would provide the raw material and then purchase the end products.

For successful implementation of the scheme, it will be entrusted to the Anganwadi Workers, the Mukhya Sevikas and CDPOs of the Department of Women and Child Development.

In order to finalize the modalities, a pilot of the scheme is started in Canacona and Valopi and some SHGs have already been formed. The Department of Rural Development has initiated discussions at various levels to fine tune the modalities of the scheme. It is expected that the scheme will gain momentum during 2004-05. The scheme is expected to benefit over 1 lakh women, around 10,000 disabled and 10,000 Senior Citizens.

c. Goa Bazar

To supplement the Swabhiman scheme, the Hon'ble Chief Minister has also announced in the Budget 2003-04 setting up of Goa Bazar on the lines of Dilli Haat in New Delhi where artisans and others would be free to sell their products subject to certain quality specifications. The Bazar would comprise outlets for products manufactured by artisans and other craftsmen in Goa and would give a tremendous boost to traditional arts and crafts and the rural economy. In addition, there will be a Food Court. The Department of Rural Development is in the process of acquiring land for the purpose and designing appropriate architecture.

d. Marketing support to the Swarozgaries

To provide marketing support to the Swarozgaries, the Rural Development Department assists the beneficiaries to participate not only in local exhibitions but also at the State / National / International trade fairs. A web site www.ruralbazar.nic.in has been launched by the Rural Development Department to provide marketing support and give a global market to the products. During the year 2003-04, swarozgaries participated in 10 exhibition cum sale events and sold products worth over Rs. 11 lakh.

e. National Old Age Pension Scheme

The scheme provides payment of monthly pension of Rs.75/- to destitute persons of age 65 years and above and whose annual income is less than Rs.1500/-. On account of awareness camps organized by the DRDAs the number of beneficiaries under the scheme has increased from 2170 in 2000-01 to 3805 in 2003-04 upto November 2003.

f. National Family Benefit Scheme

Under the scheme financial assistance of Rs.10000/- is given to a family in the event of death of its primary breadwinner (in the age group of 19-64 years). The amount is paid to the surviving member of the household of the deceased. In case of death of an unmarried adult then the term 'household' would include minor brothers / sisters. During the period 2000-01 to 2003-04 (upto November) 690 families have received assistance under the scheme.

g. Balika Samrudhi Yojana (BSY)

The Balika Samrudhi Yojana is an initiative of the Government of India as a part of long term strategy to change attitude and behavioural patter of the community towards a girl child and also the mother. The scheme also aims at improving enrolment and retention of girl students in the school, rise the age at marriage of girls and assisting to undertake income generating activities. The scheme envisages a post birth grant of Rs.500/- to a girl child and educational benefits ranging from Rs.300/- to Rs.1000/- per annum depending on the standard in which the child is studying. From 2000-01 to 2003-04 (upto November) 1230 beneficiaries have been covered under the scheme.

2. SETTLEMENT AND LAND RECORDS

a. Resurvey of entire State

As the last survey was conducted more than 30 years back, the Government of Goa has taken a decision to carry out fresh survey of entire State. The Government has appointed a committee for evaluation of the tenders and upon submission of the report, the contract was awarded to M/s Theovel Surveys, Bangalore and the work has started from 1st December 2003. The computerization of cadastral maps of two talukas viz. Bardez and Mormugao has been completed. The computerization of City Survey Records has been completed and the certified computerized copies of plan and property cards are being issued from respective City Survey Offices. In order to avail central assistance for resurvey, a proposal for Rs.10 crore has been submitted to the Government of India. The Government of India has approved and sanctioned undertaking of resurvey in one taluka initially on pilot basis and has released Rs.60 lakh.

b. Digitization of Cadastral Maps

Digitization of Cadastral Maps is in progress. A proposal for central assistance of Rs.4.50 crore for completion of digitization of cadastral maps of 10 talukas has been sent to the Government of India and approval is awaited. However, the project is being implemented from State funds.

c. Strengthening of Administration

New offices of Superintendent of Surveys & Land Records and Inspector of Surveys & Land Records, Margao have been opened for the convenience of the public. It is a modern and well-furnished computerized office. In order to bring the administration closer to the people, the issuing of plans and revenue matters of Bardez taluka have now been shifted to the office of Inspector of Survey & Land

records , Mapusa. Three posts of Inspector of Survey & Land Records have already been filled-up and three additional offices at Ponda, Bicholim and Quepem will be opened shortly.

d. DHARANI

Computerization of land records in Goa has been named as 'Dharani'. The objective is to project Goa as a model for replication of Dharani at the national level. Details of this project are covered under Information Technology Chapter.

e. Computerization of office of Director of Settlement & Land Records

As a part of programme of computerization of Revenue Department, the Directorate of Settlement & Land Records has also been computerized. Presently, cash collection, inward / outward, establishment matters, accounts matters, office management and procedures have been computerized.

2. PANCHAYATS

a. Supply of Garbage removal equipment

As announced in the Budget Speech 2003-04, action has been initiated by the Zilla Panchayats for purchase of garbage removal equipment. Pre-qualified tenders have been finalized and equipment are likely to be purchased by March 2004.

b. Untied grants to Zilla Panchayats

As announced in the Budget 2003-04, the Government has formulated a scheme to provide untied grants to Zilla Panchayats for development of the Zilla Panchayat constituencies @ Rs. 4 lakh per constituency. The Government has already released an amount of Rs.100 lakh to the North Goa Zilla Panchayat and Rs.80 lakh to the South Goa Zill Panchayat upto December 2003.

c. Awards to best performing Panchayats

As announced in the Budget 2003-04, the Government has introduced a new scheme to give awards to the best performing Panchayats. Under this scheme it is proposed to give awards to 12 best performing panchayats on 19th February 2004, which is the Panchayat Day.

d. District Planning Committee

In exercise of the powers conferred by section 239 of the Goa Panchayat Raj Act, 1994 (Goa Act 14 of 1994), the Government of Goa has constituted the District Planning Committee for the North Goa District and South Goa District. The Committee comprises Chief Minister, MLAs, ZP members, Municipality members, Sarpanchas etc. The Adhakshya of ZPs are the Chairpersons of DPC's and the Chief Executive Officers are the Member Secretaries.

As per the provisions of DPC, every Panchayat has to prepare a development plan and submit it to the Zilla Panchayat by 31st January every year. The Zilla Panchayat shall prepare every year a development plan for the District after incorporating the development plans of the Panchayats and submit it to the District Planning Committee. Every Municipal Council shall prepare every year a development plan and submit it to the District Planning Committee. The District Planning Committee shall consolidate the plans prepared by the Zilla Panchayat, Panchayats and Municipal Councils in the district and prepare a draft development plan for the district as a whole.

The District Planning Committee shall in preparing the draft development plan :

(a) have regard to,

  (i) the matters of common interest between the Zilla Panchayat, Panchayats and Municipal Councils in the district including spacial planning, sharing of water and other physical and natural resources, the integrated development of infrastructures and environmental conservation;

  (ii) the extent and type of available resources whether financial or otherwise;

(b) consult such institutions and organizations as the Government may, by order, specify.

The Chairperson of the District Planning Committee shall forward the development plan, as recommended by the Committee to the Government of Goa.

OVERVIEW

The Government is creating awareness in rural areas about various rural development schemes and help the people to avail the benefits. This has enabled to form various Self Help Groups and increase the number of beneficiaries under different schemes. It is pertinent to note that the number of beneficiaries under different schemes has almost doubled during the last two years. Formation of SHGs is gaining momentum and success story of Devi Sateri Self Help Group given in this chapter is highly encouraging. The Government is also in the process of finalizing a new scheme called 'SWABHIMAN' for women, the mentally and physically disabled and senior citizens to enable them to be independent and earn a regular, decent income. The scheme envisages forming SHGs providing training, inculcating professionalism and developing brand identity for marketing their products. A pilot of the scheme has started in Canacona and Valpoi through Anganwadi workers and experience of which would enable the Government to formulate a complete scheme for total implementation. The Government has also decided to set up Goa Bazar on the lines of Dilli Haat in New Delhi to provide infrastructure, training facilities and marketing support to swarojgaries and upgrade their skills. Land for the purpose is being acquired and architectural designs are being finalized.

The Government has already initiated resurvey of entire State as the last survey was conducted over three decades ago. The DHARANI project for total computerization of land records is progressing very well. Digitization of cadastral maps is in progress. The Government has provided untied funds to Zilla Panchayats for various developmental works. Funds have also been provided to ZPs for purchase of garbage equipment. The Government has instituted a new scheme for providing Awards to best performing Panchayats in the State.

CHAPTER 11

Science, Technology and Environment

The Government of Goa has taken a number of environmental upgradation measures like prevention and control of plastic litter comprising of used plastic bags and used pet bottles, ban on smoking in public places including vehicles, protection of ozone layer etc. Progress made in this regard is as follows.

1. PREVENTION AND CONTROL OF PLASTIC

a. Statutory measures

(i) The Goa Non Biodegradable Garbage (Control) Act 1996 / Rules (1997) has been enacted and enforced from 1.1.98. The Act / Rules provide a comprehensive framework to collect, manage, treat and dispose off non biodegradable / bio degradable garbage in an orderly and eco- friendly manner.

(ii) Manufacture / import / sale / use of plastic carry bags having a thickness of less than 40 microns has been prohibited with effect from 1.7.2002. The objective is to prevent the proliferation of 'thin / cheap' plastic bags, which are indiscriminately littered all over.

(iii) Under section 5 of Environment (Protection) Act, 1986 vide notification dated July 2002 and 10th January 2003 the following areas have been designated as 'Plastic Free Zones' in the State.

- Wild Life Sanctuaries at Kotigao, Bondla and Mahaveer
- Old Goa Heritage Complex of Churches and Monuments
- Miramar - Caranzalem Beach
- Parks and Gardens in the city of Panaji
- Panaji Tourism Jetty
- Dona Paula Jetty
- Roads connecting the Raj Bhavan to Gaspar Dias Circle (Miramar)

Accordingly, directions have been issued to various authorities like the Chief Conservator of Forests, Director of Tourism, Panaji Municipal Corporation, village Panchayat of Old Goa, concerned Divisional Magistrates and officer in charge of Panaji and Old Goa Police Stations to prohibit the carrying, use and sale of non-biodegradable PET bottles and plastic carry bags in the "Plastic Free Zones".

b. Collection scheme for PET bottles/ Plastic bottles

The Directorate of Science, Technology and Environment in association with Pepsi, Coca-Cola, Bisleri & Goa Small Industries Association (GSIA) has promoted a scheme for the collection of used PET bottles as well as used unsoiled plastic bags which otherwise are rampantly littered all over. The scheme envisages establishment of centres for collection, compaction and recycling of used PET bottles and unsoiled used plastic bags. Briefly, the scheme involves the collection of used PET bottles at the rate of Rs. 0.15 per bottle of less than 1 litre capacity and Rs. 0.30 per bottle for bottles of more than l litre capacity. Likewise, unsoiled used plastic carry bags are given a remuneration of Rs. 6.50 per kg. The used PET bottles and plastic bags collected at the collection centres are transported to the compaction centre where the same are 'pulverised / compacted' for despatch to the recycling units located outside the State. The revenue

generated by selling the recyclable material shall in turn be distributed to the collection centres and for acquisition of used PET bottles and unsoiled used plastic carry bags.

The scheme is fully operational in Panaji, Calangute and Colva. Efforts are on to extend the scheme to other municipal areas such as Mapusa, Margao, Vasco , Bicholim and Ponda. The multinationals such as Pepsi, Coca-Cola, Bisleri as well as the Corporation of the City of Panaji (CCP) have volunteered to run the collection centres and the compaction centres in collaboration with the GSIA. Five collection centres, three in Panaji and one each at Calangute and Colva are run through Pepsi and CCP.

Similarly, a compaction centre for used PET bottles managed by the GSIA with the active participation of Coca-Cola, is fully operational in Sancoale Industrial Estate. Bisleri and Pepsi have volunteered to operate the collection centres proposed at Miramar and Mapusa , and the sites for the same have already been allotted. Further, the DSTE and Coca-Cola have disbursed a grant of Rs. 3.62 lakh, to the GSIA, for upgrading the existing compaction centre at Sancoale and establishing one more Compaction Centre at Tivim Industrial Estate to serve the needs of North Goa District. Locations of collection / compaction centres and institutions associated with it are given table 11.1.

c. Awareness drives

Since the enactment of the Goa Non-Biodegradable Garbage (Control) Act, 1996, the Government has undertaken several awareness drives through press, advertising agencies, schools, NGOs, etc. Frequent meetings have been held at the level of the Chief Secretary to involve the enforcement authorities, interested NGOs, Department of Information and prominent citizens to coordinate the implementation of the Act as well as to undertake awareness and educational activities. Deserving projects involving training of local authorities in 'treatment / disposal' of garbage, public awareness campaigns for eco-friendly disposal of household wastes and state wide plastic waste collection drives are promoted through NGOs and educational institutions. A major awareness project was launched for a period of two years through one of the prominent NGOs, in the financial years 2001-02 and 2002-03. The achievements of this project are briefly summarized as under :

i) Preparation of a set of easy to understand digests on different Garbage Control Regulations.

ii) Preparation of a status report on the present level of implementation of Garbage Control Regulations.

iii) Setting up of experimental earthworm composting bins for different categories of users.

iv) Preparation of a manual for in-house construction of earthworm bins, which have proven to be successful. It is proposed to take up the installation of a large number of such vermi-composting bins in localities such as hotels, resorts, factories and schools.

v) The information collected from individual reports of different Municipal Councils / Panchayats was brought out in the form of "The Goa Garbage Times". Over 1,000 copies of the said Garbage Times were circulated to relevant authorities and interested per-

Table 11.1: Locations of collection / compaction centres and Institutions associated with collection scheme for PET bottles / Plastic bottles

Collection centres	Managed and run through
Panaji (Pato) / Colva (operational)	Pepsi
Panaji (at Ferry jetty and Campal) (operational)	Corporation of City of Panaji
Calangute (Operational)	Bisleri
Mapusa / Miramar (To start shortly)	Bisleri
Margao (awaiting allotment of land)	Pepsi
Compaction centres	Managed and run through
Sancoale (Operational)	GSIA & Coca- Cola
Tivim (Being established)	GSIA & DSTE

sons involved in disposal and treatment of Garbage.

vi) The second issue of "the Goa Garbage Times" carried the drawings of various vermi-composting bins designed for single family dwellings, hotels, housing colonies and institutions. Over 1,000 copies of the same were also circulated to all Municipalities, Panchayats and other interested parties.

vii) The CCP was assisted with the supply of 25,000 earthworms for their Garbage disposal site at Curca.

2. BAN ON SMOKING IN PUBIC PLACES/ VEHICLES

Since, the enactment of the Goa Prohibition of Smoking and Spitting Act, 1997, the Government has undertaken several awareness drives through press, advertising agencies, seminars, schools, NGOs etc. Special monthly monitoring drives conceived by the Government were formulated and communicated to certain Departments for regular compliance on a yearly basis for enforcement of various provisions under the said Act. Frequent meetings have been held at the level of the Chief Secretary with the enforcement authorities and interested NGOs to take stock periodically of the said drives. Notifications have been issued for declaring more places of public use or work under the said Act .

3. ENFORCEMENT

The cases booked and fines collected as reported by various enforcement agencies under the Goa Non Biodegradable Garbage (Control) Act, 1996 and the Goa Prohibition of Smoking and Spitting Act, 1997 are given in table 11.2.

4. OZONE DEPLETING SUBSTANCES (REGULATION & CONTROL) RULES, 2000

Depletion of Ozone Layer has been a cause of concern all over the world owing to its adverse effects on natural eco-system and human health. India being one of the signatories to the Montreal Protocol, is committed to phase out Ozone Depleting Substances (ODS) and for this purpose the Government has notified the Ozone Depleting Substances (Regulation & Control) Rules, 2000 under the Environment (Protection) Act, 1986. At the State level, the Small Industries, Government of India (SISI), Ministry of Small Scale Industries , Government of India (Margao), the Department of Science Technology and Environment (DSTE), Government of Goa (Saligao) and the Department of Industries , Trade & Commerce (DITC), Government of Goa (Panaji) are jointly involved in making the concerned 'firms/manufacturing' units aware of the statutory obligations. The International Day for Preservation of Ozone Layer (16th September) is used as a focal point for such awareness drives.

ODS are used in refrigeration, air conditioning , fire fighting, foam blowing, solvents, cleaning applications, spray products, etc. The steps taken so far by DSTE are summarized as under:

- Constitution of a standing Committee under the ODS Rules, 2000
- Identification of enterprises who are involved in 'manufacturing/consuming' and service enterprises engaged in servicing refrigeration and air conditioning equipment.
- Persuasion of the concerned units to register with the SISI as provided under the ODS Rules, 2000, through media coverage, special registration camps and door to door campaign (2003-04)

Table 11.2: No. of cases booked and fine collected under different Acts.

	Cases booked	Fine collected
1. The Goa Non Biodegradable Garbage (Control) Act, 1996	1580	Rs. 1,75,250
2. The Goa Prohibition of Smoking and spitting Act, 1997	3404	Rs. 3,50,700

- Organisation of a poster competition entitled " For a Green Tomorrow Stop Black Trade in ODS" (2002-03)

5. REMOTE SENSING

A major Remote Sensing Application project in collaboration with the Department of Space (GOI) namely the Integrated Mission for Sustainable Development (IMSD) (Goa State) has been completed. The entire State has been mapped at 1:50,000 scale with the help of latest satellite imageries separately for all themes such as land use, land cover, forestry, agriculture, settlements, mining , transportation network , water resources, slope rainfall, socio-economic status etc. On integration of the said thematic maps an action plan has been prepared for gearing all the development activities in a sustainable manner with the active participation of Collectorates, BDO's and other developmental agencies.

6. 91ST SESSION OF INDIAN SCIENCE CONGRESS

Goa for the first time participated in the 91st Session of Indian Science Congress held between 3rd to 7th January 2004 through the Directorate of Science, Technology and Environment. In the exhibition, Goa bagged first prize for presenting Best Design and Best Informative stall (Box 11.1).

BOX 11.1

Goa bagged award at 'Pride of India Exposition 2004'

Goa bagged the prize for presenting Best Design and Best Informative stall in the "Pride of India Exposition 2004" at the 91st session of India Science Congress held in Chandigarh. Goa participated through the State Department of Science and Technology for the first time in the Indian Science Congress held between 3rd to 7th January 2004. Goa pavilion projected the State as the best in the country depicting achievements in all spheres particularly socio economic, ecological wealth, tourism, heritage, science and technology. A model of the proposed sewerage treatment plant for Panaji and the Sky Bus Project were displayed. Other projects included the micro scale chemical experimenting equipment by the Goa University, research experiments from Goa Engineering College and Higher Secondary School students.

It is only the Goa stall which had participation of students with their science projects, which attracted maximum attention from the visitors. This is in tune with the policy of the Government to lay more emphasis on the younger generation by giving them appropriate exposure and opportunities to excel. The brief details of projects presented by the students of the Dempo Higher Secondary School, Miramar, Goa in the exhibition are given below.

(i) *Trignometer – the complex version of simplicity:* Shri Yatin Kamat presented a project based on trigonometric principles which can calculate heights and distance of any object. Though there are so many other instruments that find heights and distances, they cost very high and the fast electronic ones cost lakhs of rupees. Shri Yatin claims that the Trignometer prepared by him costs only Rs.350/- and it gives the same accurate results and it is by far the most economical instrument used for measuring heights and distances. According to him, it is an original concept that has not been taken from any book or any other source. More over, it does not require electricity nor it is an electronic device. It can be transported from place to place more easily. In the exhibition, Shri Yatin got an opportunity to explain his project to Shri R.A. Mashelkar, DG, CSIR who aptly listened to his explanation and complimented.

(ii) *Electro-vigil (Security Device); Audio-visual train arrival warning system- Automatic Railway Crossing gate; Automatic Headlight Dimmer:* Shri S.Venkatesh presented three projects in the exhibition.

- The electro-vigil project can be used in jails and in houses as a security warning device in case of jail breaks etc. The circuit is designed in such a way that if any prisoner tries to cut or break one of the bars the circuit gets activated an alarm rings. This can be used in houses to avoid robberies by cutting window grills, doors etc.

- The audio-visual train arrival warning system works on the weight of the train. As per the circuit prepared by Shri Venkatesh, few switches are placed below the normal railway track which are activated by the weight of the train and the warning system rings giving a clear indication to the road users that the gate is going to close in a few min-

utes and soon the train is going to arrive. The gate will also close automatically once the train passes some distance. According to him, this will prevent train mishaps.

ι The low cost automatic head light dimmer is designed in such a way that head light of a vehicle will function automatically and starts dimming allowing easy movement of the vehicles.

(iii) *Electro-static precipitator:* Miss Soumy Sriram and Miss Amruta Kunkolienkar presented a project which can control industrial air pollution by using a device known as 'Electro-static Precipitator'. It also remedies reducing of soil and water pollution. According to them, it is the cheapest method that can be used to eliminate pollutants from the atmosphere.

7. ZONING ATLAS FOR SITING OF INDUSTRIES IN GOA

The scheme funded by Central Pollution Control Board, New Delhi envisages preparation of Zoning Atlas for siting of Industries in Goa. The basic objective of preparing this report is to zone / classify the land vis-à-vis environmental assets and identify locations for siting of Industries involving the least possible environmental damage as also to select industries that are compatible with available sites. For siting of industries, both economic and environmental factors are considered.

8. NATIONAL (NATURAL) RESOURCES INFORMATION SYSTEM (NRIS)

The Directorate of Science, Technology and Environment has undertaken a project entitled "National (Natural) Resources Information System" (NRIS) which entails the preparation of a comprehensive and systematic computerized Geographical Information System on natural resources coupled with socio economic information. The project is implemented in collaboration with the Space Applications Centre (SAC), Ahmedabad. The NRIS is visualized as a network of GIS based databases or Nodes at the levels of Country, States and Districts. The NRIS - Goa node is proposed to be commissioned shortly with the active support of all line departments who will be the major beneficiaries of this system.

9. PILOT PLASMA PYROLYSIS PLANT

The Ministry of Science & Technology , Government of India have sanctioned development of a Pilot Plasma Pyrolysis Plant for the usage, incineration / disposal of plastic waste in the State of Goa. This plant will be set up at Goa Medical College and Hospital , Bambolim. The project is being implemented through the Institute of Plasma Research Facilitation centre of Industrial Plasma Technologies, Gandhinagar, Gujarat. The work is under progress and expected to be completed shortly.

10. STATE OF ENVIRONMENT REPORT

The Directorate of Science, Technology and Environment has initiated the process of preparing a 'State of Environment Report' for Goa with the help of Tata Environmental Research Institute and Ministry of Environment & Forests, Government of India. The report will cover the issues like coastal ecosystem- Estuarine and riverine stretches, forest ecosystem and agro ecosystem.

11. GOA STATE POLLUTION CONTROL BOARD

0 Pollution status in Goa

The Goa State Pollution Control Board has been constituted on 01.07.1988 under the Water (Prevention and Control of Pollution) Act, 1974. Prior to that in the erstwhile Union Territory of Goa, Daman and Diu, the Central Board for the Prevention and Control of Pollution was performing the functions of the State of Goa. A section office was established by the Central Board at Ponda – Goa for the purpose. In the meanwhile, the section office of the Central Board was shifted to Baroda on 4th April, 1988, to reconstitute it as a West Zonal Office. All the relevant records pertaining to Goa were transferred to the Goa State Pollution Control Board on 12th September, 1988. The status of industries is given in table 11.3 .

Fortunately, the industrial policy chartered by the Government of Goa has not allowed proliferation of highly polluting and hazardous waste producing industries. The majority of industrial units (90%) are of small scale and non-polluting. All the industries presently operating have installed adequate pollution control systems as per the requirement mandated under the relevant Pollution Control Acts. As such, the pollution control status in Goa is satisfactory as compared to other States.

Table 11.3: No. of Industries by type and No. of motor vehicles

Sr. No.		As on January, 2003
1.	No. of SSI units	6679
2.	No. of MSI units	148
3.	No. of LSI units	2
4.	No. of Beneficiation plants	21
5.	No. of STAR Hotels	32
6.	No. of Washing plants (iron ore)	2
7.	No. of Pig Iron Plants	2
8.	No. of Coke Plants	2
9.	No. of motor vehicles*	360699

** as on 2002*

There are 6 industries in the 17 category highly polluting industries. All the 6 industries have provided adequate measures to control water and air pollution.

(ii) Ambient Air Quality Monitoring under NAAQM

It is well known that the emissions of oxides of nitrogen and sulphur and particulate matter from different sources cause hazard not only for human beings but also on all living and non living materials such as buildings and monuments. Hence, the monitoring and control of such emissions becomes essential. Central Pollution Control Board has allotted air monitoring programme to all the State Pollution Control Boards under the National Ambient Air Quality Monitoring Programme. Goa State Pollution Control Board is also monitoring the ambient air quality at Panaji and Vasco cities. Under this programme, air samples are regularly collected for the analysis of particulate matter in air and oxides of sulphur and nitrogen at Panaji and Vasco cities. These samples are analysed in the Board's laboratory.

(iii) Monitoring of Indian National Aquatic Resources (MINARS)

Adequate resources are the most important part of the living thing on the earth. Inherently water is multiple use resource. The main uses of water are the public water supply, outdoor bathing and recreation, fisheries and wildlife propagation and irrigation. From the users point of view, the term water quality is defined as those physical, chemical or biological characteristics of water by which the user evaluates the acceptability of water. CPCB under MINARS programme has directed the State Pollution Control Boards to monitor the concerned water bodies. In this regard and in the interest of public health, Goa State Pollution Control Board has undertaken the monitoring of water bodies which are used to supply the water for different parts of the State. Following are the locations monitored regularly (monthly or quarterly basis).

i) Khandepar river at Opa.

ii) Assonora river at Assonora

iii) Valvanti river at Sanquelim

iv) Madei river at Valpoi

v) Kalna river at Chandel

vi) Talpona river at Canacona

vii) Sanguem river at Salaulim dam

All the industrial units generating industrial effluent are operating with Effluent Treatment Plant to treat the effluent to bring it within the prescribed standards. All the industries generating process emissions have also installed air pollution control appliances.

(iv) Implementation of Hazardous Waste (Management and Handling) Rules, 1989 and amendment

At present Goa State Pollution Control Board has identified 48 hazardous waste generating indus-

tries and authorisations have been granted for 46 industries till June, 2003 (since inception of rules).

(v) Implementation of Bio-medical Waste (Management and Handling) Rules, 1988

The authorization status of hospitals in Goa is as follows :

No. of applications received	128
No. of authorizations issued	106
No. of applications pending for clarification	14
No. of applications under scrutiny	08

(vi) Noise Pollution Regulation & Control Rules, 2000

The Board is monitoring noise levels where specific complaints of noise pollution are received and action is being taken accordingly, which include shifting of the industries also. In pursuance of clause © of rule (2) of the Noise Pollution Regulation & Control Rules, 2000, the Government of Goa has designated (1) District Magistrates (2) Director General of Police (3) Deputy Collector and Sub-Divisional Magistrates (4) Superintendent of Police and Sub-Divisional Police Officers as 'authority' for the maintenance of ambient air quality standards in respect of noise under the said rules with effect from 21.09.2000 vide Notification No.2/51/2000-II D(G) Vol. I.

(vii) Laboratory of GSPCB

The Board has set up a new laboratory of its own for carrying out the functions of the Board effectively. It is made partly functional since June, 2004 in its new premises at Patto Plaza and facilities are available for analyzing most of the parameters in respect of water, waste water and air samples. The Board intends to purchase and install Gas Chromatograph and Atomic Absorption Spectrophotometer for determining the heavy metals and pesticides etc. in water and waste water, so that the laboratory is made fully equipped to analyse the air, water and waste water samples for all the parameters notified by CPCB.

OVERVIEW

The Government has taken series of environmental upgradation measures in the State. Manufacture/import/sale/use of plastic carry bags having a thickness of less than 40 microns has been prohibited with effect from 1.7.2002. Certain areas have been designated as 'Plastic Free Zones' in the State and instructions have been issued to the concerned authorities to prohibit use of plastic carry bags, pet bottles etc. in such areas. The Directorate of Science, Technology and Environment has evolved a scheme for the collection of used PET bottles as well as used unsoiled plastic bags which otherwise are rampantly littered all over. Accordingly collection and compaction centres have been set up in different parts of the State. In order to ensure strict implementation of ban on smoking in public places/vehicles, progress is regularly reviewed at the level of Chief Secretary with the enforcement authorities and interested NGOs.

Fortunately, the industrial policy chartered by the Government of Goa has not allowed proliferation of highly polluting and hazardous waste producing industries. The majority of industrial units (90%) are of small scale and non-polluting. All the industries presently operating have installed adequate pollution control systems as per the requirement mandated under the relevant Pollution Control Acts. As such, the pollution control status in Goa is satisfactory. There are 6 industries in Goa in the 17 category highly polluting industries. All the 6 industries have provided adequate measures to control water and air pollution.

CHAPTER 12
Other Sectors

1. SPORTS

The implementation of the "New Sports Policy" and the long term development programme in coaching through the sports authority of Goa has been initiated. The work of the proposed " Sports Medicine Centre" at Peddem, Mapusa has already been started and is in progress. Besides, preliminary estimates for a new Sports Hostel for Girls at Sports Complex, Peddem Mapusa, have been prepared. The Directorate has launched 5 more new games during the current financial year viz. Karate, Malla- khamba, Yoga-Sana, Throw Ball, and Rolling skating.

The Skating Rink at Youth Hostel at Miramar is nearing completion. The Gymnasium at Ponda has been renovated. Works of Government school play grounds at various places viz. Agacaim, Cansaulim, Betalbatim, Colva, Utorda, Mandrem , Devbag, Benaulim, Shellar, Karve, St. Cruz are progressing in full swing.

Goa has won medals at National level in chess, athletics, Karate, Special Olympica, Taekwondo, Judo, Baseball, swimming, soft ball, skating, yatching, tennis ball, cricket, shooting ball etc. Sports persons have performed creditably at the 49th National School games in Swimming, Judo, chess and baseball.

2. GOA GAZETEER

The preliminary work of collection and compilation of data for the scheme " Goa History and Places of Interest" is in progress along with the work of standardisation/collection of the place names other than revenue villages, in Roman and Devnagari scripts.

During the year 2002-03 one more publication the VIIIth Volume in the series under the scheme "Source Material for the History of the Freedom Movement of Goa" dealing with the trials of prominent freedom fighters/Satyagrahis (1946-1961) by the then Territorial Military Court was brought out and the work pertaining to remaining volumes is in progress. Considering the usefulness and importance of the Gazetteers at the district and national level, priority is to bring out the new revised edition as the earlier one was brought out in 1979. This volume is to be brought out based on a uniform central gazetteer pattern and as per the guidelines adopted by the All India Gazetteer Chief Editor's conference held from time to time.

3. JUDICIARY

a. Fast Track Courts

Government has been instrumental in setting up all 5 Fast Track Courts in Goa , 2 being in South Goa (Margao) and 3 in North Goa viz. 2 in Panaji and 1 in Mapusa. As on June 2003, about 22456 and 5639 civil and criminal cases are pending in different courts in North and South Goa respectively

b. Goa State Legal Services Authority

During the year, in all 1733 matters were taken before the Lok Adalats in all talukas of North and South Goa . 484 cases have been disposed off. Similarly, 25 legal Literacy camps were held during the year 2003 in all the talukas specially in rural areas to bring legal awareness amongst the general public. A workshop/training programme on the subject of "Good Governance " with special emphasis on skills of writing proper judicial orders for Revenue Officers of the level of Deputy Collectors to joint Mamlatdars was organized on 22nd and 23rd November 2003. One Revenue Lok Adalat was held in November 2003 and 34 cases of Mundkar, MCA, Tenancy and uu/sec 133 P.C were referred, out of which 5 cases have been disposed off. In all 58 matters were referred before Legal Aid Clinics and all the clients were satisfied with the advice given by the Panels. In South Goa,Permanent Lok Adalats were held from May to August 2003 and 77 pre-litigation Bank matters

were referred out of which 45 Bank matters have been disposed off. In all 69 family matters were taken up before Counselling and Conciliation centre out of which 7 matters have been settled by conciliation.

c. Infrastructure Development for Judiciary

The land admeasuring 2290 sq.mts. belonging to Communidade is being acquired for the construction of Annexe building of the District Court and for housing Additional Court at Margao .

4. STATIONERY & PRINTING

The Government Printing Press is the oldest Institute of the Government dating back to the 19th Century. During the year 2003-04 the press has introduced billing system on all Indenting Officers for supply and printing of stationery thus cutting down wasteful expenditure and proper use of the stationery by indenting departments.

5. LEGAL METROLOGY

One of the main statutory functions of the Department is to ensure correctness of weights & measures , weighing & measuring instruments used in trade, commerce & industrial establishments, control on commodities sold in the package form. For this purpose it is required to maintain Laboratories viz. Secondary & working standard Laboratories with working standard electronic scale (digital). Four sets of electronic working standard of 50 kg, 20 kg, 2 kg, and 200 gms have been purchased. Meetings to educate public in respect of their right towards weights and measures will be conducted in Panchayats and in Consumer Forums. During the year 321, cases have been booked

6. CRAFTSMEN TRAINING

Achievement during the year 2003-04 is as follows:

- New trade courses like hair and skin care, painter (general) , multi skilled diploma in Industrial technology (mechanical, electrical and electronics, are introduced in ITI Mapusa. Awareness programmes are conducted at ITI Margao on Leadership development, vocational career guidance under Super School Complex and personality development .

- Exercise of conducting model lesson with the idea “Each one Teach one” in Principles of Teaching (POT) subject was launched from this academic year 2003.

- “Production Oriented Training Scheme” (POTS) was launched in all 11 Government run ITIs in the State as per the norms of DGET/NCVT who desired that the trainees get “On the job” experience in performing Industrial skills acquired in the ITIs.

- During 2003-04, 1916 trainees were enrolled in 11 Government ITIs. As against 1980 target while under apprenticeship Training scheme 650 seats have been located in 80 establishments in 31 trades out of which 408 seats are utilised for training in different industries.

OVERVIEW

A ‘Sports Medicine Centre’ is being established at Peddem Mapusa. Preliminary estimates have been prepared for a new Sports Hostel for Girls also at Peddem, Mapusa. Goa has won medals at various national level sports competitions. In order to clear pending cases in various courts, five fast track courts have been set up in the State. Lok Adalats in the State are quite active and during the year, 1733 matters were taken before the Lok Adalats. The Government is implementing the relevant statutory measures to ensure correctness of weights & measures, weighing & measuring instruments used in trade, commerce & industrial establishments. New trade courses have been introduced in different ITIs.

CHAPTER 13
National Institutions In Goa

The tiny State of Goa has some important national institutions, which have a significant bearing on the socio-economic development of the State. Some important national institutions located in Goa are National Institute of Oceanography, Mormugao Port Trust, Institute of Petroleum Safety, Health and Environment Management, National Centre for Antarctic and Ocean Research and Goa Ship Yard Limited. An attempt is made in this chapter to give an overview of these institutions.

1. NATIONAL INSTITUTE OF OCEANOGRAPHY

a. The Institute

The National Institute of Oceanography (NIO), one of the 38 laboratories of the Council of Scientific and Industrial Research (CSIR) was founded in Dona Paula, Goa on 1st January 1966 with regional centers at Mumbai, Kochi and Visakhapatnam. It was established as a natural outcome of the International Indian Ocean Expedition (1962-65), and was structured along the best traditions of any marine laboratory of that period. IIOE was a major peer effort of UNESCO of coordination of 20 countries with 40 research ships to understand the least known Indian Ocean. The four year expedition generated voluminous data and many interesting observations.

From a small group of IIOE participants, NIO today has grown to be the largest marine research institute in the Indian Ocean Region. Since its inception the Institute has been performing creditably in various disciplines of oceanography and has kept abreast with international developments.

b. Mission

The mission of the Institute is:
"to continuously improve our understanding of the seas around us and to translate this knowledge to benefit all".



Research in open and coastal waters remains the core of the institute's activities. The multidisciplinary research programmes carried out by the Institute fall in four broad areas: coastal environment, engineering & technology, ocean processes and ocean resources.

The research experience and expertise of the Institute is also extensively utilized by both public and private sector organizations through consultancy and contract services in environmental impact assessments, harbour & port development, geological and engineering surveys for feasible sites & submarine pipeline routes for seawater intake / discharge; effluent disposal; seabed surveys, pollution monitoring, resources evaluation, design and development of instruments, marine archaeological exploration and several other aspects connected with the sea.

Presently about 570 scientific, technical and administrative staff contributes to the institute's endeavors. Average annual budget / funds handled by the Institute is Rs. 150 million excluding variable revenue of Rs. 90 to 160 million generated from the consultancy and contract research projects.

c. Research Areas

Areas of importance	NIO's interest	Programmes
Coastal Zone	q Integrated Coastal Management	§ Coastal engineering & services, § Ecology of coastal zone sites § Predictive modeling, § Awareness programmes.
Special features of our surrounding seas	q Biogeochemistry q Upwelling q Coastal currents q Blooms	§ Biogeochemistry § Blooms & bio-invasion
Climate	q Monsoons q Paleoclimate	§ Physics of land-ocean-air
Food & ecology security	q Sustainability issues q Biodiversity mapping	§ Productivity mapping § Dynamics of ecosystems § Assessment of diversity
Biological processes	q Modeling biological processes in our seas	§ Microcosm & microcosm work § Large Marine Ecosystems § Flux studies
Novel ecosystems & marine protected areas	q Adaptation strategies to extreme environs	§ Biology of hydrothermal vents § AUV and ROV design § Observing & sampling extreme environs § Advisories on protection means
Bioactive molecules and Biotechnology	q Inventory of bioactive molecules q Gene mapping of bioactive molecules q Genetic engineering of bioactive molecules q Bio-remediation q Center for special bio-active organisms	§ Studies on interaction between bioactivity & environment § Genes of bioactive organisms § *In-vitro synthesis* of molecules § Cell biology § Marine genomics, proteomics, § Special applications in biology
Pollution & health hazards	q Bio-protection & bio-invasion q Predictive modeling	§ Red tides § Ballast water § PCBs § Pathways of heavy metals in marine life

Areas of importance	NIO's interest	Programmes
Operational oceanography	q Indian Ocean GOOS q Satellite oceanography q ARGO program q Ships q Buoys	§ Algorithms for coastal waters § Data assimilation programme § Long term fixed monitoring stations § Networking with neighbours § Colour & bio-optical studies
Non-living Resources	q EEZ q Legal Continental Shelf (LCS) q Gas hydrates	§ Survey and mapping programme for minerals § Geological inputs for the LCS § Niche areas in hydrate studies
Plate tectonics	q Ridges q Marine Seismology	§ Indian ridge & Back arc basins
Societal issues	q Ocean awareness q Stakeholders issues q Networking q Sustainability issues	§ Informing citizens of present know § Predictive modeling § Broadcasting the ridges observations on TV § Beach & coral watch programmes § Visualization as a decision tool § Marine archaeology

d. Infrastructure backup

- Modern laboratories
- Designated National Oceanographic Data Center (RNODC)
- In-house instrumentation
- Library with International databases
- Intra LAN & connectivity VSAT Internet
- Coastal Research Vessel Sagar Shukti
- Campus wide 500 nodes LAN
- Processing of satellite imageries
- In-house Instrumentation

e. Significant achievements

- Seabed exploration of polymetallic nodules in 4 million sq.km area in the Central Indian Ocean (water depth 4 - 6 km) and identification of mine sites, secured sovereign mining rights for India and the 'Pioneer Investor' status from the United Nations.
- A survey of shallow offshore areas of Konkan coast has indicated ilmenite deposits of about 12.5 million tones.
- Prepared maps for major portion of the Indian continental shelf on bathymetric, sedimentological, geochemical, geomorphological and magnetic anomaly
- A few models covering land-sea and air-sea interactions and ocean dynamics have been developed to understand variability of monsoon.
- Carried out physical, chemical, biological, geological and geophysical studies of the Exclusive Economic Zone (EEZ) of India and other areas in the Indian Ocean, which have led to better understanding of the ocean processes and of previously undiscovered resources.

- More than 700 species of marine flora and fauna have been studied for antiviral, anti-fungal and CNS depressant activities. A few have greater potential for development into drugs.

- Fungal enzymes for bio-bleaching for paper industries have been identified.

- The mussel hydrolysate exhibited anti-HIV properties.

- High-resolution shallow seismic reflection data from the continental shelf sediments off western India revealed methane rich horizons. Potential sub-surface methane estimates in this gas charged area are of the order of 2.6 Tg.

- Was the lead organization in organizing the First Indian Scientific Expedition to Antarctica and active participant in the subsequent expeditions.

- NIO has released the technical know-how for several marine instruments. A self-recording and transmitting tide gauge has also been developed and commercialized.

2. MORMUGAO PORT TRUST

Mormugao Port, one of the oldest Indian ports on the west coast of India with a fine natural harbour has been relentlessly serving the Nation for over a century. As per the Major Port Trust Act, 1963, a Board of Trustees was constituted with effect from 1st July 1964 for the administration and management of the Port.

The port is situated in the State of Goa on the western coast of India. The port is approximately 370 Kms. South of Mumbai and 575Kms North of Cochin. It lies in the latitude 15° 25' North and Longitude 73° 47' east and is located at the mouth of river Zuari.

Mormugao Port ever since its inception as the seventh Major Port of India in 1963, constantly strives to be a totally customer centric organization consistently delivering Value-Plus services to all its clients by offering faster turn round times, lower costs to port users, being adaptable to every requirement and consistently offering customers quality and reliable service. As a result, Mormugao Port Trust plays a key role in the growth of the trade and also the region's economy.

Attaining ISO 9001: 2000 certification is yet another milestone in the port's continual improvement and commitment towards maintaining a high standard of performance which shall translate into reliable and top rate services through excellent infrastructure and scientific streamlining of operation to its clientele in particular and the maritime community at large.

a. Vessel Traffic Management System (VTMS)

To provide a highly sophisticated computerised radar system for vessel traffic control and safety in the navigational channel and other areas of Port waters.

Approach channel

i)	length of outer channel	4.6 kms
ii)	width of channel	250m
iii)	Depth	13.7 m below chart datum
iv)	Turning basins (2nos)	DIA -480m Depth 13.1m below chart datum
v)	Tidal range	
	a) springs	2.3 m
	b) Neaps	1.0 m

The approach channel to berth is marked by lighted buoys in addition to other navigational aids prescribed in the Indian Hydrographic Chart Nos 2020 & 2078 and B.A. Chart nos. 492 and 1509. One radio Beacon "RACON" works round the clock in all weathers at Aguada Light House displaying two dashes and one dot (.) on the ships radar in X- band. The Racon operates in conjunction with other ordinary X- Band Navigational Radar with 3 cms X-band 93000 NHZ 95000NHZ .

b. Communication

Signal station functions round the clock with telephone services and equipped with VHF channel 16,14,12,11 & 10 having call sign GOA PORT. Telefax no is 0832 520184 mobile phone no 98-221 / 02653. INMARSAT I.D.For Tel 762093016 ; FAX 7620 93017 ; DATA 76209018. To contact on INMARSAT one has to dial 00-873 (Indian Ocean Code) and then the I.D Number.

c. Pilotage

Pilotage is compulsory for inward and outward movement of ships and movement between berths / moorings within the harbour. Pilotage service is provided round the clock with prior advice to the Harbour Master through agents. During monsoon season pilotage is restricted to day light hours only.

d. Tugs

Four tractor tugs i.e. 1 of 30 ton and 2 of 45 ton bollard pull and one ASD tug of 45 ton bollard pull are available.

e. Ore Loading facilities in stream through transhippers

3 transhippers, each with an average loading capacity of about 15,000 TPD are operated by private exporters for up topping large size ore carriers that are initially loaded by MOHP at Berth No. 9 to the permissible draft in mid stream and loading of large carriers exclusively by transhippers.

f. Mooring Dolphins

3 Mooring Dolphins capable of accommodating Panamax vessels of about 70,000 DWT are available in mid stream to handle ores and other bulk cargo by ship's own gears. The draft available at Mooring Dolphins is West No. 1 Dolphin Draft 11.00 m between No. 1 and 2 . Dolphins draft 12.5 m and between no 2 & 3 dolphins draft 12.5 m.

g. Bulk Liquid discharging facilities

Specialised facilities are available at B. No. 8 for handling petroleum products and other liquid cargoes like phosphoric acid, caustic soda, ammonia, molasses etc. phosphoric acid handling facilities are also provided at general cargo berth Nos 10 and 11.

h. Inland Transport Facilities

Mormugao port is connected through road with the rest of the country through two national highways NH-17 and NH 4A connecting major cities like Mumbai and Bangalore. A new bye pass 4-lane road connecting to Verna Junction by skirting the Vasco town is under construction. The broad gauge railway system of the port serves the general cargo berths. An Electronic Data Interchange (EDI) system has been installed on EDIFACT standard as suggested by the Ministry of Commerce to speed up the transactions of documents . 66 port users have already registered for the EDI system and the port has started transacting 10 documents electronically through this system.

i. Export and Import through MPT

Commodity-wise overseas export of cargo during the year 2002-03 and traffic handled at the MPT during the period 2000-01 to 2002-03 are given in table 13.1 and 13.2 respectively.

Table 13.1: Overseas export through MPT

(Rs. in Crore)

Sr. No.		
1	Iron Ore	1035.18
2	Iron Ore Pellets	62.54
3	Manganese Ore Group	7.34
4	Molasses	4.56
5	Alumina	115.52
6	Containerised cargo	362.77
7	Others	337.66
	Total	**1925.57**

Table 13.2: Traffic handled at the MPT from 2000-01 to 2002-03

(Quantity in lakh tonnes)

Commodity	2000-01	2001-02	2002-03
(a) Export			
Iron ore including pellets	154.57	179.11	186.64
Other ores	1.25	0.69	0.68
Alumina	1.40	1.24	1.66
Containerised cargo	0.25	0.37	0.61
Other cargo	1.91	1.68	3.10
Total Export	**159.38**	**183.09**	**192.69**
(b) Import			
P.O.L. product	9.73	11.34	11.28
Prosphoric acid	1.38	2.54	2.71
Fertilizers	1.24	1.59	1.56
Iron ore & pellets	2.06	0.55	0.02
Coal and coke items	18.78	26.77	24.63
Containerised cargo	0.19	0.21	0.29
Other cargo	2.36	3.00	3.31
Total Imports	**35.74**	**46.00**	**43.80**
(c) Transhipment	**1.16**	**0.19**	**-**
Grand Total (a + b + c)	**196.28**	**229.28**	**236.49**

3. GOA SHIPYARD

The Goa Shipyard is an Public Sector Undertaking under the Ministry of Defence, Government of India. The Institute established on 29th September 1967 under the Companies Act, 1957 is located in Vasco da Gama, Goa. Major achievements of the organization are building of indigenously designed Extra-Fast Petrol Vessels and other state-of-art vessels like Missile Craft, Sail Training Ship, Extra Fast Attack Craft etc.

4. NATIONAL CENTRE FOR ANTARCTIC AND OCEAN RESEARCH (NCAOR)

The Department of Ocean Development (DOD) was created in July, 1982 to function as a nodal department for organizing, coordinating and promoting Ocean development activities in the country in line with the Ocean Policy Statement. National Centre for Antarctic and Ocean Research (NCAOR) is a Research and Development Institution established on 25th May, 1998 in Vasco-da-Gama, Goa as an autonomous Institute of the DOD.

a. Mission Mandate

"To plan, promote, co-ordinate and execute the entire gamut of Polar Science and Logistic activities of the country in order to ensure a perceptible and influential presence of India in Antarctica and uphold our interests in the global framework of nations in the Southern continent and the surrounding oceans".

b. Aims and Objectives

- Undertake, aid, promote, guide and co-ordinate scientific research in the fields of polar science (Antarctic & Arctic) and Southern Ocean Oceanography.
- Serve as a nodal agency responsible for the overall coordination and implementation of the Indian Antarctic Programme.
- Establish and maintain research base in Antarctica and provide all operational and logistical support.



- Encourage co-operative research both at National and International levels, especially in the frontier areas of science, which are emerging and have application potentials.

- Devise, formulate and initiate programmes of strategic importance in the realm of polar science, which will create a knowledge base for future commercial and thematic interests of India in Antarctic, Arctic and Southern Oceans.

- Effectively interact and liaison with scientific committee on Antarctic Research (SCAR), Council of Managers of National Antarctic Logistics and Operations (SCALOP), and other international bodies in order to keep the Indian Antarctic Programme at par with international scenario and ensure a perceptible presence of India in the global framework of nations in Antarctica.

- Establish a singular research laboratory setup with such facilities which are of specialized nature or otherwise not easily accessible elsewhere. Some of these are deep drilling technology, low temperature ultra-clean laboratories for the preservation and analysis of ice cores, instrument facilities like mass spectrometer (MS) and ICP-MS for isotopic and trace metal analysis, microcosm laboratory for biological investigations, remote sensing tools and techniques for application of advanced satellite technology for studies of the polar regions.

- Develop predictive models for dynamic processes controlled and driven by Antarctica and the surrounding oceans.

- Co-ordinate, address and monitor all environmental issues arising out of the Environmental Protocol to the Antarctic Treaty.

- Develop a complete database inventory and repository for Antarctic and Arctic science and logistics including establishing a polar museum and library.

- Create a complete warehousing facility and inventory of entire Antarctic Logistics and procurements.

- Provide a fertile platform to generate specialized human resource capabilities for polar science and logistics through training, fellowships, awards, workshops, publication, exchange of scientists etc. both at national and international levels.

- Cruise planning and management of the oceanographic research vessels engaged by the Department of Ocean Development and National Centre for Antarctic and Ocean Research.

- Take up scientific and technological projects of the Department of Ocean Development as and when required.

c. Infrastructure

To implement the mandate as well as to ensure the efficient functioning of the Institute, NCAOR has been organised into different scientific, technical and management divisions / sections, as follows :

- National Antarctic Programme comprising the divisions of Antarctic Science and Expedition Logistics.

- Scientific wing comprising the Research and Development Group.

- The Legal Continental Shelf Programme.

- Ocean Survey and Services Group.

- Programme Planning and Evaluation Division.

- National Antarctic Data Centre and Information Services.

- Paleoclimatic Research Division - Antarctic Ice Core Laboratory.

- Polar Remote Sensing Division.

- Polar Environment & Ecology Division.

- Southern Ocean Oceanography Division.

- Management services, comprising the Administrative, Establishment, Finance and Purchase and Stores Sections.

d. Major Scientific Programmes of NCAPR

- Legal Continental Shelf Programme.
- Collaborative Research Programmes between NCAOR and KBCAOS :

i. Realistic simulations of Antarctic circumpolar current and other circulation features of Southern Indian Ocean.

ii. Inter Comparison of semptner's model (1976) with that of Winton three layer sea-ice model (2000) and configuring the GCM for Antarctica.

iii. Effects of through flow in the Indonesian Archipelago region of connectivity between the Pacific and Indian Oceans on the Southern Indian Oceans.

- Maintenance of Oceanographic Research Vessel (ORV) Sagar Kanya
- Antarctic Logistics

5. INSTITUTE OF PETROLEUM SAFETY, HEALTH & ENVIRONMENT MANAGEMENT

The Institute was established in 1989 with the primary objective of promoting standards of Safety, Health & Environment Management in Petroleum Sector in India. 'HSE' has advanced substantially during the past few years. The emphasis has shifted from injury preservation to loss control to integration of 'HSE' function in the management system. Keeping this trend the Institute is developed on modern risk management philosophy.

The Institute is located in a picturesque site spread over 101.5 hectares with an exclusive sea front of 1.15 kms, at Betul village, in close proximity to the well-known Mobor Beach and 58 kms south of the Dabolim Airport.

The main functions are Risk Management, Safety Management, Occupational Health services, Environment Management and Information services. It offers various courses like

- **Advance courses:** includes Internal Diploma in Safety Management, Risk Management, HSE and Loss control Management, ISO- 14001 Environment Lead Auditors, safety in process plants , incident command and control system, modern safety Management course, accredited safety auditors course , occupational health Hazard and safety
- **Basic courses:** include training in survival crafts, coxswain, HSSE and Loss control, regulatory overview, onsite hydrogen sulphide safety, distance learning diploma in environment, HSE in seismic well logging operations and handling of explosives
- **Pre basic courses**: include basic training courses in safety onshore, personnel survival technique, abandonment simulation exercises , training in first aid, fire team course for onshore personnel.



CHAPTER 14

Demystifying Financial Processes

Certain aspects relating to the financial processes such as the budget may not be clearly understood by many people, who therefore may not comprehend their implication. Since transparency in Government implies the full participation of the people, an attempt has, therefore, been made in this chapter to demystify some important processes for the benefit of readers.

1. VALUE ADDED TAX (VAT)

VAT is a multi-point sales tax with set-off for tax paid on purchases. It is collected in instalments at each transaction in the production distribution system. It does not have cascading effect due to the system of deduction or credit mechanism.

WHAT IS A VAT?

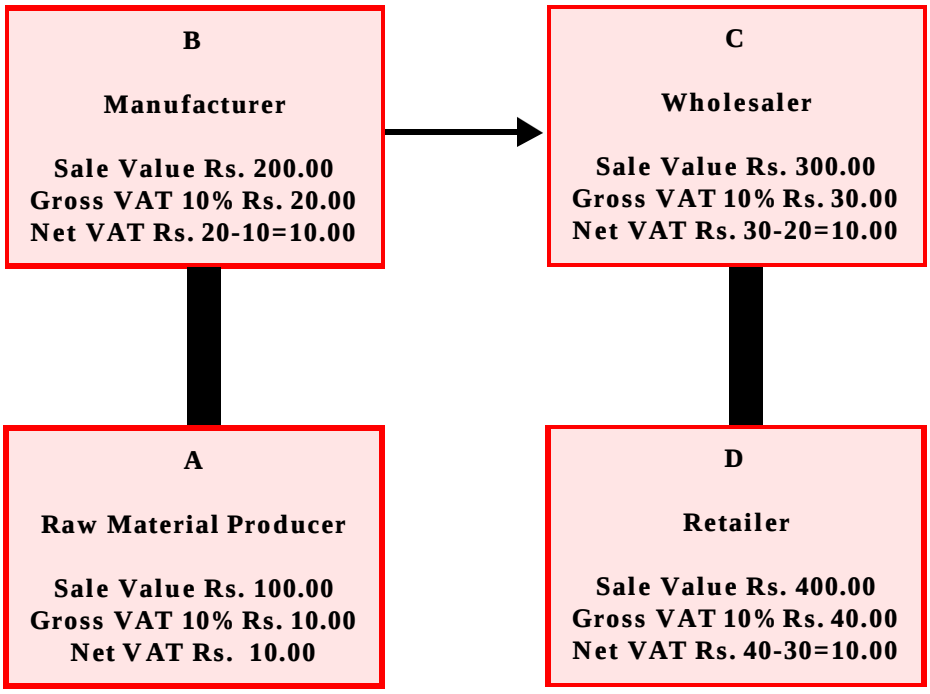
- VAT is a multi-point sales tax with set-off for tax paid on purchases.
- It is collected in instalments at each transaction in the production distribution system.
- It does not have cascading effect due to the system of deduction or credit mechanism.
- It is a tax on consumption. The final and total burden of the tax is fully and exclusively borne by the domestic consumer of goods and services.
- It being a tax on domestic consumption, no VAT is charged on goods exported.
- It is an alternative mechanism of collection of tax.
- It is in many respects equivalent to a last point retail sales tax.

It is a tax on consumption. The final and total burden of the tax is fully and exclusively borne by the domestic consumer of goods and services. It being a tax on domestic consumption, no VAT is charged on goods exported. It is an alternative mechanism of collection of tax. In many respects it is equivalent to a last point retail sales tax. Value added tax is, therefore, a multi-stage sales tax levied as a proportion of the value added (i.e. sales minus purchases, which is equivalent to wages plus profits).

To illustrate, let us assume dealer A to be producer, B to be manufacturer, C to be wholesaler and D to be retailer. Dealer A sells his produce at Rs.100 and pays tax at the rate of 10 per cent. As dealer A is a producer of primary product his inputs could be assumed to be zero. The sale price of Rs.100 would be the purchase price of dealer B, who is a manufacturer. This dealer would use wages, salaries, other manufacturing expenses and to all this he would add interest and his own profit. Assume that after adding all these costs his sale price is Rs.200. On this sale price the gross tax (at the rate of 10%) would be Rs.20. As dealer A has already paid tax on Rs.100, dealer B would get credit for this tax. Therefore, his net VAT liability would be Rs.20 minus Rs.10. That is, dealer B would pay Rs.10 only. Similarly, the sale price of Rs.300 by dealer C would have net VAT liability of Rs.10 (Rs.30 - Rs.20 = Rs.10) and the sale price of Rs.400 by Dealer D would also have net VAT liability of Rs.10 (Rs.40 - Rs.30 = Rs.10). This illustration indicates that the VAT is collected at each stage of production and distribution process and, in principle, its burden falls on final consumers only. Thus, it is a broad-based tax covering the value added of each commodity by a firm during all stages of production and distribution. (Source : Value Added Tax by Mahesh C. Purohit, Gayatri Publications, Delhi).

Illustration

Different stages of VAT



Note : Total VAT collected at four points : Rs. 10 + 10 + 10 + 10 = 40

2. BUDGET PROCEDURE

The Annual Financial Statement that is the budget, is laid before the Assembly. In brief, this contains details of the receipt and expenditure of the Government during the previous year and details of how the Government proposes to raise revenue (receipts) during the ensuing year (1st April to 31st March) and how it proposes to spend.

Along with the Annual Financial Statement the following documents are laid before the Assembly.

a. Estimates of Receipts

b. Demands for Grants (Volume I, II and III)

c. Budget at a Glance

d. Explanatory Memorandum to Plan Schemes.

The Finance Minister delivers the budget speech at the time of laying these documents. A vote on account may be taken for authorizing expenditure for a certain period till the budget is passed in the legislative assembly. Following this, the financial requirements of individual departments for the coming year are debated. These are reflected in the 'Demand for Grants' and the discussions include coverage of the utilization of past funding and the performance of the Department. At the end of each discussion the Demand is put to vote. These are known as voted expenditure.

There is also a category called charged expenditure which includes areas which are not discussed or voted in the Assembly as per the provisions of the Constitution of India such as the expenses related to Raj Bhavan and Legislature.

Consequent to passage of all the Demands for Grants, the overall budget allocations as reflected in the Annual Financial Statement is put to vote. The connected appropriation bill showing the gross budgetary provision is then put to vote and passed by the Assembly. The same is then assented to by His Excellency the Governor.

3. GOVERNMENT ACCOUNTS

The accounts of Government are kept in the following three parts.

Part I - Consolidated Fund.

Part II - Contingency Fund.

Part III - Public Account

Part I: Consolidated Fund: Under this account there are two main divisions, viz. Revenue Account and Capital Account.

(i) Revenue Account – which is further divided into two sections – Revenue Receipts and Revenue Expenditure. Revenue Receipts comprises tax revenue (such as sales tax, excise duties etc.), non-tax revenue (such as income from sale of power, sale of water, fees for services etc.), share in central taxes (such as central excise) and grants in aid from central government. Revenue expenditure mainly consists of interest payment on debt, current recurring expenditure like salaries, electricity bills, telephone bills, water bills, maintenance expenditure etc.

(ii) Capital Account – which is further divided into two sections – Capital Receipts and Capital Expenditure. Capital receipts comprises internal debt of the Government (such as loans from HUDCO, NABARD, LIC etc.), loans and advances from central government, etc. Capital expenditure mainly comprises plan expenditure for asset creation such as roads, bridges, irrigation projects, water supply and sewerage projects, power infrastructure projects etc.

Part II: Contingency Fund : Under this account, the transactions connected with the Contingency Fund established under Article 267(2) of the Constitution of India are recorded.

Part III: Public Account : Under this account, the transactions relating to "Debt", "Deposits" and "Advances", "Remittances" and "Suspense" are recorded. This include GPF, Small Savings etc.

a. Sectors and Heads of Accounts

With the capital and revenue discussions in Part I(Consolidated Fund) mentioned above, the transactions are grouped into sectors such as "Tax Revenue" and "Non-Tax Revenue". For example specific functions or services such as Education, Sports , Art and Culture, Health and Family Welfare, Water Supply, Sanitation, Housing and Urban Development, etc. in respect of Social Services are grouped in sectors for expenditure heads.

In Part III (Public Account) also, the transactions are grouped into sectors, such as "Small Savings", "Provident Funds", "Reserve Funds" etc. The sectors are sub-divided into major heads of account. In some cases, the sectors are also sub-divided into sub-sectors before their division into major heads of account.

The major heads are divided into sub-major heads in some cases and minor heads, with a number of subordinate heads generally known as sub-heads. The sub-heads are further divided into detailed heads. Under each of these heads, the expenditure is shown distributed between charged and voted. Sometimes major heads are also divided into sub-major heads before their further division into minor heads. Apart from the sectoral and sub-sectoral classification, the Major Heads, Sub / Major Heads, Minor Heads, Sub-Heads, Detailed Heads and Object-Heads together constitute a six-tier arrangement of the classification structure of the Government Accounts.

The major heads of accounts, falling within the sectors for expenditure heads, generally correspond to functions of Government while the minor heads, subordinate to them identify the programmes undertaken to achieve the objectives of the function represented by the major head. The sub-head represents the scheme, the detailed head, the sub-scheme and object head, the object level of classification.

b. Coding Pattern

(i) Major Heads

From 1st April, 1987, a four digit code has been allotted to the major heads, the first digit indicating whether the major head is a Receipt head or Revenue Expenditure head or Capital Expenditure head or a Loan head.

The first digit of code for Revenue Receipt head is either "0" or "1". Adding 2 to the first digit code of the Revenue Receipt head will give the number allotted to corresponding Revenue Expenditure head; adding another 2, the Capital Expenditure head; and another 2 the Loan head of Account. For example, for Crop Husbandry code, 0401 represents the Receipt head, 2401 represents the Revenue Expenditure head, 4401 represents Capital expenditure head and 6401 represents Loan head.

(ii) Sub-Major Heads

A two digit code has been allotted to sub-major heads under each Major head and the starting being "01". Where no sub-major head exist, it is allotted a code "00". The nomenclature "General" has been allotted code "80" so that even after fur-

ther sub-major heads are introduced, the code for "General" will continue to remain the last one.

(iii) Minor Heads

Minor heads have been allotted a three digit code starting from "001" under each sub-major / major head. Codes from "001" to "100" and few codes from "750" to "900" have been reserved for certain standard minor heads. The coding pattern for minor heads has been designed in such a way that in respect of certain minor heads having a common nomenclature under many major / sub-major heads, the same three digit code is adopted as far as possible.

Under this scheme of codification, the receipt major heads (Revenue Account) are assigned the block numbers from 0020 to 1606, expenditure major heads (Revenue Account) from 2011 to 3606, expenditure major heads (Capital Account) from 4046 to 5475, major heads under "Public Debt" from 6001 to 6004 and those under "Loans and Advances" "Inter-State Settlement" and "Transfer to Contingency Fund" from 6705 to 7999. The code number 4000 has been assigned for Capital receipt major head. The only major head "Contingency Fund" in Part -II "Contingency Fund" has been assigned the code number 8000. The major heads in the Public Account are assigned the code numbers from 8001 to 8999.

The transactions included in these accounts represent mainly the actual cash receipts and disbursements during the financial year April to March as distinguished from amounts due to or from Government during the same period. The cash basis system is, however, not entirely suitable for recording the transactions and presenting the true state of affairs of Government, commercial undertakings run on commercial principles. The detailed accounts of this class of undertakings are, therefore, maintained outside the regular accounts in proper commercial form and are subject to test-check by the Indian Audit and Accounts Department.

4. OTHER DEFINITIONS

- *Revenue Account Surplus/Deficit*: If the revenue receipt is more than the revenue expenditure then it is known as revenue surplus otherwise it is called as revenue deficit. Revenue receipt comprises tax revenue (such as sales tax, excise duties etc.), non tax revenue (such as income from sale of power, sale of water, fees for services etc.), share in central taxes (such as central excise) and grants in aid from central government. Revenue expenditure mainly consists of interest payment on debt, current recurring expenditure like salaries, electricity bills, telephone bills, water bills, maintenance expenditure etc.

- *Capital Account Surplus/Deficit*: If the capital receipt is more than the capital expenditure then it is known as capital surplus otherwise it is called as capital deficit. Capital receipt comprises internal debt of the Government (such as loans from HUDCO, NABARD, LIC etc.), loans and advances from central government etc. Capital expenditure mainly comprises plan expenditure for asset creation (such as roads, bridges, irrigation projects, water supply and sewerage projects, power infrastructure etc.), repayment of debt etc.

- *Budget Surplus/Deficit*: If the sum of 'surplus or deficit under revenue account' and 'surplus and deficit under capital account' is positive then it is known as budget surplus otherwise it is called as budgetary deficit. In other words if sum of '1' + '2' above is positive then it is known as budget surplus otherwise it is called budget deficit.

- *Fiscal Deficit(Net)*: This is a measure of net borrowing required to fund current expenditure (i.e. to meet revenue deficit) and investment expenditure (such as asset creation).

- *Primary Deficit*: This is fiscal deficit less interest payment.

5. FINANCIAL RESOURCES

Financial resources are made up of debt and non-debt resources. An increase in the size of financial resources impacts on the share of debt and non-debt resources. An increase in the share of debt resources brings forth an increase in future debt - servicing burden that is not matched by increase in non-debt resources. In this situation, future debt servicing burden is partly funded by increase in debt resources itself. Since debt resources get depleted for the purpose of servicing debt, more debt resources are required to sustain the tempo of development works. This leads to a still higher debt-servicing burden, which calls for further increase in debt resources for sustaining development expenditure. This vicious cycle will continue as long as debt resources are available in plenty. However,

savings in the economy ultimately govern the availability of debt resources. Once that limit is reached, Government will have no alternative but to moderate its borrowing programmes. This will lead to a moderation of development expenditure as well. As a result, borrowings led growth in financial resources will ultimately slow down the pace of developmental expenditure. Thus, the event that reflects an increase in the share of debt resources in total financial resources also carries with it the seeds of future slackening of development expenditure. In order to avoid this situation, it is imperative that the share of non-debt resources must increase with any increase in financial resources.

Financial resources are disbursed between development and non- development expenditure. Increase in financial resources will be accompanied with increase in their respective sizes. The increase in the size of development expenditure is a prudent application of financial resources. The increase in the size of non-development expenditure is an imprudent application of financial resources. If the increase in development expenditure is more than that of non-development expenditure, its difference is net prudent application of resources. It is this measure of net prudent application that reflects development gains deriving from increase in financial resources. It is not easy to strictly separate development from non-development expenditure. The closest one can come to is to adopt the plan-non-plan breakdown of Government expenditure, a classification much in vogue. Using this classification it follows that only a positive increase in plan expenditure net of any increase in non-plan ensures development gains from any increase in financial resources. A corollary to net increase in plan expenditure is reflected in the increase in the share of plan resources in total financial resources.

6. DEVOLUTION OF RESOURCES FROM GOVERNMENT OF INDIA

There are three types of revenue devolution from Government of India, Statutory, Plan and discretionary devolution.

Statutory devolution devolves share in Central taxes to States, general-purpose grants, specific purpose grants and reduction in debt servicing burden. This includes State share of central excise and Central Finance Commission that is a Constitutional body constituted by GOI awards this devolution every five years. The Eleventh Finance Commission has finalized its award for the period 2000-05. The Twelfth Finance Commission is in the process of finalizing its report for the award period 2005-10.

Plan devolution includes the revenue component of Central Assistance for plan schemes that is recommended by the Planning Commission. A significant portion of Central Assistance is devolved under Gadgil- Mukherjee formula. The formula distributes Central Assistance among general category States. The formula has been subjected to much criticism, as it is alleged that it does not directly relate to development requirements or requirements that are project based. Many States (including Goa) feel that this results in inadequate growth in their plan resources. Under this formula fund allocation is based on 85% population and poverty criteria, 7.5% on performance and the balance 7.5% is discretionary quota of Dy. Chairman, Planning Commission for special problems. Further, Central Assistance, formula as well as non-formula based, devolved to general category States carries a 30 percent grant and 70 percent loan. Thus, for every rupee devolved as resource from central government, 70 paisa comes as borrowings. As a result, a Central Assistance triggered increase in share of plan resources also increases the share of debt resources in total financial resources. Thus, the loan- grant composition of Central Assistance carries the germ of adverse financial position.

Discretionary devolution is on account of plan schemes of GOI that is located in States. Funds on this account are transferred to States for implementation of GOI plan schemes. These funds are referred to as Centrally sponsored and Central plan schemes and are generally in the form of revenue grants. Since these schemes belong to GOI, these supplement State's own efforts to augment its plan size. Thus, grants under these are in the nature of plan grants except that the destination of their utilisation is specified.

7. DESCRIPTION & ESTIMATION OF BUDGET- BASED SUBSIDIES

Services where subsidisation is desirable are referred to as merit goods. Here prices charged are very low or even non-existent to encourage a very large consumption of these services. A large consumption enables social and economic benefits elsewhere in the economy as these services, by virtue of their nature, carry a lot of ancillary linkages and connectivity. Services like primary education, particularly female education, primary health services, scientific research and development are few merit goods services. Services, where subsidisation is not desirable are ones that have very few ancillary linkages and connectivity. These do not justify financial losses as benefits elsewhere in the

economy do not accrue. These services are referred to as non-merit goods and include Public utilities.

What merit and non-merit goods share in common is possibility of pricing albeit in varying degrees of convenience. Apart from these two category, there exists a third category wherein pricing of services and goods is impossible. Goods and services under this category are not amenable to pricing since their consumption cannot be separated for each consumer individually. This category is referred to as Public goods category. Due to impossibility of pricing under Public Goods category and desirably low prices under merit goods category, Government has no option but to highly price non-merit good category to cross subsidise merit and public goods category.

In what follows, the budget- based subsidies have been estimated entirely for social and economic services of Government of Goa. However, instead of breaking down social and economic services into merit and non-merit goods, only certain important items of services have been isolated. These isolated items comprising both merit and non-merit segments are then dealt subjectively to arrive at the best possible options for Government. Options that will not only reduce the overall subsidy burden but also shrink the non-merit good segment in favour of merit good segment.

Items of goods and services under social and economic services only have a recurring cost that is set against the revenue earned from respective items to estimate the subsidy. Estimation of recurring cost has again been done on the lines of GOI paper. One component of recurring cost is revenue expenditure on the service asset.

Another component, depreciation allows for wear and tear of the service asset. Thus, in the current year, a portion of gross capital formation could be viewed as providing for the annual depreciation of the current assets. Hence, if the life of the asset is taken to be twenty-five years, 4 percent of cumulative capital expenditure accounts for depreciation.

Interest cost refers to cost of holding funds. To the extent borrowings constitute funds held by Government, the cost of funds is the interest payment. To the extent, revenues constitute Government funds, the cost is the interest income foregone by not investing at currently prevailing rate of interest. Borrowing and revenue cost converge when one single rate of interest is assigned to estimate interest payment and interest income. It is the effective rate of interest determined from dividing annual interest payments by outstanding debt that is taken as the all purpose interest rate. Thus, interest cost of the asset is determined by multiplying effective rate of interest with total amount of funds invested in building the asset. The total amount of funds invested in the asset is the cumulative capital expenditure on the asset.

Non-tax revenue has been taken as recoveries against recurring costs. The only exception made is in case of transport wherein Motor vehicle tax and tax on goods and passengers are included as part of recoveries.

8. CONCEPTS AND DEFINITIONS OF NATIONAL INCOME/STATE INCOME AND RELATED AGGREGATES

A. What is National income/State income ?

National income by definition is a measure in monetary terms of all the goods and services produced by an economy during a given period of time. The measure is obviously in value terms as different units of production are not directly additive. The period which the national income (also the state income) is calculated is generally a year. In measuring national income it is essential to take account of all the goods and services produced during the period whether these are marketed i.e. exchanged for money or bartered (i.e. exchanges for other goods and services) or used for self consumption (for example agricultural products consumed by the farmer). At the same time, the measure should be such that production of any particular item is not counted twice. For example, if we take the total production of iron ore and steel in the country, then we must not count that part of iron ore which is used in the production of steel, which otherwise leads to duplication. This is ensured in the procedures adopted in the estimation of national income/ state income. However, certain activities are excluded in the estimation of national income. These primarily relate to illegal activities such as smuggling, black marketing etc. In addition, the service of house wife is also excluded mainly because of the problem of measurement.

B. Factors of Production

In national income estimation the term 'Production' has a wider meaning. In general, production is synonymous with manufacturing activity in a factory or at the most production of agricultural commodities in fields. But, in national income ac-

counting such a concept becomes too narrow. For instance, if a dealer purchases wheat in Punjab mandi and sells it in Delhi, he finds that he can obtain a higher price. This process of distribution by the dealer is considered as production since it has 'added' value of wheat. The narrow concept of production also fits in this broader definition, where in any manufacturing process adds value to the existing raw material. By the use of labour and machinery, the raw materials get transformed into finished goods with greater value. The elements responsible for the creation of addition of value to the existing products are called factors of production.

C. Methodology for Compiling the Estimates of State Domestic Product

The estimates of SDP are computed on the basis of methodology and guidelines provided by the Central Statistical Organisation (CSO), Government of India and it is followed by all the States and Union Territories. The Gross State Domestic Product at factor cost by Industry of origin is estimated at current and constant prices by deducting financial inter-mediation services indirectly measured (FISM) or input banking charges. The Net State domestic Product at current and constant prices are arrived at by deducting Consumption of Fixed Capital or Depreciation, as supplied by CSO for each industry group. The estimates of current prices are arrived at by evaluating the goods and services at prices prevailing in the market. The constant price estimates are prepared by evaluating the goods and services of the current year with respect to base year price in order to estimate the effect of price changes and to measure real growth in the economy.

The total economy of the State is divided into fourteen standardised sectors, sub-grouped under the **Primary, Secondary and Tertiary** activities.

- The primary group includes commodity producing sectors like (1) Agriculture and Live Stock (2) Forestry & Logging (3) Fishing (4) Mining and Quarrying.
- The secondary group comprises (1) Manufacturing Registered (2) Manufacturing Unregistered (3) Construction (4) Electricity / Gas / Water Supply.
- The Tertiary group contain (1) Transport, storage and communication (2) Trade, Hotel & Restaurant (3) Banking & Insurance (4) Real Estates, Ownership of dwelling & Business Service (5) Public administration & (6) Other Services.

The following **three approaches** are used in etimating the State Domestic Product subject to applicability of requisite information relating to a particular sector.

- **Production approach**: The sum of value of goods and services produced within the State during a year after deducting material inputs. Production approach is followed in respect of Agriculture (including Animal Husbandry) , Forestry and Logging , Fishing, Mining & Quarrying and manufacturing registered.
- **Income Approach**: In this approach the total income is considered to be distributed among the factors of production like land in the form of rent, labour in the form of salary / wages and other benefits, capital in the form of interest and entrepreneur in the form of profits. Income approach is followed in respect of Manufacturing unregistered, Electricity, Gas and water supply, Transport, Storage & Communication , Trade, Hotel & Restaurant, Banking & Insurance , Real Estate.
- **Expenditure Approach**: In this method total expenditure incurred on final consumption of goods and services plus net investment are included. The main components of expenditure approach are : i) Expenditure on goods and services, ii) Expenditure on investment and iii) Expenditure on Government purchases.

A brief account of scope and coverage of data and methodology adopted for estimating the value added under different sectors of economy is as under:

(i) Agriculture

The estimates of domestic product for this sector are compiled by the production approach except for the operation of the government irrigation system for which the income approach is followed. Gross value of output for agricultural commodities is based on the area, production and prices data made available by the State Directorate of Agriculture. Principal crops viz. paddy, ragi, maize, groundnuts, coconut, pulses, sugarcane, fruits and vegetables along with by-products of main crops are also included. The value of inputs

and by-products are estimated based on the norms suggested by CSO. The estimated production of main live stock products viz. milk, meat and eggs are supplied by CSO. For other products the production is estimated by yield rates by CSO. The price data for milk, meat and eggs are collected from State marketing office. The value of output is worked out for agriculture and livestock separately but the value added estimates are prepared for the combined sector.

(ii) Forestry & Logging

Estimates of Domestic Product are prepared following the production approach. Gross value of output is estimated separately for (a) Industrial wood, b) Fuel wood & (c) Minor forest products. For evaluating output of industrial wood and minor forest products, prices are supplied by the Conservator of Forest. The estimates of fuel wood data are as available from the result of NSSO consumption expenditure survey.

(iii) Fishing

Estimate of GDP of this sector are prepared by following the production approach. Gross Value of Output estimated from outturn and prices of inland and marine fish as furnished by the State Fisheries Department and inputs, repair and maintenance etc. are worked out according to the standard norms of CSO.

(iv) Mining & Quarrying

The production approach is followed for obtaining the Gross Value added in this sector. The production and value of major minerals is obtained from IBM and in case of minor mineral data is obtained from the State Industry Department.

(v) Manufacturing registered

Estimates under this sector are prepared by following the production approach. The industry-wise estimates of gross value added are prepared on the basis of the summary results of Annual Survey of Industries (ASI).

(vi) Manufacturing Unregistered

The estimates of unregistered manufacturing sector other than the one relating to SSI are compiled using information on GVA per worker from the follow up surveys of the Economic Census for Directory Manufacturing Establishments (DMEs) and from the results of NSS 51st Round (1994-95) results for Non-Directory for Manufacturing Establishments (NMDs) and Own Account Enterprises (OAEs) suitably deflected to 1993-94. The estimate of workforce for this portion of unregistered manufacturing i.e. for the units neither belonging to ASI nor to SSI is obtained from the total workforce on manufacturing activities knocking out the number relating to ASI and SSI (non ASI) segments. To obtain this at two digit level of DMA, NDMA &OAE for rural and urban separately, the total work force on manufacturing minus ASI obtained above is distributed pro-rata in the ratio of employment as estimated by 51st Round of NSS Survey of Unorganised manufacturing. The estimates of current prices for the subsequent years are worked out by moving forward the industry wise estimates for the year 1993-94 with the help of indices based on State wise value of output taken from industry group 20-21, 22, 23-26, 29.

(vii) Construction

The expenditure approach is followed to compute the gross value added in the sector. Expenditure incurred on various construction activities carried out by the State Government departments is taken from the budget documents. Expenditure related to the Central Government and Private Construction are made available by the CSO.

(viii) Electricity, Gas and Water Supply

Value added information pertaining to electricity and water supply are culled out from the State budget documents. The state wise estimate of GVA in respect of gobar gas is made available by the CSO.

(ix) Transport, Storage & Communication

Income approach is adopted for obtaining the gross value added in this sector. The estimates for public and private sector are prepared separately. The public sector estimate of gross value added is obtained by analysing Annual Report of Kadamba Transport Corporation and Budget Document for River Navigation Department. For mechanised transport, base year work force is moved forward by vehicle index and value per unit of workforce is moved forward with the help of CPI (UNME) to work out the estimate. The estimate for unorganised transport is made available by the CSO. Communication service is a supra regional sector and the value added is prepared by CSO and allocated to States for adoption. The value added estimate per-

taining to 'storage' is prepared and supplied by the CSO based on the analysis of annual accounts of State Ware housing Corporation. The value added from Central Warehousing Corporation is also prepared by the CSO and allocated to States in proportion to the average capacity of central warehousing in each State. Similarly the estimate of cold storage is based on the results of Annual Survey of Industries (ASI). All India GVA from storage is allocated to different States by the CSO on the basis of percentage distribution of number of workers as in base year 1993-1994.

(x) Trade, Hotel & Restaurant

The income approach is followed under this sector. The estimates of gross value added for public and private sectors are prepared separately. The estimate of gross value added for public sector is prepared and supplied to the States by the CSO. The estimate for private organised sector comprising private corporate sector and co-operative societies engaged in trade is prepared using the results of the RBI study on company finances and information available from NABARD respectively. The All India estimate of gross value added from private organised sector of trade, hotel and restaurant is allocated to various States in proportion to work force available year wise from DGE&T. The estimate of private unorganised sector of Trade Hotels & Restaurants is worked out as a product of GVA per worker and workforce. For Trade sector, value added per worker as obtained from 1996-97 directory trade establishment survey is moved to base year 1993-94 for rural areas and in case of hotels and restaurants GVA per worker as obtained from 1993-94 enterprise survey results is used. For subsequent years the clubbed base year estimate for trade, hotel & restaurant is being moved with the help of index of state gross trading income (GTT) of commodity producing sectors. The value added from State Public Sector Undertakings is culled out from their Annual Reports.

(xi) Banking & Insurance

Income approach is followed for obtaining the value added in this sector. Being supra regional sector, value added is prepared and supplied to States by the CSO.

(xii) Real Estate, Ownership Dwelling, Business & Legal Services

The methodology followed in this sector is based on income approach. The gross rental for the base year is obtained using number of residential houses for the year 1993-94 and rent per household as per NSS 50th Round separately for rural and urban areas. The estimate of number of residential houses in urban and rural areas is worked as per growth rate observed between two decadal population censuses. The rent per household of base year is moved to subsequent years with the help of CPI (UNME) of house group prepared by CSO separately for rural and urban areas. The estimate in respect of real estate and legal services is prepared using NSS work force and value added per worker of enterprise survey 1991-92 separately for rural and urban areas. The work force is moved forward using growth rate of quinquennial employment and unemployment survey of NSS as observed between and 1993-94 and 1997-98 and value added per worker is moved forward with the help of index of CPI for agricultural labourers for rural sector and CPI (UNME) for urban sector. The estimate of Business services is supplied by the CSO.

(xiii) Public Administration

Income approach is followed for obtaining the value added in this sector. The estimation of value added from supra regional sector i.e. Central Administration is prepared by the CSO by analysing the Central Budget and supplied to States for adoption. The estimate of value added in respect of State administration is culled out from the State budget documents. The estimate of income in respect of local authorities is based on the information on compensation of employees obtained directly from the local authorities.

(xiv) Other Services

This sector covers activities pertaining to i) Educational ii) Medical & Veterinary iii) Research and Scientific iv) Sanitary v) Recreational and Entertainment vi) Tailoring vii) International and other Extra territorial bodies viii) Rest of the services.

a. Educational

The estimate of gross value added in respect of State Government Educational Institutions is prepared by taking the expenditure on salary and wages including pension of teaching staff as reflected in the State Budget documents. The value added from private institutions is estimated as 2.8% of GVA of State Government Educational Institutions and for Central school 1992-93 estimate is inflated using CPI (UNME).

b. Research and Scientific Services

The estimate of the gross value added pertaining to this sector is prepared and supplied to States by the CSO.

c. Medical & Veterinary Services

The estimate of the public part of medical and veterinary services is the sum of budget expenditure on salary and wages of doctors and other medical staff and for estimation of private part are based on number of private practitioners registered with the medical councils and the earnings per worker.

d. Sanitary Services

The public sector GVA comprising salary, wages and pension of sanitary workers is culled out from the State Budget documents. The private sector workforce is obtained after subtracting the public sector workforce from the corresponding total workforce of the sanitary services. This exercise is done for the base year only. However, for subsequent years the workforce is projected based on the growth observed in employment of municipal workers of sanitary services and value added per worker for rural and urban areas is projected separately using the total compensation of employees.

e. Recreation and Entertainment services

The estimate of value added from recreation and entertainment services other than radio and T.V. broadcasting is worked out on the basis of information on the tax rate and the total tax receipts collected from the office of the Entertainment Commissioner. The state wise estimates of T.V. and Radio are prepared and supplied to States by the CSO.

f. Tailoring Services

The estimate of value added from tailoring services is prepared using NSS results on workforce and value added per worker on manufacturing of readymade garments obtained from unorganised manufacturing survey results 1994-95. For subsequent years the estimate of workforce is projected using the inter survey exponential growth rate of quinquennial employment and unemployment surveys of NSS as observed between 1987-88 and 1993-94 and value added per worker of rural and urban sector is projected to subsequent years with the index of CPI for agricultural labourer and CPI (UNME) respectively.

g. International and other extra territorial bodies

The estimate of value added from sector is prepared and supplied to States by the CSO.

h. Rest of the services

The estimate of value added is prepared using NSS workforce and value added per worker as per enterprise survey 1991-92 separately for rural and urban areas. For subsequent years the estimate of workforce is projected using the inter survey exponential growth rate of quinquennial employment and unemployment surveys of NSS observed between 1987-88 and 1993-94 and value added per worker is moved to subsequent years using the index of CPI for agricultural labourers for rural sector and CPI (UNME) for Urban sector.

ANNEXURE 1

Index of Social and Economic Infrastructure, 1999
(Arranged in descending Order)

Sl. No.		
1	**Goa**	**200.57**
2	Punjab	187.57
3	Kerala	178.68
4	Tamil Nadu	149.10
5	Haryana	137.54
6	Gujarat	124.31
7	Maharashtra	112.80
8	West Bengal	111.25
9	Sikkim	108.99
10	Karnataka	104.88
11	Andhra Pradesh	103.30
12	Uttar Pradesh	101.23
13	Himachal Pradesh	95.03
14	Mizoram	82.13
15	Bihar	81.33
16	Orissa	81.00
17	Assam	77.72
18	Madhya Pradesh	76.79
19	Nagaland	76.14
20	Rajasthan	75.86
21	Meghalaya	75.49
22	Manipur	75.39
23	Tripura	74.87
24	Jammu & Kashmir	71.46
25	Arunachal Pradesh	69.71

Source : Eleventh Finance Commission Report 2000

ANNEXURE 2

Ranking of State/UTs on the basis of Composite Index by the National Commission on Population

		Composite Index
1	**Goa**	**83.71**
2	Pondicherry	82.70
3	Kerala	81.88
4	Chandigarh	79.68
5	Tamil Nadu	78.06
6	Delhi	78.01
7	Punjab	73.88
8	Karnataka	70.31
9	Himachal Pradesh	70.02
10	Lakshadweep	69.89
11	Mizoram	69.24
12	Daman & Diu	68.95
13	Haryana	66.80
14	Maharashtra	65.58
15	Gujarat	65.22
16	Andhra Pradesh	65.13
17	Tripura	64.17
18	Andaman & Nicobar	63.89
19	Uttaranchal	60.52
20	West Bengal	59.58
21	Sikkim	58.64
22	Assam	56.35
23	Chhatisgarh	55.44
24	Orissa	55.40
25	Manipur	54.01
26	Dadra & Nagar Haveli	51.97
27	Meghalaya	50.51
28	Madhya Pradesh	49.33
29	Nagaland	48.31
30	Uttar Pradesh	44.09
31	Arunachal Pradesh	43.58
32	Rajasthan	43.39
33	Bihar	39.01
34	Jharkhand	38.27

Source : National Commission on Population

ANNEXURE 3

Health Indicators of Goa

Sr. No.	Item	Preliberation (1961)	At the time of Statehood (1987)	As on 1.1.2004
1	Population (in lakhs)	5.9	10.08	13.97
2	Density population per Sq. Km.	163	272	363
3	Sex ratio (females per 1000 males)	1066	975	960
4	Urban population to total population (%)	14.8	32.03	49.77
5	Decadal Growth Rate (in per cent)	7.77	26.74	14.89
6	No. of Government Hospitals	17	31	32
7	No. of Beds in Government Hospitals	1098	2371	2743
8	No. of Private Hospitals	18	83	120
9	No. of Beds in private Hospitals	1405	1312	2264
10	Population served per bed	211	310	275
11	Population served per Hospital	34814	10316	9043
12	Area served per Hospital (in sq. Kms)	34.27	32.16	24.4
13	No. of Family Welfare Centres	Nil	252	261
14	No. of indoor patients treated at Govt. Hospitals (in lakhs)	0.12	0.6	1.01
15	No. of outdoor patients treated at Govt. Hospitals (in lakh)	0.55	8.32	13.27
16	Maternal Mortality Rate	1.44	0.48	0.25
17	No. of male births per 100 female births	116	106	108
18	No. of male Deaths per 100 female Deaths	107	142	151
19	Neonatal Death Rate	29.47	17.59	15
20	No. of PHCs	0	21	24
21	No. of UHCs	0	4	4
22	No. of RMDs	0	28	29
23	No. of Sub-Centres	0	166	172

Source : Directorate of Health Services, Government of Goa

ANNEXURE 4

Health Infrastructure in Goa

Sr. No.	Item	National Norms	Achievement	
			Goa	India
1.	Estimated rural general Population covered by			
	i) Sub-centre	5000	3908	5401
	ii) PHC	30000	35377	32469
	iii) CHC	120000	135000	244000
2.	Average rural area covered			
	i) Sub-centre		19.28	23.38
	ii) PHC		174.58	140.52
	iii) CHC		663.40	1054.84
3.	Average radial distance covered			
	i) Sub-centre		2.48	2.73
	ii) PHC		7.45	6.69
	iii) CHC		14.53	18.32
4.	Average number of villages covered			
	i) Sub-centre		2.09	4.46
	ii) PHC		18.95	26.81
	iii) CHC		72	201.27
5.	No. of sub-centres per PHC		9.05	6.01
6.	No. of PHC per CHC		3.80	7.51

Source: Health Statistics in India March –2002

ANNEXURE 5

Birth Rate, Death Rate and Infant Mortality Rate

Year	Birth Rate	Death Rate	Infant Mortality Rate
1987	18.24	6.72	24.88
1988	17.95	7.59	21.99
1989	17.29	6.81	21.12
1990	18.43	6.80	22.37
1991	19.50	7.46	18.76
1992	18.59	7.44	18.06
1993	17.88	7.54	18.43
1994	18.00	7.47	15.66
1995	17.82	7.07	14.39
1996	18.00	7.40	17.25
1997	18.34	7.37	16.92
1998	17.63	8.26	17.21
1999	17.40	7.47	16.42
2000(P)	17.85	7.16	13.99
2001(P)	15.95	7.46	12.40
2002(P)	16.59	7.59	12.48

Source: Directorate of Planning, Statistics & Evaluation, Government of Goa.

Note: Above vital rates are based on 100% registration of Births & Deaths in the State.

ANNEXURE 6

Per Capita Net State Domestic Product at Current Prices (New Series)

(**Rupees**)

Sl. No.				
1.	**Goa**	**8952**	**16558**	**49673**
2.	Punjab	8177	12710	25652
3.	Maharashtra	7612	12183	24736
4.	Haryana	7721	11090	24575
5.	Himachal Pradesh	5243	7870	21368
6.	Kerala	5110	7938	21310
7.	Gujarat	6343	9796	21276
8.	Tamil Nadu	5541	8943	21239
9.	Karnataka	4975	7838	18342
10.	West Bengal	5072	6756	17769
11.	Andhra Pradesh	4816	7447	17642
12.	Arunachal Pradesh	5231	8612	15785
13.	Meghalaya	4944	6720	15070
14.	Rajasthan	4883	6182	13151
15.	Madhya Pradesh	4798	6577	11718
16.	Assam	4432	5715	10951
17.	Orissa	3166	4797	10234
18.	Bihar	2966	3037	5466
19.	Chandigarh	NA	19761	NA
20.	Delhi	11373	18021	NA
21.	Pondicherry	7657	9781	NA
22.	Sikkim	5213	7550	NA
23.	Tripura	4240	5534	NA
24.	Manipur	3912	5833	NA
25.	Jammu & Kashmir	4624	6543	NA
26.	Uttar Pradesh	3937	5050	NA
27.	Jarkhand		5898	NA
28.	Chattisgarh		6539	NA
29.	Mizoram	4856	8319	NA
30.	Nagaland	5893	9129	NA
31.	Uttaranchal		NA	NA
32.	Andaman & Nicobar Islands	9729	15192	NA

P : Provisional estimates

Q : Quick estimates

NA : Not available

Note : Estimates from 1993-94 are as provided by the Directorate of Economics & Statistics of respective State Govts. & Union Territories. Estimates prior to 1993-94 have been linked by using the method of splicing using the estimates as provided by the Directorate of Economics & Statistics of respective State Governments & Union Territories. Estimates prior to 1993-94 are not available separately for Jarkhand, Chattisgarh & Uttaranchal

Source : Directorate of Economics & Statistics of respective State Governments & Union Territories.

ANNEXURE 7

Poverty Ratio in 1973-74 and 1999-2000 (percent)

Sl. No.		1973-74			1999-2000		
		Rural	Urban	Combined	Rural	Urban	Combined
1	Jammu & Kashmir	45.51	21.32	40.83	3.97	1.98	3.48
2	**Goa**	**46.85**	**37.69**	**44.26**	**1.35**	**7.52**	**4.40**
3	Chandigarh	27.96	27.96	27.96	5.75	5.75	5.75
4	Punjab	28.21	27.96	28.15	6.35	5.75	6.16
5	Himachal Pradesh	27.42	13.17	26.39	7.94	4.63	7.63
6	Delhi	24.44	52.23	49.61	10.40	9.42	8.23
7	Haryana	34.23	40.18	35.36	8.27	9.99	8.74
8	Kerala	59.19	62.74	59.79	9.38	20.27	12.72
9	Gujarat	46.35	52.57	48.15	13.17	15.59	14.07
10	Rajasthan	44.76	52.13	46.14	13.74	19.85	15.28
11	Lakshadweep	59.19	62.74	59.68	9.38	20.27	15.6
12	Andhra Pradesh	48.41	50.61	48.86	11.05	26.63	15.77
13	Dadra & Nagar Haveli	46.85	37.69	46.55	17.57	13.52	17.14
14	Mizoram	52.67	36.92	50.32	40.04	7.47	19.47
15	Karnataka	55.14	52.53	54.42	17.38	25.25	20.04
16	Andaman & Nicobar Islands	57.43	49.40	55.56	20.55	22.11	20.99
17	Tamil Nadu	57.43	49.40	54.94	20.55	22.11	21.12
18	Pondicherry	57.43	49.40	53.82	20.55	22.11	21.67
19	Maharashtra	57.71	43.87	53.24	23.72	26.81	25.02
20	West Bengal	73.16	34.67	63.43	31.85	14.86	27.02
21	Manipur	52.67	36.92	49.96	40.04	7.47	28.54
22	Uttar Pradesh	56.53	60.09	57.07	31.22	30.89	31.15
23	Nagaland	52.67	36.92	50.81	40.04	7.47	32.67
24	Arunachal Pradesh	52.67	36.92	51.93	40.04	7.47	33.47
25	Meghalaya	52.67	36.92	50.20	40.04	7.47	33.87
26	Tripura	52.67	36.92	51.00	40.04	7.47	34.44
27	Assam	52.67	36.92	51.21	40.04	7.47	36.09
28	Sikkim	52.67	36.92	50.86	40.04	7.47	36.55
29	Madhya Pradesh	62.66	57.65	61.78	37.06	38.44	37.43
30	Bihar	62.99	52.96	61.91	44.30	32.91	42.6
31	Orissa	67.28	55.62	66.18	48.01	42.83	47.15
	All India	**56.44**	**49.01**	**54.88**	**27.09**	**23.62**	**26.10**

Note : For 1999-2000

1. Poverty ratio of Assam is used for Sikkim, Arunachal Pradesh, Meghalaya, Mizoram, Manipur, Nagaland and Tripura
2. Poverty line of Maharashtra and expenditure distribution of Goa is used to estimate poverty ratio of Goa.
3. Poverty line of Himachal Pradesh and expenditure distribution on Jammu & Kashmir is used to estimate poverty ratio of Jammu & Kashmir.
4. Poverty ratio of Tamil Nadu is used for Pondicherry and Andaman and Nicobar Island
5. Urban povery ratio of Punjab is used for both rural and urban poverty of Chandigarh.
6. Poverty line of Maharashtra and expenditure distribution of Dadra & Nagar Haveli is used to estimate poverty ratio of Dadra & Nagar Haveli
7. Poverty ratio of Kerala is used for Lakshadweep.
8. Urban poverty ratio of Rajasthan may be treated as tentative

ANNEXURE 8

Total Fertility Rates, 1998 and Projected Total Fertility Rates, 2007

Sl. No.			
1	**Goa**	**1.77**	**NA**
2	Kerala	1.8	NA
3	Tamil Nadu	2.0	NA
4	Himachal Pradesh	2.14	NA
5	Andhra Pradesh	2.4	NA
6	Karnataka	2.4	2.3
7	West Bengal	2.4	NA
8	Delhi	2.4	NA
9	Arunachal Pradesh	2.52	NA
10	Punjab	2.6	NA
11	Maharashtra	2.7	2.3
12	Jammu & Kashmir	2.71	NA
13	Sikkim	2.75	NA
14	Mizoram	2.89	NA
15	Orissa	2.9	2.4
16	Gujarat	3.0	NA
17	Manipur	3.04	NA
18	Assam	3.2	2.5
19	Haryana	3.3	2.1
20	Nagaland	3.77	NA
21	Madhya Pradesh	3.9	3.4
22	Rajasthan	4.1	3.8
23	Bihar	4.3	3.3
24	Meghalaya	4.57	NA
25	Uttar Pradesh	4.6	4.4
	All India	**3.2**	**2.7**

NA : Projections for these States were not made.

Source : Draft Tenth Five-Year Plan (2002-07), Volume III, State Plans Trends, Concerns and Strategies, Planning Commission, New Delhi.

ANNEXURE 9

State-wise Infant Mortality Rate

Sl. No.	States/Union Territories	1961			1981			2001
1	**Goa**	**60**	**56**	**57**	**56**	**48**	**51**	**14**
2	Kerala	55	48	52	45	41	42	16
3	Pondicherry	77	68	73	32	35	34	21
4	Mizoram	73	65	69	51	56	53	23
5	Manipur	31	33	32	29	27	28	25
6	Andaman & Nicobar Islands	78	66	77	71	61	69	30
7	Lakshadweep	124	88	118	100	78	91	30
8	Chandigarh	53	53	53	50	47	48	32
9	Arunachal Pradesh	141	111	126	111	103	91	44
10	Jammu & Kashmir	78	78	78	NA	NA	NA	45
11	Maharashtra	96	89	92	72	76	74	49
12	Tripura	106	116	111	81	84	82	49
13	Delhi	66	70	67	55	51	54	51
14	Meghalaya	81	76	79	79	82	80	52
15	Sikkim	105	87	96	58	62	60	52
16	Tamil Nadu	89	82	86	55	51	54	53
17	West Bengal	103	57	95	75	51	54	53
18	Punjab	74	79	77	81	53	74	58
19	Karnataka	87	74	81	74	72	74	58
20	Dadra & Nagar Haveli	102	93	98	84	73	81	61
21	Gujarat	81	84	84	74	82	78	64
22	Himachal Pradesh	101	89	92	84	81	82	64
23	Andhra Pradesh	100	82	91	67	51	55	66
24	Bihar	95	94	94	62	89	75	67
25	Haryana	87	119	94	57	54	52	69
26	Assam	NA	NA	NA	96	87	92	78
27	Rajasthan	114	114	114	94	79	87	83
28	Uttar Pradesh	131	128	130	98	104	99	85
29	Madhya Pradesh	158	140	150	131	136	133	97
30	Orissa	119	111	115	129	111	125	98
31	Nagaland	76	58	68	51	52	51	NA
32	Daman & Diu	60	56	57	61	50	56	NA
	All India	**122**	**108**	**115**	**74**	**79**	**77**	**71**

Note : a) The estimates for Goa is based on 100% registration of births in the State.
b) Female and Male Infant Mortality Rate not available from 1998.
c) The estimates are not available for smaller States / Union Territories.
d) NA: Not available.

Source (excluding Goa): Office of the Registrar General of India, Ministry of Home Affairs

ANNEXURE 10

Literacy Rates for Male and Female - 1961, 1981 and 2001
(Arranged in Rank Order of 2001 for Female)

Percent

Sl.	State	1961		1981		2001	
1	Kerala	64.90	45.60	75.30	65.70	94.20	87.90
2	Mizoram	NA	NA	64.50	54.90	90.70	86.10
3	Lakshadweep	42.00	12.80	65.20	44.60	93.20	86.10
4	Chandigarh	62.60	43.10	69.00	59.30	85.70	76.70
5	**Goa**	**48.70**	**22.80**	**65.99**	**48.29**	**88.90**	**75.50**
6	Andaman & Nicobar Islands	48.80	24.50	58.70	42.10	86.10	75.30
7	Delhi	70.40	50.90	68.40	53.10	87.40	75.00
8	Pondicherry	58.90	28.70	65.80	45.70	88.90	74.10
9	Daman & Diu	44.60	26.00	65.60	47.60	88.40	70.40
10	Himachal Pradesh	37.60	11.20	53.20	31.50	86.00	68.10
11	Maharashtra	49.30	19.80	58.80	34.80	86.30	67.50
12	Tripura	35.30	12.40	51.70	32.00	81.50	65.40
13	Tamil Nadu	51.60	21.10	58.30	35.00	82.30	64.60
14	Punjab	40.70	20.70	47.20	33.70	75.60	63.60
15	Nagaland	27.20	13.00	50.10	33.90	71.80	61.90
16	Sikkim	22.40	4.90	43.90	22.20	76.70	61.50
17	Meghalaya	NA	NA	37.90	30.10	66.10	60.40
18	West Bengal	46.60	20.30	50.70	30.30	77.60	60.20
19	Manipur	53.50	18.90	53.30	29.10	77.90	59.70
20	Gujarat	0.00	0.00	54.40	32.30	80.50	58.60
21	Karnataka	42.30	16.70	48.80	27.70	76.30	57.50
22	Haryana	35.10	11.30	48.20	22.30	79.30	56.30
23	Assam	44.30	19.60	NA	NA	71.90	56.00
24	Andhra Pradesh	35.00	14.00	39.30	20.40	70.90	51.20
25	Orissa	40.30	10.10	47.10	21.10	76.00	51.00
26	Madhya Pradesh	32.20	8.10	39.50	15.50	76.80	50.30
27	Rajasthan	28.10	7.00	36.30	11.40	76.50	44.30
28	Arunachal Pradesh	53.40	24.10	28.90	11.30	64.10	44.20
29	Dadra & Nagar Haveli	17.70	5.00	36.30	16.80	73.30	43.00
30	Uttar Pradesh	31.90	8.30	38.80	14.00	70.20	43.00
31	Jammu & Kashmir	19.80	5.10	36.30	15.90	65.80	41.80
32	Bihar	35.20	8.20	38.10	13.60	60.30	33.60
	All India	**40.40**	**15.40**	**56.40**	**29.80**	**75.90**	**54.20**

Note : States/Union Territories are arranged in order of rank in 2001
N.A. : Not available
Source : Office of the Registrar General of India

ANNEXURE 11

State-wise Literacy Rates in Percentage
(Arranged in Rank Order of 2001)

Sl. No.	States/Union Territories						
1	Kerala	40.70	55.10	60.40	70.40	89.80	90.92
2	Mizoram	NA	NA	NA	59.90	82.30	88.49
3	Lakshadweep	15.20	27.20	43.70	55.10	81.80	87.52
4	**Goa**	**23.48***	**31.23****	**45.31**	**57.25**	**75.50**	**82.32**
5	Delhi	38.40	62.00	56.60	61.50	75.30	81.82
6	Chandigarh	NA	55.10	61.60	64.80	77.80	81.76
7	Pondicherry	NA	43.70	46.00	55.90	74.70	81.49
8	Andaman & Nicobar Island	25.80	40.10	43.60	51.60	73.00	81.80
9	Daman & Diu	22.90	34.90	44.80	56.70	71.20	81.09
10	Maharashtra	20.90	35.10	39.20	47.20	64.90	77.27
11	Himachal Pradesh	7.70	24.90	32.00	43.50	63.90	77.13
12	Tripura	15.50	24.30	31.00	42.10	60.40	73.66
13	Tamil Nadu	20.08	36.40	39.50	46.80	62.70	73.47
14	Gujarat	NA	NA	35.80	43.70	61.30	69.97
15	Punjab	15.20	31.50	33.70	40.90	58.50	69.95
16	Sikkim	7.30	14.20	17.70	34.10	56.90	69.68
17	West Bengal	24.00	34.50	33.20	40.90	57.70	69.22
18	Manipur	11.40	36.00	32.90	41.40	59.90	68.87
19	Haryana	NA	24.10	26.90	36.10	55.80	68.59
20	Nagaland	10.40	20.40	27.40	42.60	61.60	67.11
21	Karnataka	19.30	29.80	31.50	38.50	56.00	67.04
22	Assam	18.30	33.00	28.70	NA	52.90	64.28
23	Madhya Pradesh	9.80	20.50	22.10	27.90	44.20	64.11
24	Meghalaya	NA	NA	29.50	34.10	49.10	63.31
25	Orissa	15.80	25.20	26.20	34.20	49.10	63.31
26	Andhra Pradesh	13.20	24.60	24.60	29.90	44.10	61.11
27	Rajasthan	8.90	18.10	19.10	24.40	38.60	61.03
28	Dadra & Nagar Haveli	4.00	11.60	15.00	26.70	40.70	60.03
29	Uttar Pradesh	10.80	20.70	21.70	27.20	41.60	57.36
30	Arunachal Pradesh	NA	47.90	11.30	20.80	41.60	54.74
31	Jammu & Kashmir	NA	13.00	18.60	26.70	NA	54.46
32	Bihar	12.20	21.80	19.90	26.20	38.50	47.53
	All India	**18.30**	**28.30**	**34.45**	**43.57**	**52.20**	**65.38**

*Relates to 1950 ** Relates to 1960

Source: Office of the Registrar General of India

ANNEXURE 12

Sex Ratio (Female per thousand Males) in India (Arranged in Rank Order of 2001)

Sl. No.	States/Union Territories						
1	Kerala	1028	1022	1016	1032	1036	1058
2	Pondicherry	1030	1013	989	985	979	1001
3	Tamil Nadu	1007	992	978	977	974	986
4	Andhra Pradesh	986	981	977	975	972	978
5	Manipur	1036	1015	980	971	958	978
6	Meghalaya	949	937	942	954	955	975
7	Orissa	1022	1001	988	981	971	972
8	Himachal Pradesh	912	938	958	973	976	970
9	Karnataka	966	959	957	963	960	964
10	**Goa**	**1128**	**1066**	**981**	**975**	**967**	**960**
11	Tripura	904	932	943	946	945	950
12	Lakshadweep	1043	1020	978	975	943	947
13	Mizoram	1041	1009	946	919	921	938
14	West Bengal	865	878	891	911	917	934
15	Assam	868	869	896	910	923	932
16	Maharashtra	941	936	930	937	934	922
17	Rajasthan	921	908	911	919	910	922
18	Bihar	1000	1005	957	948	907	921
19	Gujarat	952	940	934	942	934	921
20	Madhya Pradesh	945	932	920	921	912	920
21	Nagaland	999	933	871	863	886	909
22	Arunachal Pradesh	NA	894	861	862	859	901
23	Jammu & Kashmir	873	878	878	892	896	900
24	Uttar Pradesh	998	907	876	882	876	898
25	Sikkim	907	904	863	835	878	875
26	Punjab	844	854	865	879	882	874
27	Haryana	871	868	867	870	865	861
28	Andaman & Nicobar Island	625	617	644	760	818	846
29	Delhi	768	785	801	808	827	821
30	Dadra & Nagar Haveli	946	963	1007	974	952	811
31	Chandigarh	781	652	749	769	790	773
32	Daman & Diu	1125	1169	1099	1062	969	709
	All-India	**946**	**941**	**930**	**934**	**927**	**933**

Note : States / Union Territories are arranged in order of rank in 2001.
Source : Census of India

ANNEXURE 13

State-wise Life Expectancy at Birth (Rank as in 1993-97)

(in years)

Sl. No.					
1	Kerala	68.4	72.9	73.1	73
2	**Goa***			**68 (M) 72 (F)**	
3	Punjab	63.1	67.2	67.4	68
4	Maharashtra	60.7	64.8	65.2	66
5	Tamil Nadu	56.9	63.3	63.7	64
6	Haryana	60.3	63.4	63.8	64
7	Karnataka	60.7	62.5	62.9	63
8	West Bengal	57.4	62.1	62.4	63
9	Gujarat	57.6	61.0	61.4	62
10	Andhra Pradesh	58.4	61.8	62.0	62
11	Rajasthan	53.5	59.1	59.5	60
12	Bihar	52.9	59.3	59.4	60
13	Uttar Pradesh	50.0	56.8	57.2	58
14	Orissa	53.0	56.5	56.9	57
15	Assam	51.9	55.7	56.2	57
16	Madhya Pradesh	51.6	54.7	55.2	56
	All India	**55.5**	**60.3**	**60.7**	**61**

Note : 1. The estimates are not available for smaller States / UTs
2. Madhya Pradesh includes Chhattisgarh.
3. Uttar Pradesh includes Uttaranchal
4. Bihar includes Jharkhand

* Relates to 1996-2001(Source: Directorate of Health Services, Government of Goa)

Source : SRS based abridged life tables

ANNEXURE 14

Per capita expenditure on Education and Health 1998-99 (in Rupees) Arranged in order of rank in Health expenditure

Sl. No.	State/Union Territories	Per Capita Expenditure	
		Education	Health
1	Sikkim	2355	772
2	**Goa**	**1618**	**544**
3	Arunachal Pradesh	1163	471
4	Mizoram	1383	465
5	Nagaland	825	392
6	Himachal Pradesh	1113	388
7	Delhi	853	285
8	Meghalaya	789	277
9	Punjab	739	204
10	Manipur	918	192
11	Gujarat	680	163
12	Tripura	854	162
13	Tamilnadu	646	161
14	Kerala	635	158
15	Karnataka	525	138
16	Rajasthan	525	131
17	Haryana	625	130
18	West Bengal	394	129
19	Maharashtra	580	118
20	Andhra Pradesh	385	111
21	Madhya Pradesh	339	96
22	Orissa	418	91
23	Assam	535	76
24	Uttar Pradesh	348	65
25	Bihar	249	46
	All India	**841**	**30**

Note : Per capita expenditure is estimated based on expenditure figures taken from State Finances - a study of budgets of 2000-01, RBI, December 2000

ANNEXURE 15

Per Capita Consumption of Electricity (in KwH)
(Arranged in Rank Order of 1999-2000 except for All India)

Sl. No.						
1	Daman & Diu	130.8	276.4	440.1	2335.0	3927.4
2	Dadra & Nagar Haveli	14.8	56.3	878.8	2379.0	3882.8
3	Pondicherry	214.4	263.7	592.4	867.0	931.9
4	Punjab	154.2	303.6	620.5	792.0	921.1
5	Gujarat	165.0	238.8	436.8	694.0	834.7
6	Chandigarh	363.7	309.0	686.2	795.0	823.8
7	**Goa**	**157.5**	**250.8**	**411.2**	**724.0**	**712.5**
8	Delhi	299.2	403.8	673.6	577.0	653.2
9	Haryana	115.1	209.5	367.4	504.0	530.8
10	Maharashtra	172.6	244.5	393.6	556.0	520.5
11	Tamil Nadu	126.4	186.0	295.0	468.0	484.1
12	Andhra Pradesh	55.4	101.8	233.5	346.0	391.0
13	Karnataka	119.3	146.0	272.8	340.0	380.1
14	Orissa	69.2	114.0	249.2	309.0	354.6
15	Madhya Pradesh	61.3	100.3	217.4	367.0	351.7
16	Himachal Pradesh	58.8	66.4	191.9	306.0	339.1
17	Rajasthan	55.9	99.4	191.6	301.0	334.5
18	Jammu & Kashmir	52.7	74.8	176.4	218.0	267.9
19	Kerala	79.4	112.0	171.0	241.0	261.8
20	Andaman & Nicobar Islands	27.2	42.3	109.7	210.0	222.4
21	Lakshadweep	11.2	26.8	143.6	234.0	217.9
22	West Bengal	106.1	117.0	136.2	194.0	204.4
23	Uttar Pradesh	50.0	83.1	157.4	197.0	175.8
24	Meghalaya	31.3	31.0	106.4	135.0	160.3
25	Bihar	48.0	74.1	109.9	138.0	140.8
26	Mizoram	4.3	5.6	65.0	128.0	120.7
27	Assam	24.0	33.5	92.7	104.0	95.5
28	Tripura	6.0	14.45	45.0	80.0	95.5
29	Nagaland	27.2	34.2	58.6	88.0	84.7
30	Manipur	7.7	7.9	79.5	128.0	69.5
31	Arunachal Pradesh	3.4	14.6	56.6	81.0	68.6
	All India	174.9	120.5	236.0	334.0	354.8
	Standard Deviation	86.60	108	219	553	920

Source : a) 1974-75, 19980-81, 1989-90: Statistical Extract, India, CSO publication Various issues.

b) 1996-97, 1999-2000: Annual Report (2001-02) on the Working of State Electricity Boards and Electricity Departments, Planning Commission.

ANNEXURE 16

State-wise Road Density in Kms.(Road Length per '000 Sq. Kms. of area) (Arranged in Rank Order of 1996-97)

Sl. No.					
1	Delhi	7984	10527	14256	17924
2	Chandigarh	*710	1250	14000	15377
3	Pondicherry	*3508	4286	6698	4859
4	Kerala	3106	2751	3567	3749
5	Goa	*1581	2141	2005	2245
6	Orissa	366	772	1260	1687
7	Tamil Nadu	714	1020	1523	1588
8	Tripura	386	759	1341	1405
9	Punjab	594	916	1078	1278
10	Maharashtra	316	586	730	1176
11	Nagaland	284	379	901	1107
12	Dadra & Nagar Haveli	*460	492	643	1086
13	Assam	383	760	836	872
14	Uttar Pradesh	382	520	692	868
15	West Bengal	599	642	700	850
16	Karnataka	*525	557	701	751
17	Andhra Pradesh	264	468	553	647
18	Haryana	307	542	601	637
19	Himachal Pradesh	215	369	459	542
20	Bihar	670	481	492	508
21	Manipur	392	239	314	490
22	Gujarat	221	375	419	464
23	Madhya Pradesh	162	242	321	451
24	Meghalaya	303	233	291	379
25	Rajasthan	146	212	363	378
26	Sikkim	*329	156	227	258
27	Mizoram	*43	119	179	229
28	Arunachal Pradesh	*125	152	131	168
29	Andaman & Nicobar Islands	*82	83	110	160
30	Jammu & Kashmir	40	53	56	97
31	Daman & Diu	NA	NA	NA	26
32	Lakshadweep	NA	NA	NA	31
	All India	**344**	**466**	**615**	**749**

Note : * Refers to data for 1975-76
N.A. (Not available)
Source : Basic Road Statistics, Ministry of Surface Transport - Various issues

ANNEXURE 17

Vital Rates Based on Sample Registration System by Rural and Urban Areas - India 2002

	Birth rate			Death rate		
	Total	Rural	Urban	Total	Rural	Urban
Bigger States						
Andhra Pradesh	20.6	21.0	19.2	8.0	8.8	5.4
Assam	26.6	27.5	18.3	9.2	9.6	6.0
Bihar	30.9	31.8	23.5	7.9	8.2	6.1
Gujarat	24.6	26.5	20.5	7.6	8.3	6.3
Haryana	26.5	27.5	22.5	7.0	7.2	6.3
Karnataka	22.0	23.4	18.8	7.2	7.8	5.7
Kerala	16.8	17.0	16.3	6.4	6.4	6.2
Madhya Pradesh	30.3	32.2	22.7	9.7	10.4	7.2
Maharashtra	20.2	20.5	19.8	7.3	8.3	5.6
Orissa	23.1	23.6	19.5	9.8	10.3	6.3
Punjab	20.8	21.7	18.1	7.1	7.4	6.2
Rajasthan	30.6	31.9	24.2	7.7	8.0	6.4
Tamilnadu	18.5	19.1	17.5	7.7	8.6	5.9
Uttar Pradesh	31.6	32.6	26.8	9.7	10.2	7.3
West Bengal	20.3	22.5	14.0	6.6	6.8	6.3
Smaller States						
Arunachal Pradesh	20.2	20.9	12.3	4.8	5.0	2.6
Chhatisgarh	25.0	26.5	22.6	8.7	9.7	7.2
Goa	14.0	14.4	13.5	8.3	9.0	7.2
Jaharkhand	26.4	28.5	19.0	7.9	8.6	5.8
Himachal Pradesh	20.7	21.1	16.5	7.5	7.7	5.1
Jammu & Kashmir	19.2	19.9	16.0	5.7	5.7	5.7
Manipur	16.8	17.4	15.3	4.6	4.2	5.8
Meghalaya	25.8	27.8	14.6	7.7	8.4	3.9
Mizoram	16.9	20.8	12.0	4.8	5.7	3.7
Nagaland	N.A.	N.A.	11.9	N.A.	N.A.	2.5
Sikkim	21.9	22.2	13.4	4.9	5.0	4.1
Tripura	14.9	15.3	13.3	5.7	5.6	6.2
Uttaranchal	17.0	18.1	16.2	6.4	9.0	4.4
Uts						
A & N Islands	16.8	17.6	14.6	5.5	6.1	3.8
Chandigarh	14.6	21.5	13.8	3.4	4.1	3.4
D & N Haveli	30.4	31.3	20.4	6.8	7.1	4.0
Daman & Diu	22.4	23.5	21.4	6.6	7.4	5.9
Delhi	17.2	18.2	17.1	5.1	5.7	5.1
Lakshadweep	19.3	20.2	18.3	5.2	5.3	5.1
Pondicherry	17.9	18.1	17.7	6.7	8.1	5.7
All India	**25.0**	**26.6**	**19.9**	**8.1**	**8.7**	**6.1**

Source : Office of the Registrar General of India

ANNEXURE 18

Gross State Domestic Product at Factor Cost by Industry of origin from 1993-94 to 2001-02 at Current Prices

(Rs. in lakh)

Sl. No.						2001-02 (P)
1	2	3	4	5	6	7
1	**Agriculture, Forestry & Fishing**	**35439**	**54119**	**61734**	**61669**	**62415**
1.1	Agriculture	25085	38551	47496	40848	43222
1.2	Forestry & Logging	1123	1343	1176	963	1368
1.3	Fishing	9231	14225	13062	19858	17825
2	**Mining & Quarrying**	**15165**	**31679**	**26308**	**28636**	**26074**
	SubTotal - Primary	**50604**	**85798**	**88042**	**90305**	**88489**
3	**Manufacturing**	**57365**	**151664**	**192198**	**204498**	**204658**
3.1	Registered	45453	121556	154597	163228	163356
3.2	Unregistered	11911	30108	37601	41270	41302
4	**Electricity, Gas and Water Supply**	**4368**	**11491**	**8727**	**2668**	**8554**
5	**Construction**	**10587**	**31553**	**27163**	**44000**	**49175**
	SubTotal - Secondary	**72319**	**194708**	**228088**	**251166**	**262387**
6	**Trade, Hotels and Restaurants**	**22339**	**85849**	**91735**	**116005**	**116139**
7	**Transport, Storage & Communication**	**36353**	**90737**	**98825**	**113973**	**116086**
7.1	Railways	394	1167	1812	2351	2746
7.2	Transport by other means	34582	85944	92991	107319	108424
7.3	Storage	75	132	142	161	161
7.4	Communication	1302	3494	3880	4142	4755
8	**Financing, Insurance, Real Estate & Business Services**	**30368**	**88320**	**103803**	**105707**	**111675**
8.1	Banking & Insurance	14009	51361	57258	45445	45672
8.2	Real Estate, Ownership Of dwellings and Business Services	16359	36959	46545	60262	66003
9	**Community, Social & Personal Services**	**27685**	**62094**	**65672**	**74952**	**82336**
9.1	Public Administration	10550	28129	31360	33250	35301
9.2	Other services	17135	33965	34312	41702	47035
	SubTotal - Tertiary	**116745**	**326999**	**360035**	**410637**	**426236**
10	**Total Gross State Domestic Product (GSDP)**	**239668**	**607506**	**676165**	**752108**	**777112**

Source : Directorate of Planning, Statistics & Evaluation, Government of Goa

ANNEXURE 19

Gross State Domestic Product at Factor Cost by Industry of origin from 1993-94 to 2001-02 at Constant (1993-94) Prices

(Rs. in lakh)

Sl. No.						2001-02 (P)
1	2	3	4	5	6	7
1	**Agriculture, Forestry & Fishing**	**35439**	**35333**	**38027**	**35225**	**38024**
1.1	Agriculture	25085	27837	31227	27382	29986
1.2	Forestry & Logging	1123	835	970	1071	1226
1.3	Fishing	9231	6662	5830	6772	6812
2	**Mining & Quarrying**	**15165**	**20426**	**15133**	**14691**	**13791**
	SubTotal - Primary	**50604**	**55759**	**53160**	**49916**	**51815**
3	**Manufacturing**	**57364**	**104297**	**118882**	**124813**	**134766**
3.1	Registered	45453	86427	99895	104567	104769
3.2	Unregistered	11911	17870	18987	20246	29997
4	**Electricity, Gas and Water Supply**	**4368**	**6674**	**6924**	**7208**	**8821**
5	**Construction**	**10587**	**20033**	**20280**	**26813**	**27959**
	SubTotal - Secondary	**72319**	**131004**	**146086**	**158834**	**171546**
6	**Trade, Hotels and Restaurants**	**22339**	**64610**	**52620**	**63106**	**63349**
7	**Transport, Storage & Communication**	**36353**	**42494**	**41478**	**43445**	**47140**
7.1	Railways	394	1000	1090	1137	1206
7.2	Transport by other means	34582	38401	36960	37974	40879
7.3	Storage	75	82	85	95	95
7.4	Communication	1302	3012	3343	4239	4960
8	**Financing, Insurance, Real Estate & Business Services**	**30368**	**62526**	**69308**	**72412**	**77010**
8.1	Banking & Insurance	14009	41467	46610	47169	48773
8.2	Real Estate, Ownership of dwellings and Business Services	16359	21059	22698	25243	28237
9	**Community, Social & Personal Services**	**27685**	**36794**	**38810**	**42028**	**45841**
9.1	Public Administration	10550	17119	17289	17682	18838
9.2	Other services	17135	19675	21521	24346	27003
	SubTotal - Tertiary	**116745**	**206424**	**202216**	**220991**	**233340**
10	**Total Gross State Domestic Product (GSDP)**	**239668**	**393188**	**401462**	**429741**	**456701**

Source : Directorate of Planning, Statistics & Evaluation, Government of Goa

ANNEXURE 20

Net State Domestic Product at Factor Cost by Industry of Origin from 1993-94 to 2001-02 at Current Prices

(Rs. in Lakh)

Sl. No.						2001-02 (P)
1	2	3	4	5	6	7
1	**Agriculture, Forestry & Fishing**	**31792**	**50197**	**57912**	**55802**	**56548**
1.1	Agriculture	22562	36552	45384	38751	41125
1.2	Forestry & Logging	1026	1175	999	774	1179
1.3	Fishing	8204	12470	11529	16277	14244
2	**Mining & Quarrying**	**11174**	**23675**	**20316**	**22564**	**20002**
	SubTotal - Primary	**42966**	**73872**	**78228**	**78366**	**76550**
3	**Manufacturing**	**51614**	**127465**	**163009**	**172615**	**172775**
3.1	Registered	45453	100888	129197	135551	135679
3.2	Unregistered	6160	26577	33812	37064	37096
4	**Electricity, Gas and Water Supply**	**2073**	**7186**	**4020**	**-2384**	**3502**
5	**Construction**	**10083**	**30293**	**25811**	**42329**	**47504**
	SubTotal - Secondary	**63770**	**164944**	**192840**	**212560**	**223781**
6	**Trade, Hotels and Restaurants**	**21496**	**84287**	**89998**	**114091**	**114225**
7	**Transport, Storage & Communication**	**20786**	**63195**	**69119**	**80894**	**83007**
7.1	Railways	231	1033	1473	1980	2375
7.2	Transport by other means	19477	59190	64357	75456	76561
7.3	Storage	69	122	131	149	149
7.4	Communication	1009	2850	3158	3309	3922
8	**Financing, Insurance, Real Estate & Business Services**	**26770**	**79587**	**94329**	**95300**	**101268**
8.1	Banking & Insurance	13120	47625	53327	41179	41406
8.2	Real Estate, Ownership of dwellings and Business Services	13650	31962	41002	54121	59862
9	**Community, Social & Personal Services**	**24399**	**55330**	**58224**	**67354**	**74738**
9.1	Public Administration	9119	24627	27581	29527	31578
9.2	Other services	15280	30703	30643	37827	43160
	SubTotal - Tertiary	**93451**	**282398**	**311670**	**357639**	**373238**
10	**Total Net State Domestic Product (NSDP)**	**200186**	**521215**	**582738**	**648565**	**673569**
11	**Per Capita NSDP(In Rs.)**	**16558**	**40248**	**44348**	**48582**	**49673**

Source : Directorate of Planning, Statistics & Evaluation, Government of Goa

ANNEXURE 21

Net State Domestic Product at Factor Cost by Industry of Origin from 1993-94 to 2001-02 at Constant (1993-94) Prices

Sl. No.						2001-02 (P)
1	2	3	4	5	6	7
1	**Agriculture, Forestry & Fishing**	**31792**	**32699**	**35553**	**31477**	**34276**
1.1	Agriculture	22562	26527	29899	26090	28694
1.2	Forestry & Logging	1026	736	872	968	1123
1.3	Fishing	8204	5437	4782	4419	4459
2	**Mining & Quarrying**	**11174**	**14864**	**11109**	**10751**	**9851**
	SubTotal - Primary	**42966**	**47563**	**46662**	**42228**	**44127**
3	**Manufacturing**	**51614**	**84643**	**95942**	**100189**	**110142**
3.1	Registered	45453	69405	79617	82781	82983
3.2	Unregistered	6160	15238	16325	17408	27159
4	**Electricity, Gas and Water Supply**	**2073**	**3733**	**3815**	**3963**	**5575**
5	**Construction**	**10083**	**19012**	**19204**	**25559**	**26700**
	SubTotal - Secondary	**63770**	**107388**	**118961**	**129711**	**142417**
6	**Trade, Hotels and Restaurants**	**21496**	**63356**	**51284**	**61698**	**61941**
7	**Transport, Storage & Communication**	**20786**	**21785**	**19859**	**20612**	**24307**
7.1	Railways	231	899	841	873	942
7.2	Transport by other means	19477	18233	16066	15938	18843
7.3	Storage	69	75	77	87	87
7.4	Communication	1009	2579	2875	3714	4435
8	**Financing, Insurance, Real Estate & Business Services**	**26770**	**56207**	**62677**	**65405**	**70003**
8.1	Banking & Insurance	13120	38508	43581	44052	45656
8.2	Real Estate, Ownership of dwellings and Business Services	13650	17699	19096	21353	24347
9	**Community, Social & Personal Services**	**24399**	**32158**	**33930**	**37190**	**41003**
9.1	Public Administration	9119	14872	14975	15453	16609
9.2	Other services	15280	17286	18955	21737	24394
	SubTotal - Tertiary	**93450**	**173506**	**167750**	**184905**	**197254**
10	**Total Net State Domestic Product (NSDP)**	**200186**	**328458**	**333373**	**356844**	**383798**
11	**Per Capita NSDP(In Rs.)**	**16558**	**25364**	**25371**	**26730**	**28304**

ANNEXURE 22

Gross State Domestic Product by broad sectors at current prices

(Rs. in lakh)

Year	Primary	Secondary	Tertiary	Total
1980-81	9647	11872	18232	39751
1981-82	10450	11704	21445	43599
1982-83	11013	15780	25089	51882
1983-84	12328	14767	27755	54850
1984-85	15267	19559	31781	66607
1985-86	13900	15686	35907	65493
1986-87	15866	21132	40141	77139
1987-88	16995	21878	46742	85615
1988-89	18841	26947	52002	97790
1989-90	20743	31410	60626	112779
1990-91	24985	36021	64730	125736
1991-92	30828	46806	77494	155128
1992-93	37635	55744	93867	187246

ANNEXURE 23

Gross State Domestic Product by broad sectors at constant (1980-81) prices

(Rs. in lakh)

Year	Primary	Secondary	Tertiary	Total
1980-81	9647	11872	18232	39751
1981-82	8998	10354	18052	37404
1982-83	8842	13694	19440	41976
1983-84	9156	13123	20196	42475
1984-85	9651	13119	21841	44611
1985-86	8517	11827	22951	43295
1986-87	8411	13644	23358	45413
1987-88	9254	13077	27516	49847
1988-89	9691	17501	30774	57966
1989-90	10544	18078	30961	59583
1990-91	10295	22705	35493	68493
1991-92	10389	23571	33854	67814
1992-93	11824	28920	35836	76580

ANNEXURE 24

Net State Domestic Product by broad sectors at current prices

(Rs. in lakh)

Year	Primary	Secondary	Tertiary	Total
1970-71	2371	2033	3267	7671
1971-72	2631	2236	3745	8612
1972-73	2477	2415	4384	9276
1973-74	3489	3099	5025	11613
1974-75	3646	4346	5910	13902
1975-76	3801	4452	6913	15166
1976-77	4286	5703	7155	17144
1977-78	4838	6422	8224	19484
1978-79	4807	7872	8320	20999
1979-80	5752	9957	9742	25451
1980-81	9004	9359	13183	31546
1981-82	9617	9100	15619	34336
1982-83	10032	13165	18610	41807
1983-84	11351	12098	20743	44192
1984-85	14107	16477	23581	54165
1985-86	12374	12382	26140	50896
1986-87	14171	17413	29203	60787
1987-88	14543	17458	35049	67050
1988-89	16744	22549	38847	78140
1989-90	18242	27515	46238	91995
1990-91	22229	31167	48988	102384
1991-92	27521	39788	59206	126515
1992-93	33880	47571	72994	154445

ANNEXURE 25

Net State Domestic Product by broad sectors at constant (1970-71) prices

(Rs. in lakh)

Year	Primary	Secondary	Tertiary	Total
1970-71	2371	2033	3267	7671
1971-72	2963	2116	3644	8723
1972-73	2585	2262	3944	8791
1973-74	2592	2423	4172	9187
1974-75	2724	3275	4599	10598
1975-76	3031	3469	4994	11494
1976-77	2922	3822	5106	11850
1977-78	3031	3887	5673	12591
1978-79	3075	3709	5509	12293
1979-80	3151	4524	5371	13046
1980-81	9004	9359	13183	31546
1981-82	8254	8031	12927	29212
1982-83	8046	11542	14050	33638
1983-84	8420	11068	14489	33977
1984-85	8829	10912	15553	35294
1985-86	7542	9767	16454	33763
1986-87	7450	11474	16506	35430
1987-88	7884	10737	20692	39313
1988-89	8609	15183	23957	47749
1989-90	9313	16176	24348	49837
1990-91	9005	20116	27704	56825
1991-92	9155	20791	26677	56623
1992-93	10541	25976	28410	64927

ANNEXURE 26

Index of Industrial Production
(Base 1993-94 =100)

Years	Mining Index	Manufacturing index	Combined Index
1994-95	105	133	131
1995-96	107	161	156
1996-97	90	195	185
1997-98	115	232	221
1998-99	108	238	226
1999-2000	109	285	268
2000-01	98	305	287
2001-02(P)	98	347	325

Source : Directorate of Planning, Statistics & Evaluation, Government of Goa.

ANNEXURE 27

Net Value Added by Major Industry Division

Sl. No.	NIC Code	Industry Division	Net value added (Rs. in lakh)	
			———	———
1	15	Manufacture of food products and beverages.	22419	22855
2	17	Manufacture of Textiles	6159	4876
3	24	Manufacture of chemicals and chemical products	60690	86046
4	25	Manufacture of rubber and plastic products	20286	8201
5	27	Manufacture of basic metals.	4512	6535
6	29	Manufacture of machinery & equipment n.e.c.	3299	1152
7	30	Manufacture of office accounting and computing machinery.	4787	5810
8	31	Manufacture of electrical machinery and apparatus n.e.c	6163	9623
9	-	Other industries	18733	14467
10	-	All Industries	147048	159565

Source : Directorate of Planning, Statistics & Evaluation, Government of Goa.
(P) : Provisional Figures

ANNEXURE 28

No. of Applicants on Live Register of Employment Exchange

Sl. No		Number of applicants					
		———	———	———	———	———	———
1	Below Matriculate	28692	26998	25707	24629	19766	20987
2	Matriculates	33622	31873	30892	31491	33664	34131
3	HSSC	34314	32513	31149	29237	29199	28832
4	Graduates	12452	12148	12246	12896	16346	15843
5	Post-Graduates	1070	1075	1109	1371	1759	1847
6	Diploma holders	1638	1844	1915	2346	2870	3019
7	**Total**	**111788**	**106451**	**103018**	**101970**	**103604**	**104659**

Source : Regional Employment Exchange, Government of Goa.

* Figures are provisional

ANNEXURE 29

Taluka-Wise Deposits Mobilised 2002-2003

	Deposits (Rs. in crore)	% age Distribution	Per Capita Deposits (Rs.)*
Salcete	3127.02	29.33	120369
Tiswadi	3577.11	33.55	224398
Bardez	1955.17	18.34	86127
Mormugao	837.82	7.86	57846
Ponda	400.10	3.75	26739
Quepem	252.35	2.37	34135
Bicholim	193.09	1.81	21273
Canacona	107.46	1.00	24472
Sanguem	80.02	0.75	12478
Pernem	76.76	0.72	10663
Sattari	55.50	0.52	9470
Total	10662.40	100.00	79333

Source: Lead Bank, Goa.

***** Population figures used are estimated

ANNEXURE 30

Taluka wise Credits Advanced 2002-2003

	Deposits (Rs. in crore)	% age Distribution	Per Capita Deposits (Rs.)*
Tiswadi	1405.10	43.05	88144
Salcete	855.13	26.20	32917
Mormugao	286.06	8.76	19751
Bardez	352.00	10.79	15506
Ponda	158.37	4.85	10584
Bicholim	59.36	1.82	6540
Sanguem	53.48	1.64	8339
Quepem	43.64	1.34	5903
Pernem	19.33	0.59	2685
Canacona	13.91	0.43	3168
Satari	17.32	0.53	2955
Total	**3263.70**	**100.00**	**24284**

Source: Lead Bank, Goa.

***** Population figures used are estimated

ANNEXURE 31

Distribution of Vehicles by Type as on 31.3.2003

Sl. No.		No. of vehicles	Percentage to the total
1.	Motor cycles on hire	5682	1.45
2.	Motor cycles and scooters	280252	71.60
3.	Private cars and jeeps	63301	16.17
4.	Motor cabs (taxis)	7537	1.93
5.	Goods vehicles	23770	6.07
6.	Buses / mini buses (including KTC)	3571	0.91
7.	Tractors	447	0.11
8.	Auto rickshaws	3200	0.82
9.	Government vehicles	3670	0.94
	Total	**391430**	**100.00**

Source: Directorate of Transport, Government of Goa.

ANNEXURE 32

No. of New Vehicles Registered During 1992-93 to 2002-03

Sl. No.		No. of Vehicles
1	1992-93	15180
2	1993-94	17644
3	1994-95	20888
4	1995-96	25756
5	1996-97	26683
6	1997-98	26294
7	1998-99	25390
8	1999-2000	29869
9	2000-01	25779
10	2001-02	27189
11	2002-03	33106

Source: Directorate of Transport, Govt. of Goa.

ANNEXURE 33

Tourist Arrivals in Goa 1985-2003

Year	Number of Tourists			Annual Growth (%)
	———	———	———	
1985	682545	92667	775212	-
1986	736548	97533	834081	7.6
1987	766846	94602	861448	3.3
1988	761859	93076	854935	-0.7
1989	771013	91430	862443	0.9
1990	776913	104330	881323	2.2
1991	756786	78281	835067	-5.6
1992	774568	121442	896010	7.3
1993	798576	170658	969234	8.2
1994	849404	210191	1059595	9.3
1995	878487	229218	1107705	4.5
1996	888914	237216	1126130	1.7
1997	928925	261673	1190598	5.7
1998	953212	275047	1228259	3.2
1999	960114	284298	1244412	1.3
2000	976804	291709	1268513	1.9
2001	1120242	260071	1380313	8.8
2002	1325296	271645	1596941	15.7
2003 (P)	1738330	291408	2029738	27.1

Source : Department of Tourism, Government of Goa

ANNEXURE 34

Arrivals by Charters
(October to May)

	No. of Flights	
1985-86	24	3568
1986-87	26	4401
1987-88	25	5419
1988-89	83	9705
1989-90	107	9266
1990-91	41	5815
1991-92	121	17102
1992-93	259	39871
1993-94	299	58369
1994-95	313	59881
1995-96	337	75694
1996-97	282	73172
1997-98	350	88817
1998-99	356	90635
1999-2000	405	94289
2000-2001	419	116992
2001-2002	279	76410
2002-2003	384	94350
2003-2004(Oct to Jan)	269	69815

Source: Department of Tourism, Government of Goa.

ANNEXURE 35

Foreign Tourist Arrivals 2003(P)

Country	Actuals	Percenrge
U.K	125623	43.11
Germany	15797	5.42
Finland	30226	10.37
Switzerland	6415	2.20
Sweden	3826	1.31
Italy	3236	1.11
Austria	1120	0.38
France	6423	2.20
U.S.A.	3628	1.25
Canada	3246	1.11
Australia	2387	0.82
Japan	896	0.31
Holland	1300	0.45
Denmark	1248	0.43
New – Zealand	1172	0.40
Ireland	1101	0.38
South Africa	803	0.28
Russia	8587	2.95
Portugal	5296	1.82
Others	69078	23.70
Total	**291408**	**100.00**

Source: Department of Tourism, Government of Goa.
(P) : Provisional Figures.

ANNEXURE 36

Number of Educational Institutions and Enrolment

Sl. No.		2000-01		2001-02		2002-03	
		No. of Institutions	Enrol ment	No. of Institutions	Enrol ment	No. of Institutions	Enrol ment
1	Primary	1268	97457	1267	94028	1268	85960
2	Middle	440	72726	443	73594	436	74687
3	Secondary	365	64178	367	63821	367	64365
4	Higher Secondary	80	21039	81	22868	81	18545
5	University (General education)	21	14680	23	14365	25	14894
6	University (Professional education)	20	4627	22	4897	23	5758
7	Technical & Professional	29	7305	29	6974	28	7475
	Total	**2223**	**282012**	**2232**	**280547**	**2228**	**271684**

Source: Directorate of Education, Government of Goa.

ANNEXURE 37

Drop Out Rates 2001-02

Sr. No.					
A	I – V				
1		1997-98	5.41	12.75	8.95
2		1998-99	0.15	8.72	4.20
3		1999-2000	5.83	11.50	8.58
4		2000-2001	3.00	9.94	6.37
5		2001-2002	3.48	8.12	5.73
B	I - VIII				
1		1997-98	6.20	12.23	9.12
2		1998-99	4.70	11.26	7.88
3		1999-2000	7.14	13.26	10.11
4		2000-2001	5.73	13.25	9.36
		2001-2002	(-) 1.30	9.89	4.19
C	I - X				
1		1997-98	46.49	46.71	46.60
2		1998-99	41.69	42.45	42.06
3		1999-2000	43.22	42.41	42.83
4		2000-2001	44.23	43.46	43.85
5		2001-2002	42.92	41.47	42.21

Source : Directorate of Education, Government of Goa.

ANNEXURE 38

Distribution of Patients of Communicable Diseases

Sl. No.	———	No. of patients 2003	
		Examined	Found positive & Treated
1.	Tuberculosis *	10713	2241
2.	Leprosy (surveyed)	-	313
3.	Malaria	278647	11370
4.	Filaria	18293	10
5.	Sexually Transmitted Disease#	5177	147
6.	AIDS / HIV. $	9705	877
7.	Japanese Encephalitis	NIL	Nil

Source : Directorate of Health Services, Government of Goa.

* Relates to year 2002-03

Relates to 2002

$ Jan to October 2003

ANNEXURE 39

Rain Fall in Goa

Year	in mm.
1989	2,625.0
1990	2,623.0
1991	2,152.3
1992	2,778.2
1993	2,558.3
1994	2,894.4
1995	3,555.6
1996	2,880.9
1997	3,366.9
1998	3,078.9
1999	3,680.4
2000	3,511.6
2001	2,128.1
2002	2,270.4
2003	2,682.9

Source: Goa Observatory, Government of Goa

ANNEXURE 40
Agricultural Production

Sl. No.			Estimated Production		
			———	———	———
1	2	3	4	5	6
1	Rice	Tonne	142123	126523	134767
2	Ragi	,,	499	315	301
3	Maize	,,	800	800	800
4	Pulses	,,	9652	8368	7453
5	Groundnut	,,	3151	2527	4263
6	Sugarcane	,,	72750	70565	65470
7	Cashewnut	,,	21335	22976	23109
8	Coconut	Million nuts	125	125	122
9	Arecanut	Tonne	2500	2500	2500
10	Mango	,,	18225	17228	17532
11	Banana	,,	14745	15482	15743
12	Pineapple	,,	4500	4500	4500
13	Vegetable	,,	70467	70467	70467
14	Other fruits	,,	35000	35000	35000

Source: Directorate of Agriculture, Government of Goa.

ANNEXURE 41

Land Use Pattern 2002-2003

Sl. No.		Area in hectares	Percentage to total
1.	Total reporting area	3,52,943	100.00
2.	Area under forest	1,25,473	35.55
3.	Land not available for cultivation	37,137	10.52
4.	Other uncultivated land	57,043	16.16
	a) Permanent pastures and other grazing land	1,305	0.37
	b) Land under miscellaneous tree crops and groves not included in net area sown	580	0.16
	c) Cultivable waste including fallow land	55,158	15.63
5.	Net area sown	1,33,290	37.77
6.	Area sown more than once	30,556	8.65
7.	Gross cropped area	1,63,846	46.42

Source: Directorate of Agriculture, Government of Goa.

ANNEXURE 42

Livestock and Poultry

Sl. No.		Number		Percentage variation
		———	———	
1	Cattle	99598	87978	-11.67
2	Buffaloes	44674	40222	-9.97
3	Sheep	212	288	35.85
4	Goats	15510	12799	-17.48
5	Horses & ponies	15	5	-66.67
6	Pigs	89852	105402	17.31
7	Dogs	60479	78838	30.35
8	Rabbit	-	183	-
9	Total Livestock	310340	325715	4.95
10	Poultry	928985	789504	-15.01

Source: Livestock Census, 1992 & 1997.

ANNEXURE 43

Area under Forests

Sl. No.		Area in sq. kms.
1	Tiswadi	1.78
2	Bardez	Nil
3	Pernem	13.43
4	Bicholim	8.08
5	Satari	280.99
6	Ponda	50.12
7	Sanguem	569.25
8	Quepem	114.91
9	Canacona	185.82
10	Mormugao	Nil
11	Salcete	Nil

Source: Forest Department, Govt. of Goa.

ANNEXURE 44

Forest Area by Species

Sl. No.		
1	Teak	9507
2	Eucalyptus	5204
3	Bamboo	223
4	Cashew	11196
5	Rubber	851
6	Acrocarpus	226
7	Casuarina	768
8	Others	11446
	Roadside plantation	1041
	Canal Bank plantation	1256

Source: Forest Department, Govt. of Goa.

ANNEXURE 45

Forest Produce 2002-03

Sl. No.		
1	Timber M3 (Government)	252.477
2	Firewood M3 (Government)	4023.123
3	Bamboo Nos. (Government)	42650
4	Poles (Nos.)	620
5	Canes (Nos.)	140

Source: Forest Department, Govt. of Goa.

ANNEXURE 46

Trend in Fish Catch 1997-2003

Sl. No.		Annual Fish catch in tonnes		
		———	———	———
1	1997	91277	3270	94547
2	1998	67236	3474	70710
3	1999	60075	3365	63440
4	2000	64563	3570	68133
5	2001	69386	3749	73135
6	2002	67563	3684	71247
7	2003	90990	4321	95311

Source: Directorate of Fisheries, Govt. of Goa.

ANNEXURE 47

Export of Marine Product 1997-2003

Sr. No.		Export of marine Products	
		(in tonnes)	(Rs. in lakh)
1	1997	14248	6608
2	1998	6175	2910
3	1999	9054	3491
4	2000	10732	3357
5	2001	7714	3007
6	2002	15594	5620
7	2003	10288	3273

Source: Directorate of Fisheries, Govt. of Goa.

ANNEXURE 48

Distribution of the Co-operative Societies

Sl. No.		As on 31st March	
		———	———
l.	Goa State Co-operative Bank	1	1
2.	Urban Co-operative Banks	7	7
3.	Urban Credit Co-operative Societies	364	120
4.	Consumer Co-operative Societies	100	96
5.	Dairy Co-operative Societies	159	158
6.	Farming Co-operative Societies	4	6
7.	Fisheries Co-operative Societies	8	7
8.	Housing Co-operative Societies	1114	1166
9.	Industrial Co-operative Societies	55	39
10.	Marketing Co-operative Societies	9	9
11.	Poultry Co-operative Societies	2	2
12.	Processing Co-operative Societies	4	5
13.	Service Co-operative Societies	88	87
14.	Transport Co-operative Societies	3	3
15.	Pani Vatap & Agricultural Requisites societies	47	47
16.	General Co-operative Societies	9	11
17.	Salary Earners Co-operative Credit Societies		254
18.	Labour Co-operative Societies		14
19.	Sangh & Union Federation		3
	Total	**1974**	**2035**

Source: Registrar of Co-operative Societies, Government of Goa.

ANNEXURE 49

Important Characteristics of Co-operative Societies

Sr. No		As on 31st March			
		———	———	———	———
1	No. of societies	1848	1897	1974	2035
2	No. of members	551707	556494	649803	755379
3	Share Capital (in million)	629.07	662.2	916.96	1116.65
4	Deposits (in million)	11287.25	12737.3	14110.25	15389.96
5	Working Capital (in million)	16729.87	19067.2	21425.40	22595.72
6	Advances (in million)	4989.87	5294.2	6986.15	5578.08
7	No. of Societies in profit	1442	1508	1502	1490
8	Amount of profit (in million)	121.32	155.2	253.34	264.45
9	No. of Societies in loss	357	325	170	292
10	Amount of loss (in million)	13.94	338.6	246.08	267.20
11	No. of Societies on no profit no loss basis	49	64	302	253
12	Loans outstanding				10245.81
13	Sales				1753.03

Source: Registrar of Co-operative Societies, Government of Goa.

ANNEXURE 50

Change in Occupational Structure

Sl. No.		Number		Decadal growth (%)
		———	———	
I	Cultivators	68636	50663	-26.2
2	Agricultural labourers	44775	36150	-19.3
3	Workers in household industry	9835	14094	+43.3
4	Other Workers	289490	421658	+45.6
5	**Total workers**	**412736**	**522565**	+26.6
6	Non-workers	757057	821433	+8.5

Source: Population Census, Registrar General, India.

ANNEXURE 51

Approved Outlays and Actual Expenditure from Third Five Year Plan (1962-66) to Ninth Five Year Plan (1997-02)

(Rs. in crore)

Sl. No.		Approved Outlay	Actual Expend.	Percentage utilisation
1	Third Five Year Plan (1962-66)	22.23	18.23	81.99
2	Plan holidays (1966-69)	24.02	19.83	82.56
3	Fourth Five Year Plan (1969-74)	39.51	41.94	106.15
4	Fifth Five Year Plan (1974-78)	61.67	61.65	99.97
5	Plan holidays (1978-80)	57.50	55.63	96.74
6	Sixth Five Year Plan (1980-85)	225.89	224.42	99.35
7	Seventh Five Year Plan (1985-90)	418.75	438.61	104.74
8	Plan holidays (1990-92)	300.00	294.26	98.09
9	Eighth Five Year Plan (1992-97)	969.05	843.43	87.04
10.	Ninth Five Year Plan (1997-2002)	1595.09	1389.46	87.10

Note: Figures for the period 1962 to 1987 pertains to UT Goa, Daman & Diu.
Source : Directorate of Planning, Statistics & Evaluation, Government of Goa.

ANNEXURE 52

Tenth Five Year Plan 2002-07
Sector/sub-sector-wise projected outlays

(Rs. in lakh)

Sr. No.	Head/sub-Head of Development	Tenth Plan (2002-07) Outlays
1	2	3
I.	**AGRICULTURE & ALLIED SERVICES**	**15834.00**
1	Crop Husbandry.....	2405.00
2	Soil & Water Conservation.....	6450.00
3	Animal Husbandry.....	1300.00
4	Dairy Development.....	300.00
5	Fisheries.....	780.00
6	Forestry and Wild Life.....	2500.00
7	Food, Storage & Warehousing.....	15.00
8	Agriculture Research & Education	74.00
9	Marketing & Quality Control.....	10.00
10	Cooperation.....	2000.00
II.	**RURAL DEVELOPMENT**	**8365.00**
1	Swarnjayanti gram Swarojgar Yojana	75.00
2	DRDA Administration	150.00
3	Sampoorna Grammeen Rojgar Yojana	400.00
4	Other Rural Dev. Programme (Incl. NSAP)	140.00
6	Land Reforms.....	600.00
7	Community Dev. & Panchayats.....	7000.00
III.	**SPECIAL AREAS PROGRAMME (WGDP)**	**1800.00**
IV.	**IRRIGATION & FLOOD CONTROL**	**22290.00**
1	Major & Medium Irrigation	17540.00
2	Minor Irrigation.....	2700.00
3	Command Area development.....	1250.00
4	Flood Control including anti sea Erosion.....	800.00
V.	**ENERGY**	**40585.00**
1	Power.....	40000.00
2	Non - Conventional sources of energy	500.00
3	Integrated Rural Energy Programme	85.00
VI.	**INDUSTRY & MINERALS.....**	**11640.00**
1	Village & Small Industries.....	5000.00
2	Industries other than V.S.I.....	1500.00
3	Information Technology	5000.00
4	Mining.....	140.00

Sr. No.	Head/sub-Head of Development	Tenth Plan (2002-07) Outlays
1	2	3
VII.	**TRANSPORT.....**	**39283.78**
1	Ports & Light Houses.....	25.00
2	Roads & Bridges.....	35240.00
3	Road Transport.....	1095.00
4	Inland Water Transport.....	800.00
5	Railways.....	523.78
6	Sky Bus Metro	500.00
7	Civl Aviation	1100.00
VIII.	**SCIENCE, TECHNOLOGY & ENVIRONMENT**	**475.00**
1	Scientific Research incld. S&T..	175.00
2	Ecology & Environment.....	300.00
IX.	**GENERAL ECONOMIC SERVICES.....**	**15975.00**
1	Secretariat Economic Services	450.00
2	Tourism.....	15000.00
3	Survey & Statistics including Computer Centre and Gazetteers...	405.00
4	Weights & Measures.....	120.00
X.	**SOCIAL SERVICES.....**	**152652.22**
1	Education, Sports & Culture.....	39025.00
	a) General Education.....	20380.00
	b) Technical Education.....	6350.00
	c) Art & Culture.....	6280.00
	d) Sports & Youth Services.....	6015.00
2	Medical & Public Health.....	13135.00
3	Sewerage & Water Supply.....	59496.22
4	Housing (including IAY).....	2960.00
5	Urban Development including Fire Services.....	19650.00
6	Information & Publicity.....	300.00
7	Welfare of SC & OBC's.....	400.00
8	Labour & Labour Welfare.....	3186.00
9	Social Security & Welfare.....	14050.00
10	Nutrition.....	450.00
XI.	**GENERAL SERVICES.....**	**11100.00**
1	Stationery & Printing.....	150.00
2	Public Works.....	5000.00
3	Judicial Administration.....	350.00
4	Accounts.....	100.00
5	Jails	950.00
6	Finance	4500.00
7	Notary Services	50.00
	GRAND TOTAL.....	**320000.00**

ANNEXURE 53

Declining Central Assistance to State Plan

(Rs. in crore)

Sl. No.		Approved Outlay	Central Assistance	
				plan size
1	1987-88	79.75	68.18	85.49
2	1988-89	92.00	86.00	93.48
3	1989-90	110.00	100.00	90.91
4	1990-91	130.00	110.00	84.62
5	1991-92	170.00	42.00	24.71
6	1992-93	152.50	52.00	34.10
7	1993-94	170.00	43.00	25.29
8	1994-95	185.00	50.34	27.21
9	1995-96	211.00	55.70	26.40
10	1996-97	250.50	61.83	24.68
11	1997-98	230.56	60.15	26.09
12	1998-99	291.34	59.27	20.34
13	1999-2000	281.19	57.60	20.48
14	2000-01	332.00	64.38	19.39
15	2001-02	460.00	71.45 (P)	15.53
16	2002-03	586.00	71.45 (P)	12.19
17	2003-04	650.00	71.45 (P)	10.99

Source : Resource estimate statements of Finance Department, Government of Goa,

ANNEXURE 54

Consumer Price Index for Industrial Workers
(Base year 1982 = 100)

Year	Goa	India
1	2	3
1990	195	186
1991	223	212
1992	257	237
1993	285	252
1994	311	278
1995	339	306
1996	373	334
1997	416	358
1998	451	405
1999	482	424
2000	520	441
2001	555	458

Source: Labour Bureau, Simla.

ANNEXURE 55

Consumer Price index for Urban Non-Manual Employees (Base 1991= 100)

Year	Goa	India
1	2	3
1991	100	100
1992	114	112
1993	123	119
1994	132	131
1995	142	144
1996	153	156
1997	164	168
1998	174	186
1999	182	197
2000	192	207
2001	202(P)	218

Source : Labour Bureau, Simla.

Note: Base arithmetically shifted to 1991 for Goa to enable comparison.

ANNEXURE 56

Human Development Index for Selected Developed and SAARC Countries

HDI Rank	Name of the country	Life expec-tancy at birth 2001	Adult literacy rate (% age 15 and above) 2001	Com-bined primary, secon-dary & tertiary gross enrol-ment ratio(%) 2000-01 b	GDP per capita (PPP US $) 2001	Life expect-ancy index	Edu-cation index	GDP index	Human Develo-pment index (HDI) value 2001
1	2	3	4	5	6	7	8	9	10
	Developed Countries								
8	Canada	79.2	99.0d	94eg	27130	0.90	0.97	0.94	0.937
7	USA	76.9	99.0d	94e	34320	0.86	0.97	0.97	0.937
9	Japan	81.3	99.0d	83e	25130	0.94	0.94	0.92	0.932
13	U.K.	77.9	99.0d	112ef	24160	0.88	0.99	0.92	0.930
4	Australia	79.0	99.0d	114ef	25370	0.90	0.99	0.92	0.939
18	Germany	78.0	99.0d	89eg	25350	0.88	0.96	0.92	0.921
63	Russian Federation	66.6	99.6d	82e	7100	0.69	0.93	0.71	0.779
1	Norway	78.7	99.0d	98e	29620	0.90	0.99	0.95	0.944
	SAARC Countries								
127	India	63.3	58.0	56eg	2840	0.64	0.57	0.56	0.590
139	Bangladesh	60.5	40.6	54	1610	0.59	0.45	0.46	0.502
136	Bhutan	62.5	47.0pq	33h	1833o	0.62	0.42	0.49	0.511
86	Maldives	66.8	97.0	79	4798lo	0.70	0.91	0.65	0.751
143	Nepal	59.1	42.9	64	1310	0.57	0.50	0.43	0.499
144	Pakistan	60.4	44.0	36	1890	0.59	0.41	0.49	0.499
99	Sri Lanka	72.3	91.9	63eg	3180	0.79	0.82	0.58	0.730
	World	66.7	. .	64	7376	0.70	0.75	0.72	0.722

Source: Human Development Report, 2003 (UNDP)

b Data refers to the 2000/01 school year.
d For purpose of calculating the HDI, a value of 99.0% was applied.
e. Preliminary UNESCO estimates subject to further revision.
f. For purpose of calculating the HDI, a value of 100.0% was applied.
g Data refer to a year other than that specified.
h Data refers to the 1999/2000 school year.
l. Estimate based on regression.
o Preliminary world bank estimate subject to further revision.